Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Abstract

This paper uses graduate admissions to study dynamic selection under costly evaluation. With a privacy-protecting synthetic data protocol, I relate acceptance to success measures based on job placements. I estimate structural models of an admissions committee's decision problem capturing first-round filtering and matriculation, test whether observed decision rules are optimal given the number of application files read, and quantify gains from deviating to the models' decision rules. The committee's decisions align with many applicants' estimated success probabilities, but some applicant characteristics are misweighted. Correctly weighting these characteristics significantly increases the admitted cohort's average success probability without reading more files.

Keywords: Costly Evaluation, Dynamic Optimization, Education Economics, Graduate Admissions, Optimal Selection, Structural Estimation, Synthetic Data

JEL Codes: C61, C81, D81, I23

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1 Introduction

How do decision-makers select among a set of options in which each option's true quality is unobserved and costly to evaluate in terms of time and effort? In many such problems, decision-makers rely on a coarse, low-cost screening of all the options, followed by a precise, high-cost evaluation of the subset of options that survive the screening. Moreover, if the selected options have agency over whether to match with the decision-maker, final selections are often staggered over time through a deferred process to ensure the number of matches meets a target number.

Admissions and hiring are two prominent examples of such selection problems. There, committees first screen all the applications to filter out unqualified applicants, then read a subset of promising applications in full detail, then potentially interview the most promising applicants, and finally make early and late offers, depending on how many early offers are accepted. Other examples of selection with costly evaluation, and in some cases deferred offers, include conducting multi-stage clinical trials for medicines, messaging virtually on dating apps before meeting in real life, and simulating product prototypes on a computer before physically building them.

A fundamental question in such settings is how decision-makers should interpret the available information at each stage of evaluation to select either the best subset of options or all options above a quality threshold. A related question is whether observed decision rules reflect optimal dynamic selection given evaluation costs and other constraints, or whether those rules systematically deviate from optimal behavior. One could imagine a hiring committee undervaluing a skill or discriminating against a demographic. Moreover, if there are deviations, which decision rules should decision-makers implement instead? Answering these questions requires moving beyond identification of the determinants of success and explicitly modeling the decision problem.

Graduate admissions is a meaningful setting to study these questions, as evidenced by economics papers from Krueger and Wu (2000) to Bai, Esche, MacLeod, and Shi (2022). Graduate programs, especially PhD programs, invest at least a few years of time, along with tens of thousands of dollars of funding, into students' development. In addition, high-ranking academic placements boost a program's reputation, and so increase its ranking from sources such as U.S. News. Better students also lead to better coauthoring opportunities for faculty, and since people with graduate degrees often find themselves in influential positions, such as professorships, studying how best to admit them is important for programs and society. In particular, if schools are undervaluing applicants with certain characteristics (e.g. from a particular demographic), future applicants with such characteristics may receive opportunities they would not have otherwise. For example, while Chetty, Deming, and Friedman (2026) focus on undergraduate admissions, they find that applicants from high-income families have higher admissions rates at Ivy-Plus colleges even if their standardized test scores are no higher than those of less fortunate applicants.

Moreover, graduate admissions is a setting with costly evaluation. Admissions committees observe for all applicants a limited set of objective characteristics, such as standardized test scores, but typically lack the time to read every file in detail. As such, applicants are initially filtered based on coarse signals, with more intensive evaluation reserved for the subset of applicants with promising objective characteristics. Final decisions may also involve waitlists or other forms of deferred acceptance, as uncertainty about applicant quality and matriculation status resolves over the period between the application and admissions deadlines. These features make graduate admissions well-suited for modeling dynamic selection under costly evaluation. Doing so for one economics PhD program, I find that while the committee's decisions are much better than random chance and align with many applicants' estimated success probabilities, the committee misweights some applicant characteristics, and correctly weighting them would significantly increase the admitted cohort's average success probability without reading any more files.

This paper addresses two specific research questions. Both require first determining which *ex-ante* factors in application files determine admission and success in graduate school. Following the literature, success may be defined in multiple ways, including program completion, and job placement outcomes. I study these factors as comprehensively as possible by extracting all available data from application files, including recommendation letters, application essays, and CVs. My data consist of all applicants to a research university's economics PhD program from 2015 to 2019, for which I observe both the application files and the outcomes. In Appendix B.1, I supplement these data with the objective characteristics of all applicants to that university's economics, government, history, linguistics, and psychology PhD programs from 2020 to 2023 to compare admissions across programs. Because these data are confidential, I implement a privacy-protecting research protocol that uses synthetic data for model development and validation (see Section 2). This protocol also serves as a template for researchers who wish to work with sensitive data in general. It also has a side benefit, namely that I had to commit to my specifications on the synthetic data and could adjust them only sparingly after seeing results from the actual data.

The first research question asks whether admissions committees optimize or make mistakes when admitting applicants. Specifically, if a factor predicts admission but not success, the committee may be overvaluing it, whereas if a factor predicts success only, the committee may be undervaluing it. For instance, the committee may be putting too much or too little weight on elite undergraduate institutions, standardized test scores, or other applicant characteristics. To determine whether the committee is optimizing, it is necessary to model the committee's optimization problem, estimate the parameters, and formally test the hypothesis of optimization, which to date, no other paper has done. To make such models tractable, per Bai et al. (2022), I reduce the committee's portfolio choice problem (i.e. from N applicants, accept the optimal subset of N_A applicants) to a cutoff rule problem (rank the N applicants by their expected success probabilities and select the N_A above the cutoff). I estimate multiple structural models in increasing order of complexity, modeling such features as first-round filtering and matriculation. That said, PhD program quality also affects success (Angrist, Diederichs, & Ellison 2026), so it is important to account for that lest the model encourage the committee to put more weight on a variable that only predicts success by way of predicting admission into a highly-ranked program.

Perhaps as Simon (1956) suggests, if committees are not optimizing, they may be satisficing (i.e. aiming for a satisfactory result), namely because reading every application file would require too much effort.[1] This idea lends itself to the second research question of whether committees can make better decisions without having to expend additional effort, and if so, how much better of a cohort they can admit. To answer this question, I use the structural models to calculate counterfactual deviation gains from adopting the optimal admissions strategy, while respecting the committee's time constraint of not being able to read every application file in detail. For inference, I perturb the success parameter estimates about their asymptotic distribution (Krinsky & Robb 1986) when comparing the expected success probability for the cohort admitted by the model's decision rule vs. the committee's. I then show that the fraction of perturbations in which the model's rule does not improve on the committee's is an asymptotically valid p-value. Perhaps committees have other priorities than maximizing success probabilities, such as admitting a diverse class, but I show how the models can be extended to factor in those priorities.

The structural estimates show that for many applicants, the success probabilities implied by the committee's decisions align with the success probabilities estimated from outcomes. However, the committee misweights some applicant characteristics, and the deviation gains test rejects the null hypothesis of no gains. Holding fixed the number of files the committee read in each round and the number of applicants it admitted in each year, so that the counterfactual committee only re-ranks the applicants, adopting the model's decision rule increases the acceptance set's average success probability by about 9 percentage points. When the models account for matriculation, so that the relevant set is the matriculation set, the average success probability increases by about 10 percentage points. Because the committee's own subjective score for each application enters the models as a proxy for information in the files that my variables miss, these gains are unlikely to reflect private information. Consistent with this hypothesis, the committee's actual matriculants succeeded at a rate no higher than their estimated success probabilities, so if the committee has information that the model lacks, it does not appear to be using it. In Monte Carlo simulations calibrated to the actual data, the same test finds significant gains when the simulated committee misweights applicant characteristics but not when it behaves optimally (see Appendix D). That said, any changes in the actual data-generating process across cohorts, however minor, will reduce the magnitude of the deviation gains when applied in practice. Therefore, 9–10 percentage points for the 2015–2019 application years is not a gain that this committee could have achieved using success probability parameters estimated from before those years.

To determine which variables the committee misweights, I reweight one variable at a time in the committee's estimated decision rule, holding the remaining weights fixed, and calculate the gain from the best reweighting. No individual variable accounts for more than three percentage points of the gain, so the committee did not make any large mistakes. The committee underweighted age, work experience, and English proficiency as measured by TOETL/IELTS scores.

[1] Satisficing is distinct from rational inattention (see Section 1.1), which assumes that decision-makers are optimizing and so embeds the choice of how much effort to allocate into an optimization framework.

In additon, in application essays, positive language was underweighted, whereas economics-specific language was overweighted. The committee also penalized applicants who listed more coding languages on their CVs even though coding did not negatively impact success. Meanwhile, the committee overweighted elite U.S. undergraduate institutions, though only for the acceptance set, as accepted applicants from elite U.S. universities matriculated at lower rates. However, most of these individual misweightings are not statistically significant, and they may not hold going forward. For example, if applicants start using AI to write their essays, then it is unlikely that any essay-specific results from 2015–2019 will hold in the present.

Moreover, the committee's constraint on how many application files it reads does not affect the success of its admitted cohort. I first graph the model's decision rule's average success probability of the acceptance and matriculation sets as a function of the number of files read. Reading every application file rather than the roughly 40 percent the committee actually read raises the model's admitted cohort's average success probability by only about one percentage point. The committee's own decision rule, meanwhile, yields nearly the same average success probability regardless of how many files it reads. Given that the university in this study is not necessarily representative of all graduate programs, some of this paper's results might not generalize to other settings. However, the methods are generalizable to admissions, hiring, and any other setting where decision-makers must decide which options to select based on *ex-ante* information.

1.1 Related Literature

While there have been earlier papers on graduate education, Ehrenberg and Mavros (1995) were the first to include predictors of success found in application files. They analyze Cornell PhD students, including economics, and find that verbal but not quantitative GRE scores are predictive of program completion. Subsequent research implies the reason is likely that their sample is restricted to only the applicants who attended Cornell, nearly all of whose quantitative GRE scores were concentrated in the top percentiles. Two years later, Attiyeh and Attiyeh (1997) use a sample across 48 universities and multiple programs from 1990 to 1991 to study determinants of admission. They find that all else equal, committees prioritized underrepresented minorities.

Krueger and Wu (2000) and Grove and Wu (2007) then study admission and success at a single top economics department in 1989, including all applicants in their success regressions and tracking placement outcomes for non-matriculants. They find that admissions committee scores, GRE scores, and the prominence of recommendation letter writers predict good placements, but there is a lot of uncertainty. Finally, Athey, Katz, Krueger, Levitt, and Poterba (2007) analyze the determinants of success for Chicago, Harvard, MIT, Princeton, and Stanford economics PhD students from 1990 to 1999, adding graduate course grades into their models. While GRE scores and other *ex-ante* factors predict good placements, they do not when controlling for grades.

Stock, Finegan, and Siegfried (2006) begin a series of papers by compiling a 586 student dataset of Fall 2002 economics PhD matriculants. They start by looking at attrition rates in the first two years and find that low GRE scores predict attrition, but RAships and shared department office space help programs retain students. Siegfried and Stock (2007) compile another, larger dataset of everyone who earned a U.S. economics PhD from 1966 to 2003 and observe that top liberal arts colleges feed economics PhD programs at disproportionally high rates.

Stock, Finegan, and Siegfried (2009a) and Stock, Finegan, and Siegfried (2009b) follow up on the progress of the students in Stock et al. (2006) and find similar results but emphasize that there is a lot of uncertainty in predicting completion after five years. Stock, Siegfried, and Finegan (2011) and Stock and Siegfried (2014) track the students after eight years and summarize their multi-paper study, concluding that while some factors like high GRE scores, a liberal arts undergraduate institution, and financial support predict completion, uncertainty remains very high. Finally, Stock and Siegfried (2015) add more recent years to Siegfried and Stock (2007)'s larger dataset, noting that foreign undergraduate institutions have become more common over time.

Meanwhile, Grove, Dutkowsky, and Grodner (2007) study Syracuse economics PhD students, who have different characteristics than students attending top-ranked programs. They also incorporate factor analysis into their regressions, finding that different attributes predict success at different stages. For example, GRE scores predict passing comprehensive exams, whereas research motivation (a factor from their factor analysis) predicts program completion. Schlauch and Startz (2018) study economics PhD candidates from top 50 programs posting their CVs for the 2016–2017 job market and find that while RA experience predicts attending a highly-ranked PhD program, a masters degree does not. This result tracks with the increasing prevalence of predocs in economics. Meanwhile, Jones, Schuhmann, Soques, and Witman (2020) do not run regressions but rather survey individuals involved in admissions at 132 economics PhD programs. They find that higher-ranked programs care about different factors than lower-ranked programs. For example, they care more about undergraduate rankings. Also, while recommendation letters matter for admission, programming skills usually do not, though they may matter for success.

Bai et al. (2022) represent the frontier of economics graduate admissions research. They study economics PhD applicants at a single top program from 2013 to 2019 and like Krueger and Wu (2000) collect data on admissions and placement outcomes for all applicants. They start by proving that a committee's portfolio choice problem simplifies to a cutoff rule problem, which is more tractable. I build on this result by directly modeling the cutoff rule problem and estimating the parameters, which is necessary to make claims as to whether committees are optimizing. In addition, they perform textual analysis of recommendation letters and find that subjective factors, such as letters, matter more for both admission and success more than objective factors, such as GREs. As such, the models in this paper incorporate subjective as well as objective factors.

This paper also relates to a broader methodological literature on optimal dynamic selection with costly evaluation, as well as a literature on synthetic data and privacy protection in empirical economics. Within the former literature, earlier papers develop classic models of sequential search where agents face uncertainty about option quality and must decide when to stop searching given search costs. McCall (1970) and Mortensen (1970) model job search as a dynamic stopping problem in which workers optimally accept offers that exceed a reservation wage. Weitzman (1979) extends this logic to settings in which decision-makers can acquire information about multiple alternatives at a cost, which he denotes as "Pandora's Problem." Together, these papers establish that when information is costly, optimal decision rules are often threshold-based.

A more recent group of papers studies costly evaluation outside of search problems. In models of rational inattention, agents are fully optimizing but face constraints on their ability to process information. Sims (2003) introduces rational inattention as a framework in which information acquisition is costly, leading agents to optimally trade off decision accuracy with information costs. Matějka and McKay (2015) show that rational inattention can generate probabilistic discrete choice rules in line with multinomial logit models, while Caplin and Dean (2015) provide a revealed-preference characterization of optimal information acquisition that separates rational inattention from actual mistaken decisions. These models differ from Simon (1956)'s concept of satisficing, which proposes heuristic, "good enough" decision rules. In contrast, rational inattention models retain the mathematical framework of rationality but endogenize information acquisition. Notably, unlike this paper, none of these papers estimate their models on data.

This paper differs from both rational inattention and satisficing. I take the information environment faced by decision-makers as given, since admissions committees, at least in this framework, do not choose how much information is available at each stage of evaluation. Instead, they interpret the information they observe at each stage and decide how to act on it. The contribution of this paper is to model this dynamic selection problem explicitly, estimate parameters governing the resulting decision rules, and determine whether observed behavior is optimal conditional on respecting the committee's constraints. By doing so, this paper makes optimality testable and allows for the computation of counterfactual decision rules and deviation gains.

The second methodological literature addresses synthetic data and privacy protection. Rubin (1993) introduces synthetic data as a tool for protecting confidentiality by replacing sensitive microdata with simulated observations drawn from an estimated data-generating process. Before synthetic data, privacy protection often involved masking, or adding random noise to the data. Masking, however, induces a trade-off between privacy and information loss (Little 1993), and the rows in a masked dataset still correspond to real subjects. More recent work, such as Abowd and Schmutte (2015), emphasizes the challenges posed by restricted access to data. Meanwhile, Koenecke and Varian (2020) propose that researchers publicly release synthetic data, such as those generated by the Synthetic Data Vault (Patki, Wedge, & Veeramachaneni 2016), when they cannot release the actual data. Their proposed protocol differs from this paper's protocol, as this protocol is designed for a setting where the *researcher* cannot access the actual data.

The rest of the paper proceeds as follows. Section 2 describes the data and synthetic data protocol. Section 3 presents regression-based evidence of the determinants of acceptance, matriculation, and success, and shows why comparing the determinants of acceptance with those of success is insufficient to test optimality. Section 4 develops and estimates structural models of admissions that incorporate first-stage filtering and matriculation probabilities. Section 5 implements the deviation gains test, which rejects optimality. Section 6 concludes.

2 Data

This paper uses two datasets: one consisting of all applicants to a research university's economics PhD program from 2015 to 2019, and another of all applicants to that university's economics, government, history, linguistics, and psychology programs from 2020 to 2023. Unlike the graduate school dataset, the economics department dataset contains subjective variables (i.e. those derived from textual documents such as CVs and recommendation letters) and program outcomes data (e.g. post-program job placements). Therefore, I estimate this paper's structural models on the economics department dataset only. However, the graduate school dataset, which still contains objective variables and each applicant's admission status, is useful for making regression-based comparisons across programs. All results from the graduate school dataset are in Appendix B.1. In this section, I describe the variables in the economics department dataset, and I describe the synthetic data protocol, which serves as a template for conducting research on confidential data.

2.1 Variables

There are three types of variables: dependent variables, objective independent variables, and subjective independent variables. Table 6 of Appendix A contains a full list of the variables. The dependent variables consist of measures of both admission and success. For admission, such measures include binary variables of whether the applicant makes it past the first-round filtering stage, whether they are accepted into the program, whether they are waitlisted, whether they attend the university's open house, whether they matriculate at the university, and whether they attend (and were therefore accepted at) a different university's PhD program. I also have the U.S. News American and global rankings of the program they attended. If they attended a U.S. program, I use their American ranking, whereas if they attended a non-U.S. program, I use their global ranking times 3/8. The reason is that the maximum American ranking is 147, the maximum global ranking is 400, and I assign 150 to anyone who did not attend a PhD program. This procedure ensures that all U.S. and non-U.S. programs are ranked on the same scale.

For success in economics, I have a binary variable of whether they complete the program, along with variables for the type and ranking of their eventual job placement. Placement types include academic, government, consulting, and technology. For academic placements, I use the department's IDEAS/RePEc American and global rankings with a plus five penalty for business schools (Krueger & Wu 2000). For postdocs, I use a 2/3 to 1/3 weighted average of the university's ranking and the maximum ranking, which is 138 for American and 340 for global. For non-tenure-track positions, I use a 1/3 to 2/3 weighted average. For unranked tenure-track positions, I assign the maximum ranking plus one; for unranked postdocs and non-tenure-track positions, I assign 150 for American and 350 for global. Ultimately, for U.S. placements, I use the American ranking, whereas for non-U.S. placements, I use the global ranking times 3/7.

Meanwhile, for non-academic placements, I construct academia-equivalent rankings shown in Appendix A. Following Krueger and Wu (2000), the most prestigious non-academic placements (World Bank, IMF, Federal Reserve Board of Governors, European Central Bank, Amazon Science, Google Research, and Microsoft Research) are given a ranking of 40, with lower rankings down to 135 for less prestigious placements.[2] Per Eberhardt, Facchini, and Rueda (2026), economics PhD graduates, even those with Nobel laureate advisors, are increasingly taking non-academic jobs, especially in technology. As in Krueger and Wu (2000) and Bai et al. (2022), to avoid selection bias, I use publicly available LinkedIn profiles, graduate program websites, personal websites, PandaInUniv (2026), and García Guzmán (2026)'s *The Econ PhD placements dataset* to obtain outcomes data for all applicants, not just matriculants. The minority of applicants whose outcomes data are not available receive the lowest possible ranking of 150.

Regarding independent variables from the application files, I distinguish objective variables from subjective variables. Objective variables are those variables that the application software compiles into a spreadsheet for the committee's convenience, whereas subjective variables are those that the applicant uploads as pdf files that cannot be easily converted into a spreadsheet.[3] Because the admissions committee does not have time to read everyone's file, it uses the objective variables to filter out a predetermined percentage of the applicants before reading any files. For example, it filters out applicants with low quantitative GRE scores. For this reason, the structural models in this paper treat objective variables separately from subjective ones.

Objective variables include demographics; such as application year, age, gender, and a region of the world categorical variable, in which the categories are U.S., International Asian, and International Other. There are also performance variables, such as verbal, quantitative, and analytical GRE percentiles; undergraduate GPA; TOEFL/IELTS scores for non-native English speakers; two binary variables for having a graduate degree and work experience respectively; and a categorical variable for the type of undergraduate institution (elite U.S. university, elite U.S. liberal arts college, other U.S. college/university, elite foreign university, and other foreign university), where elite is always defined as top 50. Moreover, recommender variables, namely the number of professor (Krueger & Wu 2000) and prolific recommenders the applicant has, measure how well-connected the applicant is. A prolific recommender is one who writes a letter for at least one other applicant in the sample, which Bai et al. (2022) find predictive of acceptance and success.

Meanwhile, subjective variables in economics come from recommendation letters, application essays, CVs, and supplemental forms. I use textual processing algorithms on the letters to

[2] I use an average of the rankings provided by ChatGPT 5.5, Claude Opus 4.7, and Gemini 3 in temporary chat mode. The prompt and a link to screenshots of the outputs are available in Appendix A.

[3] Bai et al. (2022) define objective variables as "verifiable performance measures" of applicant quality and subjective variables as committee member assessments. For most variables, our definitions coincide. While they do not categorize demographics, such as gender and nationality, I categorize demographics as objective variables.

measure the overall word count and rates of positive, negative, standout, ability, research, grindstone, teaching, communal, and agentic words (Bai et al. 2022). From application essays, the algorithms extract the word count; rates of positive, negative, economics, and research words; and whether the applicant mentioned a professor's name from this university. The last three are my analogues of Grove et al. (2007)'s *Specific Topic*, *Mention Paper*, and *Specific Member*.[4] From CVs, there are the word count, a binary variable for whether the applicant has a publication or working paper, and the number of programming languages the applicant listed. Jones et al. (2020) find that economics committee members, especially at high-ranking programs, do not put much weight on coding skills. But given the increasing prevalence of empirical research that uses techniques like webscraping and machine learning, along with the increasing use of AI tools in economics, such skills have become more commonplace. There is also a supplemental form, in which the applicant lists the number of math courses they took, as well as the names of advanced math courses such as real analysis. I addition, I have a categorical variable of the applicant's indicated field of interest (e.g. econometrics, macroeconomics, etc.). Finally, for applicants whose files the committee read, I have the subjective score from 0 (worst) to 10 (best) that their assigned reader gave them relative to the scores that reader gave other applicants. This score is a proxy for information that the committee can observe but I cannot. The committee score is available for only the applicants whose files were read, and the other subjective variables from the textual documents are available for only about one-fifth of applicants. Per the 2015–2019 committee chair, the availability of these documents is mostly random, though the fraction of Round 1 survivors is higher for the applicants with available documents than for the full application pool.

2.2 Synthetic Data Protocol

In this section, I describe the synthetic data protocol used to protect the applicants' privacy, which I employ in conjunction with a university administrative office that is responsible for stewarding the data. While a handful of recent papers, such as Stanley and Totty (2024) and Carr, Wiemers, and Moffitt (2025) use synthetic data, the synthetic data in those papers had already been made available to researchers by the U.S. Census Bureau. In contrast, this paper's protocol covers all the steps from creation of the synthetic data to obtaining the final results on the actual data.

The steps of the protocol are as follows. First, the administrative office receives the data from the graduate admissions office and individual departments, such as economics. They remove all direct identifiers, such as names and email addresses, and run algorithms I provide to extract variables from textual documents, such as recommendation letters. Second, they use Patki et al. (2016)'s publicly-available Synthetic Data Vault (SDV) to generate the synthetic data. The SDV

[4] Grove et al. (2007)'s *Specific Topic* is a binary variable that equals 1 if the essay contains at least one economics topic. *Mention Paper* is a binary variable that equals 1 if the essay mentions any paper (including papers like undergraduate term papers that are not publications or working papers), and is designed to indicate a "demonstrated interest in doing economics research." Finally, *Specific Member* is the same as my binary specific professor variable.

estimates a Gaussian copula to capture correlations between variables and then simulates a new, synthetic dataset from the copula using a random number generator. Crucially, the rows in the synthetic dataset do not correspond to real subjects, which prevents identity disclosure (Chapelle & Falissard 2023). The SDV works best when the final versions of the variables are already created (e.g. turning a list of country names into a categorical variable), so without seeing any actual data, I work with the office to create the variables before implementing the SDV.

However, there remains the possibility of attribute disclosure. For example, if an applicant from an unusual country has an unusually low GRE percentile, information about that applicant may leak into the synthetic dataset. As such, if any cell in the synthetic data has fewer than five other cells from that variable with the same value, the variable's observations are binned and in each bin, replaced with the bin's sample mean. Only at this point do I receive the synthetic data. I use the data to finalize my estimation code, which I then give to the administrative department. The department takes that code, runs it on the actual data, and returns the parameter estimates.

3 Regression Analysis

In this section, I present regression results for the economics department dataset with dependent variables for acceptance, matriculation, and success. For success, I have two dependent variables. The first variable is a binary variable for a top 100 or top 100 equivalent placement. *Top 100 Placement* = 1 for applicants who attained a top 100 U.S. placement, an international placement that maps to a top 100 U.S. placement, or a non-academic placement with a top 100 academia-equivalent ranking. Per Table 7 in Appendix B, *Top 100 Placement* = 1 for 19.71% of applicants vs. 34.18% of matriculants, indicating that the committee's matriculation set is much better than random chance would predict. The second variable is *Placement Rank*, which ranges from 1 to 150. For applicants whose placements I could not find, I set *Top 100 Placement* to 0, *Placement Rank* to 150, and *Missing Placement Rank* to 1. Likewise, for applicants whose PhD program I could not find, I set *Program Rank* to 150 and *Missing Program Rank* to 1.

Meanwhile, for the subjective variables in recommendation letters, I use principal component analysis to reduce the dimensionality of the letters part of the state space from 9 to 2. The loadings are in Table 1, and they show that PC1 can be interpreted as the degree to which the letters contain words from the dictionaries in general, with an emphasis on positive and ability words. Per Table 8 in Appendix B, the correlation between PC1 and the rate of positive words is 0.8114, and that between PC1 and the rate of ability words is 0.7486. Meanwhile, PC2 can be interpreted as the degree to which the letters are factual and unflattering. Its loadings are positive for negative, research (largest positive loading), grindstone, and teaching words and negative for positive, standout (largest negative loading), ability, communal, and agentic words.

The regression results are in Tables 2a and 2b.[5] I find that different independent variables best predict each dependent variable. For *Accept*, *Committee Score* is the best predictor; a one point increase in *Committee Score* makes an applicant 7.88 percentage points more likely to be accepted, and including it increases the Pseudo-R^2 from 0.2957 to 0.8108. However, other significant predictors of acceptance include demographics, *GRE Quantitative Percentile*, *Work Experience*, and *CV Coding*. Interestingly, *CV Coding*'s average marginal effect (AME) is negative, meaning that for each additional coding language an applicant indicates on their CV, they are 0.44 percentage points less likely to be accepted. Despite the reduced sample size for *CV Coding*, it is still significant at the 10% level. Perhaps having more coding languages on a CV is associated with being from more of an industry background, which the committee might find less appealing. Meanwhile, women are 1.63 percentage points more likely to be accepted, and applicants from East or South Asia are 2.14 percentage points less likely to be accepted. While *Female* and *International Asian* do not significantly predict success, I will show that the committee's decision rule with respect to these two demographics does not negatively affect the average success probability of the acceptance or matriculation set. Therefore, if this decision rule is part of a diversity objective, it is nonetheless still consistent with a success-maximizing objective.

Meanwhile, predictors of acceptance often negatively predict matriculation for applicants who are accepted. For example, *International Asian*'s AME is significantly negative for *Accept* but significantly positive for *Matriculate*. In contrast, *GRE Quantitative Percentile*'s AME is significantly positive for *Accept* but significantly negative for *Matriculate*. Presumably, applicants with high GRE quantitative scores have more outside acceptances and so are less likely to matriculate at this university. However, *Work Experience* is significantly positive for both *Accept* and *Matriculate*, whereas *CV Coding* is significantly negative for both. That said, the best predictor of matriculation is whether the accepted applicant attended the department's open house. Those who attended were 26.15 percentage points more likely to matriculate than those who did not. Attending the open house is a positive signal that the applicant is interested in this university.

Finally, the best predictor of success is *Program Rank*. Applicant characteristics equal, attending a PhD program ranked 10 places better (i.e. 10 places closer to rank 1) increases the probability of a top 100 placement by 1.1 percentage points and improves the expected placement rank (i.e. toward rank 1) by 0.8 places. However, other significant predictors include *Winsorized Age*, *TOEFL/IETLS*, *Work Experience*, and *Essay Positive Terms (%)*.[6] Curiously, a high *Undergraduate GPA* significantly predicts failure when controlling for *Program Rank*. If GPA does not provide much signal, which is plausible because GPAs are not standardized across institu-

[5] In Tables 9a and 9b in Appendix B, I present the regression results on synthetic data. For some variables, the synthetic estimates are similar to the actual estimates, but the synthetic estimates are generally less significant.

[6] The age distribution of applicants drops off sharply past the early 30s to the point that a quadratic *Age* term is unidentified. To avoid extrapolating *Age*'s positive AME to the small fraction of much older applicants, I winsorize *Age* at its 95th percentile, which is 33 years old. Also, I use percentiles to convert TOEFL's 0–120 scores to IELTS's 1–9 scores, ensuring that both tests are on the same scale when combining them into one variable.

tions, but admissions committees in general select positively on them, then applicants with equal characteristics who were selected into the same program tier may exhibit a negative correlation between GPA and success. However, this paper's structural models are meant for pre-selection decision-making, so I follow Bai et al. (2022) and remove GPA after this section.

In Table 10 of Appendix B, I present lasso logistic regression results with 5-fold cross-validation for variable selection. For acceptance and matriculation, lasso selects most of the variables, but it selects fewer variables for success. In the table's sixth column, with *Top 100 Placement* as the dependent variable and controlling for *Committee Score* and *Program Rank/Missing Program Rank*, lasso selects the following ten variables: *Winsorized Age*, *Undergraduate GPA*, *TOEFL/IELTS*, *Work Experience*, *Letters Terms (%) PC1*, *Essay Economics Terms (%)*, *Essay Positive Terms (%)*, *Missing Committee Score*, *Program Rank*, and *Missing Program Rank*.

Readers familiar with Table 18 of Bai et al. (2022) may notice that their success regressions without controlling for PhD program rank fit substantially better than this paper's success regressions without *Program Rank* and *Missing Program Rank*. It is only after including *Program Rank* and *Missing Program Rank* that the Pseudo-R^2s in Table 2a become comparable to those in Bai et al. (2022), though it is worth noting that adding PhD program rank controls does not futher increase their Pseudo-R^2s by much. This pattern occurs because they define success as attaining a top 10 or top 20 assistant professorship. It is very hard to attain such a professorship without attending a top-ranked PhD program. Per their Table 17, 83% of top 10 assistant professor placements in their data came from top 10 economics PhD programs, an additional 8% came from non-economics PhD programs which may have also been top 10 in their disciplines, and only one applicant attained a top 10 placement from a PhD program ranked lower than this paper's university. It is also the case that variables in application files, such as GRE scores, predict acceptance at top PhD programs. Therefore, their application file variables predict placement indirectly by predicting program rank, and adding program indicators contributes little because the file has already revealed most of that information. In contrast, while attending a PhD program is necessary for a top 100 or top 100-equivalent placement, attending a top-ranked program is not. Graduates of programs across many tiers attain it, in part because non-academic research institutions (e.g. IMF, Federal Reserve, etc.) account for many of this paper's top 100 placements. As such, application file variables do not indirectly predict success as they do in Bai et al. (2022), making it necessary to control for program rank to increase the Pseudo-R^2 in Table 2a.

I briefly address inference vs. prediction, as well as sample selection. Both are important when running regressions with success measures as dependent variables. Regarding inference vs. prediction, omitted independent variables make it challenging to obtain consistent estimates in a success regression. For example, I do not observe the effort that students put in after matriculating. Even effort, let alone any other variables about the applicants that I do not observe, can bias success parameter estimates upward if effort is correlated with the variables corresponding to those parameters and correlated with success irrespective of those variables.

However, there are reasons for reassurance. First, I incorporate *Committe Score* as an in-

Table 1: Principal Component Loadings

	PC1	PC2
Positive	0.5044	-0.1769
Negative	0.1479	0.3759
Standout	0.2115	-0.5082
Ability	0.4667	-0.0177
Research	0.1214	0.6418
Grindstone	0.3879	0.3075
Teaching	0.3708	0.0743
Communal	0.1284	-0.1983
Agentic	0.3768	-0.1324
Proportion of Variance	0.2763	0.1456

dependent variable, which proxies variables that the committee observes and I do not. I also follow Bai et al. (2022) and include as a proxy of effort the number of "grindstone terms," such as depend* and diligen*, in recommendation letters. In addition, in this paper, prediction matters as much as inference. Mullainathan and Spiess (2017) distinguish $\hat{\beta}$ (e.g. the effect on educational outcomes of hiring an additional teacher) from $\hat{y}$ (e.g. which teacher to hire) problems. For graduate admissions, if *Work Experience* predicts success, then the committee should consider it regardless of whether the committee knows the mechanism through which it affect success.

Moreover, the determinants of success conditional on applying (see Krueger and Wu 2000, and Grove and Wu 2007) are different from those conditional on matriculating (see Athey et al. 2007). Bai et al. (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples, a theoretical finding consistent with the significance of the quantitative GRE being weaker in studies on matriculants only. This problem is akin to height not predicting success in a sample of professional basketball players. Like Bai et al. (2022), I collect placement outcomes for all applicants, not just those who attended this university. But there are other measures of success, such as course grades and passing comprehensive examinations, that are only available for matriculants, and that have been used in papers like Athey et al. (2007).

As such, for these measures, one can implement a Heckman (1979) correction. Notationally, let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$. y is a measure of success, x are variables in applications, and d is matriculation. The first step is to run a probit regression of d on z and compute the inverse Mills ratio $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$. The second step is to run OLS or heckprobit of y on $(x, \hat{\lambda})$ to obtain $\hat{\beta}$. For a Heckman instrument, one needs a variable that affects d but not y. An example is a variable measuring the applicant's enthusiasm about the program, such as whether they included a specific professor's name in their application essay. In Section 4, I discuss the implementation of Heckman corrections in a subset of this paper's structural models.

In addition, in Section B, I implement a Wald test of whether the determinants of acceptance are the same across this university's graduate programs, and I find that they are not the same.

Table 2a: Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.3826	-0.5049	-0.2577	0.0230	0.0301	0.0885	1.8038	1.7607	-0.6618
	(0.2365)	(0.4703)	(0.6853)	(0.1708)	(0.1711)	(0.1887)	(4.2583)	(4.2504)	(3.6266)
2017	-0.0954	-1.3247***	0.1357	0.0829	0.0711	0.1460	2.7746	3.0235	0.7473
	(0.2564)	(0.4881)	(0.6732)	(0.1707)	(0.1721)	(0.1934)	(4.2295)	(4.2188)	(3.7157)
2018	-0.2206	-3.7609***	0.9476	0.0624	-0.0392	0.0142	2.7265	5.2685	4.9770
	(0.2648)	(0.6518)	(0.6200)	(0.1761)	(0.1807)	(0.2011)	(4.4414)	(4.5011)	(3.8786)
2019	-0.1059	-3.1670***	0.5909	0.0594	0.0144	0.0366	5.7085	6.6750	5.2241
	(0.2653)	(0.7114)	(0.6855)	(0.1821)	(0.1830)	(0.2052)	(4.6400)	(4.6266)	(4.1424)
Winsorized Age	-0.0218	-0.0080	-0.1567**	0.0240	0.0242	0.0420*	0.3596	0.3632	-0.1277
	(0.0342)	(0.0592)	(0.0778)	(0.0207)	(0.0205)	(0.0236)	(0.5457)	(0.5410)	(0.4839)
Female	0.3759**	0.8784**	-0.3445	-0.0880	-0.1026	-0.0256	1.8085	2.0921	-1.6375
	(0.1657)	(0.3813)	(0.3794)	(0.1126)	(0.1130)	(0.1252)	(2.8776)	(2.8750)	(2.5117)
U.S.	-0.4982*	-0.0681	1.0348*	0.1167	0.1396	0.1520	-0.6372	-0.9759	-0.7059
	(0.2624)	(0.4538)	(0.6098)	(0.1903)	(0.1909)	(0.2236)	(4.7990)	(4.7913)	(4.3464)
International Asian	-1.1569***	-1.1507***	1.1559**	0.1695	0.1949	-0.1005	-7.8212**	-8.2008**	3.3007
	(0.2129)	(0.4146)	(0.4904)	(0.1379)	(0.1395)	(0.1546)	(3.5528)	(3.5708)	(3.1577)
GRE Verbal Percentile	0.0264***	0.0079	-0.0107	0.0049	0.0030	0.0007	-0.1958**	-0.1470*	-0.0330
	(0.0054)	(0.0130)	(0.0118)	(0.0031)	(0.0032)	(0.0036)	(0.0787)	(0.0806)	(0.0735)
GRE Quantitative Percentile	0.1329***	0.0589**	-0.1033**	0.0245***	0.0184**	0.0047	-0.8498***	-0.7196***	-0.2428*
	(0.0202)	(0.0275)	(0.0464)	(0.0077)	(0.0075)	(0.0084)	(0.1567)	(0.1578)	(0.1460)
GRE Analytic Percentile	0.0076**	0.0094	-0.0119	-0.0020	-0.0027	-0.0021	0.0656	0.0821	0.0160
	(0.0039)	(0.0085)	(0.0095)	(0.0025)	(0.0025)	(0.0029)	(0.0669)	(0.0671)	(0.0618)
Undergraduate GPA	1.2726*	1.0509	-0.7209	-0.3976	-0.4722	-0.8687*	4.8040	6.9311	17.2597**
	(0.6975)	(1.2453)	(1.2760)	(0.3912)	(0.3848)	(0.4442)	(10.1461)	(10.0615)	(7.7507)
TOEFL/IELTS	0.4236	-0.1165	-0.4434	0.1778	0.1437	0.3240*	-2.2504	-1.2960	-7.6837*
	(0.2924)	(0.5976)	(0.5525)	(0.1571)	(0.1559)	(0.1757)	(4.3852)	(4.3431)	(3.9285)
Elite U.S. University	0.7328***	-0.1822	-0.4980	-0.0159	-0.0853	-0.1523	-2.3790	-0.3552	-0.3450
	(0.2774)	(0.5918)	(0.5799)	(0.2133)	(0.2133)	(0.2344)	(5.2934)	(5.2869)	(4.4943)
Elite U.S. Liberal Arts College	0.0167	0.5446	0.9907	0.0731	0.0803	-0.1010	-3.9470	-3.8047	-3.4006
	(0.4568)	(0.6896)	(0.8447)	(0.3370)	(0.3361)	(0.3936)	(8.8571)	(8.7916)	(8.2332)
Elite Foreign University	0.1244	-0.5507	-0.0673	0.1210	0.0751	0.0969	-2.7705	-1.5414	-2.2142
	(0.5319)	(0.9725)	(1.1224)	(0.3239)	(0.3259)	(0.3526)	(7.4748)	(7.4679)	(6.1380)
Other Foreign University	0.5323*	0.6328	0.7373	0.1483	0.1091	0.1587	-6.7584	-5.7186	-9.1260**
	(0.3022)	(0.6505)	(0.6347)	(0.1964)	(0.1970)	(0.2216)	(4.9679)	(4.9620)	(4.3486)
Missing Undergraduate Institution	0.3260	0.2482	-0.6851	0.2292	0.2082	0.1770	-7.9862	-7.5382	-8.1758*
	(0.3111)	(0.6016)	(0.7260)	(0.2017)	(0.2022)	(0.2345)	(5.1133)	(5.1129)	(4.5085)
Graduate Degree	0.1726	-0.2058	-0.2954	-0.0406	-0.0921	-0.1835	-3.2803	-2.0242	2.6215
	(0.1788)	(0.3845)	(0.5017)	(0.1231)	(0.1273)	(0.1370)	(3.1662)	(3.2087)	(2.7452)
Work Experience	0.7785***	0.7519**	1.1183**	0.2298*	0.1772	0.2839**	-3.5942	-2.2559	-6.4380**
	(0.1915)	(0.3400)	(0.5291)	(0.1253)	(0.1270)	(0.1422)	(3.2005)	(3.2301)	(2.9093)
#Professor Recommenders	-0.0377	-0.1797	0.3587	-0.0472	-0.0513	-0.0547	0.8765	0.9972	0.7371
	(0.1079)	(0.2434)	(0.3115)	(0.0749)	(0.0754)	(0.0862)	(2.0492)	(2.0495)	(1.7160)
#Prolific Recommenders	0.2208***	0.2081	-0.2993	0.1011*	0.0899*	0.0600	-1.4623	-1.1967	0.1726
	(0.0811)	(0.1721)	(0.2060)	(0.0528)	(0.0533)	(0.0585)	(1.3680)	(1.3670)	(1.1704)
Missing GRE	-1.5399	1.0141		-0.4515	-0.3258	0.0174	11.5202	8.5434	-3.7106
	(0.9821)	(1.1356)		(0.3510)	(0.3544)	(0.3797)	(8.6385)	(8.6602)	(6.1802)
Missing Undergraduate GPA	-0.3432	-0.2798	0.0798	-0.1010	-0.0639	-0.0513	2.4296	1.5256	-0.8710
	(0.2210)	(0.4413)	(0.4943)	(0.1565)	(0.1585)	(0.1756)	(3.9808)	(3.9842)	(3.3572)
Missing TOEFL/IELTS	0.4719**	0.3743	-0.0734	-0.1261	-0.1447	-0.0065	4.0029	4.3290	-2.3792
	(0.2062)	(0.4223)	(0.5891)	(0.1267)	(0.1282)	(0.1443)	(3.3209)	(3.3244)	(2.9657)
Committee Score		4.2419***	-0.2608		0.0943**	0.0059		-2.5088**	0.6884
		(0.4534)	(0.2250)		(0.0456)	(0.0488)		(1.1860)	(1.0132)
#Math Courses	-0.0949	-0.2706	0.0755	0.0626	0.0662	0.0008	-2.3828	-2.4550	0.9116
	(0.0773)	(0.2829)	(0.1680)	(0.0759)	(0.0766)	(0.0885)	(1.8033)	(1.8149)	(1.5082)
Advanced Math	0.6333**	1.3229	-1.3706**	-0.1932	-0.2078	-0.2275	3.0400	3.4570	1.9957
	(0.3036)	(0.9203)	(0.5621)	(0.2605)	(0.2639)	(0.2949)	(6.4446)	(6.4377)	(5.5819)
Letters Length/100	0.1358***	0.1084	-0.0910	0.0067	-0.0018	-0.0150	-0.2593	-0.0078	0.7206
	(0.0518)	(0.1424)	(0.0712)	(0.0335)	(0.0339)	(0.0388)	(0.8254)	(0.8325)	(0.7084)
Letters Terms (%) PC1	0.1507	0.2939	0.4716**	0.0362	0.0251	0.1260	1.6082	1.8374	-0.5612
	(0.0923)	(0.2912)	(0.2051)	(0.0732)	(0.0748)	(0.0915)	(2.0561)	(2.0540)	(1.8186)
Letters Terms (%) PC2	-0.0913	-0.4980	0.2724	-0.1226	-0.1199	-0.1420	0.7118	0.5616	0.1155
	(0.1464)	(0.4143)	(0.2604)	(0.1162)	(0.1177)	(0.1272)	(2.7992)	(2.8081)	(2.3450)
CV Length/100	0.0361	0.0559	-0.1986	0.0301	0.0274	0.0352	-1.0456	-1.0120	-0.8208
	(0.0617)	(0.1186)	(0.1337)	(0.0562)	(0.0563)	(0.0593)	(1.3224)	(1.3130)	(1.0252)
CV Coding	-0.0439	-0.2369*	-0.2911**	0.0716	0.0710	0.0992	-1.9601	-1.8941	-2.0142
	(0.0765)	(0.1381)	(0.1390)	(0.0532)	(0.0532)	(0.0654)	(1.5459)	(1.5358)	(1.3384)
CV Publication/Working Paper	0.6162*	0.8309	-0.1668	0.2940	0.2834	0.1310	-11.5166	-10.8849	-4.6204
	(0.3421)	(0.9326)	(0.5756)	(0.2998)	(0.3036)	(0.3383)	(7.2873)	(7.3131)	(6.7001)
Essay Length/100	0.1747*	0.2225	0.1776	-0.0008	-0.0008	-0.0711	-0.4758	-0.5351	1.8425
	(0.0933)	(0.3920)	(0.1536)	(0.0751)	(0.0764)	(0.0821)	(1.8648)	(1.8721)	(1.6240)
Essay Specific Professor	0.3842	0.4170	0.6845	0.2144	0.1774	0.2872	-6.8688	-5.9104	-7.5240
	(0.2908)	(0.7014)	(0.5119)	(0.2407)	(0.2410)	(0.2789)	(6.0223)	(5.9982)	(5.2718)
Essay Economics Terms (%)	-0.0169	0.0070	0.2576**	-0.1126**	-0.1150**	-0.0822	3.6540***	3.7277***	2.1702*
	(0.0547)	(0.0866)	(0.1159)	(0.0533)	(0.0531)	(0.0618)	(1.2540)	(1.2496)	(1.1179)
Essay Research Terms (%)	-0.0018	-0.0459	-0.1868	0.0697	0.0642	0.0150	-3.7084**	-3.6795**	-1.0575
	(0.0736)	(0.1690)	(0.1451)	(0.0633)	(0.0639)	(0.0674)	(1.6220)	(1.6209)	(1.3903)
Essay Positive Terms (%)	-0.0202	-0.0700	-0.0504	0.0724	0.0772*	0.0977*	-0.9256	-0.9929	-0.7266
	(0.0512)	(0.1044)	(0.1331)	(0.0453)	(0.0458)	(0.0542)	(1.1634)	(1.1496)	(0.9752)
Essay Negative Terms (%)	-0.0151	-0.0886	0.0019	-0.0386	-0.0454	-0.0472	0.8164	0.9436	0.9507
	(0.0770)	(0.1557)	(0.1071)	(0.0663)	(0.0663)	(0.0748)	(1.6317)	(1.6259)	(1.3985)
Missing Committee Score		-0.5168			-0.3458**	-0.2199		9.2234***	2.8589
		(0.6383)			(0.1397)	(0.1519)		(3.4582)	(3.0258)
Missing Application Pdf	-1.0107***	0.1926	-0.7407	0.1461	0.2686	0.2416	-5.1611	-8.4497	-6.8690
	(0.3188)	(0.8088)	(0.6084)	(0.2442)	(0.2479)	(0.2763)	(6.0328)	(6.1404)	(5.4325)
Open House			1.8384***						
			(0.4115)						
Program Rank						-0.0087***			0.1813***
						(0.0021)			(0.0404)
Missing Program Rank						-3.1432***			89.4721***
						(0.4077)			(5.6947)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2957	0.8108	0.2877	0.0308	0.0353	0.2430	0.0089	0.0097	0.1065

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 2b: Logistic Regression AMEs

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0274	-0.0094	-0.0367	0.0035	0.0046	0.0112	0.8694	0.8485	-0.3038
	(0.0171)	(0.0085)	(0.0976)	(0.0262)	(0.0261)	(0.0238)	(2.0523)	(2.0482)	(1.6650)
2017	-0.0068	-0.0246***	0.0193	0.0127	0.0108	0.0184	1.3374	1.4571	0.3431
	(0.0184)	(0.0090)	(0.0958)	(0.0262)	(0.0263)	(0.0244)	(2.0386)	(2.0331)	(1.7054)
2018	-0.0158	-0.0699***	0.1348	0.0096	-0.0060	0.0018	1.3142	2.5391	2.2847
	(0.0189)	(0.0086)	(0.0857)	(0.0270)	(0.0276)	(0.0254)	(2.1405)	(2.1686)	(1.7791)
2019	-0.0076	-0.0589***	0.0841	0.0091	0.0022	0.0046	2.7516	3.2169	2.3982
	(0.0190)	(0.0109)	(0.0970)	(0.0279)	(0.0279)	(0.0259)	(2.2355)	(2.2284)	(1.8977)
Winsorized Age	-0.0016	-0.0001	-0.0223**	0.0037	0.0037	0.0053*	0.1734	0.1751	-0.0586
	(0.0024)	(0.0011)	(0.0108)	(0.0032)	(0.0031)	(0.0030)	(0.2629)	(0.2606)	(0.2222)
Female	0.0269**	0.0163**	-0.0490	-0.0135	-0.0157	-0.0032	0.8717	1.0082	-0.7517
	(0.0118)	(0.0068)	(0.0534)	(0.0173)	(0.0172)	(0.0158)	(1.3867)	(1.3851)	(1.1534)
U.S.	-0.0356*	-0.0013	0.1472*	0.0179	0.0213	0.0192	-0.3071	-0.4703	-0.3241
	(0.0188)	(0.0084)	(0.0864)	(0.0292)	(0.0291)	(0.0282)	(2.3132)	(2.3090)	(1.9951)
International Asian	-0.0828***	-0.0214***	0.1644**	0.0260	0.0298	-0.0127	-3.7699**	-3.9523**	1.5152
	(0.0148)	(0.0073)	(0.0691)	(0.0212)	(0.0213)	(0.0195)	(1.7114)	(1.7195)	(1.4490)
GRE Verbal Percentile	0.0019***	0.0001	-0.0015	0.0008	0.0005	0.0001	-0.0944**	-0.0708*	-0.0151
	(0.0004)	(0.0002)	(0.0017)	(0.0005)	(0.0005)	(0.0005)	(0.0379)	(0.0388)	(0.0337)
GRE Quantitative Percentile	0.0095***	0.0011**	-0.0147**	0.0038***	0.0028**	0.0006	-0.4096***	-0.3468***	-0.1114*
	(0.0014)	(0.0005)	(0.0063)	(0.0012)	(0.0011)	(0.0011)	(0.0757)	(0.0762)	(0.0670)
GRE Analytic Percentile	0.0005**	0.0002	-0.0017	-0.0003	-0.0004	-0.0003	0.0316	0.0395	0.0073
	(0.0003)	(0.0002)	(0.0013)	(0.0004)	(0.0004)	(0.0004)	(0.0322)	(0.0323)	(0.0283)
Undergraduate GPA	0.0910*	0.0195	-0.1026	-0.0610	-0.0721	-0.1096**	2.3156	3.3404	7.9232**
	(0.0499)	(0.0235)	(0.1820)	(0.0600)	(0.0587)	(0.0558)	(4.8911)	(4.8499)	(3.5564)
TOEFL/IELTS	0.0303	-0.0022	-0.0631	0.0273	0.0219	0.0409*	-1.0847	-0.6246	-3.5273*
	(0.0209)	(0.0111)	(0.0776)	(0.0241)	(0.0238)	(0.0221)	(2.1145)	(2.0935)	(1.8033)
Elite U.S. University	0.0524***	-0.0034	-0.0708	-0.0024	-0.0130	-0.0192	-1.1467	-0.1712	-0.1584
	(0.0198)	(0.0111)	(0.0816)	(0.0327)	(0.0326)	(0.0296)	(2.5514)	(2.5479)	(2.0632)
Elite U.S. Liberal Arts College	0.0012	0.0101	0.1409	0.0112	0.0123	-0.0127	-1.9025	-1.8336	-1.5611
	(0.0327)	(0.0126)	(0.1197)	(0.0517)	(0.0513)	(0.0496)	(4.2692)	(4.2369)	(3.7802)
Elite Foreign University	0.0089	-0.0102	-0.0096	0.0186	0.0115	0.0122	-1.3354	-0.7428	-1.0164
	(0.0381)	(0.0182)	(0.1597)	(0.0497)	(0.0497)	(0.0445)	(3.6022)	(3.5986)	(2.8173)
Other Foreign University	0.0381*	0.0118	0.1049	0.0228	0.0167	0.0200	-3.2576	-2.7560	-4.1894**
	(0.0216)	(0.0119)	(0.0890)	(0.0301)	(0.0301)	(0.0279)	(2.3945)	(2.3912)	(1.9955)
Missing Undergraduate Institution	0.0233	0.0046	-0.0975	0.0352	0.0318	0.0223	-3.8495	-3.6329	-3.7531*
	(0.0222)	(0.0111)	(0.1014)	(0.0309)	(0.0308)	(0.0296)	(2.4639)	(2.4635)	(2.0698)
Graduate Degree	0.0123	-0.0038	-0.0420	-0.0062	-0.0141	-0.0231	-1.5811	-0.9755	1.2034
	(0.0128)	(0.0071)	(0.0712)	(0.0189)	(0.0194)	(0.0172)	(1.5257)	(1.5461)	(1.2606)
Work Experience	0.0557***	0.0140**	0.1591**	0.0353*	0.0271	0.0358**	-1.7325	-1.0872	-2.9554**
	(0.0135)	(0.0064)	(0.0700)	(0.0192)	(0.0194)	(0.0178)	(1.5432)	(1.5570)	(1.3344)
#Professor Recommenders	-0.0027	-0.0033	0.0510	-0.0072	-0.0078	-0.0069	0.4225	0.4806	0.3384
	(0.0077)	(0.0045)	(0.0437)	(0.0115)	(0.0115)	(0.0109)	(0.9881)	(0.9881)	(0.7878)
#Prolific Recommenders	0.0158***	0.0039	-0.0426	0.0155*	0.0137*	0.0076	-0.7048	-0.5767	0.0792
	(0.0058)	(0.0031)	(0.0285)	(0.0081)	(0.0081)	(0.0074)	(0.6595)	(0.6589)	(0.5373)
Missing GRE	-0.1102	0.0188		-0.0693	-0.0497	0.0022	5.5529	4.1174	-1.7034
	(0.0706)	(0.0212)		(0.0538)	(0.0541)	(0.0479)	(4.1627)	(4.1727)	(2.8358)
Missing Undergraduate GPA	-0.0246	-0.0052	0.0114	-0.0155	-0.0098	-0.0065	1.1711	0.7352	-0.3999
	(0.0158)	(0.0082)	(0.0703)	(0.0240)	(0.0242)	(0.0221)	(1.9187)	(1.9201)	(1.5412)
Missing TOEFL/IELTS	0.0338**	0.0070	-0.0104	-0.0194	-0.0221	-0.0008	1.9295	2.0863	-1.0922
	(0.0147)	(0.0078)	(0.0839)	(0.0194)	(0.0195)	(0.0182)	(1.6007)	(1.6020)	(1.3613)
Committee Score		0.0788***	-0.0371		0.0144**	0.0007		-1.2091**	0.3160
		(0.0045)	(0.0322)		(0.0069)	(0.0062)		(0.5716)	(0.4649)
#Math Courses	-0.0068	-0.0050	0.0107	0.0096	0.0101	0.0001	-1.1485	-1.1832	0.4185
	(0.0055)	(0.0053)	(0.0239)	(0.0116)	(0.0117)	(0.0112)	(0.8692)	(0.8747)	(0.6922)
Advanced Math	0.0453**	0.0246	-0.1950**	-0.0296	-0.0317	-0.0287	1.4653	1.6660	0.9161
	(0.0216)	(0.0168)	(0.0793)	(0.0399)	(0.0402)	(0.0372)	(3.1061)	(3.1022)	(2.5621)
Letters Length/100	0.0097***	0.0020	-0.0129	0.0010	-0.0003	-0.0019	-0.1250	-0.0038	0.3308
	(0.0037)	(0.0026)	(0.0100)	(0.0051)	(0.0052)	(0.0049)	(0.3978)	(0.4012)	(0.3253)
Letters Terms (%) PC1	0.0108	0.0055	0.0671**	0.0056	0.0038	0.0159	0.7752	0.8855	-0.2576
	(0.0066)	(0.0054)	(0.0283)	(0.0112)	(0.0114)	(0.0115)	(0.9905)	(0.9892)	(0.8348)
Letters Terms (%) PC2	-0.0065	-0.0093	0.0387	-0.0188	-0.0183	-0.0179	0.3431	0.2706	0.0530
	(0.0105)	(0.0077)	(0.0371)	(0.0178)	(0.0179)	(0.0160)	(1.3495)	(1.3535)	(1.0765)
CV Length/100	0.0026	0.0010	-0.0282	0.0046	0.0042	0.0044	-0.5040	-0.4877	-0.3768
	(0.0044)	(0.0022)	(0.0190)	(0.0086)	(0.0086)	(0.0075)	(0.6374)	(0.6328)	(0.4706)
CV Coding	-0.0031	-0.0044*	-0.0414**	0.0110	0.0108	0.0125	-0.9448	-0.9129	-0.9246
	(0.0055)	(0.0026)	(0.0193)	(0.0082)	(0.0081)	(0.0082)	(0.7454)	(0.7403)	(0.6139)
CV Publication/Working Paper	0.0441*	0.0154	-0.0237	0.0451	0.0433	0.0165	-5.5511	-5.2459	-2.1210
	(0.0244)	(0.0173)	(0.0820)	(0.0460)	(0.0463)	(0.0427)	(3.5094)	(3.5215)	(3.0760)
Essay Length/100	0.0125*	0.0041	0.0253	-0.0001	-0.0001	-0.0090	-0.2293	-0.2579	0.8458
	(0.0067)	(0.0073)	(0.0219)	(0.0115)	(0.0117)	(0.0103)	(0.8988)	(0.9022)	(0.7451)
Essay Specific Professor	0.0275	0.0077	0.0974	0.0329	0.0271	0.0362	-3.3108	-2.8484	-3.4539
	(0.0208)	(0.0130)	(0.0724)	(0.0369)	(0.0368)	(0.0351)	(2.9044)	(2.8921)	(2.4207)
Essay Economics Terms (%)	-0.0012	0.0001	0.0366**	-0.0173**	-0.0175**	-0.0104	1.7613***	1.7965***	0.9962*
	(0.0039)	(0.0016)	(0.0162)	(0.0082)	(0.0081)	(0.0078)	(0.6048)	(0.6027)	(0.5134)
Essay Research Terms (%)	-0.0001	-0.0009	-0.0266	0.0107	0.0098	0.0019	-1.7875**	-1.7733**	-0.4855
	(0.0053)	(0.0031)	(0.0206)	(0.0097)	(0.0098)	(0.0085)	(0.7813)	(0.7808)	(0.6382)
Essay Positive Terms (%)	-0.0014	-0.0013	-0.0072	0.0111	0.0118*	0.0123*	-0.4461	-0.4785	-0.3335
	(0.0037)	(0.0019)	(0.0189)	(0.0069)	(0.0070)	(0.0068)	(0.5609)	(0.5541)	(0.4475)
Essay Negative Terms (%)	-0.0011	-0.0016	0.0003	-0.0059	-0.0069	-0.0060	0.3935	0.4548	0.4364
	(0.0055)	(0.0029)	(0.0152)	(0.0102)	(0.0101)	(0.0094)	(0.7865)	(0.7837)	(0.6420)
Missing Committee Score		-0.0096			-0.0528**	-0.0277		4.4451***	1.3124
		(0.0120)			(0.0212)	(0.0191)		(1.6677)	(1.3892)
Missing Application Pdf	-0.0723***	0.0036	-0.1054	0.0224	0.0410	0.0305	-2.4877	-4.0722	-3.1533
	(0.0227)	(0.0149)	(0.0858)	(0.0375)	(0.0379)	(0.0348)	(2.9090)	(2.9614)	(2.4962)
Open House			0.2615***						
			(0.0512)						
Program Rank						-0.0011***			0.0832***
						(0.0003)			(0.0185)
Missing Program Rank						-0.3965***			41.0729***
						(0.0512)			(2.6010)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Could I similarly implement a Wald test of whether the determinants of acceptance are the same as those of success within a program? For example, does the quantitative GRE have the same AME for acceptance as for success? Intuitively, provided the committee wants to select applicants based on a "success" model, predictors of admission only may be overvalued, and vice versa. Such a test would be not work, however. First, there may be unwanted rejections of the null. For example, the committee may value diversity if it does not hurt success probabilities too much, in which case it may favor demographics not strictly more likely to succeed. There may also be unwanted non-rejections. If there is discrimination in the job market, the committee may rationalize discriminating in the same way, which would make the test not reject the null.

But more importantly, this test would not account for the mechanics of the admissions process. As a simple example, suppose the following conditions hold. First, *Work Experience* has a large positive effect on success, and *Graduate Degree* has a smaller positive effect. Second, 25% of applicants have both characteristics and therefore the highest success probability, 25% have *Work Experience* only (second-highest), 25% have *Graduate Degree* only (third-highest), and 25% have neither (lowest). Third, this university accepts half the applicants, and the other half attend a lower-ranked program, which ensures that everyone's outcomes are observable. Fourth, unlike applicant quality, program quality does not affect outcomes, at least in this example.

Under these suppositions, an optimal committee will accept the applicants with *Work Experience* and ignore *Graduate Degree*. As such, an acceptance regression will have a degenerate *Work Experience* AME of 1 and a *Graduate Degree* AME of 0, while a success regression will have AMEs as follows: $1 >$ *Work Experience* $>$ *Graduate Degree* > 0. Also, for the acceptance AMEs to equal the success AMEs, the committee would suboptimally have to accept applicants without work experience. Therefore, unlike with, for instance, wages equaling marginal revenue product in a competitive labor market which allows for cross-regression comparisons, the acceptance and success AMEs need not be equal for the committee to be optimal. In turn, to draw conclusions about optimality, the relationship between the AMEs must be interpreted through models of the committee's selection problem, which are the basis for Section 4.

4 Structural Analysis

There are two main reasons to estimate structural models of the admissions committee's decision problem. First, in order to make any claims about whether committees are optimizing, it is necessary to define what optimization is by explicitly modeling their problem. Second, structural models can uncover feasible counterfactual deviation decision rules to help committees admit stronger applicants, and the models can quantify any deviation gains from such rules.

The question then is how to set up the models. Bai et al. (2022) prove that if all the variables in an application file can be reduced to a single number (i.e. an expected success probability),[7] then accepting the optimal subset of applicants is the same as using a cutoff rule. A cutoff rule problem is far more tractable than a portfolio choice problem because I do not have to consider all the subsets of the hundreds of the applicants in each year who can be accepted; I only have to rank the applicants and choose the ones above the cutoff. Because committees may be more selective in some years than others, I allow the cutoff probability to vary across years.

After generalizing into a two-round dynamic framework Bai et al. (2022)'s proof that the committee's portfolio choice problem reduces to a cutoff rule problem, I start by modeling the committee's problem as follows. In Round 1, the committee observes only each applicant's objective variables, such as GRE scores. Based on those variables, it can transition the applicant to Round 2, or it can issue an immediate rejection. If the applicant survives Round 1, the committee then takes the time to read the application file and access all variables, both objective and subjective (e.g. recommendation letters). Finally, it accepts or rejects the applicant.

Rather than setting up a two-round model, why do I not rank the applicants based on their success probabilities conditional on all data and compare this one-round model's ranking to the committee's? The reason is that a one-round model is unfair to the committee because the committee does not have time to read everyone's files (though in the future, perhaps committees can use textual processing algorithms or local LLMs to "read" files quickly and generate additional objective variables without too much effort). Moreover, two-round models are useful not just for graduate admissions but for any other setting with first-stage filtering, such as hiring, etc.

I start with two different two-round models: dynamic and static. In Round 2 of the dynamic model, the committee accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. In Round 1, it forms a conditional expectation about those who survive to Round 2, meaning that the Round 1 cutoff is the level of that applicant's expected Round 2 value. As such, the Round 1 cutoff can take a value greater than one. In contrast, in the static model, the committee ranks the applicants based on their objective variables, reads the files of those whose expected success probabilities are above the first cutoff probability, re-ranks the surviving applicants based on all their data, and accepts those above the second cutoff probability. I refer to this model as static because each round is a separate static problem. The static model is useful for its simplicity and interpretable cutoff probabilities in both rounds.

The dynamic model appears more sophisticated because in Round 1, it favors applicants whose objective variables predict good subjective variables. However, the static model also correctly favors those applicants. While the static Round 1 success probability parameter estimates are inconsistent because the subjective variables are omitted, the fitted values estimate the success probability averaged over the subjective variables given the objective ones. Therefore, the component of the subjective variables that is predictable from the objective variables is absorbed

[7] The assumption that committees select based on expected success may have become more realistic in light of the 2023 Supreme Court ruling against race-based affirmative action in *Students for Fair Admissions v. Harvard*.

into the estimates. Only the myopic model, which is identical in setup to the dynamic model except that the parameters for the objective variables in the distribution of the subjective variables are restricted to zero, does not favor such applicants in Round 1. The static model also better describes what real-world committees do, as they do not solve Round 2 conditional expectations.

Later in this section, when I present maximum likelihood estimation results of these models, I include the Akaike Information Criterion (AIC) $= 2[K - \log(\hat{\mathcal{L}}^R)]$, which can be used to compare each restricted model with its unrestricted counterpart. K is the number of parameters, and $\hat{\mathcal{L}}^R$ is the restricted likelihood's value; lower AIC values are preferred. AIC values are comparable only between the restricted and unrestricted specifications of the same model. The reason is that the static objective is a composite likelihood over decisions and placement outcomes, with every outcome entering once in each of the Rounds 1 and 2 success blocks, whereas the dynamic likelihoods also include the densities of the subjective variables conditional on the objective variables.

Still later in this section, I present dynamic and static models in which not every accepted applicant matriculates, and which are also derived from two-round portfolio choice models, albeit models in which the objective is to maximize the sum of the success probabilities of the matriculants rather than the acceptances. In Round 2, the committee still accepts the applicants who survived Round 1 and have success probabilities greater than the Round 2 cutoff probability. No weight is placed on matriculation probabilities in Round 2 because while likely matriculants are more likely to produce a positive payoff, they are also more likely to take up cohort spots and so are more "expensive." However, weight is placed on matriculation probabilities in Round 1 because if the committee chooses to read the files of only the applicants who are most likely to succeed, it could find itself in a situation in Round 2 in which the only accepted applicants who matriculate are those with strong objective variables and weak subjective variables. This situation becomes increasingly likely the lower-ranked the committee's university is.

In Appendix F, I present three-round models that include waitlists. The reason for these waitlist models is that the university in this study is highly-ranked enough that it does not consider matriculation probabilities in Round 1. However, it still considers matriculation probabilities in terms of which applicants receive early vs. late acceptances. In Round 2 of these models, the committee does not accept or reject the surviving applicants. Rather, it early sends them an early acceptance, or it waitlists them. It then sees which of the early accepted applicants matriculate and which do not. Finally, in Round 3, it accepts or rejects the waitlisted applicants. The key dynamic is that the committee favors likely matriculants in Round 2, in addition to applicants with high expected success probabilities. It does so to avert a Round 3 situation in which it has barely filled up its class, applicants it could have accepted in Round 2 chose other schools, and now it has to accept weaker applicants. Because I do not have enough years' worth of data to identify this model's parameters, I present only estimation results on simulated data.

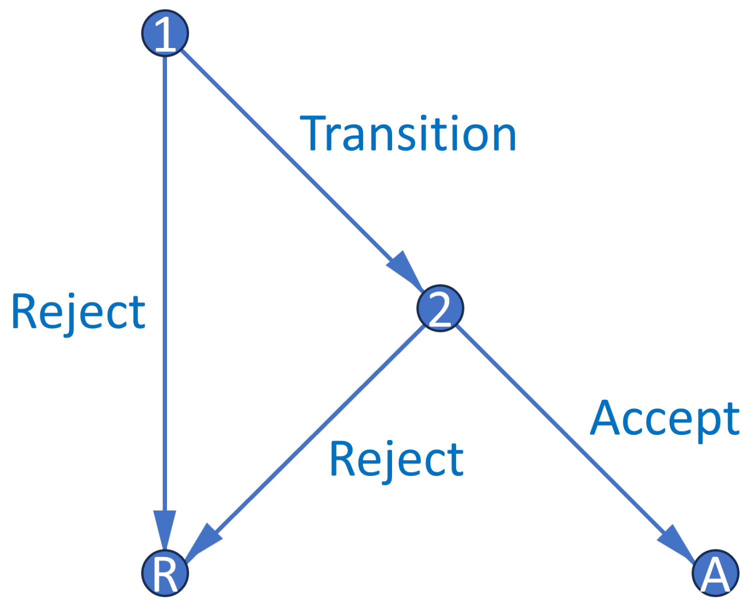


Figure 1: Decision Graph

4.1 Portfolio Choice to Cutoff Rule

Bai et al. (2022) prove that selecting the best N_A applicants out of N is equivalent to selecting the N_A whose success probability exceeds a cutoff.[8] Let x be a vector of objective variables (e.g. GRE scores), and z be a vector of subjective variables (e.g. variables from recommendation letters). Also, let d_n denote the accept or reject decision for the nth applicant, and y_n denote that applicant's outcome, where $S =$ success and $F =$ failure. The committee solves a constrained optimization problem, in which it maximizes the sum of the success probabilities of the accepted applicants under the constraint that it cannot accept more than N_A applicants. For simplicity, suppose there is one round in one year, and all accepted applicants matriculate:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A), \text{ such that } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \tag{1}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned} \mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\left\{N_A - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_n = R)\right]\right\} \\ &= \sum_{n=1}^{N} \left[P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N). \end{aligned} \tag{2}$$

[8] To relax the assumption that the committee only cares about success, its objective can be generalized to maximizing the sum of a convex combination of success probabilities and other factors, such as diversity.

Suppose without loss of generality that the applicants are sorted in decreasing order such that $P(y_1 = S|x_1, z_1) \geq \cdots \geq P(y_N = S|x_N, z_N)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^* \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$. That is, in this setup, λ^* is set identified.[9] Then the solution is to accept all applicants whose success probability exceeds λ^* and reject the rest, which means that λ^* is a cutoff probability:

$$\forall n \in \{1, \ldots, N\},\, d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^*. \end{cases} \tag{3}$$

If $P(y_n = S|x_n, z_n) = \lambda^*$, then the committee is indifferent between acceptance and rejection, which means that in the event of a tie at the margin, the committee will randomly accept enough applicants to reach N_A acceptances. Also, while λ^* is set identified in this setup, I show in the proof of Theorem 2 in the next subsection how to point identify it with admissions data by modeling a committee that compares each applicant's success probability to a single-valued cutoff, and then by exploiting the observed committee's exact selectivity. Finally, given a large number of applicants, λ^* can be interpreted as the marginal accepted applicant's success probability. Therefore, I denote it going forward as $P(y^* = S)$, or the success probability of that applicant.

This version of the portfolio choice problem contains just one round. But the committee's actual problem has two rounds: first-round filtering on x only, along with second-round final selections on x and z. In Appendix E.1, I generalize, for both the dynamic and static setups, the one-round portfolio choice problem to a two-round portfolio choice problem, and show that it also reduces to a two-round cutoff rule problem. The Round 2 cutoff rule problem is:

$$\max_{(d_{2,1},\ldots,d_{2,N_T}) \in \{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A. \tag{4}$$

Its solution with $\lambda_2^* = P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$ is:

$$\forall n \in \{1, \ldots, N_T\},\, d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^*. \end{cases} \tag{5}$$

I show in Appendix E.1 that under an independence assumption, the set-identified interval for λ_2^* collapses to a point, denoted $\overline{\lambda_2^*}$. Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n, z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\widetilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{6}$$

[9] If there were a continuum of applicants as in Bai et al. (2022)'s proof, the set's length would be 0, and so λ^* would be point identified. But all the models in this paper have N applicants, as there are in the data.

Its solution with $\lambda_1^* = L(y_1^* = S) \in \left[EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T})\right]$ is:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{7}$$

That said, it is possible to add the preference shocks directly to the portfolio choice problem and carry them through the derivation, in which case the result is exactly the cutoff rule problems in this section. For notational simplicity, I do not do so throughout this paper, but I prove in Appendix H that for the one-round portfolio choice problem without matriculation (i.e. the portfolio choice problem I first presented), the preference shocks can be carried through the derivation:

Theorem 1. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. See Appendix H.

4.1.1 Dynamic Model

Figure 3 portrays the dynamic model, whose notation is as follows. $r \in \{1,2\}$ is the round, x is a vector of objective variables, z is a vector of subjective variables, t is the year, $\bar{k}$ is this university's U.S. News program rank (k is the applicant's observed future program rank), and ε_r is a type 1 extreme value (T1EV) shock with scale σ. The shocks let the model fit the data, as a real-world committee will not always accept the applicants with the highest success probabilities.

But what do the shocks represent? Mechanically, ε_r is the part of the committee's decision that the observables $(x,z,t,\bar{k})$ do not explain. It is convenient to call ε_r a preference shock, but the label should not be read as pure idiosyncratic taste. The shock reflects my ignorance of what the committee conditioned on, not randomness in the committee's behavior, and is therefore a residual defined relative to the observables. As such, anything the committee responded to that is absent from the observables is folded into ε_r, which means that what ε_r contains is an empirical matter. The more of the committee's information I observe, the closer ε_r comes to genuine taste, and the more I do not observe, the more ε_r stands in for private information that the committee acted on. This gap matters most when such variables also predict success, since the model then omits determinants of success. I discuss private information and its implications in Section 5.

Starting in Round 2, the committee chooses $d_2 \in \{\text{Accept}, \text{Reject}\}$. If it accepts an applicant, it gets the applicant's success probability $P(y = S|x,z,t,\bar{k})$; the probability of succeeding at this university is a function of applicant characteristics, the year, and this university's quality. Whereas if the committee rejects the applicant, it gets the Round 2 cutoff probability $P(y_2^* = S|t)$, which is the success probability of t's marginal accepted applicant, whose success probability is conditional on $(x,z,t,\bar{k})$. There are various measures of success, from completing the program to obtaining a job placement above a threshold ranking. The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,\bar{k},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{8}$$

$$u_2(x,z,t,\bar{k},d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + P(y_2^*=S|t)\mathbb{1}(d_2=R).$$

Meanwhile in Round 1, the committee chooses $d_1 \in \{\text{Transition}, \text{Reject}\}$. If it transitions an applicant to Round 2, it gets $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$, which is the expected value of Round 2, where z is integrated out conditional on (x,t). However, if it rejects the applicant, it gets $L(y_1^*=S|t)$, or the expected Round 2 value of t's marginal transitioned applicant. Because this value can exceed 1, it cannot be interpreted as a probability. The Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{9}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T) + L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$
$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(\sum_{\tilde{d}_2 \in \{A,R\}} e^{u_2(x,\tilde{z},t,\bar{k},\tilde{d}_2)/\sigma}\right) dF(\tilde{z}|x,t).$$

Each round's conditional choice probabilities (CCPs) then have the closed-form expression:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\tilde{d}_r \in D_r} \exp(u_r(x,(z),t,\bar{k},\tilde{d}_r)/\sigma)}. \tag{10}$$

To estimate this model, the first step is to set up the unrestricted likelihood function. I parameterize the unrestricted CCPs (P_{d_1},P_{d_2}), success probability (P_y), and cutoffs ($L_{y_1^*}$,$P_{y_2^*}$) with the following flexible exponential and logistic specifications containing parameters θ:

$$P_{d_1} = P\big(d_1 = T\big|x,t;\theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1+\exp(f_T(x,t)\theta_T)}, \quad P_{d_2} = P\big(d_2 = A\big|x,z,t;\theta_A\big) = \frac{\exp(f_A(x,z,t)\theta_A)}{1+\exp(f_A(x,z,t)\theta_A)} \tag{11}$$

$$P_y = P\big(y=S\big|x,z,t,\bar{k};\theta_S\big) = \frac{\exp(f_S(x,z,t,\bar{k})\theta_S)}{1+\exp(f_S(x,z,t,\bar{k})\theta_S)} \tag{12}$$

$$L_{y_1^*} = P\big(y_1^*=S\big|t;\theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*}), \quad P_{y_2^*} = P\big(y_2^*=S\big|t;\theta_{S_2^*}\big) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1+\exp(f_{S_2^*}(t)\theta_{S_2^*})}. \tag{13}$$

Letting F_j be the conditional CDF for each z_j, I parameterize the integral in Equation (9) as:

$$\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\,\cdot\,\right] dF_1(\tilde{z}_1|\tilde{z}_2,\ldots\tilde{z}_J,x,t) \cdots dF_J(\tilde{z}_J|x,t). \tag{14}$$

Depending on whether each z_j is continuous, binary, or multinomial, I use either: $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$, $P(z_j=1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$, or $P(z_j=\ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}$ for $\ell = 1,\ldots,L_j-1$, respectively.

Let N_T be the number of N total applicants who survive to Round 2. Also, T = transition, A = accept, R = reject, S = success, and F = failure. Finally, z^c = *Committee Score*, whereas $z^{\neg c}$ = all the other subjective variables. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}$$
$$\prod_{n=N_T+1}^{N} \left(1 - P_{d_{1,n}}(\theta_T)\right) \prod_{n=1}^{N} f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (15)$$

I estimate the restricted likelihood by substituting the restricted CCPs for the unrestricted ones. To save computation time, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R). There is also a complication with the data, namely that *Committee Score* is unobservable for the $N - N_T$ of N applicants who were rejected in Round 1. The missing data pose a problem when estimating $f(z_n | x_n, t_n; \theta_F)$ in the likelihood because *Committee Score* is part of z. Because I cannot obtain *Committee Score* for everyone, I tried estimating the following Heckman-corrected likelihood:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \Phi(x_n, t_n; \theta_H) f\left(z_n | x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \left(1 - \Phi(x_n, t_n; \theta_H)\right)$$
$$f\left(z_n^{\neg c} | x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right) \left(1 - P_{d_{1,n}}(\theta_T)\right) \prod_{n=1}^{N} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (16)$$

The probit selection block is $\Phi(x_n, t_n; \theta_H)$, and the inverse Mills ratio is $\frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}$. Unfortunately, there are no available Heckman instruments that affect selection but not the distribution of z (year dummies affect z as well as selection). Therefore, identification must come from the nonlinearity of the inverse Mills ratio. This identification is weak, and the corrected results are nearly identical to the uncorrected ones, so the uncorrected likelihood is my preferred specification.

After estimating the unrestricted likelihood, I estimate the restricted likelihood $\mathcal{L}^R$ by substituting in the model's CCPs, which are functions of the success probability (θ_S) and structural $(\sigma, \theta_{S_1^*}, \theta_{S_2^*})$ parameters. Because I can observe whether the committee read each applicant's file and whether the committee accepted each applicant, I identify the cutoff parameters $(\theta_{S_1^*}, \theta_{S_2^*})$ for each year by how selective the committee is that year. Without loss of generality, every time $d_2 = A$, the term $\frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma) + \exp(P(y_2^*=S|t)/\sigma)}$ enters $\mathcal{L}^R$. This term decreases in $P(y_2^* = S|t)$, so every time an applicant is accepted, it puts downward pressure on $P(y_2^* = S|t)$, considering the objective is to maximize the likelihood. Intuitively, the committee's selectivity identifies the cutoff probability. Also, if the committee engages in sorting, or selecting the applicants with the highest success probabilities $P(y = S|x, z, t, \bar{k})$, that improves the likelihood. Formally:

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. See Appendix H.

When estimating this model, it is necessary to use public outcomes data to track success for all applicants, including those who did not matriculate at this university. Bai et al. (2022) prove that success parameter estimates will be biased toward zero if the sample is restricted to matriculants only. The reason is that all matriculants must have been accepted, and accepted applicants tend to have high values of the determinants of success. For example, conditional on matriculating, the relationship between the quantitative GRE score and success is weakened because nearly all matriculants have very high scores (analogous to height not predicting success in the NBA).

Along with tracking success for all applicants, it is also necessary to track which, if any, graduate program each applicant attended in order to avoid assuming that success depends only on applicant characteristics. For instance, a rejected applicant who succeeded at some other university might not have succeeded at this one, and so the committee may not have made a mistake by rejecting them. If a variable predicts success only through its effect on matriculation at a higher-ranked university, then this university should not use that variable in admissions. In estimation, I use each applicant's observed future program rank k in the success probability block and this university's fixed program rank in the restricted conditional choice probabilities.

In Table 3, I present the dynamic model's estimated cutoffs with 95% confidence intervals, as well as the estimated T1EV scale parameter σ and the AIC. While the Round 1 cutoffs are not probabilities, they and the Round 2 cutoffs show that the committee was similarly selective across years. $\hat{\sigma}$ is also small, meaning that the committee's decisions are relatively deterministic with respect to the variables in the application files. A larger set of the unrestricted and restricted parameter estimates is in Tables 16 and 17 of Appendix C. I also present estimation results on simulated data in Tables 26 and 27 of Appendix D, which illustrate features I embed in the simulated data-generating process, namely that the simulated committee's weights are more aligned with the success probability parameters in the optimal simulated dataset than the suboptimal one.

Table 3: Restricted Dynamic Cutoffs

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.3731	36.58%	
	(0.3067, 0.4539)	(29.81%, 43.92%)	
2016	0.3795	37.21%	
	(0.3025, 0.4761)	(29.20%, 45.99%)	
2017	0.4126	40.27%	
	(0.3563, 0.4778)	(34.52%, 46.30%)	
2018	0.3836	37.68%	
	(0.3271, 0.4497)	(31.79%, 43.97%)	
2019	0.3887	38.14%	
	(0.3268, 0.4624)	(31.89%, 44.81%)	
Sigma			0.0020
			(0.0013)
Observations = 2329	Parameters = 529	LL = -14131.0	AIC = 29319.9

4.1.2 Static Model

I now present the static model. Relative to the dynamic model, this model omits z in the Round 1 success probability specification, meaning that z's parameter estimates are inconsistent (though crucially, the fitted values are still correct). In addition, for any multiyear model, such as those in Appendix G, it is necessary for the Round 1 value function to be dynamically linked to the Round 2 value function, meaning that the static framework cannot work across years. However, this model has two advantages. First, it more accurately describes how real-world committees behave. Second, the Round 1 cutoff can be interpreted as a probability and compared to the Round 2 cutoff probability. The static model's Round 2 value function is the same as the dynamic's:

$$v_2(x,z,t,\bar{k},\varepsilon_2) = \max_{d_2\in\{A,R\}} u_2(x,z,t,\bar{k},d_2)+\varepsilon_2(d_2), \text{ such that:} \tag{17}$$

$$u_2(x,z,t,\bar{k},d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A)+P(y_2^*=S|t)\mathbb{1}(d_2=R).$$

But unlike in the dynamic model, if the committee transitions an applicant in Round 1, it gets $P(y=S|x,t,\bar{k})$, or the applicant's success probability unconditional on z, which approximates the expected Round 2 value in the dynamic model (see Appendix E.1) and may differ from $P(y=S|x,z,t,\bar{k})$. Meanwhile, if the committee rejects the applicant, it gets $P(y_1^*=S|t)$, or the success probability of t's marginal transitioned applicant, whose success probability is conditional on x only. $P(y_1^*=S|t)$ can be less or greater than $P(y_2^*=S|t)$ depending on how much the committee values its time vs. admitting the best possible applicants. The Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1)+\varepsilon_1(d_1), \text{ such that:} \tag{18}$$

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T)+P(y_1^*=S|t)\mathbb{1}(d_1=R).$$

The restricted CCPs are the same as in the dynamic model:

$$P(d_r|x,(z),t,\bar{k}) = \frac{\exp(u_r(x,(z),t,\bar{k},d_r)/\sigma)}{\sum_{\tilde{d}_r\in D_r}\exp(u_r(x,(z),t,\bar{k},\tilde{d}_r)/\sigma)}. \tag{19}$$

Rather than write out the specifications for the unrestricted CCPs, success probabilities, and cutoff probabilities, I reference Equations (11)–(13). That said, there are two differences from the dynamic framework. First, there are two estimated success probabilities (P_{y_1}, P_{y_2}), and second, the Round 1 cutoff is now a probability rather than a level. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N}\left(1-P_{d_{1,n}}(\theta_T)\right)$$
$$\prod_{n=1}^{N}\left[P_{y_n}(\theta_{S_1})P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)}\left[(1-P_{y_n}(\theta_{S_1}))(1-P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}. \tag{20}$$

The static model's estimated cutoff probabilities, $\hat{\sigma}$, and AIC are in Table 4, and the unrestricted and restricted estimates are in Table 18 and 19 of Appendix C (with simulated estimates in Tables 28 and 29 of Appendix D). Both rounds' cutoffs are probabilities,[10] and in all five years, the Round 2 cutoff is greater than the Round 1 cutoff. This ordering is expected because both forecasts are scaled to the same outcomes over the same applicant pool, Round 2 accepts about a tenth of that pool, and Round 1 transitions between one third and three fifths of it. The committee read 59% of the files in 2018 vs. 35–43% in the other years, but 2018's Round 1 cutoff is not lower than the other years' cutoffs because the committee's decisions by themselves identify only which applicants fell above and below the cutoff. The success probabilities are necessary to identify the cutoff's level, but they do so relatively weakly, as the confidence intervals show.

Table 4: Restricted Static Cutoffs

	R1 Cutoff Probability	R2 Cutoff Probability	Sigma
2015	29.36%	39.20%	
	(23.20%, 36.38%)	(32.75%, 46.05%)	
2016	29.17%	40.53%	
	(22.21%, 37.27%)	(33.55%, 47.92%)	
2017	37.06%	41.69%	
	(32.10%, 42.29%)	(34.83%, 48.90%)	
2018	38.12%	39.30%	
	(33.98%, 42.44%)	(31.80%, 47.33%)	
2019	34.04%	39.62%	
	(28.67%, 39.85%)	(32.54%, 47.16%)	
Sigma			0.0064
			(0.0035)
Observations = 2329	Parameters = 80	LL = -2992.0	AIC = 6144.1

The myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t, dependence parameters among the components of z, and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. The myopic estimates on actual data are in Tables 20 and 21 of Appendix C, and those on simulated data are in Tables 30 and 31 of Appendix D. The myopic model is the one model where the restricted AIC is worse than the unrestricted AIC, which is evidence that the committee is not myopic.

4.2 Portfolio Choice with Matriculation

In the portfolio choice problem in Section 4.1, as well as in Bai et al. (2022), there is an assumption that every accepted applicant will matriculate. But in practice, many accepted applicants do

[10] The cutoffs are with respect to restricted success probabilities. The average restricted success probability of the acceptance set is greater than its average unrestricted counterpart because the restricted model assumes the accepted applicants are likeliest to succeed. As such, the restricted success probability parameter estimates shift accordingly.

not matriculate and instead attend another university. For committees at the very top universities, matriculation probabilities may be less of a concern, as accepted applicants who do not matriculate likely do so for idiosyncratic reasons (e.g. choosing Harvard over MIT). But at the majority of universities, the accepted applicants with the highest success probabilities will often have relatively low matriculation probabilities, as the lower-ranked universities were likely safety options from the outset. The question then is whether the committee should select applicants based on matriculation probabilities as well as success probabilities, or whether it should still accept the applicants with the highest success probabilities irrespective of matriculation.

As such, I generalize the portfolio choice problem to incorporate matriculation probabilities. The committee now accepts the subset of applicants with the highest sum of success times matriculation probabilities subject to a capacity constraint on the expected number of matriculants. For each applicant n, the committee makes an acceptance or rejection decision given:

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n,z_n)P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A) \leq N_M, \tag{21}$$

where each applicant's matriculation decision m is to either matriculate or decline, $P(m = M|x,z)$ is each applicant matriculation probability, and N_M is the maximum number of matriculants. Due to the linearity of expectations, the objective function is equivalent to the expected sum of the success probabilities of the accepted applicants who matriculate. Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_n = A$, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}
\mathcal{L}(d_1,\dots,d_N;\lambda) &= \sum_{n=1}^{N} P(y_n = S|x_n,z_n)P(m_n = M|x_n,z_n)\mathbb{1}(d_n = A) \\
&\quad + \lambda\left\{N_M - \left[\sum_{n=1}^{N} P(m_n = M|x_n,z_n) - \sum_{n=1}^{N} P(m_n = M|x_n,z_n)\mathbb{1}(d_n = R)\right]\right\} \\
&= \sum_{n=1}^{N} \big[P(y_n = S|x_n,z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\big]P(m_n = M|x_n,z_n) \\
&\quad + \lambda\left[N_M - \sum_{n=1}^{N} P(m_n = M|x_n,z_n)\right].
\end{aligned} \tag{22}$$

From the Lagrangian, it is clear that if there is a cutoff rule solution, the solution is on the success probabilities alone. While applicants with higher matriculation probabilities are more valuable in terms of the objective, they are also more expensive in terms of the capacity constraint on the number of matriculants. Intuitively, if the committee were to prioritize applicants with higher matriculation probabilities, it would not be able to give as many acceptances.

Moreover, the only way to get a cutoff rule solution that incorporates matriculation probabilities would be to have the constraint be on the number of acceptances. That is, the constraint would be $\sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$, the Lagrangian would equal $\sum_{n=1}^{N} \big[P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)\mathbb{1}(d_n = A) + \lambda \mathbb{1}(d_n = R)\big] + \lambda(N_A - N)$, the applicants would be ranked relative based on $P(y_n = S|x_n, z_n)P(m_n = M|x_n, z_n)$, and likely matriculants would receive more acceptances.

Crucially, however, because not every accepted applicant will matriculate, this exact problem does not have a cutoff rule solution. For example, suppose $N = 2$, $N_M = 1$, $P(y_1 = S|w_1) = 0.9$, $P(m_1 = M|w_1) = 0.5$, $P(y_2 = S|w_2) = 0.89$, and $P(m_2 = M|w_2) = 0.89$. Since $0.5 + 0.89 > 1$, the committee can accept only one applicant. The cutoff rule solution would be to accept Applicant 1, but Applicant 2 would yield a higher value of the objective since $0.89 \times 0.89 > 0.9 \times 0.5$. But fortunately, any exceptions to the cutoff rule solution disappear once N becomes sufficiently large; that is, when there is a continuum of applicants as in Bai et al. (2022)'s setup.

Theorem 3. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. See Appendix H.

Given this result, it appears that committees should not factor in matriculation probabilities when making admissions decisions. In a one-round model, that is the case. However, in a two-round portfolio choice model with matriculation, the cutoff rule is such that the committee should consider matriculation probabilities in Round 1. This rule holds in particular for lower-ranked programs. For example, if a lower-ranked program's committee chooses to read only the files of applicants with perfect GRE scores, then it may have trouble filling its cohort in Round 2, considering that the best applicants are unlikely to matriculate. And even if the committee can fill its cohort, that cohort is likely to be filled with applicants who have weak subjective variables. In contrast, if the committee had considered matriculation probabilities in Round 1, it may have transitioned applicants with lower GRE scores but strong subjective variables who will actually matriculate. I provide the full derivation in Appendix E.2, but I present the dynamic portfolio choice problem and cutoff rule solution here. The Round 2 problem is:

$$\max_{(d_{2,1},\ldots,d_{2,N_T}) \in \{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M. \tag{23}$$

Edge cases aside, its solution with $\lambda_2^* = P(y_2^* = S)$ is:

$$\forall n \in \{1,\ldots,N_T\},\, d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{24}$$

Let $EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n, \widetilde{z}_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|x_n, \widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$, where is $\overline{\lambda_2^*}$ is the collapsed interval for λ_2^*. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n) \mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{25}$$

Edge cases aside, its solution with $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1, \ldots, N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{26}$$

As I show in Appendix E.2, $EV_{2,n}(x_n)$ puts considerably more weight on success than matriculation. Therefore, the matriculation-dynamic and dynamic models have similar decision rules, and the best static approximation for the matriculation-dynamic model is the static model.

4.2.1 Matriculation Models

Because matriculation probabilities do not matter in Round 2, the matriculation-dynamic and static Round 2 value functions are the same as the previous section's Round 2 value functions:

$$v_2(x, z, t, \bar{k}, \varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \bar{k}, d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{27}$$

$$u_2(x, z, t, \bar{k}, d_2) = P(y = S|x, z, t, \bar{k}) \mathbb{1}(d_2 = A) + P(y_2^* = S|t) \mathbb{1}(d_2 = R).$$

However, the Round 1 value function contains matriculation probabilities:

$$v_1(x, t, \bar{k}, \varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \bar{k}, d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{28}$$

$$u_1(x, t, \bar{k}, d_1) = \mathbb{E}[v_2(x, z, t, \bar{k}, \varepsilon_2)|x, t, \bar{k}] \mathbb{1}(d_1 = T) + L(y_1^* = S, m_1^* = M|t) \mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \bar{k}, \varepsilon_2)|x, t, \bar{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{[P(y=S|x,\widetilde{z},t,\bar{k}) - P(y_2^*=S|t)]/\sigma} + 1\right) P(m = M|x, \widetilde{z}, t) dF(\widetilde{z}|x, t).$$

The derivation of the Round 1 value function is in Appendix E.2. The object I denote by $\mathbb{E}[v_2(x, z, t, \bar{k}, \varepsilon_2)|x, t, \bar{k}]$ is the Round 1 transition payoff, as opposed to the literal conditional expectation of v_2. As such, the resulting social surplus function contains $e^{[P(y=S|x,\widetilde{z},t,\bar{k}) - P(y_2^*=S|t)]/\sigma} + 1$ instead of the usual $e^{P(y=S|x,\widetilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma}$. Due to the matriculation term $P(m = M|x, \widetilde{z}, t)$, the incorrect latter expression does not result in the same ordering of applicants as the correct former expression. The restricted CCPs are the same as in the dynamic and static models:

$$P(d_r|x, (z), t, \bar{k}) = \frac{\exp(u_r(x, (z), t, \bar{k}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x, (z), t, \bar{k}, \widetilde{d}_r)/\sigma)}. \tag{29}$$

Rather than write out the specifications for the unrestricted CCPs, success probability, and cutoff probabilities, I reference Equations (11)–(13). But the specification for the matriculation probability is: $P_m = P\big(m = M \big| x, z, t; \theta_M\big) = \frac{\exp(f_M(x,z,t)\theta_M)}{1+\exp(f_M(x,z,t)\theta_M)}$. The unrestricted likelihood is then:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1-P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)} \prod_{n=1}^{N_T} f\big(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\big) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}$$
$$(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} \prod_{n=N_T+1}^{N} \big(1-P_{d_{1,n}}(\theta_T)\big) \prod_{n=1}^{N} f\big(z_n^{\neg c} | x_n, t_n; \theta_F\big) P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (30)$$

As with the restricted dynamic likelihood, I use (θ_F^U, θ_S^U) as an initial guess of (θ_F^R, θ_S^R) for the restricted matriculation-dynamic likelihood. Also, a Heckman correction can be implemented for $f_{z|x,t}$ because *Committee Score* is missing for the applicants rejected in Round 1, but due to the lack of an available Heckman instrument, I do not implement one in my preferred specification.

The matriculation-dynamic model's estimated cutoffs, $\hat{\sigma}$, and AIC are in Table 5, and the unrestricted and restricted estimates are in Table 22 and 23 of Appendix C (with simulated estimates in Tables 32 and 32 of Appendix D). The Round 1 cutoffs are much lower in the matriculation-dynamic model than in the dynamic model, but that is because of the expression for $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$. Both rounds' cutoffs and σ are well-identified in all five of the years.

Table 5: Restricted Matriculation-Dynamic Cutoffs

	R1 Cutoff Level	R2 Cutoff Probability	Sigma
2015	0.0071	36.35%	
	(0.0026, 0.0194)	(29.47%, 43.84%)	
2016	0.0071	38.25%	
	(0.0028, 0.0180)	(31.05%, 46.00%)	
2017	0.0105	40.65%	
	(0.0041, 0.0267)	(35.15%, 46.39%)	
2018	0.0095	36.59%	
	(0.0038, 0.0240)	(30.01%, 43.71%)	
2019	0.0095	37.41%	
	(0.0035, 0.0255)	(31.04%, 44.26%)	
Sigma			0.0020**
			(0.0010)
Observations = 2329	Parameters = 568	LL = -14260.9	AIC = 29657.8

Nevertheless, discussions with members of this university's committee indicate that the committee does not consider matriculation probabilities in Round 1 because this university is ranked highly enough that in general, it can fill its cohort with applicants who have strong objective and subjective variables. However, this university has at times considered matriculation probabilities when making admissions decisions; it would ignore matriculation probabilities entirely only in a world of perfect information about each applicant's intention. Suppose, for example, that every time the committee accepts an applicant, it immediately hears back from that applicant whether the applicant will matriculate. Then in Round 2, the committee would start with the applicant with the highest success probability, immediately hear back, proceed to the applicant with the second-highest success probability, and continue until the cohort is filled.

But in the real world, committees do not immediately hear back from applicants. Therefore, if a committee does not consider matriculation probabilities when giving early non-waitlisted acceptances, by the time it hears back from the accepted applicants (most of which will not matriculate since success and matriculation probabilities are negatively correlated), relatively good applicants who would have matriculated will have moved on to other universities. As a result, the committee may have to settle for weaker applicants than if it had considered matriculation probabilities. This dynamic motivates the models of the waitlist process in this section.[11] Because this dynamic is beyond the scope of the portfolio choice problem to cutoff rule framework, I do not write a three-round portfolio choice problem with this dynamic. Rather, the waitlist models extend the existing Bellman equations to three rounds and incorporate the dynamic by endogenizing the cutoff probabilities to make them a function of matriculation probabilities. More details about the waitlist models, as well as estimation results on simulated data, are in Appendix F.

5 Optimality Test

Suppose the committee adopts the dynamic, static, myopic or matriculation-dynamic model's decision rule. Will the average success probability of the acceptance or matriculation set increase, and if so, by how much? To answer these questions, I solve the model with $\sigma = 0$ in the conditional choice probabilities (CCPs), compute the average success probabilities of the model's acceptance or matriculation set, and compare the probabilities to those of the committee's acceptance or matriculation set. When I solve the model, I plug in the unrestricted success probability parameter estimates, not the restricted estimates. The reason is that the restricted estimates may be biased to rationalize suboptimal behavior (see Anderson, Rosen, Rust, and Wong 2025). For example, if the committee puts too much weight on the quantitative GRE score, then its restricted parameter estimate will be biased up since the restricted CCPs assume optimality.

For the dynamic model, I rank the applicants in each year by the expected Round 2 social surplus function $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\tilde{z}|x,t)$,[12] transition the number of transitioned applicants in the data, rank the survivors by $P(y=S|x,z,t,\bar{k})$, and accept the number of accepted applicants in the data. In contrast, for the static model, I rank the applicants in Round 1 by $P(y=S|x,t,\bar{k})$. After that, the algorithm is the same as the dynamic model's. Once I have determined which applicants each model accepts, I compute their average success probability, pooled across years, and compare it to the average success probability of the applicants that the committee's rule selects. Determining which applicants the committee's

[11] In addition, the committee can adjust the number of late acceptances it gives based on how many early matriculants it receives. Doing so helps it attain the target number of matriculants not only in expectation but in reality.

[12] For the myopic model, the expected social surplus is $\int_{\tilde{z}} \sigma \log\left(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}\right) dF(\tilde{z}|t)$.

decision rule selects is straightforward. In Round 1, I rank the applicants by $P(d_1 = T|x,t)$, and in Round 2, I rank the survivors by $P(d_2 = A|x,z,t)$. The number of acceptances is the same.

For the matriculation model, the algorithm is more complicated because the purpose of these models is to evaluate the committee's performance based on the applicants who matriculate at the university. Accepting strong applicants who do not matriculate due to getting offers from higher-ranked university is not accomplishing anything. For the matriculation-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\left(e^{[P(y=S|x,\tilde{z},t,\bar{k}) - P(y_2^*=S)]/\sigma} + 1\right) P(m=M|x,\tilde{z},t) dF(\tilde{z}|x,t)$, transition the number of transitioned applicants in the data, rank the surviving applicants by $P(y=S|x,z,t,\bar{k})$, and accept them in descending order, each matriculating with probability $P(m=M|x,z,t)$ until the number of matriculants in the data matriculate. I then compute the average success probability of the matriculation set across matriculants and across 500 simulations, and compare it to that of the committee's set. As I explain in Section 4, the best static approximation of the matriculation-dynamic model is the static model, so I also run the matriculation set's algorithm for the static model's optimal decision rule.

Suppose I find deviation gains. Could committees have incorrect beliefs about which applicants are most likely to succeed? Also, if so, can they do a better job admitting applicants by changing the weights they put on some of the variables? If there are gains in the dynamic, static, or myopic models, the matriculation rate of the model's (stronger) acceptance set may be less than that of the committee's (weaker) acceptance set. In that case, the estimated gains would be an upper bound. However, the matriculation model accepts those liklier to matriculate to prevent losing good applicants with weaker observables by rejecting them before reading their files.

In addition, the test can credit the committee only for information I observe. For example, the committee may personally know a recommendation letter writer, or the letter may contain information that algorithms cannot detect, such as exactly how rigorous an applicant's masters program was. The test evaluates both the committee's and the model's acceptance or matriculation set with the same measure, the unrestricted success probability $P(y = S|x,z,t,\bar{k})$, which is estimated from the observables. If the committee acts on a success-predicting variable I do not observe, the measure cannot detect that variable, so any edge the committee's private information provides goes unmeasured. The end result would be an overstated deviation gain.

The protection against this limitation is the comprehensiveness of the data. The application files I observe are the same files the committee read, and from those files, I construct (in addition to the objective variables x) measures of mathematical preparation, the substance of the application essay, the content and tone of the recommendation letters, and the standing of the recommenders. Most importantly, because z includes the committee score, conditioning the suc-

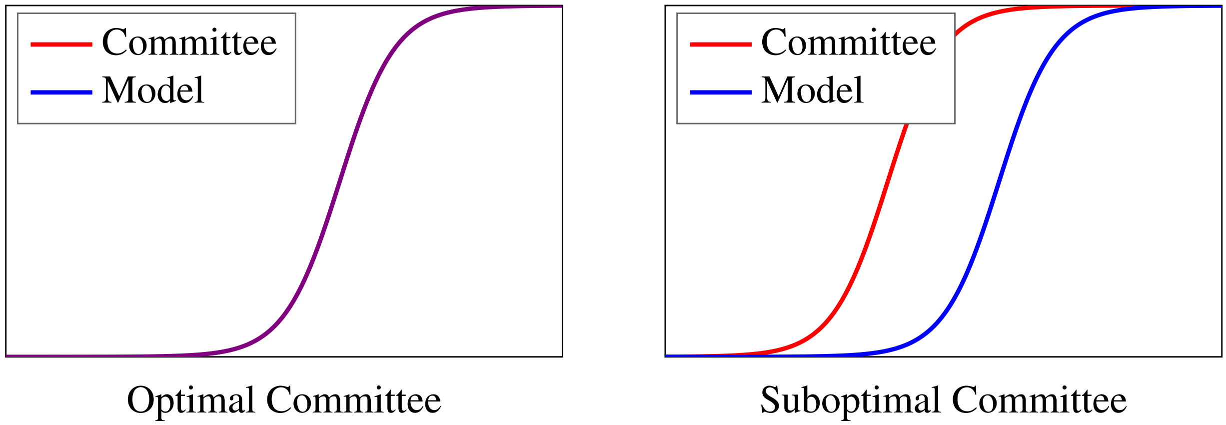


Figure 2: Deviation Gains CDFs

cess probability on z absorbs much of the residual private information.[13] In this way, I effectively merge human and data-driven approaches. While humans have access to more information and can use their intuition to evaluate less quantifiable characteristics, they may also fall prey to human biases that models do not. Nonetheless, no set of observables is guaranteed to span the committee's information set, so the deviation gains test is best read as conditional on the observed variables closely reconstructing what the committee saw. That said, the realized success rates of the committee's observed acceptance and matriculation sets are no higher than the average success probabilities of the committee's decision rule's acceptance and matriculation sets, meaning that any private information the committee had did not measurably help the committee.

A separate consideration is sampling error. There are over 30 success probability parameters, so it is possible for the models to overfit the data. Ideally, I would test the performance of each model's decision rule on a holdout set. For example, I could use 2015–2018 success probability parameter estimates to select the model's acceptance or matriculation set for 2019, compute the average success probability of that set with 2019 success probability parameter estimates, and repeat this procedure holding out each of 2015, 2016, 2017, and 2018. However, there are too few observations in any single year (e.g. 2019) to reliably estimate the success probability parameters. Even a three years of training, two years of testing split's sample sizes are too small. Therefore, I follow Anderson et al. (2025) by using a randomization test to address sampling error.

If I assume for simplicity that the model is a two-round model that chooses the same transition set of applicants as the committee, then the following inequality holds:

Theorem 4. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the*

[13]In Table 10 of Appendix B, *Missing Committee Score* survives the lasso with a negative sign (and has an insignificant AME of -0.0277 in Table 2b of Section 3). As such, application variables equal, applicants who were filtered out in Round 1 might have been less likely to succeed. In Round 1, the committee could have observed each applicant's country of origin and undergraduate institution, as well as their letter writers' names. Aside from that, it did not observe anything not in my dataset, at least not until Round 2, where my dataset contains *Committee Score*.

committee's decision rule. By optimality:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}. \quad (31)$$

Proof. See Appendix H.

In this inequality, I am careful to assume that the committee's Round 1 decision rule is the same as the model's. Without this assumption, it is possible in rare cases for the sign to flip if, for example, the committee favors high-upside applicants in Round 1, and it turns out that favoring such applicants ultimately helps it attain a better set of accepted applicants (see Appendix H).

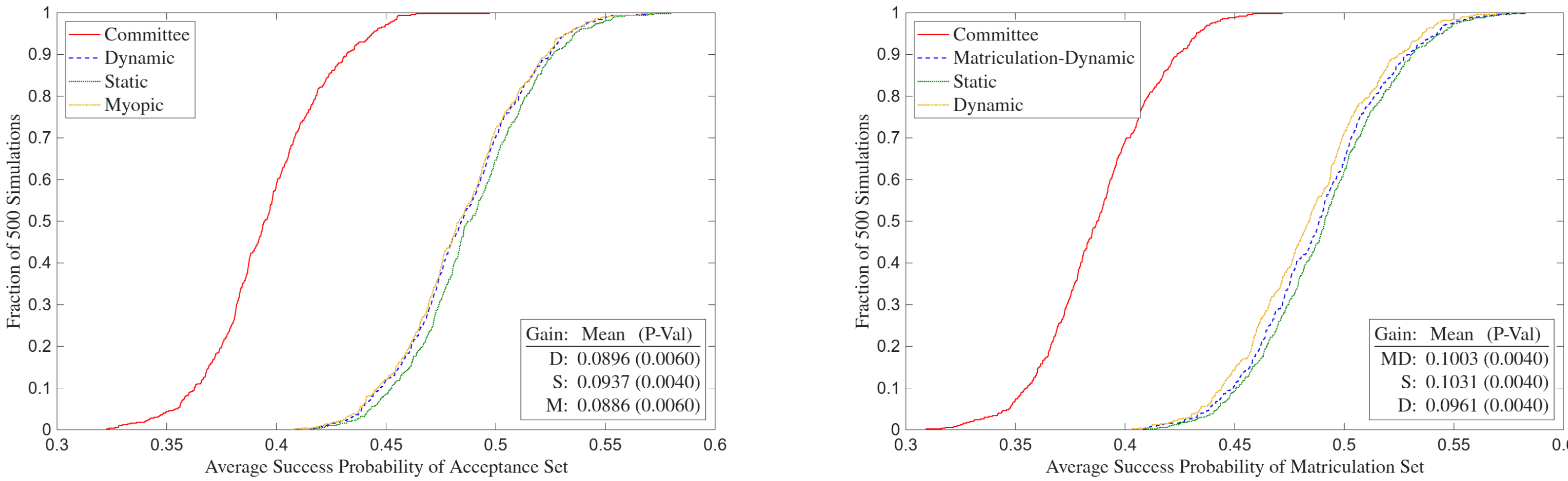
Regardless, for a perturbation $\widetilde{\theta_{S,t}^U}$ about $\hat{\theta}_S^U$'s asymptotic distribution, the sign can flip:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$< \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}. \quad (32)$$

Given that the sign can flip if the committee optimizes or comes close to it, I can use such perturbations of the success parameters to test whether the deviation gains are robust to sampling error. Following Krinsky and Robb (1986), across $B = 500$ perturbations, I plot the CDFs of the average success probabilities of the acceptance and matriculation sets of the dynamic, static, myopic, and matriculation-dynamic models, as well as those of the sets implied by the committee's decision rule. While the acceptance and matriculation sets are selected with the actual parameter estimates, the average success probabilities are computed with the perturbed estimates.

Figure 2 illustrates what the CDFs look like in general. When a committee switches from a suboptimal decision rule to the optimal decision rule, the actual decision rule's CDF shifts right to match the model's CDF. I provide a demonstration on the simulated data, where per Figures 7 and 8 in Appendix D, in the dataset where the committee is optimizing, the models' CDFs are virtually identical to the committee's. However, in the dataset where the committee is not optimizing, all the models' CDFs stochastically dominate the committee's CDF.

In addition, I prove that with a large number of perturbations, the fraction of perturbations in which the committee's set has a weakly higher average success probability than the model's set is a p-value for the null hypothesis that the committee's average success probability is at least as high as the model's. This proof defines a p-value in the standard way; if the null hypothesis is true, the p-value is the probability of getting a positive deviation gain at least as large as the one I observe. As the test should, it does not reject equality of the models' CDFs vs. the committee's CDF in the optimal simulated dataset, but it always rejects equality of the CDFs at the 1% level in the suboptimal dataset. Formally, the theorem that the fraction of perturbations is a p-value is:



(a) Acceptance Set

(b) Matriculaton Set

Figure 3: Deviation Gains

Theorem 5. *For large B, the fraction of* $\tilde{\theta}^U_{S,b}$ *with gain* $g(\tilde{\theta}^U_{S,b}) \leq 0$ *is a p-value for* $H_0\colon g_0 \leq 0$.

Proof. See Appendix H.

Figure 3 reveals that the average success probability of the model's acceptance set is about 9 percentage points higher than that of the committee's, and the average success probability of the model's matriculation set is about 10 percentage points higher. There is little variation across models, though as expected, the myopic model is slightly worse than the dynamic model for the acceptance set, and the dynamic model is slightly worse than the matriculation-dynamic model for the matriculation set. The p-values are also statistically significant at the 1% level.

Given that there are meaningfully large deviation gains, the next step is to determine which variables are responsible for the gains. For simplicity, I work with the static models for this attribution exercise. I start with the Rounds 1 and 2 conditional choice probability parameter weights for each variable. One variable at a time, I hold all the other variables' weights fixed and choose the Rounds 1 and 2 weights that maximize the average success probability of the acceptance or matriculation set. Once those weights are chosen, I recalculate the average success probability using the 500 perturbations and plot the results in Figure 4. The bar shows the average of those average success probabilities across the perturbations, and the whiskers extend to the 5th and 95th percentiles.[14] Also, the bars are dark blue for previously-underweighted variables where the reweighting increases their average in the acceptance or matriculation set. In contrast, the bars are pink for previously-overweighted variables where the reweighting decreases their average. Among the objective and subjective variables, a variable qualifies for inclusion in the figure if

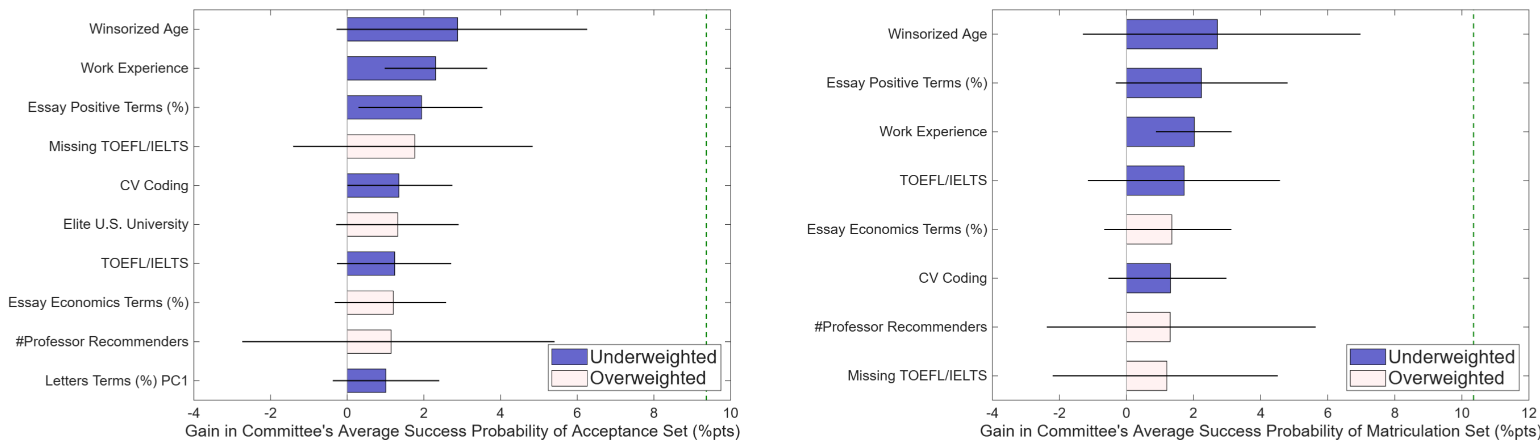
[14]Unlike a 95% confidence interval, the whiskers do not extend to the 2.5th and 97.5th percentiles, as the null hypothesis of the committee's average success probability being at least as high as the model's is one-sided.

its bar extends to at least one percentage point. For each of the variables in Figure‘4, Table 24 of Appendix 4 contains their average values for the full sample, the committee's acceptance or matriculation set, the single-variable optimized set, and the joint optimized set.

The bars are similar for both the acceptance and matriculation sets; all bars for the matriculation set are present for the acceptance set, only two bars are present for the acceptance set only, and no bars change colors across sets. Starting with the matriculation set, the bars fall into four categories. The first and most pronounced category is experience; older applicants and those with work experience are more likely to succeed, and the committee does not give them enough credit. The second category is English proficiency. Applicants who score low on the TOEFL/IELTS are less likely to succeed. Moreover, the applicants with the best signal are non-native speakers (i.e. those with *Missing TOEFL/IELTS* $= 0$) who nonetheless scored high on the exam. The third category is the subtext of the application essays. Applicants with a high percentage of positive words in their essays are more likely to succeed, and those with a high percentage of economics words (i.e. words like microeconomics, macroeconomics, econometrics, etc.) are less likely to succeed. Finally, the fourth category relates to applicants from with more of an industry background. In particular, applicants with a high number of coding languages on their CVs and with at least one non-professor recommender are undervalued. That said, *#Professor Recommenders* and *Missing TOEFL/IELTS* have wide whiskers, so they should be discounted accordingly. Finally, *Elite U.S. University* is an interesting case, as it shows up for the acceptance set but not the matriculation set. Applicants from an elite U.S. university are more likely to be accepted and not more likely to succeed.[15] However, they are also less likely to matriculate if accepted, so accepting them in greater numbers does not hurt the matriculation set's average success probability.

There are two additional points worth making. First, as expected, there is substantial overlap between the variables that survive the lasso in the sixth column of Table 10 in Appendix B and the variables that show up in Figure 4. Second, the AMEs in Table 2b of Section 3 hint at the attribution results. For example, *Work Experience* has an AME of 0.0140 for *Accept* vs. 0.0358 for *Success*, and Figure 4 shows that it is underweighted. However, there are other variables in Table 2b whose *Accept* AMEs do not line up with their *Success* AMEs, yet are neither under nor overweighted. For example, *Female* has a positive *Accept* AME and *International Asian* has a negative *Accept* AME, both variables have insignificant *Success* AMEs, and neither variable shows up in Figure 4. The reason is that there are enough strong applicants with *Female* $= 1$ and *International Asian* $= 0$ that the committee's decision rule does not affect the average success probability of the acceptance or matriculation sets. So if these two *Accept* AMEs resulted from a diversity objective, that objective was nonetheless also consistent with a success probability-maximizing objective. In contrast, *CV Coding* has a negative *Accept* AME, an insignificant *Success* AME, and it shows up as undervalued in Figure 4. There is no way to know from reading

[15] Bai et al. (2022) find that attending an Ivy Plus undergraduate institution predicts success. However, unlike their university, this paper's university is less likely to receive applications from the strongest Ivy Plus students.



(a) Acceptance Set

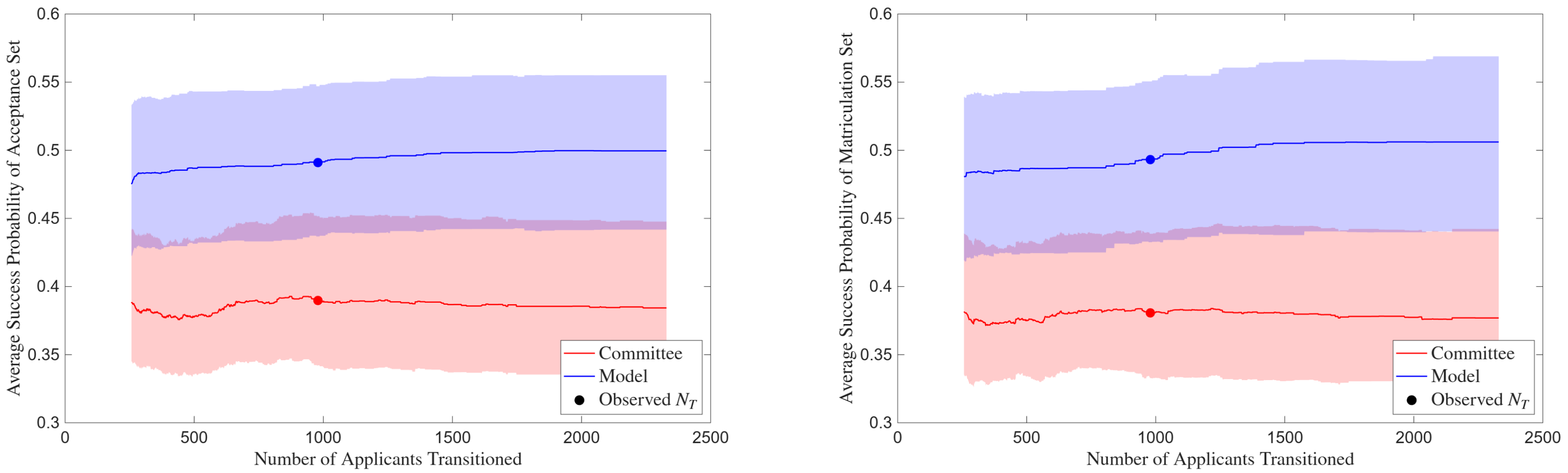
(b) Matriculaton Set

Figure 4: Deviation Gains Attribution

Table 2b that it would work out that way, which demonstrates why a model is necessary to relate acceptance AMEs to success AMEs when drawing any conclusions about optimality.

Table 2b also shows that *GRE Quantitative Percentile*'s AME is positive and significant until I control for *Program Rank* and *Missing Program Rank*, at which point it is no longer significant. Controlling for *Program Rank* is necessary since the quantitative GRE may only be positively correlated with success because a high percentile is often required to get into a good PhD program, which in turn is often required to attain a good placement. In that case, it would be incorrect to recommend selecting based on the quantitative GRE. However, *GRE Quantitative Percentile*'s insignificant AME does not necessarily mean that selecting on it is incorrect. For instance, this university filters out applicants with low percentiles. It is possible that despite the insignificant AME, such applicants would not do well at this university. The reason is that there are very few applicants at top PhD programs with low percentiles, so the data cannot identify well what would happen if someone with a low percentile attended a top program. Also, Figure 4 shows that there is no evidence that the committee is overweighting *GRE Quantitative Percentile*.

Finally, in Figure 5, I present the results from a counterfactual in which I allow N_T, which is the number of files the committee chooses to transition to Round 2, to vary within each year. Assuming the committee reads every accepted applicant's file, the minimum value of N_T is N_A. And the maximum value of N_T is the total number of applicants N. This counterfactual seeks to answer the question of whether reading more files would help the committee attain an acceptance or matriculation set with a higher average success probability. I find little meaningful evidence that reading more files helps. The committee's line is effectively flat across files, and while the model's line is slightly increasing and concave (indicating positive but diminishing returns to reading more files), the increase in files read all the way from N_A to N_T only slightly increases the average success probabilities. Therefore, reading more files is less effective than reweighting



(a) Acceptance Set

(b) Matriculaton Set

Figure 5: Benefit of Reading More Files

the information attained from the objective variables and the files already read.

6 Conclusion

This paper studies how decision-makers select among options whose true quality is uncertain and costly to evaluate. Graduate admissions is a natural setting for such a study, as it features a low-cost filtering process followed by a high-cost review of the application files that survive the filter. Because not all accepted applicants matriculate, the committee must also consider matriculation probabilities when making admissions decisions. By structurally modeling the committee's decision problem, I develop the first framework in the economics graduate admissions literature that links application information to admissions decisions and post-program success. This framework makes optimality testable, and gains from deviating to the optimal decision rule quantifiable.

Before estimating structural modes, I start with regressions of acceptance, matriculation, and success dependent variables on information in application files from a research university's economics PhD program in 2015–2019. I find that different independent variables best predict each dependent variable. The 0 to 10 score the committee gives every applicant whose file it read best predicts acceptance, and attending the university's open house best predicts matriculation if accepted. I use program completion and placement rank to measure success, and my primary success measure is a binary variable that equals 1 if the applicant completed a PhD program and attained a top 100 or top 100-equivalent placement. The best predictor of success is the rank of the PhD program the applicant attended, though other variables like age, work experience, and TOEFL/IELTS scores also predict success. To avoid selection bias, I use publicly-available information from the Internet to track PhD program and placement oucomes for all applicants, not just those who matriculated at this university. I also regress acceptance on application file variables using a second dataset of this university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023, and I find that economics admissions is more data-driven, based on variables such as GRE scores, than admissions in the other programs.

There are papers in the graduate admissions literature dating back to Krueger and Wu (2000) that qualitatively compare the determinants of acceptance to those of success, but I show that while such comparisons can provide useful information, they are insufficient to determine whether an admissions committee is optimizing. If I were to test whether the average marginal effects (AMEs) of variables in application files, such as GRE scores, are the same in an acceptance regression vs. a success regression, I would not be testing optimality. As I show at the end of Section 3, equal AMEs are neither necessary nor sufficient for optimization. Therefore, any optimality tests must be based on structural models of the committee's selection problem that account for evaluation costs and multi-stage decision-making. For this reason, I develop and estimate such models of the committee's problem in order of increasing complexity.

I start with two-round dynamic, static, and myopic models based on cutoff rules. In these models, there is a Round 1 filter based on only objective variables like GRE scores, followed by a Round 2 final decision based on objective and subjective variables, such as those constructed by textual analysis of recommendation letters. I then develop a two-round matriculation-dynamic model, in which not every accepted applicant matriculates, and the committee considers matriculation probabilities in Round 1. The reason for doing so is to ensure that the committee can fill its cohort with applicants who do not have particularly weak subjective variables. I derive all these models from dynamic portfolio choice problems that the committee faces, and in Section 4 and Appendix E, I solve for the cutoff rule solutions to those problems. I estimate the models on this university's data and back out the committee's Rounds 1 and 2 cutoffs in each year.

In Appendix F, I develop three-round models that I estimate on simulated data, since five years of actual data are too few for identification. Round 1 in these models is the same as in the dynamic and static models without matriculation, but in Round 2, the committee can accept or waitlist the survivors of Round 1. Finally, in Round 3, the committee gives additional acceptances off the waitlist and rejects the remaining waitlisted applicants. In Round 2, the committee is more likely, all else equal, to give early acceptances to likely matriculants. The reason is to ensure that it does not have to give too many late acceptances to weaker applicants from the waitlist. Ideally, a commitee would be able to uncover variables that predict matriculation without also predicting failure. This situation is analogous to the small-market Oakland Athletics baseball team in *Moneyball* using analytics to find good players *they could afford* (Lewis 2004).

After estimating the structural models, I use the estimates to compare each two-round model's decision rule to the committee's decision rule. For each model, I simulate the acceptance and matriculation processes, the latter based on estimated matriculation probabilities, to attain two acceptance sets and two matriculation sets: one each for the committee's decision rule and one each for that model's decision rule. I select the model's acceptance and matriculation sets using the unrestricted success probability parameter estimates, but to guard against sampling error, I calculate the average success probability of each set using perturbations of the estimates about their asymptotic distribution. I also prove that the fraction of perturbations where the model's set's average success probability does not exceed the committee's is a p-value for the null hypothesis that the model's set's average success probability is no greater than that of the committee. In a simulated dataset where the committee is optimizing, this randomization test does not reject, whereas in a different simulated dataset where the committee is making mistakes, the test rejects.

I find that the model's acceptance and matriculation sets exhibit statistically significant deviation gains of 9–10 percentage points relative to the committee's sets. For example, the average success probability of the committee's matriculation set is 37%, whereas that of the model's matriuclation set is 47%. These gains can be attributed to experience (in the form of *Age* and *Work Experience*, English proficiency (*TOEFL/IELTS* scores), application essay subtext (a higher rate of positive terms predicts success, and a higher rate of economics jargon terms predicts failure), and industry backgrounds (*CV Coding* predicts rejection but not failure). However, it is important to note that a gain of 9–10 percentage points would be unlikely in practice. The determinants of success may have changed since 2015–2019, and the randomization test is not fully robust to imprecise success probability parameter estimates yielding misleading conclusions about which variables predict success. I am developing a procedure to correct upward bias in deviation gains resulting from any such overfitting and will implement it in a future version of this paper.

I also run a counterfactual in which I allow the number of files the committee chooses to read in Round 2 to vary. The goal is to determine whether reading more files will result in a higher average success probability of the acceptance and matriculation sets. However, I find that there is no evidence that reading more files would benefit the committee. It would benefit the model following the optimal decision rule, but still not by much, and there are diminishing returns to reading more files. In a different setting, however, where there is evidence that reading more files would benefit the committee, I could use committee's acceptance or matriculation sets in a given year to put a dollar value on success in that year. Assume that it takes h hours for a committee member to read an application file, and also that each committee member makes an hourly wage of w. Therefore, reading n files has a monetary cost of $c = n \times h \times w$. If I were to plot c vs. average success probability, then assuming positive but diminishing returns to reading more files, the plot would be increasing and convex. If N_T is the number of files the committee actually reads, and I assume the committee is reading the optimal number of files, then c divided by the average success probability attained at N_T files is the value the committee puts on success.

In addition to its contributions to the graduate admissions literature and to literatures on dynamic selection and costly evaluation, this paper contributes to the literature on synthetic data and privacy protection. Recent economics papers use synthetic data provided by the U.S. Census Bureau, but this paper is the first to design and implement a comprehensive synthetic data protocol from the construction of the synthetic data to the final estimation results. As a result, it serves as a template for researchers who wish to work with confidential data that they are not allowed to access directly, which can help them make discoveries they could not have otherwise.

More broadly, this paper provides a tractable framework for studying selection under costly evaluation. By doing so, it advances the graduate admissions literature from identifying variables that predict acceptance and success to testing whether decision rules are optimal conditional on time and effort constraints. This study is based on a single institution, so the specific empirical results reflect that setting. However, the framework appllies broadly to environments where due to evaluation costs, decision-makers must employ a multi-stage process to select among their available options. Such environments range from hiring committees to clinical trials of medicines. That said, admissions committees may pursue objectives beyond maximizing success probabilities, such as diversity or research field balance. The framework can be extended to incorporate such objectives directly into the committee's problem. In addition, a worthwhile avenue for future research would be collecting data from multiple universities to model the applicant's decision problem and compute equilibrium admissions outcomes for applicants across universities.

Ultimately, the models in this paper can help decision-makers make better selection decisions based on *ex-ante* information when faced with uncertainty and costly evaluation. Crucially, decision-makers can improve selection outcomes without having to increase evaluation effort. Instead, they can use the paper's structural framework and optimality tests to refine how they use that information, and consequently do the best possible job selecting from their available options.

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Appendices

A Variables, Rankings, and Dictionaries

This table contains a list of the variables in this paper. Variables available only for the economics program are bold, and those available for all but economics are italic. *#Math Courses* and *Advanced Math* became objective variables starting in 2020 due to a change in the university's online application system. PhD program rankings are from U.S. News, and placement rankings are from IDEAS/RePEc. The dictionaries for the subjective independent variables are from Madera, Hebl, and Martin (2009), Young and Soroka (2012), and Bai et al. (2022). These dictionaries list word roots (e.g. diligen*), and I match words against the roots directly. A word counts for a dictionary if it begins with the same letter as, and contains, one of that dictionary's roots, which credits inflected and derived forms (e.g. diligent, diligence) to their root. Each dictionary is counted independently against the full document, a word counts at most once per dictionary, and a word containing roots from several dictionaries counts in each of those dictionaries.

Table 6: Variables

(a) Dependent

Admission	Success
$\mathbb{1}$(Transition)	**$\mathbb{1}$(Complete Program)**
$\mathbb{1}$(Accept)	**$\mathbb{1}$(Academic Placement)**
$\mathbb{1}$(Waitlist)	**$\mathbb{1}$(Government Placement)**
$\mathbb{1}$(Open House)	**$\mathbb{1}$(Consulting Placement)**
$\mathbb{1}$(Matriculate)	**$\mathbb{1}$(Tech Placement)**
$\mathbb{1}$(Attend Program)	**$\mathbb{1}$(Other Placement)**
Program Rank	**$\mathbb{1}$(No Placement)**
	Placement Rank
	$\mathbb{1}$(Top 100 Placement)

(b) Objective Independent

Demographic	Performance
Year	GRE Verbal Percentile
$\mathbb{1}$*(Covid Year)*	GRE Quantitative Percentile
Winsorized Age	GRE Analytic Percentile
$\mathbb{1}$(Female)	Undergraduate GPA
$\mathbb{1}$(U.S.)	TOEFL/IELTS
$\mathbb{1}$*(U.S. African/Hispanic/Native)*	$\mathbb{1}$(Elite U.S. University)
$\mathbb{1}$*(U.S. Asian)*	$\mathbb{1}$(Elite U.S. Liberal Arts College)
$\mathbb{1}$*(U.S. Other)*	$\mathbb{1}$(Other U.S. University/College)
$\mathbb{1}$(International Asian)	$\mathbb{1}$(Elite Foreign University)
$\mathbb{1}$(International Other)	$\mathbb{1}$(Other Foreign University)
	$\mathbb{1}$(Graduate Degree)
	$\mathbb{1}$(Work Experience)
	#Professor Recommenders
	#Prolific Recommenders
	Concentration Dummy Variables

(c) Subjective Independent

Score/Recommendation Letters	Supplement/CV/Essay
Committee Score	**#Math Courses**
Letters Length	**$\mathbb{1}$(Advanced Math)**
Positive Terms (%) (YS 2012)	**CV Length**
Negative Terms (%) (YS 2012)	**CV Coding**
Standout Terms (%) (BEMS 2022)	**CV $\mathbb{1}$(Publication/Working Paper)**
Ability Terms (%) (BEMS 2022)	**Essay Length**
Research Terms (%) (BEMS 2022)	**Essay $\mathbb{1}$(Specific Professor)**
Grindstone Terms (%) (BEMS 2022)	**Essay Economics Terms (%)**
Teaching Terms (%) (BEMS 2022)	**Essay Research Terms (%)**
Communal Terms (%) (BEMS 2022)	**Essay Positive Terms (%)**
Agentic Terms (%) (MHM 2009)	**Essay Negative Terms (%)**
	Field Dummy Variables

Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)

- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Academia-Equivalent Rankings (Consulting/Technology/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Dictionaries (Recommendation Letters)

- **Positive/Negative Words:** Young and Soroka (2012)'s Lexicoder Sentiment Dictionary
- **Standout, Ability, Research, Grindstone, Communal Words:** Bai et al. (2022)
- **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera et al. 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
- **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop. (Note: I constructed this dictionary because it is unavailable in Bai et al. 2022.)

Dictionaries (Application Essays/CVs)

- **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
- **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
- **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
- **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

A.1 Academia-Equivalent Prompt and Link to Rankings

I am writing a research paper that entails using data in application files to predict success in economics PhD programs, where success is defined as the quality of the first post-PhD job placement. For academic jobs, I use RePEc rankings. For non-academic jobs, I will follow the same procedure as Krueger and Wu (2000):

"Another issue concerns the ranking of the various job placements. For our analysis, we assumed that one goal of admissions and graduate training in economics is to place students in leading research jobs, although we recognize that different departments have different objectives. In ranking academic institutions, we relied primarily on the ranking of the top 100 economics departments based on faculty publications in elite journals produced by Scott and Mitias (1996). Students who were placed in business school jobs were given a rank equal to their university's economics department ranking plus five. (A lower rank signified a more prestigious job placement.) The World Bank, IMF, and Federal Reserve Board received a ranking equivalent to the 40th best economics department. Consulting jobs (e.g., NERA, Abt, DRI) were given a rank equal to the 120th best economics department. Finally, applicants who could not be found were assigned a rank of 150, the worst rank we gave, and treated as censored observations in much of the analysis."

I am attaching an Excel file with the RePEc rankings, as well as job titles for Government, Consulting, Tech, and Other Non-Academia. In each of the latter four sheets, please fill in the "Academia-Equivalent Ranking" column with the ranking you deem most appropriate. Keep it as consistent as possible with Krueger and Wu, meaning that the best non-academic placements should receive a ranking of 40. Krueger and Wu deem those placements to be the World Bank, IMF, and Federal Reserve Board (that is, the Board of Governors and not the regional banks). However, because the economics profession has changed since 2000, you can give additional placements a ranking of 40. Limit any rankings of 40 to the Government sheet and the "Tech Research Labs" part of the Tech sheet. Be very strict about which, if any, additional placements you give a 40, since Krueger and Wu gave only three 40s, each of which were to very prestigious placements. You can give at most three additional 40s in Government, and you can give at most three 40s in Tech.

Regarding rankings other than 40 that you can give, you can give 80, 120, and 135. The logic behind 80 is that it is halfway between 40 and 120, and the logic behind 135 is that it is halfway between 120 and 150. Do not give any placements a rank of 150; 150 is reserved for applicants who cannot be found. 40 is for placements in the World Bank tier, 80 is for placements between the World Bank and NERA tiers, 120 is for placements in the NERA tier, and 135 is for placements that are below the NERA tier. If you ever give a Consulting or Other Non-Academia placement a ranking better than 120, make sure you have a very sound reason for doing so because Krueger and Wu capped rankings for consulting placements at 120. Please post all your academia-equivalent rankings in the chat as well, as I will save a screenshot of the chat.

A.2 Synthetic Data Example

As a demonstration of synthetic data, I use a Gaussian copula to simulate 1,000 observations of the following independent variables: *GRE Quantitative Percentile* (continuous), *Gender* (binary), *Demographic* (categorical with three demographics), *Advanced Math* (binary), and *Committee Score* (continuous). Then I use the standard logistic function to simulate binary dependent variables for transitioning past the first-round filter, getting waitlisted, getting accepted into the program, matriculating at the program if accepted, and obtaining a successful placement above a threshold ranking. I also generate the (continuous) rank of the PhD program each applicant attends, which is constant for the applicants with $Matriculate = 1$ and equals 150 for applicants who do not attend a PhD program (for these applicants, $Success = 0$). The top frame of Figure 6 contains the first ten observations of this simulated dataset. *Committee Score* is missing for all applicants who did not survive the first-round filter, which is a feature of the actual data.

I then pass this dataset through another Gaussian copula and use that copula to generate a synthetic dataset, the first ten observations of which are in the bottom frame of the figure. The means, variances, and correlations of the variables are similar in both datasets, with only the correlations in the synthetic dataset slightly attenuated. However, none of the rows of the simulated dataset are present in the synthetic dataset. In addition, important statistical properties; such as *Accept* equaling 1 only when *Transition* equals 1, *Demographic 1* and *Demographic 2* never both equaling 1, and *Committee Score* missing when *Transition* equals 0; are preserved.

Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
1	1	0	0	0	89	0	0	0	0	5	150
1	0	1	0	1	96	0	1	0	0	6	23
0	0	0	0	0	88	0	1	0	1	·	150
0	0	0	0	0	80	1	1	0	1	·	150
0	0	0	0	1	99	0	1	0	0	·	29
0	0	0	0	0	81	0	0	1	1	·	23
1	0	1	1	1	92	0	1	0	0	3	37
1	1	1	1	1	93	0	0	0	1	3	37
0	0	0	0	0	83	1	0	1	0	·	48
1	1	0	0	0	87	0	1	0	1	5	34

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Transition	Waitlist	Accept	Matriculate	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score	Program Rank
0	0	0	0	0	77	0	0	0	1	·	30
0	0	0	0	1	89	0	0	0	0	·	25
0	0	0	0	0	85	1	0	0	0	·	150
0	0	0	0	0	76	0	1	0	1	·	51
1	1	0	0	0	88	0	1	0	0	8	150
1	0	1	0	0	88	0	0	0	1	5	150
1	1	0	0	0	84	0	0	1	0	2	20
0	0	0	0	0	82	0	1	0	0	·	35
0	0	0	0	0	70	1	0	0	0	·	150
1	1	0	0	1	93	0	1	0	0	5	57

Figure 6: Synthetic Data Example

B Additional Regression Results

Table 7: Economics Department Summary Statistics

	Full Sample		Transition = 1		Accept = 1		Matriculate = 1	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Transition	0.4204	0.4937	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
Waitlist	0.3324	0.4712	0.7686	0.4220	0.0743	0.2628	0.0896	0.2877
Accept	0.1103	0.3134	0.2625	0.4402	1.0000	0.0000	1.0000	0.0000
Matriculate	0.0339	0.1811	0.0807	0.2725	0.3074	0.4623	1.0000	0.0000
Open House	0.0353	0.1847	0.0817	0.2741	0.3267	0.4702	0.5522	0.5010
Attend Program	0.6024	0.4895	0.6752	0.4685	0.7704	0.4214	1.0000	0.0000
Complete Program	0.5578	0.4968	0.6210	0.4854	0.6654	0.4728	0.6709	0.4729
Top 100 Placement	0.1971	0.3979	0.2472	0.4316	0.3074	0.4623	0.3418	0.4773
Program Rank	88.9133	57.4823	78.5844	57.1361	63.3828	54.1845	37.0000	0.0000
Missing Program Rank	0.4199	0.4937	0.3422	0.4747	0.2490	0.4333	0.0000	0.0000
Placement Rank	127.7505	34.9306	123.0222	37.2570	117.2696	40.5746	117.4430	37.5728
Missing Placement Rank	0.4444	0.4970	0.3830	0.4864	0.3346	0.4728	0.3291	0.4729
2016	0.2052	0.4040	0.1696	0.3754	0.1790	0.3841	0.2025	0.4045
2017	0.2014	0.4011	0.1788	0.3833	0.1829	0.3873	0.1772	0.3843
2018	0.1928	0.3946	0.2686	0.4435	0.2062	0.4054	0.2532	0.4376
2019	0.2027	0.4021	0.2084	0.4064	0.2179	0.4136	0.2152	0.4136
Winsorized Age	25.9910	3.1479	26.2635	3.1823	26.0817	3.0652	26.3291	3.2806
Female	0.3912	0.4881	0.4147	0.4929	0.4397	0.4973	0.4177	0.4963
U.S.	0.2357	0.4245	0.2380	0.4261	0.2996	0.4590	0.3038	0.4628
International Asian	0.4809	0.4997	0.5230	0.4997	0.3424	0.4754	0.4304	0.4983
GRE Verbal Percentile	68.0129	22.5869	76.8421	18.3703	79.5703	18.3047	76.9744	19.4041
GRE Quantitative Percentile	88.4550	11.6330	93.1362	5.4135	93.5391	4.4614	92.3077	5.7554
GRE Analytic Percentile	50.0613	27.5744	56.3653	26.5466	60.3398	26.0880	57.4615	26.6481
Undergraduate GPA	3.6307	0.3026	3.6645	0.3003	3.7227	0.2610	3.7068	0.2880
TOEFL/IELTS	7.4376	0.5587	7.5641	0.5834	7.7011	0.6667	7.4464	0.6137
Elite U.S. University	0.1267	0.3327	0.1736	0.3790	0.2257	0.4188	0.1772	0.3843
Elite U.S. Liberal Arts College	0.0296	0.1696	0.0347	0.1832	0.0428	0.2028	0.0633	0.2450
Elite Foreign University	0.0386	0.1928	0.0490	0.2160	0.0350	0.1842	0.0253	0.1581
Other Foreign University	0.4049	0.4910	0.4188	0.4936	0.3619	0.4815	0.4430	0.4999
Missing Undergraduate Institution	0.2203	0.4145	0.1910	0.3933	0.1984	0.3996	0.1266	0.3346
Graduate Degree	0.5067	0.5001	0.5730	0.4949	0.5136	0.5008	0.5190	0.5028
Work Experience	0.5247	0.4995	0.5894	0.4922	0.6770	0.4685	0.7468	0.4376
#Professor Recommenders	2.6230	0.7371	2.6302	0.7165	2.5992	0.7118	2.5696	0.7104
#Prolific Recommenders	1.2602	1.0686	1.3912	1.0840	1.4825	1.0424	1.3038	1.0664
Missing GRE	0.0335	0.1800	0.0102	0.1006	0.0039	0.0624	0.0127	0.1125
Missing Undergraduate GPA	0.7853	0.4107	0.7487	0.4340	0.6770	0.4685	0.6835	0.4681
Missing TOEFL/IELTS	0.6076	0.4884	0.5935	0.4914	0.6615	0.4741	0.6456	0.4814
Missing Winsorized Age	0.0009	0.0293	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Committee Score	6.4070	1.8701	6.4070	1.8701	8.5003	0.8151	8.4854	0.7157
#Math Courses	5.7637	1.8154	5.8173	1.7904	5.6992	1.8640	5.6585	1.6219
Advanced Math	0.4860	0.5004	0.4909	0.5007	0.5680	0.4973	0.4651	0.5047
Letters Length/100	8.3474	3.5749	8.5333	3.8263	10.0514	4.2864	9.7088	3.4876
Letters Positive Terms (%)	11.9770	2.2993	11.9976	2.2077	12.2167	1.9255	12.5110	2.0904
Letters Negative Terms (%)	5.7286	1.3427	5.7216	1.3318	5.7446	1.2313	5.8938	1.2871
Letters Standout Terms (%)	0.6241	0.4020	0.6241	0.4029	0.7151	0.4509	0.7077	0.5109
Letters Ability Terms (%)	1.3326	0.6239	1.3457	0.5964	1.3796	0.5620	1.4334	0.6037
Letters Research Terms (%)	3.4837	1.3319	3.5208	1.3423	3.8548	1.1984	3.8395	1.2121
Letters Grindstone Terms (%)	1.8189	0.7734	1.8674	0.7808	1.9975	0.7588	2.1600	0.8093
Letters Teaching Terms (%)	1.7379	0.7710	1.7489	0.7759	1.7275	0.7536	1.8976	0.7577
Letters Communal Terms (%)	0.5946	0.3890	0.6019	0.3954	0.5494	0.3234	0.5630	0.3875
Letters Agentic Terms (%)	0.4768	0.3150	0.4842	0.3229	0.5039	0.3222	0.5597	0.3314
Letters Terms (%) PC1	-0.0579	1.6183	0.0000	1.5768	0.2176	1.3817	0.5697	1.5386
Letters Terms (%) PC2	-0.0271	1.1177	0.0000	1.1446	0.1002	1.1774	0.1699	1.2823
CV Length/100	4.3532	2.4237	4.3982	2.4758	4.9641	2.7175	4.8261	2.3473
CV Coding	2.8321	2.1135	2.7931	2.1000	2.8618	2.1129	2.5122	1.6450
CV Publication/Working Paper	0.2518	0.4346	0.2571	0.4377	0.3659	0.4836	0.3902	0.4939
Essay Length/100	4.5099	1.5741	4.5173	1.5805	4.7488	1.6373	4.9816	1.7932
Essay Specific Professor	0.4822	0.5003	0.5219	0.5003	0.5691	0.4972	0.6512	0.4822
Essay Economics Terms (%)	9.7284	2.6219	9.7216	2.6635	9.5303	2.7324	9.8941	3.2608
Essay Research Terms (%)	5.7975	2.0501	5.7402	2.0168	5.7313	1.8797	5.5231	1.8432
Essay Positive Terms (%)	11.8293	2.6518	11.7324	2.6650	11.4734	2.4592	11.3169	2.4455
Essay Negative Terms (%)	6.2844	1.9056	6.3233	1.9712	6.2889	2.0860	6.2651	1.9436
Missing Committee Score	0.5844	0.4929	0.0112	0.1055	0.0195	0.1384	0.0633	0.2450
Missing Application Pdf	0.8154	0.3881	0.6650	0.4722	0.5136	0.5008	0.4557	0.5012
Missing #Math Courses	0.8201	0.3842	0.6701	0.4704	0.5214	0.5005	0.4810	0.5028
Missing Letters	0.8175	0.3863	0.6691	0.4708	0.5136	0.5008	0.4557	0.5012
Missing CV	0.8210	0.3835	0.6742	0.4689	0.5214	0.5005	0.4810	0.5028
Missing Essay	0.8192	0.3849	0.6731	0.4693	0.5214	0.5005	0.4557	0.5012
Observations	2329		979		257		79	

Table 8: Economics Department Correlation Matrix

	Trans	Wait	Acc	Mat	OH	Att	Com	T100	ProR	MProR	PlaR	MPlaR	2016	2017	2018	2019	Age	Fem	US	IntAsn
Transition	1																			
Waitlist	0.8083	1																		
Accept	0.4136	-0.1908	1																	
Matriculate	0.2200	-0.0994	0.5320	1																
Open House	0.2192	-0.1104	0.5496	0.5400	1															
Attend Program	0.1266	0.0767	0.1209	0.1522	0.0833	1														
Complete Program	0.1085	0.0889	0.0763	0.0427	0.0154	0.9124	1													
Top 100 Placement	0.1073	0.0778	0.0977	0.0682	0.0498	0.4025	0.4412	1												
Program Rank	-0.1531	-0.0909	-0.1565	-0.1693	-0.0999	-0.8635	-0.7942	-0.4106	1											
Missing Program Rank	-0.1341	-0.0860	-0.1219	-0.1594	-0.0856	-0.9548	-0.8749	-0.4018	0.9044	1										
Placement Rank	-0.1153	-0.0802	-0.1057	-0.0553	-0.0666	-0.5176	-0.5673	-0.9004	0.4999	0.5032	1									
Missing Placement Rank	-0.1051	-0.0835	-0.0778	-0.0435	-0.0164	-0.9084	-0.9957	-0.4431	0.7924	0.8708	0.5698	1								
2016	-0.0752	-0.0883	-0.0229	-0.0013	-0.0126	-0.0194	0.0073	-0.0005	0.0077	0.0178	-0.0069	-0.0073	1							
2017	-0.0480	-0.0732	-0.0162	-0.0113	0.0162	0.0076	0.0138	0.0096	-0.0025	-0.0064	-0.0040	-0.0138	-0.2552	1						
2018	0.1637	0.1695	0.0120	0.0287	0.0009	0.0101	0.0078	0.0096	-0.0118	-0.0233	-0.0076	-0.0034	-0.2483	-0.2454	1					
2019	0.0121	-0.0050	0.0133	0.0058	-0.0045	0.0145	-0.0565	-0.0081	0.0123	0.0017	0.0166	0.0543	-0.2562	-0.2532	-0.2464	1				
Winsorized Age	0.0738	0.0157	0.0102	0.0201	0.0122	-0.0260	-0.0179	0.0398	0.0474	0.0304	-0.0148	0.0229	0.0102	-0.0091	-0.0128	0.0160	1			
Female	0.0411	0.0255	0.0350	0.0102	0.0328	-0.0068	-0.0179	-0.0189	0.0074	0.0168	0.0007	0.0198	0.0066	-0.0163	0.0142	0.0358	-0.1468	1		
U.S.	0.0046	-0.0393	0.0530	0.0301	0.0318	-0.1069	-0.1165	-0.0310	0.0900	0.1075	0.0538	0.1141	0.0083	0.0263	-0.0073	-0.0409	-0.1537	-0.0886	1	
International Asian	0.0717	0.1739	-0.0976	-0.0189	-0.0210	0.1744	0.1736	0.0416	-0.1725	-0.1816	-0.0646	-0.1707	-0.0210	-0.0162	-0.0107	0.0407	-0.0407	0.1108	-0.5345	1
GRE Verbal Percentile	0.3349	0.2346	0.1825	0.0746	0.1228	0.0524	0.0343	0.0479	-0.0681	-0.0536	-0.0548	-0.0358	-0.0272	-0.0158	0.0260	0.0343	-0.1056	-0.0155	0.2891	-0.1077
GRE Quantitative Percentile	0.3448	0.2791	0.1559	0.0623	0.0605	0.1957	0.1930	0.0980	-0.2102	-0.1965	-0.1287	-0.1899	0.0061	0.0095	0.0116	-0.0255	-0.0622	-0.0398	-0.1080	0.2269
GRE Analytic Percentile	0.1960	0.1198	0.1330	0.0505	0.0710	-0.0468	-0.0700	-0.0083	0.0359	0.0504	0.0058	0.0695	-0.0542	0.0113	0.0537	0.0379	-0.1533	0.0022	0.3950	-0.2554
Undergraduate GPA	0.0517	-0.0095	0.0747	0.0322	0.0440	0.0329	0.0179	-0.0166	-0.0382	-0.0336	0.0056	-0.0179	-0.0272	-0.0058	0.0260	0.0109	-0.0853	0.0693	0.0418	-0.0425
TOEFL/IELTS	0.1251	0.1060	0.0898	0.0017	0.0413	-0.0200	-0.0127	0.0342	-0.0053	0.0226	-0.0431	0.0108	-0.0018	0.0047	0.0176	-0.0092	-0.1177	0.0969	0.0119	-0.0524
Elite U.S. University	0.1203	0.0775	0.1048	0.0285	0.0769	-0.0151	-0.0196	-0.0167	-0.0171	0.0108	0.0124	0.0179	-0.0305	-0.0271	0.0070	0.0392	-0.0769	0.0095	0.2082	-0.0410
Elite U.S. Liberal Arts College	0.0256	-0.0020	0.0274	0.0372	0.0205	-0.0081	-0.0280	-0.0038	-0.0103	0.0104	-0.0012	0.0272	0.0178	0.0133	-0.0405	-0.0125	-0.0639	0.0260	0.1177	-0.0567
Elite Foreign University	0.0459	0.0691	-0.0066	-0.0130	-0.0062	0.0218	0.0305	0.0071	-0.0445	-0.0261	0.0027	-0.0314	0.0084	0.0215	-0.0246	-0.0290	-0.0129	0.0447	-0.1008	0.1325
Other Foreign University	0.0241	0.0431	-0.0309	0.0146	-0.0327	0.0571	0.0564	0.0311	-0.0260	-0.0496	-0.0454	-0.0512	-0.0228	-0.0085	-0.0772	0.1586	0.1709	-0.0105	-0.4231	0.2355
Missing Undergraduate Institution	-0.0601	-0.0583	-0.0185	-0.0424	-0.0215	0.0169	0.0435	0.0180	-0.0124	-0.0198	-0.0227	-0.0458	0.0300	-0.0008	0.1946	-0.2576	-0.0011	0.0092	-0.0364	-0.0761
Graduate Degree	0.1131	0.1254	0.0049	0.0046	0.0248	0.0529	0.0706	0.0226	-0.0399	-0.0548	-0.0328	-0.0698	0.0443	0.1379	-0.1556	-0.0729	0.3704	-0.0010	-0.2492	0.1402
Work Experience	0.1103	0.0043	0.1074	0.0834	0.0484	-0.0723	-0.0789	0.0371	0.0722	0.0851	-0.0148	0.0795	-0.0017	-0.0002	-0.0056	0.0093	0.4141	-0.0229	0.0870	-0.2765
#Professor Recommenders	0.0083	0.0374	-0.0114	-0.0136	-0.0257	0.0606	0.0466	-0.0073	-0.0472	-0.0547	0.0069	-0.0514	-0.0791	0.0186	0.0329	0.0390	-0.2398	-0.0114	-0.0330	0.1402
#Prolific Recommenders	0.1044	0.0794	0.0733	0.0076	0.0374	0.1042	0.1084	0.0632	-0.1156	-0.1054	-0.0567	-0.1070	-0.0262	0.0330	0.0430	0.0062	0.0200	-0.0017	-0.2072	0.1686
Missing GRE	-0.1102	-0.1318	-0.0579	-0.0217	-0.0358	-0.0731	-0.0649	-0.0322	0.0715	0.0786	0.0303	0.0641	0.0118	-0.0042	-0.0123	0.0071	0.0718	0.0122	-0.0303	-0.0502
Missing Undergraduate GPA	-0.0759	-0.0361	-0.0929	-0.0464	-0.0501	0.0261	0.0292	0.0119	-0.0105	-0.0213	-0.0191	-0.0269	-0.0372	-0.0582	-0.0334	0.2064	0.1637	-0.0095	-0.3206	0.1579
Missing TOEFL/IELTS	-0.0246	-0.0503	0.0389	0.0146	0.0438	-0.1211	-0.1296	-0.0660	0.1027	0.1244	0.0753	0.1260	-0.0248	-0.0372	-0.0241	0.0552	-0.1342	-0.0207	0.4256	-0.2472
Missing Winsorized Age	-0.0250	-0.0231	-0.0103	-0.0055	-0.0063	-0.0361	-0.0329	-0.0145	0.0312	0.0345	0.0187	0.0328	0.0214	0.0218	-0.0143	-0.0148	0.0000	-0.0235	-0.0163	-0.0282
Committee Score	0.0000	-0.3453	0.5999	0.3028	0.3395	0.0873	0.0554	0.0493	-0.1117	-0.0842	-0.0463	-0.0594	-0.0390	-0.0215	0.0691	0.0682	-0.0598	-0.0014	-0.0064	-0.0812
#Math Courses	0.0107	0.0311	-0.0215	-0.0177	-0.0467	0.0368	0.0367	0.0132	-0.0467	-0.0422	-0.0177	-0.0375	0.0291	0.0473	0.0642	-0.1633	-0.0764	-0.0104	0.0306	0.0164
Advanced Math	0.2226	0.1304	0.2298	0.1071	0.1558	0.0463	0.0406	0.0031	-0.0742	-0.0510	-0.0116	-0.0420	-0.0182	-0.0752	0.0408	0.0398	-0.0349	-0.0054	-0.0328	0.0827
Letters Length/100	0.0189	-0.1069	0.1781	0.0827	0.0792	0.0261	0.0189	0.0146	-0.0368	-0.0238	-0.0114	-0.0197	0.0075	-0.0049	-0.0155	0.0170	0.0376	0.0367	0.0330	-0.0790
Letters Positive Terms (%)	0.0161	-0.0760	0.0674	0.0780	0.0058	-0.0500	-0.0268	0.0708	0.0375	0.0280	-0.0454	0.0186	0.0464	-0.0001	-0.0522	0.0536	0.0432	-0.0145	0.1376	-0.0883
Letters Negative Terms (%)	-0.0093	-0.0545	0.0077	0.0413	-0.0479	-0.0158	-0.0610	-0.0737	0.0245	0.0241	0.0592	0.0574	0.0186	0.0081	0.0824	-0.0272	0.0807	-0.0036	0.1392	-0.0830
Letters Standout Terms (%)	0.0003	-0.1493	0.1464	0.0698	-0.0346	0.0222	0.0131	0.0878	-0.0031	-0.0252	-0.0422	-0.0153	0.0733	-0.0587	-0.0863	0.0900	0.0507	-0.0141	0.0120	-0.0394
Letters Ability Terms (%)	0.0375	-0.0301	0.0487	0.0543	0.0006	-0.1785	-0.1573	-0.0137	0.1340	0.1765	0.0371	0.1502	0.0512	0.0014	0.0153	-0.0218	0.0866	-0.0015	0.1424	-0.1930
Letters Research Terms (%)	0.0499	-0.0879	0.1800	0.0897	0.0944	0.0439	0.0375	0.0492	-0.0771	-0.0588	-0.0934	-0.0376	-0.0062	-0.0162	0.1051	0.0225	0.0258	0.0245	0.1273	-0.0539
Letters Grindstone Terms (%)	0.1125	-0.0320	0.1493	0.1481	0.0630	-0.1067	-0.1156	-0.0128	0.0728	0.1050	0.0347	0.1104	-0.0133	-0.0563	-0.0136	0.0294	0.1012	0.0839	0.1233	-0.1591
Letters Teaching Terms (%)	0.0258	-0.0022	-0.0087	0.0696	0.0224	-0.1522	-0.1563	0.0241	0.1552	0.1493	0.0234	0.1579	0.0595	-0.0270	-0.0398	0.0459	0.1921	0.0369	0.0349	-0.1064
Letters Communal Terms (%)	0.0338	0.0517	-0.0749	-0.0272	-0.0216	-0.0020	0.0357	-0.0500	0.0132	0.0135	0.0186	-0.0394	0.0324	0.0711	-0.0603	-0.0627	0.0382	-0.0136	-0.1655	0.0499
Letters Agentic Terms (%)	0.0421	0.0589	0.0557	0.0884	0.0913	-0.0509	-0.0716	0.0030	0.0147	0.0225	0.0016	0.0686	0.0927	-0.0091	0.0022	-0.0789	-0.0172	0.1191	0.1148	-0.0943
Letters Terms (%) PC1	0.0130	-0.0283	0.0593	0.0870	0.0251	-0.0585	-0.0577	0.0109	0.0481	0.0525	0.0028	0.0547	0.0310	-0.0050	-0.0129	0.0043	0.0514	0.0207	0.0601	-0.0803
Letters Terms (%) PC2	0.0088	-0.0033	0.0396	0.0383	0.0466	-0.0062	-0.0170	-0.0190	-0.0053	0.0063	0.0007	0.0174	-0.0204	-0.0007	0.0628	-0.0096	0.0192	0.0150	0.0516	-0.0323
CV Length/100	0.0067	-0.0429	0.0959	0.0421	0.0772	0.0199	0.0202	0.0299	-0.0124	-0.0185	-0.0335	-0.0222	0.0320	-0.0248	-0.0129	-0.0118	0.0394	0.0249	0.0341	-0.0336
CV Coding	-0.0067	-0.0225	0.0102	-0.0320	-0.0333	0.0280	0.0338	0.0297	-0.0250	-0.0246	-0.0377	-0.0365	-0.0225	-0.0306	0.0089	0.0368	-0.0633	-0.0004	-0.0209	0.0151
CV Publication/Working Paper	0.1587	0.0464	0.2207	0.1422	0.1135	0.0666	0.0560	0.0432	-0.0707	-0.0633	-0.0408	-0.0569	-0.0284	-0.0575	0.0197	0.0603	0.0894	-0.0215	-0.0475	0.0269
Essay Length/100	0.0017	-0.0422	0.0591	0.0712	0.0195	0.0473	0.0287	0.0026	-0.0530	-0.0452	-0.0053	-0.0274	0.0204	-0.0137	-0.0110	0.0253	0.0388	-0.0133	-0.0100	0.0078
Essay Specific Professor	0.2518	0.1247	0.2312	0.1775	0.1550	0.0489	0.0361	0.0306	-0.0633	-0.0470	-0.0396	-0.0374	-0.0440	-0.0641	0.0419	0.0752	0.0149	-0.0044	-0.0712	0.0804
Essay Economics Terms (%)	-0.0009	0.0317	-0.0293	0.0156	-0.0042	-0.0358	-0.0452	-0.0431	0.0338	0.0270	0.0456	0.0426	-0.0102	0.0036	0.0191	-0.0257	-0.0088	-0.0026	-0.0373	0.0596
Essay Research Terms (%)	-0.0101	0.0042	-0.0059	-0.0277	-0.0234	0.0838	0.0766	0.0144	-0.0690	-0.0720	-0.0305	-0.0802	-0.0296	0.0235	-0.0081	0.0445	-0.0767	0.0498	-0.0432	0.0665
Essay Positive Terms (%)	-0.0132	0.0211	-0.0442	-0.0405	-0.0379	-0.0035	0.0102	0.0332	0.0043	0.0060	-0.0127	-0.0110	-0.0123	-0.0010	0.0151	0.0029	-0.0450	0.0208	-0.0019	0.0002
Essay Negative Terms (%)	0.0074	0.0139	-0.0041	-0.0057	-0.0264	0.0107	0.0005	-0.0066	-0.0026	-0.0143	0.0085	-0.0030	0.0015	0.0056	-0.0294	0.0257	0.0215	-0.0251	-0.0063	0.0323
Missing Committee Score	-0.9903	-0.7997	-0.4037	-0.1981	-0.2043	-0.1226	-0.1054	-0.1012	0.1477	0.1297	0.1106	0.1020	0.0748	0.0433	-0.1554	-0.0105	-0.0656	-0.0471	-0.0078	-0.0793
Missing Application Pdf	-0.3301	-0.2162	-0.2739	-0.1737	-0.1677	-0.0565	-0.0516	-0.0230	0.0840	0.0596	0.0243	0.0514	0.0528	0.0955	-0.0508	-0.1152	-0.0056	-0.0041	0.1078	-0.1112
Missing #Math Courses	-0.3326	-0.2159	-0.2739	-0.1654	-0.1709	-0.0585	-0.0547	-0.0293	0.0872	0.0610	0.0309	0.0544	0.0498	0.0902	-0.0516	-0.1115	-0.0077	-0.0025	0.1021	-0.1108
Missing Letters	-0.3273	-0.2105	-0.2771	-0.1755	-0.1700	-0.0545	-0.0491	-0.0230	0.0815	0.0574	0.0229	0.0489	0.0502	0.0986	-0.0509	-0.1130	-0.0081	-0.0063	0.1052	-0.1060
Missing CV	-0.3261	-0.2095	-0.2752	-0.1661	-0.1568	-0.0590	-0.0528	-0.0248	0.0842	0.0615	0.0230	0.0525	0.0571	0.0893	-0.0528	-0.1156	-0.0142	-0.0020	0.1142	-0.1198
Missing Essay	-0.3233	-0.2066	-0.2726	-0.1770	-0.1718	-0.0533	-0.0476	-0.0253	0.0819	0.0560	0.0253	0.0474	0.0508	0.0912	-0.0448	-0.1101	-0.0035	-0.0099	0.1058	-0.1106
Observations	2329																			

	GREV	GREQ	GREA	GPA	T/I	EUni	ELA	EFor	OFor	MUg	Grad	Work	Prof	Pro
GRE Verbal Percentile	1													
GRE Quantitative Percentile	0.2114	1												
GRE Analytic Percentile	0.5519	-0.0386	1											
Undergraduate GPA	0.0835	0.0793	0.0863	1										
TOEFL/IELTS	0.2772	0.0337	0.2465	0.0031	1									
Elite U.S. University	0.1155	0.0303	0.1320	-0.0917	0.0035	1								
Elite U.S. Liberal Arts College	0.0805	0.0260	0.1074	0.0216	0.0113	-0.0665	1							
Elite Foreign University	0.0335	0.0642	-0.0303	0.0047	0.0366	-0.0764	-0.0350	1						
Other Foreign University	-0.1253	0.0709	-0.2141	-0.0002	-0.0057	-0.3141	-0.1441	-0.1654	1					
Missing Undergraduate Institution	-0.0414	-0.0056	-0.0322	0.0301	-0.0164	-0.2024	-0.0929	-0.1066	-0.4384	1				
Graduate Degree	-0.1043	0.0039	-0.1313	-0.0877	0.0107	-0.1458	-0.0808	0.0597	0.1649	0.0831	1			
Work Experience	0.0369	-0.1144	0.1031	0.0121	0.0128	0.0083	0.0294	-0.0367	-0.0399	0.0163	0.1219	1		
#Professor Recommenders	-0.0051	0.1028	-0.0399	0.0557	-0.0180	-0.0592	0.0069	0.0270	0.0136	-0.0023	-0.0631	-0.2477	1	
#Prolific Recommenders	-0.0282	0.1814	-0.1224	-0.0547	0.0052	-0.0311	-0.0497	0.0492	0.1487	-0.0014	0.1173	-0.0724	0.2047	1
Missing GRE	-0.0081	-0.0129	0.0026	-0.0117	-0.0034	0.0080	-0.0325	0.0122	-0.0028	-0.0183	-0.0073	0.0434	-0.0375	-0.0163
Missing Undergraduate GPA	-0.2161	0.0599	-0.3214	0.0000	-0.0211	-0.2096	-0.1183	0.0940	0.3226	0.0180	0.1345	-0.0097	0.0036	0.1459
Missing TOEFL/IELTS	0.0917	-0.1772	0.2285	0.0072	0.0000	0.2876	0.1352	-0.0396	-0.3778	-0.1160	-0.1265	0.0345	-0.0031	-0.2002
Missing Winsorized Age	-0.0442	-0.0172	0.0000	0.0000	0.0000	-0.0112	-0.0051	-0.0059	0.0057	0.0198	0.0289	0.0279	0.0150	-0.0071
Committee Score	0.0408	0.0378	0.0481	0.0800	0.0726	0.0496	-0.0133	-0.0126	-0.0104	0.0234	-0.0609	0.0246	0.0279	0.0849
#Math Courses	0.0307	-0.0093	0.0315	-0.0128	0.0409	0.0339	-0.0032	-0.0111	-0.0406	-0.0053	-0.0392	0.0041	-0.0020	-0.0002
Advanced Math	0.0181	0.1157	-0.0207	0.0099	0.0072	0.0701	0.0337	0.0150	0.0042	-0.0763	-0.0478	-0.0561	0.0363	0.1345
Letters Length/100	0.0531	-0.0110	0.0579	-0.0050	0.0487	0.0012	0.0635	0.0250	-0.0216	-0.0030	-0.0020	0.1001	-0.0420	0.0333
Letters Positive Terms (%)	0.0349	0.0533	0.1164	-0.0099	-0.0573	-0.0018	0.1339	-0.0416	-0.0931	-0.0256	-0.0638	0.0769	-0.1049	-0.1867
Letters Negative Terms (%)	0.0764	-0.0665	0.0632	-0.0612	-0.1270	0.0379	0.0199	0.0261	-0.0506	-0.0449	0.0424	0.0943	-0.1391	0.0507
Letters Standout Terms (%)	0.0100	0.0078	0.0742	0.0477	0.0066	-0.1116	0.0846	0.0073	0.0514	-0.0690	-0.0211	0.0198	0.0226	0.0132
Letters Ability Terms (%)	0.0423	0.0487	0.1273	0.0808	-0.0476	-0.0047	0.0583	0.0035	-0.1331	0.0026	0.0081	0.1310	-0.1811	-0.2310
Letters Research Terms (%)	0.1004	-0.0220	0.1052	-0.0448	0.0235	0.0296	0.0482	-0.0391	-0.0767	0.0477	0.0037	0.1626	-0.2141	-0.0945
Letters Grindstone Terms (%)	0.0209	-0.0701	0.0946	0.0185	-0.0524	0.0140	0.1687	-0.0368	-0.1797	0.0350	0.0451	0.2148	-0.1778	-0.1497
Letters Teaching Terms (%)	-0.0334	-0.0692	0.0481	0.0218	-0.0055	-0.0565	0.0274	-0.0166	0.0065	-0.0288	0.0670	0.1777	-0.0771	-0.1383
Letters Communal Terms (%)	-0.0011	0.0538	0.0001	0.0203	-0.1205	-0.0344	0.0260	0.0362	-0.0034	0.0305	0.0770	-0.0646	0.1095	-0.0050
Letters Agentic Terms (%)	0.0638	-0.0830	0.0517	-0.0681	0.0347	0.0740	0.1425	-0.0270	-0.1247	0.0009	-0.0336	0.0006	-0.1529	-0.0772
Letters Terms (%) PC1	0.0219	-0.0120	0.0601	0.0043	-0.0258	-0.0053	0.0708	-0.0127	-0.0645	-0.0019	0.0041	0.0764	-0.0732	-0.0963
Letters Terms (%) PC2	0.0285	-0.0245	0.0213	-0.0207	-0.0088	0.0366	-0.0004	-0.0110	-0.0449	0.0199	0.0192	0.0790	-0.0774	-0.0236
CV Length/100	0.0120	-0.0114	0.0448	0.0014	0.0399	-0.0027	0.0241	-0.0241	0.0074	-0.0123	0.0314	0.1054	-0.0738	0.0126
CV Coding	-0.0125	0.0021	-0.0083	-0.0131	0.0023	0.0122	0.0125	-0.0072	0.0100	-0.0158	0.0065	-0.0100	-0.0106	0.0452
CV Publication/Working Paper	-0.0141	0.0741	-0.0197	-0.0054	-0.0400	0.0292	0.0109	-0.0006	0.0358	-0.0256	0.0282	0.0825	-0.0292	0.0846
Essay Length/100	-0.0314	-0.0020	-0.0472	-0.0304	-0.0084	-0.0051	-0.0053	0.0103	0.0145	-0.0109	0.0979	0.0273	0.0028	0.0398
Essay Specific Professor	-0.0046	0.1188	-0.0416	0.0123	0.0222	0.0562	-0.0091	-0.0067	0.0521	-0.0577	0.0065	-0.0138	0.0279	0.1057
Essay Economics Terms (%)	-0.0069	0.0022	-0.0280	0.0313	-0.0115	0.0065	-0.0007	0.0222	-0.0436	0.0058	0.0017	-0.0049	-0.0217	0.0205
Essay Research Terms (%)	-0.0242	0.0132	-0.0050	0.0206	-0.0017	0.0157	0.0134	0.0160	-0.0295	-0.0003	-0.0478	-0.0345	0.0073	0.0095
Essay Positive Terms (%)	-0.0149	0.0057	0.0096	0.0287	0.0103	0.0155	-0.0262	-0.0164	-0.0213	-0.0069	-0.0358	-0.0021	0.0141	-0.0244
Essay Negative Terms (%)	-0.0046	0.0103	-0.0007	-0.0384	-0.0316	0.0534	0.0249	0.0123	-0.0148	-0.0414	-0.0002	0.0003	-0.0086	0.0500
Missing Committee Score	-0.3358	-0.3437	-0.1972	-0.0516	-0.1281	-0.1189	-0.0273	-0.0479	-0.0179	0.0551	-0.1108	-0.1049	-0.0094	-0.1045
Missing Application Pdf	0.0270	-0.1797	0.0709	-0.0220	0.0179	-0.0650	-0.0082	-0.0079	-0.0674	0.0874	0.0263	0.0656	-0.0632	-0.1400
Missing #Math Courses	0.0188	-0.1817	0.0657	-0.0200	0.0155	-0.0670	-0.0039	-0.0105	-0.0646	0.0844	0.0275	0.0646	-0.0636	-0.1423
Missing Letters	0.0263	-0.1765	0.0686	-0.0209	0.0219	-0.0674	-0.0092	-0.0091	-0.0632	0.0848	0.0274	0.0645	-0.0622	-0.1461
Missing CV	0.0255	-0.1806	0.0760	-0.0193	0.0171	-0.0579	-0.0109	-0.0110	-0.0757	0.0915	0.0297	0.0556	-0.0596	-0.1431
Missing Essay	0.0263	-0.1746	0.0683	-0.0222	0.0160	-0.0660	-0.0101	-0.0042	-0.0649	0.0881	0.0230	0.0668	-0.0601	-0.1415

	MGRE	MGPA	MT/I	MAge	Score	NMath	AMath	LetN	LPos	LNeg	LSta	LAbi	LRes	LGri	LTea	LCom	LAge
Missing GRE	1																
Missing Undergraduate GPA	0.0625	1															
Missing TOEFL/IELTS	0.0714	-0.2810	1														
Missing Winsorized Age	0.1575	0.0153	0.0236	1													
Committee Score	-0.0451	-0.0466	0.0188	0.0000	1												
#Math Courses	0.0170	-0.0632	0.0423	0.0004	-0.0232	1											
Advanced Math	-0.0417	-0.0261	-0.0030	-0.0092	0.1388	0.2255	1										
Letters Length/100	-0.0082	-0.0084	0.0034	0.0007	0.1631	0.0027	0.0502	1									
Letters Positive Terms (%)	0.1234	-0.0626	-0.0402		0.0587	-0.0070	-0.0916	0.0015	1								
Letters Negative Terms (%)	0.0151	-0.0109	0.1318		0.0330	0.0100	-0.0104	0.0546	0.1546	1							
Letters Standout Terms (%)	0.1382	0.0456	-0.0033		0.1475	0.0194	-0.1303	0.0427	0.4324	-0.0327	1						
Letters Ability Terms (%)	-0.0190	-0.0979	0.0358		0.0943	-0.0408	-0.1419	0.0162	0.5004	0.1896	0.2105	1					
Letters Research Terms (%)	-0.0446	0.0039	0.0552		0.1776	0.0127	0.0874	0.2071	0.0820	0.1569	-0.0520	0.0863	1				
Letters Grindstone Terms (%)	0.0905	-0.0876	0.0740		0.1127	-0.0523	-0.0447	0.0523	0.3502	0.1286	0.0678	0.3655	0.2281	1			
Letters Teaching Terms (%)	-0.0047	-0.0060	-0.0189		-0.0163	-0.0878	-0.1488	0.1290	0.3650	-0.0016	0.1177	0.3161	0.0976	0.2383	1		
Letters Communal Terms (%)	0.1483	-0.0116	-0.1435		-0.1044	-0.0440	-0.0754	-0.1683	0.2329	-0.0224	0.0070	0.0719	-0.1124	0.1381	0.0216	1	
Letters Agentic Terms (%)	0.0548	-0.1457	0.1325		0.0353	0.1087	0.0595	0.0830	0.3284	0.0143	0.1124	0.3027	0.0256	0.2276	0.2610	0.0207	1
Letters Terms (%) PC1	0.0261	-0.0426	0.0213	0.0004	0.0492	-0.0199	-0.1116	0.0803	0.8114	0.2386	0.3964	0.7486	0.2082	0.6157	0.5816	0.2083	0.5580
Letters Terms (%) PC2	-0.0271	-0.0068	0.0497	0.0003	0.0365	-0.0202	0.0655	0.1600	-0.2292	0.4519	-0.5572	-0.0082	0.7168	0.3168	0.0333	-0.2524	-0.1306
CV Length/100	-0.0228	-0.0163	-0.0019	0.0002	0.0962	0.0312	0.0677	0.2051	-0.0026	-0.0135	0.0985	0.0481	0.1497	0.0613	0.0839	-0.1494	0.1032
CV Coding	-0.0094	0.0346	0.0194	-0.0002	0.0740	0.0970	0.0863	0.1100	-0.0178	0.0284	0.0407	-0.0115	0.2123	0.0344	-0.0499	-0.0274	0.0278
CV Publication/Working Paper	-0.0289	0.0128	-0.0457	-0.0064	0.1599	-0.0381	0.2648	0.1565	0.1046	0.0451	0.1148	0.0639	0.2120	0.1178	0.1343	-0.0182	0.0086
Essay Length/100	-0.0057	0.0249	0.0041	0.0001	0.0474	-0.0164	0.0317	0.1083	-0.0279	0.0331	-0.0042	-0.0206	0.0524	-0.0092	-0.0672	-0.0119	-0.0283
Essay Specific Professor	-0.0406	0.0133	-0.0322	-0.0091	0.1428	0.0022	0.4462	0.0738	-0.0047	0.0772	-0.0022	-0.0215	0.0483	0.0270	0.0159	-0.0317	0.0294
Essay Economics Terms (%)	0.0164	-0.0006	0.0160	0.0000	0.0035	0.1185	0.0515	-0.0282	-0.0572	-0.0354	-0.0553	0.0057	0.1233	0.0995	0.0289	0.0579	0.0002
Essay Research Terms (%)	-0.0019	-0.0008	0.0049	-0.0003	0.0376	0.0664	0.0430	0.0499	-0.0016	0.0032	0.0595	-0.0279	0.2357	-0.0866	-0.0744	-0.0378	-0.0300
Essay Positive Terms (%)	0.0013	-0.0200	0.0169	-0.0005	-0.0277	0.0540	-0.0383	-0.1339	0.0887	-0.0558	-0.0026	-0.0028	-0.0436	-0.0223	0.0400	-0.0516	-0.0216
Essay Negative Terms (%)	-0.0132	0.0141	0.0353	0.0003	0.0270	0.0719	0.0827	-0.0128	0.0174	0.1960	0.0757	-0.0206	-0.0427	-0.0338	-0.0425	-0.0131	0.0363
Missing Committee Score	0.1134	0.0768	0.0216	0.0247	0.0000	-0.0153	-0.2077	-0.0137	-0.0089	0.0003	0.0162	-0.0200	-0.0288	-0.0786	0.0211	-0.0473	-0.0157
Missing Application Pdf	0.0578	-0.0143	0.0482	0.0140	-0.1544	0.0263	-0.6598	0.0467									
Missing #Math Courses	0.0623	-0.0108	0.0517	0.0137	-0.1618	0.0268	-0.6547	0.0402	-0.0241	-0.0133	0.0534	-0.0279	-0.0512	0.0813	0.0401	-0.0056	0.0053
Missing Letters	0.0571	-0.0142	0.0437	0.0139	-0.1570	0.0336	-0.6607	0.0470									
Missing CV	0.0558	-0.0205	0.0536	0.0137	-0.1568	0.0332	-0.6449	0.0394	-0.0109	-0.0393	-0.0461	0.0037	-0.0206	0.0649	0.0421	0.0770	0.1614
Missing Essay	0.0564	-0.0119	0.0429	0.0138	-0.1489	0.0146	-0.6606	0.0474	-0.0201	-0.0545	-0.0171	0.0164	-0.0382	0.0205	-0.0490	-0.0453	-0.0824

	LPC1	LPC2	CVN	CVCod	CVPap	EssN	EssProf	EssEcon	EssRes	EssPos	EssNeg	MScore	MPDF	MNMath	ML	MCV	MEss
Letters Terms (%) PC1	1																
Letters Terms (%) PC2	-0.0204	1															
CV Length/100	0.0820	0.0721	1														
CV Coding	0.0147	0.1218	0.2274	1													
CV Publication/Working Paper	0.1158	0.0865	0.2482	0.1413	1												
Essay Length/100	-0.0332	0.0460	0.0706	0.1168	0.0772	1											
Essay Specific Professor	-0.0084	0.0360	0.0795	0.1038	0.2923	0.0701	1										
Essay Economics Terms (%)	0.0166	0.1110	0.0016	0.0578	0.0391	-0.0593	0.0049	1									
Essay Research Terms (%)	-0.0307	0.0886	0.0185	0.1414	0.0621	0.0651	-0.0097	0.1593	1								
Essay Positive Terms (%)	0.0132	-0.0494	-0.0413	-0.0248	-0.0291	-0.1956	-0.0535	-0.0689	-0.0250	1							
Essay Negative Terms (%)	0.0140	-0.0083	0.0279	0.0566	0.0102	0.0716	0.0666	-0.1217	-0.1206	-0.0714	1						
Missing Committee Score	-0.0015	-0.0054	-0.0070	0.0098	-0.1443	0.0100	-0.2336	-0.0045	0.0027	0.0134	-0.0106	1					
Missing Application Pdf	0.0322	0.0218	0.0165	-0.0164	-0.4566	0.0042	-0.6494	-0.0023	-0.0250	-0.0327	0.0183	0.3105	1				
Missing #Math Courses	0.0343	0.0164	0.0076	-0.0239	-0.4531	0.0007	-0.6399	0.0044	-0.0279	-0.0345	0.0283	0.3149	0.9843	1			
Missing Letters	0.0324	0.0219	0.0199	-0.0202	-0.4599	0.0066	-0.6462	0.0039	-0.0208	-0.0261	0.0113	0.3076	0.9929	0.9769	1		
Missing CV	0.0419	0.0187	0.0168	-0.0168	-0.4653	0.0026	-0.6299	-0.0129	-0.0385	-0.0266	0.0242	0.3061	0.9814	0.9650	0.9740	1	
Missing Essay	0.0253	0.0200	0.0134	-0.0246	-0.4464	0.0042	-0.6578	-0.0023	-0.0253	-0.0331	0.0185	0.3034	0.9871	0.9710	0.9798	0.9709	1

Table 9a: Economics Department Logistic Regression Results (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.2091	-0.1311	0.1291	0.0997	0.0892	0.1020	-0.8042	-0.5831	-1.3927
	(0.2288)	(0.2577)	(0.6478)	(0.1812)	(0.1814)	(0.1970)	(4.5726)	(4.5702)	(4.1436)
2017	-0.3355	-0.5461**	0.6767	0.2827	0.2903	0.1565	-12.4495***	-12.2002***	-6.3639
	(0.2390)	(0.2627)	(0.6543)	(0.1761)	(0.1767)	(0.1925)	(4.4305)	(4.4347)	(3.9262)
2018	0.2889	-0.2993	0.2947	0.1627	0.2192	0.1272	-9.3709**	-9.2465**	-6.0098
	(0.2056)	(0.2212)	(0.5517)	(0.1806)	(0.1827)	(0.1972)	(4.4278)	(4.4749)	(3.9975)
2019	-0.1933	-0.3208	-0.0662	0.0116	0.0023	-0.1488	-5.3768	-4.7378	1.0400
	(0.2227)	(0.2490)	(0.6069)	(0.1927)	(0.1933)	(0.2061)	(4.4927)	(4.5026)	(3.9171)
Winsorized Age	-0.0508**	-0.0414*	0.0065	-0.0141	-0.0150	-0.0094	0.4466	0.4383	0.2496
	(0.0220)	(0.0243)	(0.0600)	(0.0170)	(0.0170)	(0.0185)	(0.4147)	(0.4141)	(0.3766)
Female	-0.1826	-0.1535	0.0691	0.1351	0.1379	0.2109*	-1.5097	-1.6305	-4.7144*
	(0.1446)	(0.1590)	(0.4006)	(0.1167)	(0.1168)	(0.1259)	(2.9357)	(2.9318)	(2.6789)
U.S.	0.0002	-0.0549	0.0998	0.0198	0.0241	-0.0123	2.8625	2.5697	4.2450
	(0.2240)	(0.2468)	(0.6663)	(0.1794)	(0.1801)	(0.1997)	(4.4691)	(4.4669)	(4.1868)
International Asian	-0.1230	-0.0646	-0.1205	0.2014	0.1972	0.1700	-0.7475	-0.9433	1.1783
	(0.1688)	(0.1838)	(0.5028)	(0.1394)	(0.1392)	(0.1541)	(3.3754)	(3.3732)	(3.1231)
GRE Verbal Percentile	-0.0025	-0.0001	0.0008	0.0004	0.0001	-0.0021	-0.0668	-0.0623	0.0181
	(0.0034)	(0.0036)	(0.0089)	(0.0026)	(0.0026)	(0.0029)	(0.0661)	(0.0663)	(0.0595)
GRE Quantitative Percentile	-0.0028	-0.0005	-0.0125	0.0145**	0.0143**	-0.0003	-0.5982***	-0.5916***	-0.0184
	(0.0075)	(0.0080)	(0.0195)	(0.0065)	(0.0065)	(0.0069)	(0.1570)	(0.1569)	(0.1376)
GRE Analytic Percentile	0.0041	0.0019	-0.0010	0.0009	0.0011	0.0019	0.0181	0.0156	-0.0008
	(0.0031)	(0.0033)	(0.0074)	(0.0024)	(0.0024)	(0.0026)	(0.0587)	(0.0587)	(0.0529)
Undergraduate GPA	-0.4230	-0.3157	0.9077	0.4075	0.3986	0.4627	-2.7870	-2.7400	-5.8113
	(0.3862)	(0.4536)	(1.8572)	(0.3984)	(0.3990)	(0.4326)	(9.4800)	(9.4624)	(8.8952)
TOEFL/IELTS	0.2329	0.2776	-0.2073	-0.0682	-0.0625	-0.0142	2.4237	2.2404	-0.4376
	(0.2154)	(0.2202)	(0.4100)	(0.1509)	(0.1507)	(0.1670)	(3.7070)	(3.7021)	(3.1595)
Elite U.S. University	-0.0473	-0.1157	-0.2546	0.3145	0.3119	0.3760*	-6.7562	-6.6352	-7.1987
	(0.2618)	(0.2943)	(0.7160)	(0.2054)	(0.2062)	(0.2230)	(5.3790)	(5.3875)	(4.8870)
Elite U.S. Liberal Arts College	-0.6642	-0.6053		0.0069	-0.0044	-0.0958	-2.2643	-2.2283	3.3486
	(0.5448)	(0.5744)		(0.3656)	(0.3659)	(0.3964)	(8.9098)	(8.8713)	(7.5578)
Elite Foreign University	-0.1797	-0.1798	0.5156	-0.0538	-0.0692	-0.0298	-1.0302	-0.9315	-0.2765
	(0.3900)	(0.4341)	(1.2147)	(0.3273)	(0.3294)	(0.3495)	(8.2453)	(8.2793)	(7.1254)
Other Foreign University	-0.0406	-0.1089	0.6613	0.1055	0.1125	0.0046	-6.0523	-6.2650	-0.6409
	(0.2078)	(0.2262)	(0.5235)	(0.1688)	(0.1689)	(0.1840)	(4.2123)	(4.2076)	(3.8417)
Missing Undergraduate Institution	-0.0012	0.0377	-0.6367	0.1037	0.0966	0.0297	-5.9887	-5.9936	-2.3404
	(0.2217)	(0.2413)	(0.5829)	(0.1844)	(0.1842)	(0.1977)	(4.5947)	(4.5865)	(4.1622)
Graduate Degree	-0.0846	-0.1354	-0.0100	0.1515	0.1611	0.1906	-1.4756	-1.6341	-2.0725
	(0.1458)	(0.1626)	(0.3834)	(0.1166)	(0.1165)	(0.1252)	(2.8643)	(2.8619)	(2.5933)
Work Experience	0.2547*	0.2149	-0.2181	0.0750	0.0805	0.1443	0.4084	0.3542	-2.7442
	(0.1463)	(0.1588)	(0.3573)	(0.1161)	(0.1160)	(0.1250)	(2.8682)	(2.8651)	(2.6268)
#Professor Recommenders	0.0194	-0.0565	0.1806	0.0038	0.0074	-0.0302	-0.2230	-0.1761	1.2584
	(0.0950)	(0.1021)	(0.2725)	(0.0759)	(0.0761)	(0.0794)	(1.9087)	(1.9104)	(1.7006)
#Prolific Recommenders	-0.0388	-0.0071	0.0168	0.0027	-0.0000	0.0325	0.3327	0.3170	-1.1079
	(0.0671)	(0.0713)	(0.1685)	(0.0521)	(0.0522)	(0.0575)	(1.3245)	(1.3242)	(1.2261)
Missing GRE	0.3002	0.0736		0.3136	0.3314	0.2674	-12.0035	-11.9570	-11.5860
	(0.3757)	(0.3872)		(0.2858)	(0.2857)	(0.3297)	(8.1036)	(8.0998)	(8.0641)
Missing Undergraduate GPA	0.1997	0.1720	0.2910	-0.1321	-0.1290	-0.0816	5.2305	5.1674	4.3351
	(0.1863)	(0.2016)	(0.5443)	(0.1391)	(0.1389)	(0.1534)	(3.6167)	(3.6124)	(3.3094)
Missing TOEFL/IELTS	-0.1740	-0.0792	-0.3009	0.1948	0.1815	0.2286*	-1.5099	-1.4121	-3.1183
	(0.1609)	(0.1741)	(0.4256)	(0.1280)	(0.1281)	(0.1380)	(3.1162)	(3.1192)	(2.7616)
Committee Score		0.0372	-0.1004		0.0465	0.0330		-1.6627	-0.4906
		(0.0423)	(0.1060)		(0.0505)	(0.0558)		(1.1655)	(1.1645)
#Math Courses	0.0632	0.0756	-0.1683	-0.1269	-0.1305*	-0.1764**	1.1826	1.2073	3.0509**
	(0.0687)	(0.0733)	(0.1601)	(0.0786)	(0.0780)	(0.0863)	(1.7287)	(1.7209)	(1.5010)
Advanced Math	0.0222	-0.0336	0.9873	-0.1639	-0.1630	-0.2842	1.1644	1.2290	6.1671
	(0.2590)	(0.2747)	(0.7209)	(0.3004)	(0.3013)	(0.3256)	(6.4828)	(6.4875)	(5.6351)
Letters Length/100	-0.0254	-0.0128	-0.0575	-0.1049***	-0.1057***	-0.1218***	1.6052**	1.6308**	2.1010***
	(0.0333)	(0.0361)	(0.0885)	(0.0381)	(0.0376)	(0.0428)	(0.7873)	(0.7869)	(0.6934)
Letters Terms (%) PC1	0.0280	-0.0073	0.0381	0.0574	0.0643	0.0566	1.1364	1.1816	1.0356
	(0.0779)	(0.0821)	(0.2097)	(0.0828)	(0.0834)	(0.0930)	(1.8933)	(1.9000)	(1.6894)
Letters Terms (%) PC2	-0.2328*	-0.2369*	-0.4921	0.0802	0.0718	0.1318	-2.8589	-2.7586	-5.0669**
	(0.1226)	(0.1255)	(0.3028)	(0.1205)	(0.1211)	(0.1250)	(2.6915)	(2.7005)	(2.1872)
CV Length/100	0.0971**	0.0795	0.1022	-0.0358	-0.0336	-0.0095	2.4879**	2.5172**	0.5354
	(0.0492)	(0.0508)	(0.1340)	(0.0548)	(0.0551)	(0.0641)	(1.2406)	(1.2458)	(1.0685)
CV Coding	-0.0198	-0.0473	-0.0834	0.0688	0.0720*	0.0795*	-1.6529	-1.6778	-1.0250
	(0.0411)	(0.0448)	(0.1069)	(0.0434)	(0.0434)	(0.0474)	(1.0435)	(1.0451)	(0.8904)
CV Publication/Working Paper	0.3079	0.3005	1.3491*	0.1111	0.1183	0.1111	2.7524	2.4246	5.5955
	(0.2852)	(0.2962)	(0.7475)	(0.3264)	(0.3272)	(0.3482)	(7.3505)	(7.3749)	(6.2211)
Essay Length/100	0.0679	0.0790	-0.0558	-0.0667	-0.0636	-0.0694	0.5523	0.4398	0.3231
	(0.0770)	(0.0831)	(0.1489)	(0.0780)	(0.0794)	(0.0871)	(1.7254)	(1.7321)	(1.3973)
Essay Specific Professor	-0.1756	-0.2494	-0.1115	-0.0547	-0.0524	0.0009	1.2639	1.2219	-3.1507
	(0.2564)	(0.2714)	(0.6302)	(0.2774)	(0.2798)	(0.3022)	(6.3588)	(6.3580)	(5.5747)
Essay Economics Terms (%)	-0.0123	-0.0094	-0.0406	-0.0091	-0.0082	-0.0222	0.2276	0.2960	0.6769
	(0.0542)	(0.0572)	(0.1334)	(0.0539)	(0.0547)	(0.0624)	(1.3179)	(1.3189)	(1.1799)
Essay Research Terms (%)	0.0303	0.0066	-0.0464	0.1042	0.1084	0.1176	-1.8679	-1.9069	-1.8170
	(0.0656)	(0.0702)	(0.1578)	(0.0703)	(0.0705)	(0.0723)	(1.5218)	(1.5245)	(1.2596)
Essay Positive Terms (%)	0.0180	0.0074	0.3699***	0.0871*	0.0868*	0.0947	-1.5542	-1.4775	-1.1629
	(0.0474)	(0.0484)	(0.1213)	(0.0511)	(0.0517)	(0.0609)	(1.1518)	(1.1568)	(1.1006)
Essay Negative Terms (%)	0.0333	0.0312	-0.0731	-0.0304	-0.0280	-0.0168	0.5011	0.4716	0.2911
	(0.0314)	(0.0351)	(0.0769)	(0.0359)	(0.0359)	(0.0382)	(0.8608)	(0.8614)	(0.7100)
Missing Committee Score		-4.3141***			0.2635**	0.4129***		-0.9440	-6.1957**
		(0.4312)			(0.1267)	(0.1388)		(3.0258)	(2.8364)
Missing Application Pdf	-0.9321***	-0.0588	0.9473	0.0574	-0.0560	-0.1100	4.5136	4.8399	5.6614
	(0.2534)	(0.2753)	(0.6793)	(0.2628)	(0.2678)	(0.2943)	(5.9238)	(6.0503)	(5.4160)
Open House			1.0643						
			(0.8268)						
Program Rank						-0.0049**			0.1176***
						(0.0019)			(0.0386)
Missing Program Rank						-2.5925***			94.5255***
						(0.3273)			(5.6183)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.0552	0.2804	0.1402	0.0221	0.0245	0.1935	0.0040	0.0042	0.0953

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 9b: Economics Department Logistic Regression AMEs (Synthetic)

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.0185	-0.0100	0.0201	0.0136	0.0122	0.0120	-0.3764	-0.2729	-0.6112
	(0.0203)	(0.0197)	(0.1005)	(0.0248)	(0.0248)	(0.0233)	(2.1401)	(2.1390)	(1.8189)
2017	-0.0297	-0.0417**	0.1052	0.0387	0.0396	0.0185	-5.8265***	-5.7099***	-2.7929
	(0.0212)	(0.0199)	(0.1005)	(0.0241)	(0.0241)	(0.0227)	(2.0748)	(2.0770)	(1.7228)
2018	0.0256	-0.0229	0.0458	0.0223	0.0299	0.0150	-4.3857**	-4.3275**	-2.6375
	(0.0182)	(0.0168)	(0.0856)	(0.0247)	(0.0250)	(0.0233)	(2.0732)	(2.0958)	(1.7555)
2019	-0.0171	-0.0245	-0.0103	0.0016	0.0003	-0.0176	-2.5164	-2.2174	0.4564
	(0.0197)	(0.0189)	(0.0944)	(0.0264)	(0.0264)	(0.0243)	(2.1013)	(2.1061)	(1.7192)
Winsorized Age	-0.0045**	-0.0032*	0.0010	-0.0019	-0.0020	-0.0011	0.2090	0.2051	0.1095
	(0.0020)	(0.0019)	(0.0093)	(0.0023)	(0.0023)	(0.0022)	(0.1941)	(0.1938)	(0.1654)
Female	-0.0162	-0.0117	0.0107	0.0185	0.0188	0.0249*	-0.7066	-0.7631	-2.0690*
	(0.0128)	(0.0121)	(0.0621)	(0.0160)	(0.0159)	(0.0149)	(1.3745)	(1.3728)	(1.1767)
U.S.	0.0000	-0.0042	0.0155	0.0027	0.0033	-0.0015	1.3397	1.2026	1.8630
	(0.0198)	(0.0189)	(0.1035)	(0.0246)	(0.0246)	(0.0236)	(2.0914)	(2.0905)	(1.8379)
International Asian	-0.0109	-0.0049	-0.0187	0.0276	0.0269	0.0201	-0.3498	-0.4415	0.5171
	(0.0150)	(0.0140)	(0.0782)	(0.0191)	(0.0190)	(0.0182)	(1.5798)	(1.5789)	(1.3706)
GRE Verbal Percentile	-0.0002	-0.0000	0.0001	0.0001	0.0000	-0.0002	-0.0313	-0.0291	0.0080
	(0.0003)	(0.0003)	(0.0014)	(0.0004)	(0.0004)	(0.0003)	(0.0309)	(0.0310)	(0.0261)
GRE Quantitative Percentile	-0.0002	-0.0000	-0.0019	0.0020**	0.0020**	-0.0000	-0.2800***	-0.2769***	-0.0081
	(0.0007)	(0.0006)	(0.0030)	(0.0009)	(0.0009)	(0.0008)	(0.0734)	(0.0734)	(0.0604)
GRE Analytic Percentile	0.0004	0.0001	-0.0002	0.0001	0.0002	0.0002	0.0085	0.0073	-0.0004
	(0.0003)	(0.0003)	(0.0011)	(0.0003)	(0.0003)	(0.0003)	(0.0275)	(0.0275)	(0.0232)
Undergraduate GPA	-0.0375	-0.0241	0.1411	0.0558	0.0544	0.0546	-1.3043	-1.2824	-2.5504
	(0.0342)	(0.0346)	(0.2875)	(0.0545)	(0.0545)	(0.0510)	(4.4373)	(4.4292)	(3.9043)
TOEFL/IELTS	0.0206	0.0212	-0.0322	-0.0093	-0.0085	-0.0017	1.1343	1.0486	-0.1921
	(0.0191)	(0.0168)	(0.0637)	(0.0207)	(0.0206)	(0.0197)	(1.7354)	(1.7331)	(1.3866)
Elite U.S. University	-0.0042	-0.0088	-0.0396	0.0431	0.0426	0.0444*	-3.1620	-3.1054	-3.1593
	(0.0232)	(0.0225)	(0.1109)	(0.0281)	(0.0282)	(0.0263)	(2.5179)	(2.5220)	(2.1436)
Elite U.S. Liberal Arts College	-0.0588	-0.0462		0.0009	-0.0006	-0.0113	-1.0597	-1.0429	1.4696
	(0.0483)	(0.0439)		(0.0500)	(0.0500)	(0.0468)	(4.1696)	(4.1517)	(3.3177)
Elite Foreign University	-0.0159	-0.0137	0.0801	-0.0074	-0.0095	-0.0035	-0.4822	-0.4360	-0.1214
	(0.0346)	(0.0332)	(0.1891)	(0.0448)	(0.0450)	(0.0412)	(3.8590)	(3.8749)	(3.1271)
Other Foreign University	-0.0036	-0.0083	0.1028	0.0144	0.0154	0.0005	-2.8325	-2.9321	-0.2813
	(0.0184)	(0.0173)	(0.0813)	(0.0231)	(0.0231)	(0.0217)	(1.9702)	(1.9680)	(1.6858)
Missing Undergraduate Institution	-0.0001	0.0029	-0.0989	0.0142	0.0132	0.0035	-2.8028	-2.8051	-1.0271
	(0.0196)	(0.0184)	(0.0902)	(0.0252)	(0.0251)	(0.0233)	(2.1499)	(2.1460)	(1.8264)
Graduate Degree	-0.0075	-0.0103	-0.0015	0.0207	0.0220	0.0225	-0.6906	-0.7648	-0.9095
	(0.0129)	(0.0124)	(0.0596)	(0.0160)	(0.0159)	(0.0147)	(1.3406)	(1.3395)	(1.1381)
Work Experience	0.0226*	0.0164	-0.0339	0.0103	0.0110	0.0170	0.1911	0.1658	-1.2044
	(0.0129)	(0.0121)	(0.0551)	(0.0159)	(0.0158)	(0.0147)	(1.3422)	(1.3408)	(1.1533)
#Professor Recommenders	0.0017	-0.0043	0.0281	0.0005	0.0010	-0.0036	-0.1044	-0.0824	0.5523
	(0.0084)	(0.0078)	(0.0423)	(0.0104)	(0.0104)	(0.0094)	(0.8933)	(0.8941)	(0.7465)
#Prolific Recommenders	-0.0034	-0.0005	0.0026	0.0004	-0.0000	0.0038	0.1557	0.1484	-0.4862
	(0.0059)	(0.0054)	(0.0262)	(0.0071)	(0.0071)	(0.0068)	(0.6197)	(0.6196)	(0.5384)
Missing GRE	0.0266	0.0056		0.0429	0.0453	0.0316	-5.6177	-5.5961	-5.0848
	(0.0333)	(0.0296)		(0.0391)	(0.0390)	(0.0389)	(3.7950)	(3.7934)	(3.5412)
Missing Undergraduate GPA	0.0177	0.0131	0.0452	-0.0181	-0.0176	-0.0096	2.4479	2.4184	1.9026
	(0.0165)	(0.0154)	(0.0847)	(0.0190)	(0.0190)	(0.0181)	(1.6942)	(1.6921)	(1.4526)
Missing TOEFL/IELTS	-0.0154	-0.0060	-0.0468	0.0267	0.0248	0.0270*	-0.7067	-0.6609	-1.3685
	(0.0143)	(0.0133)	(0.0659)	(0.0175)	(0.0175)	(0.0162)	(1.4590)	(1.4604)	(1.2133)
Committee Score		0.0028	-0.0156		0.0064	0.0039		-0.7782	-0.2153
		(0.0032)	(0.0165)		(0.0069)	(0.0066)		(0.5455)	(0.5111)
#Math Courses	0.0056	0.0058	-0.0262	-0.0174	-0.0178*	-0.0208**	0.5535	0.5650	1.3390**
	(0.0061)	(0.0056)	(0.0248)	(0.0107)	(0.0106)	(0.0101)	(0.8095)	(0.8059)	(0.6591)
Advanced Math	0.0020	-0.0026	0.1534	-0.0224	-0.0223	-0.0335	0.5450	0.5752	2.7066
	(0.0229)	(0.0210)	(0.1122)	(0.0411)	(0.0411)	(0.0384)	(3.0339)	(3.0362)	(2.4720)
Letters Length/100	-0.0022	-0.0010	-0.0089	-0.0144***	-0.0144***	-0.0144***	0.7512**	0.7632**	0.9221***
	(0.0029)	(0.0028)	(0.0137)	(0.0052)	(0.0051)	(0.0050)	(0.3694)	(0.3693)	(0.3050)
Letters Terms (%) PC1	0.0025	-0.0006	0.0059	0.0079	0.0088	0.0067	0.5318	0.5530	0.4545
	(0.0069)	(0.0063)	(0.0325)	(0.0113)	(0.0114)	(0.0110)	(0.8858)	(0.8889)	(0.7413)
Letters Terms (%) PC2	-0.0206*	-0.0181*	-0.0765*	0.0110	0.0098	0.0156	-1.3380	-1.2911	-2.2237**
	(0.0108)	(0.0095)	(0.0462)	(0.0165)	(0.0165)	(0.0147)	(1.2609)	(1.2651)	(0.9607)
CV Length/100	0.0086**	0.0061	0.0159	-0.0049	-0.0046	-0.0011	1.1643**	1.1781**	0.2350
	(0.0043)	(0.0039)	(0.0210)	(0.0075)	(0.0075)	(0.0076)	(0.5802)	(0.5825)	(0.4688)
CV Coding	-0.0018	-0.0036	-0.0130	0.0094	0.0098*	0.0094*	-0.7736	-0.7852	-0.4499
	(0.0036)	(0.0034)	(0.0167)	(0.0059)	(0.0059)	(0.0056)	(0.4887)	(0.4894)	(0.3909)
CV Publication/Working Paper	0.0273	0.0230	0.2097*	0.0152	0.0162	0.0131	1.2881	1.1348	2.4557
	(0.0252)	(0.0226)	(0.1128)	(0.0447)	(0.0447)	(0.0411)	(3.4400)	(3.4515)	(2.7307)
Essay Length/100	0.0060	0.0060	-0.0087	-0.0091	-0.0087	-0.0082	0.2585	0.2058	0.1418
	(0.0068)	(0.0063)	(0.0231)	(0.0107)	(0.0108)	(0.0103)	(0.8075)	(0.8107)	(0.6133)
Essay Specific Professor	-0.0156	-0.0190	-0.0173	-0.0075	-0.0072	0.0001	0.5915	0.5719	-1.3828
	(0.0227)	(0.0207)	(0.0978)	(0.0380)	(0.0382)	(0.0357)	(2.9760)	(2.9757)	(2.4463)
Essay Economics Terms (%)	-0.0011	-0.0007	-0.0063	-0.0012	-0.0011	-0.0026	0.1065	0.1385	0.2971
	(0.0048)	(0.0044)	(0.0207)	(0.0074)	(0.0075)	(0.0074)	(0.6168)	(0.6173)	(0.5179)
Essay Research Terms (%)	0.0027	0.0005	-0.0072	0.0143	0.0148	0.0139	-0.8742	-0.8925	-0.7975
	(0.0058)	(0.0054)	(0.0245)	(0.0096)	(0.0096)	(0.0085)	(0.7127)	(0.7141)	(0.5530)
Essay Positive Terms (%)	0.0016	0.0006	0.0575***	0.0119*	0.0118*	0.0112	-0.7274	-0.6915	-0.5104
	(0.0042)	(0.0037)	(0.0182)	(0.0070)	(0.0071)	(0.0072)	(0.5394)	(0.5418)	(0.4832)
Essay Negative Terms (%)	0.0030	0.0024	-0.0114	-0.0042	-0.0038	-0.0020	0.2345	0.2207	0.1278
	(0.0028)	(0.0027)	(0.0120)	(0.0049)	(0.0049)	(0.0045)	(0.4029)	(0.4032)	(0.3117)
Missing Committee Score		-0.3296***			0.0360**	0.0487***		-0.4418	-2.7191**
		(0.0330)			(0.0173)	(0.0162)		(1.4166)	(1.2447)
Missing Application Pdf	-0.0826***	-0.0045	0.1472	0.0079	-0.0076	-0.0130	2.1124	2.2652	2.4846
	(0.0225)	(0.0210)	(0.1064)	(0.0360)	(0.0366)	(0.0347)	(2.7720)	(2.8317)	(2.3765)
Open House			0.1654						
			(0.1278)						
Program Rank						-0.0006**			0.0516***
						(0.0002)			(0.0169)
Missing Program Rank						-0.3060***			41.4849***
						(0.0381)			(2.4608)
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	237	2329	2329	2329	2329	2329	2329

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 10: Economics Department Lasso Logistic Regression Results

Dependent Variable	Accept	Accept	Matriculate	T100 Placement	T100 Placement	T100 Placement	Placement Rank	Placement Rank	Placement Rank
2016	-0.1107								
2017		-0.4928							
2018		-2.4456	0.0186						
2019		-1.7752							
Winsorized Age				0.0084	0.0047	0.0184			
Female	0.2674	0.5589							
U.S.	-0.1959							0.0477	
International Asian	-0.8686	-0.8651	0.1271				-0.5295	-0.7927	
GRE Verbal Percentile	0.0227	0.0047	-0.0015	0.0015			-0.0085		
GRE Quantitative Percentile	0.1027	0.0174	-0.0273	0.0179	0.0125		-0.2706	-0.2243	-0.0070
GRE Analytic Percentile	0.0056	0.0035							
Undergraduate GPA	0.9812	0.7135				-0.1316			0.2561
TOEFL/IELTS	0.3055		-0.2943	0.0300		0.0934	-0.6903	-0.5850	-2.2692
Elite U.S. University	0.4024								
Elite U.S. Liberal Arts College		0.1112							
Elite Foreign University		-0.4025							
Other Foreign University	0.0790	0.0940	0.1998						-0.2265
Missing Undergraduate Institution			-0.0036						
Graduate Degree	0.1178								
Work Experience	0.6006	0.4068	0.0404	0.0961	0.0470	0.1976			-2.0809
#Professor Recommenders									0.0922
#Prolific Recommenders	0.1534	0.0301	-0.0870	0.0526	0.0363		-0.0789	-0.0358	
Missing GRE	-0.6534			-0.0697					
Missing Undergraduate GPA	-0.2348	-0.1148							
Missing TOEFL/IELTS	0.1447			-0.1243	-0.1295		1.8402	2.2548	
Committee Score		3.4243			0.0353			-0.4899	
#Math Courses	-0.0553	-0.0354							
Advanced Math	0.4302	0.7352							
Letters Length/100	0.1196	0.0587							
Letters Terms (%) PC1	0.0746	0.1485	0.1258			0.0102			
Letters Terms (%) PC2	-0.0162	-0.2901							
CV Length/100	0.0219	0.0298		0.0014			-0.0286	-0.1179	
CV Coding	-0.0038	-0.1754		0.0165	0.0076		-0.2540	-0.4154	-0.0998
CV Publication/Working Paper	0.5563	0.6436		0.0892			-0.0987		
Essay Length/100	0.1202	0.0295							
Essay Specific Professor	0.3403	0.3958	0.0674						
Essay Economics Terms (%)	-0.0016			-0.0405	-0.0325	-0.0104	0.4473	0.6570	0.2117
Essay Research Terms (%)		-0.0278						-0.1506	
Essay Positive Terms (%)	-0.0049	-0.0382		0.0192	0.0140	0.0267			
Essay Negative Terms (%)		-0.0417							
Missing Committee Score		-0.8416			-0.2615	-0.0714		4.0146	0.7971
Missing Application Pdf	-1.0365								
Open House			1.0982						
Program Rank						-0.0082			0.1385
Missing Program Rank						-2.2557			19.4273
Sample	All	All	Accept = 1	All	All	All	All	All	All
Observations	2329	2329	257	2329	2329	2329	2329	2329	2329
Pseudo-R^2	0.2841	0.8007	0.1281	0.0196	0.0213	0.2265	0.0206	0.0278	0.2693
CV Pseudo-R^2	0.2441	0.7605	0.0596	0.0082	0.0129	0.2199	0.0121	0.0183	0.2638

B.1 Graduate School Regression Results

The main dataset in this paper consists of all 2,329 applicants to the university's economics PhD program from 2015–2019. However, I also have another dataset consisting of all 6,222 applicants to the university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023. Unlike the economics department dataset, the graduate school dataset does not contain subjective variables or program outcomes data. Therefore, I estimate this paper's structural models on the economics department dataset only. However, the graduate school dataset, which still contains objective variables and each applicant's admission status, is necessary for making comparisons across programs. The graduate school dataset also contains an ethnicity variable and concentration variables for the non-economics programs. Compared with fields of interest in economics, concentrations are more binding because applicants cannot costlessly change their concentration after entering the program. Summary statistics and correlations for the graduate school data are in Tables 11 and 12; the economics program has the highest acceptance rate at 11.21%, whereas the government program has the lowest acceptance rate at 4.30%.

In Table 13, I present logistic regressions of acceptance on the objective variables for all five programs.[16] The left subtable contains the regression coefficients, whereas the right subtable contains the average marginal effects (AMEs). Most notably, relative to the other programs, economics places more weight on undergraduate GPA and the quantitative GRE. A one point increase in an applicant's quantitative GRE percentile increases their acceptance probability by over one percentage point. As Bai et al. (2022) find, having a prolific recommender, defined as one who also recommended other applicants in the sample, matters as well. Going from zero to three prolific recommenders increases an applicant's acceptance probability by over six percentage points. This result provides evidence of academic network effects in the social sciences.

There are a few more results worth describing. First, across programs, acceptance rates were lowest in Spring 2021, which was the year that Covid reduced department funding. Second, all else equal, economics favors younger applicants, and economics, government, and history favor women. Third, government and history favor underrepresented minorities, whereas economics disfavors applicants from East and South Asia. Fourth, relative to a non-elite U.S. undergraduate institution, an elite institution matters more than a graduate degree or work experience. Fifth, taking a high number of math courses matters for economics. Note, however, that these results are descriptive and by themselves do not support any normative claims about admissions. In addition, the Pseudo-R^2 values never exceed 0.3 for any program. However, they are highest for economics at 0.2677, meaning that economics admissions is more data-driven than the other programs, at least with respect to the observable variables. That said, some of the other programs have admissions criteria, such as interviews, that economics does not. Although I cannot observe interview results, interviews may explain some of the variation in acceptances in those programs.

[16] In Table 14, I present the regression results on synthetic data. The synthetic estimates are more often than random chance similar to the actual estimates, but the synthetic estimates are generally less statistically significant.

I also run lasso with 5-fold cross-validation for variable selection. Per Table 15, lasso selects more variables for economics than for any other program. The last three variables in this table are missing variable dummies. Any time an applicant's GRE percentiles, undergraduate GPA, or TOEFL/IELTS scores are missing, I impute the variable's sample mean for that observation and encode the missing variable dummy equal to 1 (Krueger & Wu 2000). Allowing lasso to select one but not the other (see Bai et al. 2022) has a meaningful interpretation. For example, when lasso selects GPA but not its missing dummy for economics, it means that GPAs, as opposed to TOEFL/IELTS scores where lasso selects the missing dummy, are missing at random.

Finally, I implement a Wald test of whether the determinants of acceptance are the same across programs. The test's setup is as follows. Because the samples across programs are different, I assume that $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$; that is, conditional on the independent variables X, the dependent variable for economics Y^E is independent of its non-economics counterpart Y^{NE}. Also, suppose the dependent variable is binary; for example, accepted vs. rejected. As such, I parameterize the economics and non-economics outcome probabilities with the inverse logit functional form: $P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$ and $P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$.

Notably, I separate the intercept θ_0 from the slope θ_1, since the goal is to test for equal slopes, not equal intercepts. That said, I can add slopes I do not test to θ_0. Then the likelihood is:

$$\mathcal{L}_N^U\left(y|x;\theta\right) = \prod_{n=1}^{N_E} P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right)^{Y_n^E}\left[1 - P\left(Y_n^E = 1\middle|x_n^E;\theta^E\right)\right]^{1-Y_n^E}$$
$$\prod_{n=1}^{N_{NE}} P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right)^{Y_n^{NE}}\left[1 - P\left(Y_n^{NE} = 1\middle|x_n^{NE};\theta^{NE}\right)\right]^{1-Y_n^{NE}}. \quad (33)$$

Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. The Wald test statistic is therefore $\mathrm{W} = g(\hat{\theta})'[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})']^{-1}g(\hat{\theta}) \sim \chi_Q^2$, where Q is the number of restrictions; that is, the number of determinants I test at a time. Note that the null hypothesis is of equal AMEs, not equal logit coefficients, since only AMEs can be compared across samples (Mood 2010).

If the dependent variable were continuous (e.g. placement rank), I would parameterize its economics and non-economics distributions as: $f\left(y_n^E\middle|x_n^E;\beta^E,\sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}}\exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$ and $f\left(y_n^{NE}\middle|x_n^{NE};\beta^{NE},\sigma_S^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}}\exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y|x;\beta,\sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E\middle|x_n^E;\beta^E,\sigma_E^2\right)\prod_{n=1}^{N_{NE}} f\left(y_n^{NE}\middle|x_n^{NE};\beta^{NE},\sigma_{NE}^2\right). \quad (34)$$

Since linear coefficients can be compared across samples, the test is simpler (and can also be done with a likelihood-ratio test). Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. Under homoskedasticity (i.e. $\forall n, \sigma_n^2 = \sigma^2$), the Wald statistic is $\mathrm{W} = g(\hat{\beta}_1)'[\nabla_{\beta,\sigma^2}g(\hat{\beta}_1)\hat{V}(\hat{\beta},\hat{\sigma}^2)\nabla_{\beta,\sigma^2}g(\hat{\beta}_1)']^{-1}g(\hat{\beta}_1) \sim \chi_Q^2$.

Returning to Table 13b, the first column contains the economics acceptance AMEs, the sixth column contains the non-economics acceptance AMEs, and the Wald test resoundingly rejects the null hypothesis of equal AMEs with $p < 0.0001$. Therefore, the differences in AMEs for economics vs. non-economics are highly significant. To ensure that this test does not over-reject equal AMEs in general, I simulate two datasets, which for simplicity are for only the economics program and two non-economics programs. In each dataset, I simulate acceptance and success dependent variables. In the first dataset, the means, standard deviations, and correlations are similar to the actual data, meaning that the determinants of acceptance and success are different across programs.[17] In the second dataset, where the test is not supposed to reject equal AMEs, I modify the correlations, so the non-economics determinants are the same as those for economics.

In Table 25 of Appendix D, I present Wald test results on the simulated data. In the dataset where the data-generating parameters are such that the true AMEs of the determinants of acceptance and success are the same, the test does not reject the null; I obtain $p = 0.6843$ for the acceptance dependent variable and $p = 0.6993$ for the success dependent variable. However, in the dataset where the determinants are different, the test rejects with $p < 0.0001$. Therefore, on the simulated data, the Wald test works properly, rejecting resoundingly when it should and not rejecting when it should not. Because the simulated dataset has 5,000 observations, which is comparable to the number of observations in the actual data, the test is sufficiently powerful.

As I explain in Section 3, I cannot use the Wald test framework to compare the determinants of acceptance to those of success within programs as a test of optimality. However, I can answer additional questions within this framework, as long as those questions are not about optimality. For example, I can test if there are different determinants of different types of success. Specifically, applicants may need different skills at different stages of the program; Athey et al. (2007) find that the quantitative GRE matters for course grades, whereas research experience matters more for placements. Statistically testing this hypothesis may be useful for committees who must weigh the early vs. late stages when deciding which applicants to accept. I can also test if the determinants of admission or success are different for U.S. vs. international applicants (see Krueger and Wu 2000). International applicants may have different levels of preparation or research interests, and be more likely to complete the program for lack of outside options and return to their native country after graduation. Finally, I can test if the determinants have changed over time. Reasons for such changes may include Covid and other macro/geopolitical factors, and for economics in particular, a shift toward elite predocs and Federal Reserve RAships.

[17]For some programs, the initial post-program job placement matters less career-wise than it does for economics, which is worth considering when interpreting Wald test results for success across programs.

Table 11: Graduate School Means by Program

(a) All Applicants

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

(b) Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

Table 12: Graduate School Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Table 13: Graduate School Logistic Regression Results

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	-0.4332*	-0.5752*	-0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Winsorized Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2971)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5394	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7201)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0199**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7219)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3943	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts	1.1173***	0.6276*	1.0698**	-0.6088	2.3151**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4759	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4972	0.1495	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4781*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1790)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
MissingTOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2677	0.1525	0.1982	0.1871	0.2475

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Winsorized Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0334	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0393)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0006**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008**	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0208)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0987**	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0221**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0259*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 14: Graduate School Logistic Regression Results (Synthetic)

(a) Coefficients

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.2028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Winsorized Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.6866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	-0.2650	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0192	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Winsorized Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0268	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0006)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1281**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0221)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1771***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0692**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0286***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	37.4658	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 15: Graduate School Lasso Logistic Regression Results

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R^2	0.2297	0.0912	0.0592	0.0850	0.0537

C Additional Structural Results

Table 16: Unrestricted Dynamic Estimates

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{MSE}$	1.6467***						
	(0.0370)						
Obs | Param | LL | AIC	2329	582	-14135.6	29435.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 17: Restricted Dynamic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-5.6768**	-0.9860***	0.3731	-0.5503***	36.58%	-0.7153*	
	(2.3401)	(0.1000)	(0.3067, 0.4539)	(0.1561)	(29.81%, 43.92%)	(0.3751)	
2016	0.0001	0.0171	0.3795	0.0269	37.21%	0.0155	
	(0.1827)	(0.1265)	(0.3025, 0.4761)	(0.2038)	(29.20%, 45.99%)	(0.2039)	
2017	0.4867**	0.1006	0.4126	0.1561	40.27%	0.1404	
	(0.2144)	(0.1085)	(0.3563, 0.4778)	(0.1786)	(34.52%, 46.30%)	(0.1795)	
2018	0.9329***	0.0277	0.3836	0.0473	37.68%	-0.0004	
	(0.1967)	(0.1208)	(0.3271, 0.4497)	(0.1943)	(31.79%, 43.97%)	(0.2015)	
2019	0.6228***	0.0411	0.3887	0.0667	38.14%	0.0232	
	(0.1963)	(0.1142)	(0.3268, 0.4624)	(0.1852)	(31.89%, 44.81%)	(0.1863)	
Winsorized Age	-0.0454*					-0.0004	
	(0.0246)					(0.0009)	
Female	-0.1026					0.0114	
	(0.1055)					(0.0091)	
U.S.	-0.3132					0.0035	
	(0.2098)					(0.0081)	
International Asian	-0.3624**					-0.0051	
	(0.1577)					(0.0070)	
GRE Verbal Percentile	0.0110***					0.0002	
	(0.0036)					(0.0003)	
GRE Quantitative Percentile	0.0580***					0.0007	
	(0.0107)					(0.0008)	
GRE Analytic Percentile	0.0026					0.0001	
	(0.0028)					(0.0001)	
TOEFL/IELTS	0.3161*					0.0007	
	(0.1703)					(0.0073)	
Elite U.S. University	0.5975***					-0.0108	
	(0.1775)					(0.0108)	
Elite U.S. Liberal Arts College	-0.4500					0.0047	
	(0.3224)					(0.0151)	
Elite Foreign University	0.2329					-0.0086	
	(0.2727)					(0.0136)	
Other Foreign University	0.4356**					0.0048	
	(0.1932)					(0.0112)	
Missing Undergraduate Institution	0.3922*					-0.0001	
	(0.2038)					(0.0098)	
Graduate Degree	-0.0531					-0.0022	
	(0.1498)					(0.0063)	
Work Experience	0.0819					0.0088	
	(0.1415)					(0.0079)	
#Professor Recommenders	0.1025					-0.0005	
	(0.0784)					(0.0037)	
#Prolific Recommenders	0.0536					-0.0002	
	(0.0598)					(0.0026)	
Missing GRE	-1.2274					0.0216	
	(0.9140)					(0.0228)	
Missing TOEFL/IELTS	0.2149*					0.0036	
	(0.1252)					(0.0061)	
Committee Score						0.0514	
						(0.0355)	
#Math Courses	-0.1443					-0.0029	
	(0.0920)					(0.0047)	
Advanced Math	0.4000					-0.0022	
	(0.2955)					(0.0108)	
Letters Length/100	0.0906**					0.0005	
	(0.0439)					(0.0020)	
Letters Terms (%) PC1	0.2131**					0.0053	
	(0.0942)					(0.0056)	
Letters Terms (%) PC2	-0.1480					-0.0045	
	(0.1963)					(0.0067)	
CV Length/100	0.0283					0.0013	
	(0.0724)					(0.0021)	
CV Coding	0.0306					-0.0044	
	(0.0981)					(0.0040)	
CV Publication/Working Paper	1.1465***					0.0006	
	(0.4362)					(0.0134)	
Essay Length/100	0.2719***					0.0080	
	(0.1044)					(0.0090)	
Essay Specific Professor	0.1374					-0.0060	
	(0.2212)					(0.0136)	
Essay Economics Terms (%)	0.0664					0.0004	
	(0.0550)					(0.0015)	
Essay Research Terms (%)	0.1364*					-0.0002	
	(0.0821)					(0.0030)	
Essay Positive Terms (%)	0.0422					-0.0004	
	(0.0718)					(0.0029)	
Essay Negative Terms (%)	0.1003					0.0008	
	(0.0766)					(0.0033)	
Missing Committee Score						-0.2001	
						(0.1292)	
Missing Application Pdf	-0.7069**					-0.0267	
	(0.2763)					(0.0180)	
Program Rank						-0.0080***	
						(0.0020)	
Missing Program Rank						-3.0773***	
						(0.4051)	
$\sqrt{\text{MSE}}$	1.7300***						
	(0.0475)						
Sigma							0.0020
							(0.0013)
Obs | Param | LL | AIC	2329	529	-14131.0	29319.9			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 18: Unrestricted Static Estimates

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.6847***	-0.5998***	-4.4553**	-0.5635**
	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(1.6180)	(0.2054)	(2.1831)	(0.2752)
2016	-0.2896	-0.0435	-0.5294	-0.0237	0.0683	0.0087	0.0762	0.0096
	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1852)	(0.0237)	(0.1883)	(0.0238)
2017	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1166	0.0149	0.1266	0.0160
	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1877)	(0.0240)	(0.1929)	(0.0244)
2018	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0112	0.0014	0.0095	0.0012
	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.1920)	(0.0246)	(0.2011)	(0.0254)
2019	0.2905	0.0437	-3.0765***	-0.1377***	0.0177	0.0023	0.0114	0.0014
	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1913)	(0.0245)	(0.1974)	(0.0250)
Winsorized Age	0.0628***	0.0094***	-0.0133	-0.0006	0.0439*	0.0056*	0.0428*	0.0054*
	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0229)	(0.0029)	(0.0235)	(0.0030)
Female	0.3919***	0.0589***	0.8909**	0.0399***	-0.0342	-0.0044	-0.0422	-0.0053
	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1232)	(0.0158)	(0.1246)	(0.0158)
U.S.	-0.1124	-0.0169	0.0293	0.0013	0.1634	0.0209	0.1597	0.0202
	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2198)	(0.0281)	(0.2219)	(0.0280)
International Asian	0.0087	0.0013	-1.1484***	-0.0514***	-0.0894	-0.0114	-0.0921	-0.0116
	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1489)	(0.0191)	(0.1540)	(0.0195)
GRE Verbal Percentile	0.0326***	0.0049***	0.0039	0.0002	0.0015	0.0002	0.0006	0.0001
	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0034)	(0.0004)	(0.0036)	(0.0005)
GRE Quantitative Percentile	0.1829***	0.0275***	0.0402	0.0018	0.0065	0.0008	0.0028	0.0004
	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0076)	(0.0010)	(0.0080)	(0.0010)
GRE Analytic Percentile	0.0152***	0.0023***	0.0078	0.0004	-0.0011	-0.0001	-0.0022	-0.0003
	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0028)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS	-0.0097	-0.0015	-0.1286	-0.0058	0.3267*	0.0418*	0.3375*	0.0427*
	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1721)	(0.0220)	(0.1752)	(0.0221)
Elite U.S. University	0.9066***	0.1363***	-0.5317	-0.0238	-0.0506	-0.0065	-0.0941	-0.0119
	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2283)	(0.0292)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College	0.3268	0.0491	0.3253	0.0146	-0.0812	-0.0104	-0.0866	-0.0110
	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3863)	(0.0495)	(0.3922)	(0.0496)
Elite Foreign University	0.0518	0.0078	-0.9107	-0.0408	0.0910	0.0117	0.0958	0.0121
	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3413)	(0.0437)	(0.3439)	(0.0435)
Other Foreign University	0.1161	0.0174	0.2293	0.0103	0.2023	0.0259	0.1627	0.0206
	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2140)	(0.0274)	(0.2168)	(0.0274)
Missing Undergraduate Institution	-0.4337**	-0.0652**	0.0505	0.0023	0.1941	0.0249	0.1768	0.0224
	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2292)	(0.0293)	(0.2321)	(0.0294)
Graduate Degree	1.0229***	0.1537***	-0.3419	-0.0153	-0.1377	-0.0176	-0.1684	-0.0213
	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1312)	(0.0168)	(0.1367)	(0.0172)
Work Experience	0.7870***	0.1183***	0.5914*	0.0265*	0.3144**	0.0403**	0.2740*	0.0347*
	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1361)	(0.0173)	(0.1424)	(0.0179)
#Professor Recommenders	-0.0211	-0.0032	-0.1896	-0.0085	-0.0570	-0.0073	-0.0622	-0.0079
	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0846)	(0.0108)	(0.0859)	(0.0109)
#Prolific Recommenders	0.1325***	0.0199***	0.1903	0.0085	0.0559	0.0072	0.0667	0.0084
	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0574)	(0.0073)	(0.0582)	(0.0073)
Missing GRE	-1.8142***	-0.2727***	1.6533	0.0740	-0.0087	-0.0011	0.0247	0.0031
	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3716)	(0.0476)	(0.3772)	(0.0477)
Missing TOEFL/IELTS	0.0947	0.0142	0.4511	0.0202	-0.0080	-0.0010	-0.0036	-0.0005
	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1419)	(0.0182)	(0.1435)	(0.0181)
Committee Score			3.7089***	0.1660***			-0.0006	-0.0001
			(0.3984)	(0.0065)			(0.0486)	(0.0062)
#Math Courses			-0.2494	-0.0112			0.0007	0.0001
			(0.2386)	(0.0108)			(0.0879)	(0.0111)
Advanced Math			1.1003	0.0492			-0.2220	-0.0281
			(0.7916)	(0.0347)			(0.2926)	(0.0370)
Letters Length/100			0.0770	0.0034			-0.0135	-0.0017
			(0.1080)	(0.0048)			(0.0385)	(0.0049)
Letters Terms (%) PC1			0.2451	0.0110			0.1278	0.0162
			(0.2489)	(0.0112)			(0.0911)	(0.0115)
Letters Terms (%) PC2			-0.3767	-0.0169			-0.1364	-0.0172
			(0.3450)	(0.0153)			(0.1257)	(0.0159)
CV Length/100			0.0391	0.0017			0.0347	0.0044
			(0.1060)	(0.0047)			(0.0592)	(0.0075)
CV Coding			-0.2228*	-0.0100*			0.1024	0.0130
			(0.1301)	(0.0060)			(0.0651)	(0.0082)
CV Publication/Working Paper			0.7611	0.0341			0.1440	0.0182
			(0.8128)	(0.0364)			(0.3359)	(0.0425)
Essay Length/100			0.2047	0.0092			-0.0718	-0.0091
			(0.3483)	(0.0157)			(0.0818)	(0.0103)
Essay Specific Professor			0.4125	0.0185			0.2894	0.0366
			(0.6138)	(0.0275)			(0.2770)	(0.0350)
Essay Economics Terms (%)			0.0154	0.0007			-0.0854	-0.0108
			(0.0814)	(0.0036)			(0.0614)	(0.0077)
Essay Research Terms (%)			-0.0751	-0.0034			0.0146	0.0018
			(0.1492)	(0.0067)			(0.0672)	(0.0085)
Essay Positive Terms (%)			-0.0689	-0.0031			0.0977*	0.0124*
			(0.0968)	(0.0043)			(0.0539)	(0.0068)
Essay Negative Terms (%)			-0.0863	-0.0039			-0.0441	-0.0056
			(0.1368)	(0.0061)			(0.0745)	(0.0094)
Missing Committee Score							-0.2179	-0.0276
							(0.1515)	(0.0191)
Missing Application Pdf			0.2614	0.0117			0.2498	0.0316
			(0.7005)	(0.0311)			(0.2756)	(0.0348)
Program Rank					-0.0084***	-0.0011***	-0.0086***	-0.0011***
					(0.0020)	(0.0003)	(0.0021)	(0.0003)
Missing Program Rank					-3.1385***	-0.4018***	-3.1438***	-0.3976***
					(0.4051)	(0.0517)	(0.4076)	(0.0514)
Obs | Param | LL | AIC	2329	133	-2964.5	6195.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 19: Restricted Static Estimates

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	-0.8781***	29.36%	-0.4389***	39.20%	-1.2521***	-0.9262**	
	(0.1628)	(23.20%, 36.38%)	(0.1432)	(32.75%, 46.05%)	(0.4466)	(0.4540)	
2016	-0.0091	29.17%	0.0554	40.53%	-0.0178	0.0421	
	(0.2466)	(22.21%, 37.27%)	(0.1767)	(33.55%, 47.92%)	(0.2477)	(0.1771)	
2017	0.3482*	37.06%	0.1035	41.69%	0.3447*	0.0698	
	(0.1963)	(32.10%, 42.29%)	(0.1727)	(34.83%, 48.90%)	(0.1964)	(0.1725)	
2018	0.3936**	38.12%	0.0040	39.30%	0.4390**	-0.0912	
	(0.1856)	(33.98%, 42.44%)	(0.1908)	(31.80%, 47.33%)	(0.1859)	(0.1872)	
2019	0.2163	34.04%	0.0175	39.62%	0.2258	-0.0617	
	(0.2017)	(28.67%, 39.85%)	(0.1791)	(32.54%, 47.16%)	(0.2018)	(0.1778)	
Winsorized Age					0.0018	0.0000	
					(0.0012)	(0.0016)	
Female					0.0118	0.0233	
					(0.0073)	(0.0146)	
U.S.					-0.0034	0.0022	
					(0.0058)	(0.0131)	
International Asian					-0.0006	-0.0302	
					(0.0041)	(0.0187)	
GRE Verbal Percentile					0.0010*	0.0001	
					(0.0005)	(0.0003)	
GRE Quantitative Percentile					0.0055*	0.0011	
					(0.0032)	(0.0011)	
GRE Analytic Percentile					0.0005*	0.0002	
					(0.0003)	(0.0002)	
TOEFL/IELTS					-0.0009	-0.0015	
					(0.0047)	(0.0144)	
Elite U.S. University					0.0252	-0.0131	
					(0.0155)	(0.0178)	
Elite U.S. Liberal Arts College					0.0087	0.0095	
					(0.0113)	(0.0216)	
Elite Foreign University					0.0013	-0.0229	
					(0.0086)	(0.0272)	
Other Foreign University					0.0038	0.0075	
					(0.0062)	(0.0180)	
Missing Undergraduate Institution					-0.0119	0.0030	
					(0.0084)	(0.0173)	
Graduate Degree					0.0306*	-0.0103	
					(0.0175)	(0.0117)	
Work Experience					0.0235*	0.0168	
					(0.0141)	(0.0130)	
#Professor Recommenders					-0.0004	-0.0057	
					(0.0025)	(0.0074)	
#Prolific Recommenders					0.0039	0.0052	
					(0.0026)	(0.0053)	
Missing GRE					-0.0566	0.0458	
					(0.0361)	(0.0532)	
Missing TOEFL/IELTS					0.0029	0.0123	
					(0.0041)	(0.0127)	
Committee Score						0.0961*	
						(0.0491)	
#Math Courses						-0.0064	
						(0.0069)	
Advanced Math						0.0287	
						(0.0244)	
Letters Length/100						0.0019	
						(0.0029)	
Letters Terms (%) PC1						0.0063	
						(0.0072)	
Letters Terms (%) PC2						-0.0103	
						(0.0104)	
CV Length/100						0.0010	
						(0.0027)	
CV Coding						-0.0054	
						(0.0043)	
CV Publication/Working Paper						0.0207	
						(0.0245)	
Essay Length/100						0.0050	
						(0.0089)	
Essay Specific Professor						0.0106	
						(0.0168)	
Essay Economics Terms (%)						0.0002	
						(0.0021)	
Essay Research Terms (%)						-0.0019	
						(0.0039)	
Essay Positive Terms (%)						-0.0014	
						(0.0025)	
Essay Negative Terms (%)						-0.0023	
						(0.0038)	
Missing Committee Score						-0.2328**	
						(0.1162)	
Missing Application Pdf						0.0079	
						(0.0185)	
Program Rank					-0.0085***	-0.0078***	
					(0.0021)	(0.0020)	
Missing Program Rank					-3.0030***	-3.1171***	
					(0.4080)	(0.4041)	
Sigma							0.0064*
							(0.0035)
Obs | Param | LL | AIC	2329	80	-2992.0	6144.1			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 20: Unrestricted Myopic Estimates

	Committee Score	Transition		Accept		Success	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	4.0405***	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**
	(0.9990)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)
2016	0.1701	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096
	(0.1966)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)
2017	0.4355**	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160
	(0.2214)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)
2018	0.8410***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012
	(0.1873)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)
2019	0.7560***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014
	(0.1987)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)
Winsorized Age		0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*
		(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)
Female		0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053
		(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)
U.S.		-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202
		(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)
International Asian		0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116
		(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)
GRE Verbal Percentile		0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001
		(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)
GRE Quantitative Percentile		0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004
		(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)
GRE Analytic Percentile		0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003
		(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)
TOEFL/IELTS		-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*
		(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)
Elite U.S. University		0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119
		(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)
Elite U.S. Liberal Arts College		0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110
		(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)
Elite Foreign University		0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121
		(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)
Other Foreign University		0.1161	0.0174	0.2293	0.0103	0.1627	0.0206
		(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)
Missing Undergraduate Institution		-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224
		(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)
Graduate Degree		1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213
		(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)
Work Experience		0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*
		(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)
#Professor Recommenders		-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079
		(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)
#Prolific Recommenders		0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084
		(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)
Missing GRE		-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031
		(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)
Missing TOEFL/IELTS		0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005
		(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001
				(0.3984)	(0.0065)	(0.0486)	(0.0062)
#Math Courses	-0.0804			-0.2494	-0.0112	0.0007	0.0001
	(0.0520)			(0.2386)	(0.0108)	(0.0879)	(0.0111)
Advanced Math	0.3560*			1.1003	0.0492	-0.2220	-0.0281
	(0.1909)			(0.7916)	(0.0347)	(0.2926)	(0.0370)
Letters Length/100	0.1225***			0.0770	0.0034	-0.0135	-0.0017
	(0.0242)			(0.1080)	(0.0048)	(0.0385)	(0.0049)
Letters Terms (%) PC1	0.0974*			0.2451	0.0110	0.1278	0.0162
	(0.0536)			(0.2489)	(0.0112)	(0.0911)	(0.0115)
Letters Terms (%) PC2	-0.0925			-0.3767	-0.0169	-0.1364	-0.0172
	(0.0856)			(0.3450)	(0.0153)	(0.1257)	(0.0159)
CV Length/100	0.0588*			0.0391	0.0017	0.0347	0.0044
	(0.0323)			(0.1060)	(0.0047)	(0.0592)	(0.0075)
CV Coding	0.0551			-0.2228*	-0.0100*	0.1024	0.0130
	(0.0417)			(0.1301)	(0.0060)	(0.0651)	(0.0082)
CV Publication/Working Paper	0.5324***			0.7611	0.0341	0.1440	0.0182
	(0.1975)			(0.8128)	(0.0364)	(0.3359)	(0.0425)
Essay Length/100	0.0763			0.2047	0.0092	-0.0718	-0.0091
	(0.0567)			(0.3483)	(0.0157)	(0.0818)	(0.0103)
Essay Specific Professor	0.1442			0.4125	0.0185	0.2894	0.0366
	(0.1767)			(0.6138)	(0.0275)	(0.2770)	(0.0350)
Essay Economics Terms (%)	0.0042			0.0154	0.0007	-0.0854	-0.0108
	(0.0314)			(0.0814)	(0.0036)	(0.0614)	(0.0077)
Essay Research Terms (%)	0.0304			-0.0751	-0.0034	0.0146	0.0018
	(0.0446)			(0.1492)	(0.0067)	(0.0672)	(0.0085)
Essay Positive Terms (%)	0.0175			-0.0689	-0.0031	0.0977*	0.0124*
	(0.0373)			(0.0968)	(0.0043)	(0.0539)	(0.0068)
Essay Negative Terms (%)	0.0429			-0.0863	-0.0039	-0.0441	-0.0056
	(0.0460)			(0.1368)	(0.0061)	(0.0745)	(0.0094)
Missing Committee Score						-0.2179	-0.0276
						(0.1515)	(0.0191)
Missing Application Pdf	-0.4006**			0.2614	0.0117	0.2498	0.0316
	(0.1850)			(0.7005)	(0.0311)	(0.2756)	(0.0348)
Program Rank						-0.0086***	-0.0011***
						(0.0021)	(0.0003)
Missing Program Rank						-3.1438***	-0.3976***
						(0.4076)	(0.0514)
$\sqrt{\text{MSE}}$	1.7525***						
	(0.0403)						
Obs | Param | LL | AIC	2329	297	-14470.7	29535.4			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 21: Restricted Myopic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	3.4796***	-0.9771***	0.3764	-0.5522***	36.54%	-1.2859**	
	(1.2648)	(0.0960)	(0.3118, 0.4544)	(0.1553)	(29.80%, 43.84%)	(0.5090)	
2016	0.0978	-0.0005	0.3762	-0.0048	36.42%	-0.0164	
	(0.1990)	(0.1320)	(0.2992, 0.4731)	(0.2135)	(28.21%, 45.51%)	(0.2130)	
2017	0.7100***	0.1077	0.4192	0.1614	40.35%	0.1384	
	(0.2332)	(0.1059)	(0.3701, 0.4749)	(0.1760)	(35.00%, 45.95%)	(0.1772)	
2018	1.0737***	0.0146	0.3820	0.0286	37.20%	-0.0305	
	(0.1730)	(0.1219)	(0.3237, 0.4506)	(0.1968)	(31.07%, 43.76%)	(0.1993)	
2019	0.8095***	0.0005	0.3766	0.0004	36.54%	-0.0654	
	(0.1948)	(0.1254)	(0.3086, 0.4596)	(0.2030)	(29.43%, 44.30%)	(0.2045)	
Winsorized Age						0.0010	
						(0.0007)	
Female						0.0132**	
						(0.0063)	
U.S.						0.0010	
						(0.0048)	
International Asian						-0.0075	
						(0.0051)	
GRE Verbal Percentile						0.0008**	
						(0.0003)	
GRE Quantitative Percentile						0.0039**	
						(0.0017)	
GRE Analytic Percentile						0.0004**	
						(0.0002)	
TOEFL/IELTS						0.0009	
						(0.0042)	
Elite U.S. University						0.0105	
						(0.0071)	
Elite U.S. Liberal Arts College						0.0064	
						(0.0086)	
Elite Foreign University						-0.0012	
						(0.0077)	
Other Foreign University						0.0066	
						(0.0059)	
Missing Undergraduate Institution						-0.0034	
						(0.0053)	
Graduate Degree						0.0194**	
						(0.0090)	
Work Experience						0.0189**	
						(0.0089)	
#Professor Recommenders						-0.0014	
						(0.0022)	
#Prolific Recommenders						0.0016	
						(0.0016)	
Missing GRE						-0.0449	
						(0.0282)	
Missing TOEFL/IELTS						0.0012	
						(0.0035)	
Committee Score						0.0686**	
						(0.0302)	
#Math Courses	-0.0627					-0.0011	
	(0.0689)					(0.0071)	
Advanced Math	0.3298					0.0023	
	(0.2533)					(0.0161)	
Letters Length/100	0.1467***					0.0023	
	(0.0317)					(0.0029)	
Letters Terms (%) PC1	0.1155					0.0016	
	(0.0772)					(0.0087)	
Letters Terms (%) PC2	-0.1020					-0.0069	
	(0.1152)					(0.0089)	
CV Length/100	0.0628					0.0022	
	(0.0407)					(0.0026)	
CV Coding	0.0466					-0.0051	
	(0.0580)					(0.0036)	
CV Publication/Working Paper	0.7501***					-0.0009	
	(0.2653)					(0.0155)	
Essay Length/100	0.1355					0.0083	
	(0.0858)					(0.0131)	
Essay Specific Professor	0.1387					0.0023	
	(0.2579)					(0.0205)	
Essay Economics Terms (%)	-0.0024					-0.0018	
	(0.0396)					(0.0021)	
Essay Research Terms (%)	0.0663					0.0023	
	(0.0596)					(0.0044)	
Essay Positive Terms (%)	0.0403					0.0011	
	(0.0476)					(0.0033)	
Essay Negative Terms (%)	0.0459					-0.0009	
	(0.0575)					(0.0048)	
Missing Committee Score						-0.1177	
						(0.1482)	
Missing Application Pdf	-0.9529***					-0.0359**	
	(0.2274)					(0.0157)	
Program Rank						-0.0078***	
						(0.0020)	
Missing Program Rank						-3.0747***	
						(0.4058)	
$\sqrt{\text{MSE}}$	1.8070***						
	(0.0436)						
Sigma							0.0027**
							(0.0012)
Obs | Param | LL | AIC	2329	244	-14536.1	29560.2			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 22: Unrestricted Matriculation-Dynamic Estimates

	Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-2.7966	-23.0601***	-3.4656***	-28.5278***	-1.2765***	-4.4553**	-0.5635**	21.0916***	3.3374***
	(2.1751)	(2.0491)	(0.2371)	(6.9081)	(0.2933)	(2.1831)	(0.2752)	(6.9868)	(1.0163)
2016	0.0915	-0.2896	-0.0435	-0.5294	-0.0237	0.0762	0.0096	0.3206	0.0507
	(0.1856)	(0.1794)	(0.0268)	(0.5264)	(0.0230)	(0.1883)	(0.0238)	(0.6530)	(0.1030)
2017	0.2354	-0.1533	-0.0230	-1.3366**	-0.0598**	0.1266	0.0160	0.9971	0.1578
	(0.2130)	(0.1726)	(0.0260)	(0.5312)	(0.0235)	(0.1929)	(0.0244)	(0.6410)	(0.1008)
2018	0.5999***	1.5429***	0.2319***	-3.6854***	-0.1649***	0.0095	0.0012	1.6496***	0.2610***
	(0.1777)	(0.1858)	(0.0251)	(0.6281)	(0.0209)	(0.2011)	(0.0254)	(0.5920)	(0.0886)
2019	0.6118***	0.2905	0.0437	-3.0765***	-0.1377***	0.0114	0.0014	1.0916*	0.1727*
	(0.1907)	(0.1833)	(0.0273)	(0.6899)	(0.0257)	(0.1974)	(0.0250)	(0.6190)	(0.0971)
Winsorized Age	-0.0621***	0.0628***	0.0094***	-0.0133	-0.0006	0.0428*	0.0054*	-0.1396*	-0.0221*
	(0.0235)	(0.0218)	(0.0032)	(0.0612)	(0.0027)	(0.0235)	(0.0030)	(0.0738)	(0.0113)
Female	-0.0543	0.3919***	0.0589***	0.8909**	0.0399***	-0.0422	-0.0053	-0.4562	-0.0722
	(0.1125)	(0.1109)	(0.0164)	(0.3672)	(0.0154)	(0.1246)	(0.0158)	(0.3550)	(0.0556)
U.S.	-0.5071**	-0.1124	-0.0169	0.0293	0.0013	0.1597	0.0202	0.9160	0.1449
	(0.2043)	(0.1894)	(0.0285)	(0.4978)	(0.0223)	(0.2219)	(0.0280)	(0.5700)	(0.0896)
International Asian	-0.7530***	0.0087	0.0013	-1.1484***	-0.0514***	-0.0921	-0.0116	1.1667**	0.1846***
	(0.1452)	(0.1396)	(0.0210)	(0.4133)	(0.0172)	(0.1540)	(0.0195)	(0.4531)	(0.0689)
GRE Verbal Percentile	0.0066*	0.0326***	0.0049***	0.0039	0.0002	0.0006	0.0001	-0.0043	-0.0007
	(0.0036)	(0.0034)	(0.0005)	(0.0131)	(0.0006)	(0.0036)	(0.0005)	(0.0115)	(0.0018)
GRE Quantitative Percentile	0.0447***	0.1829***	0.0275***	0.0402	0.0018	0.0028	0.0004	-0.1358***	-0.0215***
	(0.0111)	(0.0163)	(0.0018)	(0.0326)	(0.0015)	(0.0080)	(0.0010)	(0.0444)	(0.0064)
GRE Analytic Percentile	0.0029	0.0152***	0.0023***	0.0078	0.0004	-0.0022	-0.0003	-0.0185**	-0.0029**
	(0.0027)	(0.0026)	(0.0004)	(0.0090)	(0.0004)	(0.0029)	(0.0004)	(0.0087)	(0.0013)
TOEFL/IELTS	0.4407***	-0.0097	-0.0015	-0.1286	-0.0058	0.3375*	0.0427*	-0.5140	-0.0813
	(0.1696)	(0.1632)	(0.0245)	(0.5621)	(0.0250)	(0.1752)	(0.0221)	(0.5175)	(0.0807)
Elite U.S. University	0.5507***	0.9066***	0.1363***	-0.5317	-0.0238	-0.0941	-0.0119	-0.4342	-0.0687
	(0.1994)	(0.2103)	(0.0312)	(0.6438)	(0.0293)	(0.2319)	(0.0293)	(0.5475)	(0.0858)
Elite U.S. Liberal Arts College	-0.0711	0.3268	0.0491	0.3253	0.0146	-0.0866	-0.0110	1.0020	0.1586
	(0.3683)	(0.3402)	(0.0510)	(0.8010)	(0.0356)	(0.3922)	(0.0496)	(0.7518)	(0.1185)
Elite Foreign University	0.2902	0.0518	0.0078	-0.9107	-0.0408	0.0958	0.0121	-0.2759	-0.0437
	(0.2991)	(0.2959)	(0.0445)	(0.9288)	(0.0424)	(0.3439)	(0.0435)	(0.9777)	(0.1548)
Other Foreign University	0.4797**	0.1161	0.0174	0.2293	0.0103	0.1627	0.0206	0.4536	0.0718
	(0.2022)	(0.1969)	(0.0296)	(0.6888)	(0.0306)	(0.2168)	(0.0274)	(0.6285)	(0.0992)
Missing Undergraduate Institution	0.5097**	-0.4337**	-0.0652**	0.0505	0.0023	0.1768	0.0224	-0.8208	-0.1299
	(0.2157)	(0.1996)	(0.0300)	(0.6688)	(0.0299)	(0.2321)	(0.0294)	(0.6805)	(0.1058)
Graduate Degree	-0.0757	1.0229***	0.1537***	-0.3419	-0.0153	-0.1684	-0.0213	-0.0653	-0.0103
	(0.1279)	(0.1262)	(0.0177)	(0.3883)	(0.0171)	(0.1367)	(0.0172)	(0.4719)	(0.0746)
Work Experience	0.3458**	0.7870***	0.1183***	0.5914*	0.0265*	0.2740*	0.0347*	0.9166**	0.1450**
	(0.1345)	(0.1256)	(0.0179)	(0.3412)	(0.0154)	(0.1424)	(0.0179)	(0.4411)	(0.0667)
#Professor Recommenders	0.0618	-0.0211	-0.0032	-0.1896	-0.0085	-0.0622	-0.0079	0.3588	0.0568
	(0.0769)	(0.0839)	(0.0126)	(0.2574)	(0.0113)	(0.0859)	(0.0109)	(0.2789)	(0.0433)
#Prolific Recommenders	0.1667***	0.1325***	0.0199***	0.1903	0.0085	0.0667	0.0084	-0.2845	-0.0450
	(0.0548)	(0.0513)	(0.0077)	(0.1757)	(0.0077)	(0.0582)	(0.0073)	(0.1875)	(0.0288)
Missing GRE	-1.6087*	-1.8142***	-0.2727***	1.6533	0.0740	0.0247	0.0031		
	(0.9422)	(0.5268)	(0.0772)	(1.8092)	(0.0811)	(0.3772)	(0.0477)		
Missing TOEFL/IELTS	0.3199**	0.0947	0.0142	0.4511	0.0202	-0.0036	-0.0005	-0.0091	-0.0014
	(0.1319)	(0.1295)	(0.0194)	(0.4238)	(0.0189)	(0.1435)	(0.0181)	(0.5591)	(0.0885)
Committee Score				3.7089***	0.1660***	-0.0006	-0.0001	-0.1637	-0.0259
				(0.3984)	(0.0065)	(0.0486)	(0.0062)	(0.2099)	(0.0336)
#Math Courses	-0.0790			-0.2494	-0.0112	0.0007	0.0001	-0.0182	-0.0029
	(0.0519)			(0.2386)	(0.0108)	(0.0879)	(0.0111)	(0.1671)	(0.0265)
Advanced Math	0.2979			1.1003	0.0492	-0.2220	-0.0281	-1.1143**	-0.1763**
	(0.1889)			(0.7916)	(0.0347)	(0.2926)	(0.0370)	(0.5578)	(0.0877)
Letters Length/100	0.0858***			0.0770	0.0034	-0.0135	-0.0017	-0.0744	-0.0118
	(0.0245)			(0.1080)	(0.0048)	(0.0385)	(0.0049)	(0.0711)	(0.0113)
Letters Terms (%) PC1	0.1027*			0.2451	0.0110	0.1278	0.0162	0.3997**	0.0632**
	(0.0551)			(0.2489)	(0.0112)	(0.0911)	(0.0115)	(0.1903)	(0.0287)
Letters Terms (%) PC2	-0.0208			-0.3767	-0.0169	-0.1364	-0.0172	0.2897	0.0458
	(0.0823)			(0.3450)	(0.0153)	(0.1257)	(0.0159)	(0.2460)	(0.0388)
CV Length/100	0.0513			0.0391	0.0017	0.0347	0.0044	-0.1109	-0.0175
	(0.0329)			(0.1060)	(0.0047)	(0.0592)	(0.0075)	(0.1105)	(0.0175)
CV Coding	0.0381			-0.2228*	-0.0100*	0.1024	0.0130	-0.2901**	-0.0459**
	(0.0424)			(0.1301)	(0.0060)	(0.0651)	(0.0082)	(0.1176)	(0.0181)
CV Publication/Working Paper	0.5482***			0.7611	0.0341	0.1440	0.0182	-0.3451	-0.0546
	(0.1912)			(0.8128)	(0.0364)	(0.3359)	(0.0425)	(0.5102)	(0.0812)
Essay Length/100	0.0927			0.2047	0.0092	-0.0718	-0.0091	0.0882	0.0140
	(0.0597)			(0.3483)	(0.0157)	(0.0818)	(0.0103)	(0.1386)	(0.0220)
Essay Specific Professor	0.1411			0.4125	0.0185	0.2894	0.0366	0.6812	0.1078
	(0.1744)			(0.6138)	(0.0275)	(0.2770)	(0.0350)	(0.4976)	(0.0773)
Essay Economics Terms (%)	0.0287			0.0154	0.0007	-0.0854	-0.0108	0.2244**	0.0355**
	(0.0312)			(0.0814)	(0.0036)	(0.0614)	(0.0077)	(0.1075)	(0.0169)
Essay Research Terms (%)	0.0396			-0.0751	-0.0034	0.0146	0.0018	-0.1904	-0.0301
	(0.0426)			(0.1492)	(0.0067)	(0.0672)	(0.0085)	(0.1440)	(0.0226)
Essay Positive Terms (%)	0.0171			-0.0689	-0.0031	0.0977*	0.0124*	-0.0170	-0.0027
	(0.0345)			(0.0968)	(0.0043)	(0.0539)	(0.0068)	(0.1151)	(0.0182)
Essay Negative Terms (%)	0.0441			-0.0863	-0.0039	-0.0441	-0.0056	-0.0176	-0.0028
	(0.0443)			(0.1368)	(0.0061)	(0.0745)	(0.0094)	(0.1043)	(0.0165)
Missing Committee Score						-0.2179	-0.0276		
						(0.1515)	(0.0191)		
Missing Application Pdf	-0.5376***			0.2614	0.0117	0.2498	0.0316	-0.7440	-0.1177
	(0.1841)			(0.7005)	(0.0311)	(0.2756)	(0.0348)	(0.5803)	(0.0919)
Program Rank						-0.0086***	-0.0011***		
						(0.0021)	(0.0003)		
Missing Program Rank						-3.1438***	-0.3976***		
						(0.4076)	(0.0514)		
$\sqrt{\text{MSE}}$	1.6467***								
	(0.0370)								
Obs | Param | LL | AIC	2329	621	-14259.1	29760.1					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 23: Restricted Matriculation-Dynamic Estimates

	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-6.5631***	-4.9411***	0.0071	-0.5602***	36.35%	-0.8176**	1.5446	
	(2.5333)	(0.5082)	(0.0026, 0.0194)	(0.1594)	(29.47%, 43.84%)	(0.3643)	(5.3407)	
2016	0.2912	-0.0124	0.0071	0.0811	38.25%	0.0776	-1.2678*	
	(0.2079)	(0.1800)	(0.0028, 0.0180)	(0.1961)	(31.05%, 46.00%)	(0.1966)	(0.7295)	
2017	0.6424***	0.3850**	0.0105	0.1816	40.65%	0.1644	-0.3048	
	(0.2488)	(0.1788)	(0.0041, 0.0267)	(0.1820)	(35.15%, 46.39%)	(0.1821)	(0.8727)	
2018	0.8514***	0.2894	0.0095	0.0102	36.59%	-0.0518	1.3060*	
	(0.1924)	(0.1890)	(0.0038, 0.0240)	(0.2053)	(30.01%, 43.71%)	(0.2096)	(0.7318)	
2019	0.5883***	0.2802	0.0095	0.0456	37.41%	-0.0066	0.7687	
	(0.2154)	(0.2153)	(0.0035, 0.0255)	(0.1960)	(31.04%, 44.26%)	(0.1975)	(0.8962)	
Winsorized Age	-0.0569**					-0.0001	-0.0312	
	(0.0252)					(0.0011)	(0.0547)	
Female	-0.0776					0.0116	0.0927	
	(0.1152)					(0.0083)	(0.2112)	
U.S.	-0.3707*					-0.0033	0.4696	
	(0.2114)					(0.0084)	(0.4504)	
International Asian	-0.5051***					-0.0103	0.5968*	
	(0.1631)					(0.0083)	(0.3430)	
GRE Verbal Percentile	0.0082**					0.0000	0.0098	
	(0.0037)					(0.0002)	(0.0075)	
GRE Quantitative Percentile	0.0704***					0.0010	-0.0610*	
	(0.0105)					(0.0009)	(0.0338)	
GRE Analytic Percentile	0.0016					0.0001	0.0021	
	(0.0029)					(0.0002)	(0.0066)	
TOEFL/IELTS	0.2848					-0.0023	-0.0629	
	(0.1746)					(0.0101)	(0.2753)	
Elite U.S. University	0.6391***					-0.0098	-0.0637	
	(0.1892)					(0.0142)	(0.3925)	
Elite U.S. Liberal Arts College	-0.3528					0.0127	-0.2757	
	(0.3489)					(0.0174)	(0.6783)	
Elite Foreign University	0.1906					-0.0187	0.4536	
	(0.2875)					(0.0195)	(0.5403)	
Other Foreign University	0.5511***					0.0038	0.2515	
	(0.2049)					(0.0148)	(0.3764)	
Missing Undergraduate Institution	0.5458***					0.0033	-0.3893	
	(0.2109)					(0.0140)	(0.4067)	
Graduate Degree	-0.1826					-0.0069	0.4784	
	(0.1589)					(0.0082)	(0.3581)	
Work Experience	0.0765					0.0080	0.5662	
	(0.1486)					(0.0075)	(0.3731)	
#Professor Recommenders	0.0824					-0.0021	0.2152	
	(0.0822)					(0.0054)	(0.1592)	
#Prolific Recommenders	0.0919					-0.0002	-0.1769*	
	(0.0590)					(0.0032)	(0.1041)	
Missing GRE	-1.3931					0.0140		
	(0.8753)					(0.0192)		
Missing TOEFL/IELTS	0.2632**					0.0023	0.0851	
	(0.1311)					(0.0066)	(0.2494)	
Committee Score						0.0649**	0.5117***	
						(0.0299)	(0.1497)	
#Math Courses	-0.1660					-0.0048	-0.1664	
	(0.1024)					(0.0057)	(0.1619)	
Advanced Math	0.7224*					0.0257	0.0920	
	(0.4161)					(0.0208)	(0.5297)	
Letters Length/100	0.1095**					0.0013	-0.0122	
	(0.0471)					(0.0022)	(0.0646)	
Letters Terms (%) PC1	0.2635***					0.0089	0.4342***	
	(0.0988)					(0.0071)	(0.1642)	
Letters Terms (%) PC2	-0.3026*					-0.0108	-0.2680	
	(0.1764)					(0.0091)	(0.2167)	
CV Length/100	0.0263					0.0004	-0.0601	
	(0.0752)					(0.0023)	(0.0819)	
CV Coding	0.0417					-0.0049	-0.0663	
	(0.1086)					(0.0044)	(0.1017)	
CV Publication/Working Paper	1.3411***					0.0190	-0.2251	
	(0.3435)					(0.0192)	(0.5320)	
Essay Length/100	0.2771*					0.0065	0.1436	
	(0.1475)					(0.0103)	(0.1533)	
Essay Specific Professor	0.4040					0.0133	0.6561	
	(0.3655)					(0.0168)	(0.4106)	
Essay Economics Terms (%)	0.0881					0.0010	0.0869	
	(0.0633)					(0.0022)	(0.0809)	
Essay Research Terms (%)	0.1320					0.0005	-0.1728	
	(0.0843)					(0.0030)	(0.1284)	
Essay Positive Terms (%)	0.0538					-0.0006	0.0066	
	(0.0761)					(0.0025)	(0.0966)	
Essay Negative Terms (%)	0.0781					-0.0021	-0.0494	
	(0.0830)					(0.0035)	(0.1032)	
Missing Committee Score						-0.1920		
						(0.1290)		
Missing Application Pdf	-0.4098					0.0124	-1.5977***	
	(0.2846)					(0.0125)	(0.2218)	
Program Rank						-0.0079***		
						(0.0020)		
Missing Program Rank						-3.0808***		
						(0.4053)		
$\sqrt{\text{MSE}}$	1.7632***							
	(0.0469)							
Sigma								0.0020**
								(0.0010)
Obs | Param | LL | AIC	2329	568	-14260.9	29657.8				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 24: Deviation Gains Attribution Table

(a) Acceptance Set

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.8788	25.9910	26.2840	29.9572	29.5331	1.1674	1.0326
Work Experience	2.3065	0.5247	0.7276	0.9844	0.9650	0.5141	0.4752
Essay Positive Terms (%)	1.9410	11.7499	11.6521	12.5376	12.1183	0.7858	0.4137
Missing TOEFL/IELTS	1.7635	0.6076	0.6654	0.0739	0.4086	-1.2110	-0.5258
CV Coding	1.3478	2.8001	2.7820	3.3399	2.9415	0.6244	0.1785
Elite U.S. University	1.3179	0.1267	0.2335	0.0078	0.0389	-0.6784	-0.5848
TOEFL/IELTS	1.2400	7.4376	7.5675	7.7759	7.6827	0.5957	0.3294
Essay Economics Terms (%)	1.2043	9.7228	9.7423	9.0284	9.5388	-0.6411	-0.1828
#Professor Recommenders	1.1455	2.6230	2.6148	1.3852	2.1362	-1.6680	-0.6493
Letters Terms (%) PC1	1.0074	-0.0106	0.1374	0.4969	0.1387	0.5202	0.0018

(b) Matriculation-Set

	Gain (%pt)	Sample Mean	Committee Mean	1-Variable Optimal Mean	Joint Optimal Mean	1-Variable Difference (SD)	Joint Difference (SD)
Winsorized Age	2.7099	25.9910	26.3284	30.9159	29.7737	1.4580	1.0950
Essay Positive Terms (%)	2.2339	11.7499	11.6107	12.9136	12.2763	1.1534	0.5905
Work Experience	2.0220	0.5247	0.7938	0.9949	0.9799	0.4077	0.3725
TOEFL/IELTS	1.7179	7.4376	7.4939	7.8501	7.6128	1.0238	0.3397
Essay Economics Terms (%)	1.3503	9.7228	9.9994	9.0856	9.5738	-0.8271	-0.3821
CV Coding	1.3100	2.8001	2.6661	3.3395	2.8782	0.7618	0.2374
#Professor Recommenders	1.3005	2.6230	2.6171	1.4173	2.2255	-1.6321	-0.5312
Missing TOEFL/IELTS	1.1999	0.6076	0.6858	0.0442	0.4368	-1.3075	-0.5099

D Simulated Results

Table 25: Simulated Wald Test Results

(a) Same Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848***	-1.5822***	-28.5524***	-1.3677***	-10.4586***	-1.9105***	-10.2610***	-1.8514***
	(1.1315)	(0.0956)	(1.4963)	(0.0500)	(0.7236)	(0.1070)	(0.4815)	(0.0628)
GRE Quantitative Percentile	0.1509***	0.0146***	0.3245***	0.0155***	0.1058***	0.0193***	0.1154***	0.0208***
	(0.0131)	(0.0012)	(0.0179)	(0.0007)	(0.0088)	(0.0014)	(0.0070)	(0.0011)
Gender	0.4714***	0.0455***	0.7880***	0.0377***	-0.1384	-0.0253	-0.0497	-0.0090
	(0.1475)	(0.0140)	(0.1696)	(0.0081)	(0.1095)	(0.0200)	(0.0895)	(0.0161)
Demographic 1	-0.6287***	-0.0607***	-1.8488***	-0.0886***	-0.3486**	-0.0637**	-0.3079***	-0.0556***
	(0.2088)	(0.0200)	(0.2445)	(0.0115)	(0.1467)	(0.0267)	(0.1123)	(0.0202)
Demographic 2	0.1531	0.0148	0.3703*	0.0177*	0.1756	0.0321	0.0282	0.0051
	(0.2072)	(0.0200)	(0.1929)	(0.0091)	(0.1558)	(0.0284)	(0.1069)	(0.0193)
Advanced Math	0.2889*	0.0279*			0.3267***	0.0597***		
	(0.1612)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0216***	0.3994***	0.0191***	0.1026***	0.0187***	0.0757***	0.0137***
	(0.0413)	(0.0039)	(0.0506)	(0.0023)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			0.8376***	0.0401***			1.9483***	0.3515***
			(0.2297)	(0.0107)			(0.1236)	(0.0205)
Observations | Pseudo-R^2	5000	0.3251			5000	0.1431		
Parameters | LL | AIC	14	-1130.4	2288.8		14	-2691.4	5410.8	
Wald Statistic | DF | P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *$p<0.10$, **$p<0.05$, ***$p<0.01$.

(b) Different Determinants

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291***	-1.5784***	-8.4753***	-0.3642***	-10.4581***	-1.9104***	-3.2236***	-0.5997***
	(1.1274)	(0.0955)	(0.8673)	(0.0412)	(0.7237)	(0.1070)	(0.4018)	(0.0728)
GRE Quantitative Percentile	0.1502***	0.0145***	0.0402***	0.0017***	0.1058***	0.0193***	0.0034	0.0006
	(0.0131)	(0.0012)	(0.0128)	(0.0006)	(0.0088)	(0.0014)	(0.0062)	(0.0012)
Gender	0.4702***	0.0454***	0.5942***	0.0255***	-0.1372	-0.0251	0.0375	0.0070
	(0.1473)	(0.0140)	(0.1820)	(0.0079)	(0.1095)	(0.0200)	(0.0868)	(0.0161)
Demographic 1	-0.6246***	-0.0604***	-0.2541	-0.0109	-0.3544**	-0.0647**	-0.1152	-0.0214
	(0.2085)	(0.0200)	(0.2393)	(0.0103)	(0.1467)	(0.0267)	(0.1103)	(0.0205)
Demographic 2	0.1555	0.0150	0.0081	0.0003	0.1713	0.0313	-0.0562	-0.0105
	(0.2070)	(0.0200)	(0.2109)	(0.0091)	(0.1557)	(0.0284)	(0.1030)	(0.0192)
Advanced Math	0.2886*	0.0279*			0.3284***	0.0600***		
	(0.1611)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0217***	0.3817***	0.0164***	0.1026***	0.0187***	0.1808***	0.0336***
	(0.0413)	(0.0039)	(0.0528)	(0.0025)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			-0.0452	-0.0019			1.1393***	0.2120***
			(0.1995)	(0.0086)			(0.1219)	(0.0219)
Observations | Pseudo-R^2	5000	0.1488			5000	0.0758		
Parameters | LL | AIC	14	-1155.7	2339.5		14	-2747.2	5522.4	
Wald Statistic | DF | P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *$p<0.10$, **$p<0.05$, ***$p<0.01$.

D.1 Simulated Structural Results

In this appendix, I present structural estimation results from two simulated datasets, one in which the committee is optimizing and one in which it is not. The data-generating parameters for the means, variances, and correlations across the independent variables, as well as the parameters governing the effects of the independent variables on success, are meant to resemble those of an actual admissions dataset. In the optimal dataset, the parameters governing the effect of the independent variables on acceptance are chosen to be similar to the success parameters. In contrast, the suboptimal committee discriminates against *Demographic 1*, undervalues *Advanced Math*, and overvalues *Committee Score*. The goal of this exercise is to verify that the optimality test does not reject optimality for the optimal dataset and rejects for the suboptimal dataset.

In Tables 26 and 27, I present the unrestricted and restricted dynamic estimates, respectively. Because I have two z variables, namely *Advanced Math* and *Committee Score*, I have two PDFs to estimate. Committee score is a continuous z, and since I parameterize continuous z's with the normal distribution, I have an additional σ_F parameter that equals the square root of the distribution's mean squared error. I also let *Committee Score* depend on *Advanced Math*, so I can estimate the joint distribution of the two z's without assuming the z's are independent. As usual, I do not assume independent z's. The simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% levels.[18]

The unrestricted and restricted static estimates are in Tables 28 and 29, respectively. Although the data are simulated, Table 29 contains an interesting result. There are two years in each of the simulated datasets: one in which the committee transitions 45–50% of the applicants to Round 2, and one in which it only transitions 25–30%. Essentially, in *Year Transition More*, the committee is more concerned with admitting a strong class than saving time and effort.[19] And in that year, as expected, the Round 1 cutoff probability is lower than the Round 2 cutoff probability. That is, it is easier to move on to Round 2 than it ultimately is to get into the program. However, in the other year, which is represented by the intercept term, the Round 1 cutoff is greater than the Round 2 cutoff, meaning the $P(\text{Success}|x,t,\bar{k})$ of the marginal applicant rejected in Round 1 is greater than the $P(\text{Success}|x,z,t,\bar{k})$ of the marginal applicant accepted into the program.

How can this be? Sometimes when a committee transitions a seemingly strong applicant to Round 2, the applicant's z reveals that they are weaker than previously thought. For example, the applicant may have a negative recommendation letter. If the committee transitions too few applicants to Round 2, it can end up in an unfortunate situation where it threw out so many files that it now has to accept those weaker applicants. As a simple example, suppose there are three applicants: A, B, and C. The committee has time to read two applicants' files and space to accept one applicant. Suppose that overall, A is the best applicant and C is the worst, but A

[18] As in the actual data, *Program Rank*'s sign is negative because the best rank is 1, and the worst rank is 150.

[19] In *Year Transition More*, the applicants also have higher matriculation probabilities on average, so this variable is called *Year Transition/Matriculate More* in the matriculation models in this appendix.

is the worst in objective variables. Then the committee will reject A in Round 1 and accept B, who is actually worse than A. In the simulated data, the minimum percentage of applications the committee must read in Round 2 to avoid this situation is between 25–30%, where the situation occurs, and 45–50%, where it does not. However, this situation does not occur in the actual data.

In Tables 30 and 31, I present estimates from the myopic model. Recall that the myopic model is a special case of the dynamic model in which the parameters for x in every block of $f_{z|x,t}$ are restricted to equal zero, while the intercepts, parameters for t, dependence parameters among the components of z, and variance parameters remain unrestricted. In this model, if an objective variable is associated with z in addition to being associated with success in its own right, the committee puts no extra weight on that variable in Round 1. Finally, the matriculation-dynamic estimates are in Tables 32 and 33. As with the dynamic estimates without matriculation, the simulated data are such that *GRE Quantitative Score*, *Advanced Math*, and *Program Rank* significantly predict success at the 1% level. But *Committee Score* significantly negatively predicts matriculation at the 1% level. The rationale is that applicants with higher committee scores likely have more acceptances from other universities and so are less likely to matriculate if accepted.

I conclude this appendix by presenting the deviation gains test results on the simulated datasets. Per Figure 7, in the dataset where the committee is optimizing, the models' CDFs for the acceptance set are virtually identical to the committee's. But in the suboptimal dataset, all the models' CDFs stochastically dominate the committee's CDF, and the test always rejects with $p = 0.0060$. The reason is that the committee is discriminating against *Demographic 1*, not putting enough weight on *Advanced Math*, and putting too much weight on *Committee Score*. The results for the matriculation-dynamic and static models on the matriculation set are in Figure 8, and those results are essentially the same as the results in Figure 7. For the optimal dataset, the optimality test never rejects, and for the suboptimal dataset, it always rejects with $p = 0.0020$. In this figure, I also plot the CDF for the matriculation set of the dynamic model without matriculation. As in the actual data, the average success probabilities for the matriculation set are nearly identical regardless of whether or not the committee considers matriculation probabilities in Round 1.

Table 26: Simulated Unrestricted Dynamic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	4.8637***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***
	(0.7564)	(0.1830)	(1.3238)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)
GRE Quantitative Percentile	0.0208**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1289	0.0233	0.2078	0.0290
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)
Demographic 1	0.3691**	0.0905**	-0.3430	-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)
Demographic 2	0.0925	0.0227	-0.1806	0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)
Advanced Math			0.6870***			0.7737***	0.1396***	0.6827***	0.0954***
			(0.2024)			(0.2535)	(0.0443)	(0.1770)	(0.0237)
Committee Score						0.0783	0.0141	0.0467	0.0065
						(0.0602)	(0.0108)	(0.0672)	(0.0094)
Missing Committee Score								0.2177	0.0304
								(0.1896)	(0.0264)
Year Transition More	0.0664	0.0163	0.0932	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)
Program Rank								-0.0309***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9465***						
			(0.0673)						
Obs | Param | LL | AIC	1000	38	-2655.7	5387.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	5.2717***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***
	(0.7564)	(0.1830)	(1.2701)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)
GRE Quantitative Percentile	0.0208**	0.0051**	-0.0016	0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***
	(0.0090)	(0.0022)	(0.0144)	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)
Gender	-0.2335*	-0.0572*	-0.1023	0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278
	(0.1309)	(0.0319)	(0.2051)	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)
Demographic 1	0.3691**	0.0905**	-0.4893*	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547
	(0.1732)	(0.0421)	(0.2701)	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)
Demographic 2	0.0925	0.0227	-0.2744	0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290
	(0.1911)	(0.0468)	(0.2740)	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)
Advanced Math			0.7219***			0.3580	0.0707	0.6387***	0.0896***
			(0.1966)			(0.2375)	(0.0465)	(0.1763)	(0.0238)
Committee Score						0.2962***	0.0585***	0.0626	0.0088
						(0.0592)	(0.0106)	(0.0657)	(0.0092)
Missing Committee Score								0.2535	0.0356
								(0.1896)	(0.0264)
Year Transition More	0.0664	0.0163	0.0031	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066
	(0.1281)	(0.0314)	(0.2010)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)
Program Rank								-0.0303***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9186***						
			(0.0682)						
Obs | Param | LL | AIC	1000	38	-2696.1	5468.2					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 27: Simulated Restricted Dynamic Estimates

(a) Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-2.1626***	4.6976***	-0.3078***	0.7351	0.0503	51.26%	-9.5097***	
	(0.7577)	(1.3258)	(0.0655)	(0.6465, 0.8358)	(0.1841)	(42.30%, 60.13%)	(1.2612)	
GRE Quantitative Percentile	0.0240***	0.0028					0.1148***	
	(0.0090)	(0.0150)					(0.0137)	
Gender	-0.2129	-0.0649					0.2357**	
	(0.1303)	(0.2112)					(0.1084)	
Demographic 1	0.3565**	-0.3526					-0.3177**	
	(0.1724)	(0.2903)					(0.1549)	
Demographic 2	0.0948	-0.1805					0.0020	
	(0.1904)	(0.3010)					(0.1594)	
Advanced Math		0.6919***					0.6157***	
		(0.2026)					(0.1367)	
Committee Score							0.0496	
							(0.0383)	
Missing Committee Score							0.2183	
							(0.1819)	
Year Transition More	0.0743	0.0967	-0.0466	0.7017	0.5233**	63.96%	-0.0124	
	(0.1277)	(0.2069)	(0.0675)	(0.6137, 0.8022)	(0.2398)	(54.65%, 72.33%)	(0.1720)	
Program Rank							-0.0310***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9468***						
		(0.0673)						
Sigma								0.1390***
								(0.0189)
Obs | Param | LL | AIC	1000	29	-2683.0	5423.9				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-2.1764***	4.5383***	-0.2386***	0.7878	0.0226	50.57%	-9.4138***	
	(0.7589)	(1.2591)	(0.0803)	(0.6730, 0.9220)	(0.2017)	(40.79%, 60.30%)	(1.4009)	
GRE Quantitative Percentile	0.0242***	0.0072					0.1066***	
	(0.0090)	(0.0143)					(0.0150)	
Gender	-0.2179*	-0.0641					0.2086*	
	(0.1306)	(0.2029)					(0.1206)	
Demographic 1	0.3420**	-0.5643**					-0.5037***	
	(0.1728)	(0.2716)					(0.1798)	
Demographic 2	0.1011	-0.2640					-0.0204	
	(0.1906)	(0.2738)					(0.1843)	
Advanced Math		0.7292***					0.5164***	
		(0.1970)					(0.1483)	
Committee Score							0.1934***	
							(0.0382)	
Missing Committee Score							0.2727	
							(0.1887)	
Year Transition More	0.0706	0.0108	-0.1333*	0.6895	0.4714*	62.11%	-0.0673	
	(0.1278)	(0.2007)	(0.0712)	(0.5886, 0.8075)	(0.2663)	(51.59%, 71.59%)	(0.1760)	
Program Rank							-0.0305***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9197***						
		(0.0684)						
Sigma								0.1776***
								(0.0305)
Obs | Param | LL | AIC	1000	29	-2742.2	5542.4				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 28: Simulated Unrestricted Static Estimates

(a) Optimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-8.1374***	-1.1678***	-9.4807***	-1.3248***
	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.1381)	(0.1344)	(1.3435)	(0.1562)
GRE Quantitative Percentile	0.1236***	0.0241***	0.1372***	0.0248***	0.1079***	0.0155***	0.1150***	0.0161***
	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0133)	(0.0015)	(0.0145)	(0.0016)
Gender	0.2266	0.0441	0.1289	0.0233	0.1399	0.0201	0.2078	0.0290
	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1739)	(0.0249)	(0.1766)	(0.0246)
Demographic 1	-0.3014	-0.0587	-0.2640	-0.0476	-0.2533	-0.0364	-0.3380	-0.0472
	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2359)	(0.0338)	(0.2456)	(0.0342)
Demographic 2	0.1538	0.0299	0.1708	0.0308	-0.1179	-0.0169	-0.1354	-0.0189
	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2581)	(0.0371)	(0.2652)	(0.0371)
Advanced Math			0.7737***	0.1396***			0.6827***	0.0954***
			(0.2535)	(0.0443)			(0.1770)	(0.0237)
Committee Score			0.0783	0.0141			0.0467	0.0065
			(0.0602)	(0.0108)			(0.0672)	(0.0094)
Missing Committee Score							0.2177	0.0304
							(0.1896)	(0.0264)
Year Transition More	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0587	-0.0084	-0.0466	-0.0065
	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1690)	(0.0242)	(0.1740)	(0.0243)
Program Rank					-0.0310***	-0.0045***	-0.0309***	-0.0043***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1622.9	3307.8				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-8.2825***	-1.1913***	-9.7587***	-1.3686***
	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.1392)	(0.1341)	(1.3345)	(0.1538)
GRE Quantitative Percentile	0.1226***	0.0238***	0.0550***	0.0109***	0.1097***	0.0158***	0.1175***	0.0165***
	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0133)	(0.0015)	(0.0145)	(0.0016)
Gender	0.2575*	0.0500*	0.0098	0.0019	0.1355	0.0195	0.1984	0.0278
	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1741)	(0.0250)	(0.1766)	(0.0247)
Demographic 1	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.2776	-0.0399	-0.3897	-0.0547
	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2344)	(0.0337)	(0.2462)	(0.0343)
Demographic 2	0.0797	0.0155	0.1764	0.0348	-0.1809	-0.0260	-0.2068	-0.0290
	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2566)	(0.0369)	(0.2636)	(0.0370)
Advanced Math			0.3580	0.0707			0.6387***	0.0896***
			(0.2375)	(0.0465)			(0.1763)	(0.0238)
Committee Score			0.2962***	0.0585***			0.0626	0.0088
			(0.0592)	(0.0106)			(0.0657)	(0.0092)
Missing Committee Score							0.2535	0.0356
							(0.1896)	(0.0264)
Year Transition More	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0767	-0.0110	-0.0473	-0.0066
	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1688)	(0.0243)	(0.1757)	(0.0247)
Program Rank					-0.0304***	-0.0044***	-0.0303***	-0.0043***
					(0.0019)	(0.0002)	(0.0019)	(0.0002)
Obs | Param | LL | AIC	1000	31	-1647.5	3357.0				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 29: Simulated Restricted Static Estimates

(a) Optimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.6911***	66.62%	0.2080	55.18%	-8.0508***	-9.5892***	
	(0.1817)	(58.30%, 74.02%)	(0.2148)	(44.69%, 65.23%)	(1.1629)	(1.3344)	
GRE Quantitative Percentile					0.1061***	0.1155***	
					(0.0134)	(0.0146)	
Gender					0.1997**	0.1694	
					(0.1017)	(0.1361)	
Demographic 1					-0.2479*	-0.3121	
					(0.1351)	(0.1916)	
Demographic 2					0.0408	-0.0331	
					(0.1509)	(0.1966)	
Advanced Math						0.6628***	
						(0.1426)	
Committee Score						0.0557	
						(0.0426)	
Missing Committee Score						0.2170	
						(0.1838)	
Year Transition More	-0.8478***	46.09%	0.7486**	72.24%	-0.0606	-0.0276	
	(0.2178)	(39.63%, 52.68%)	(0.3087)	(60.18%, 81.76%)	(0.1667)	(0.1730)	
Program Rank					-0.0310***	-0.0310***	
					(0.0019)	(0.0019)	
Sigma							0.1922***
							(0.0246)
Obs | Param | LL | AIC	1000	22	-1623.7	3291.3			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.8247***	69.52%	0.2858	57.10%	-8.5784***	-8.3580***	
	(0.2158)	(59.91%, 77.69%)	(0.2418)	(45.31%, 68.13%)	(1.1959)	(1.4863)	
GRE Quantitative Percentile					0.1141***	0.0950***	
					(0.0138)	(0.0158)	
Gender					0.2291**	0.1303	
					(0.1093)	(0.1417)	
Demographic 1					-0.5672***	-0.3661*	
					(0.1465)	(0.2059)	
Demographic 2					-0.0218	-0.0705	
					(0.1599)	(0.2104)	
Advanced Math						0.5399***	
						(0.1477)	
Committee Score						0.1918***	
						(0.0412)	
Missing Committee Score						0.1703	
						(0.1949)	
Year Transition More	-1.0834***	43.57%	0.5451*	69.66%	-0.0713	-0.1035	
	(0.2489)	(37.09%, 50.27%)	(0.3249)	(57.04%, 79.87%)	(0.1664)	(0.1784)	
Program Rank					-0.0306***	-0.0305***	
					(0.0019)	(0.0019)	
Sigma							0.2211***
							(0.0320)
Obs | Param | LL | AIC	1000	22	-1662.3	3368.6			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 30: Simulated Unrestricted Myopic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.6668***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***
	(0.0894)	(0.0224)	(0.1940)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)
GRE Quantitative Percentile				0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***
				(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)
Gender				0.2266	0.0441	0.1289	0.0233	0.2078	0.0290
				(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)
Demographic 1				-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472
				(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)
Demographic 2				0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189
				(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)
Advanced Math			0.6763***			0.7737***	0.1396***	0.6827***	0.0954***
			(0.2006)			(0.2535)	(0.0443)	(0.1770)	(0.0237)
Committee Score						0.0783	0.0141	0.0467	0.0065
						(0.0602)	(0.0108)	(0.0672)	(0.0094)
Missing Committee Score								0.2177	0.0304
								(0.1896)	(0.0264)
Year Transition More	0.0480	0.0120	0.1023	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065
	(0.1265)	(0.0316)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)
Program Rank								-0.0309***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9512***						
			(0.0674)						
Obs | Param | LL | AIC	1000	30	-2666.4	5392.8					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0000	0.0000	4.7588***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***
	(0.0894)	(0.0224)	(0.1881)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)
GRE Quantitative Percentile				0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***
				(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)
Gender				0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278
				(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)
Demographic 1				-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547
				(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)
Demographic 2				0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290
				(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)
Advanced Math			0.7040***			0.3580	0.0707	0.6387***	0.0896***
			(0.1957)			(0.2375)	(0.0465)	(0.1763)	(0.0238)
Committee Score						0.2962***	0.0585***	0.0626	0.0088
						(0.0592)	(0.0106)	(0.0657)	(0.0092)
Missing Committee Score								0.2535	0.0356
								(0.1896)	(0.0264)
Year Transition More	0.0480	0.0120	0.0218	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066
	(0.1265)	(0.0316)	(0.2022)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)
Program Rank								-0.0303***	-0.0043***
								(0.0019)	(0.0002)
$\sqrt{\text{MSE}}$			1.9289***						
			(0.0684)						
Obs | Param | LL | AIC	1000	30	-2708.0	5475.9					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 31: Simulated Restricted Myopic Estimates

(a) Optimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0146	4.6803***	-0.3118***	0.7321	0.0452	51.13%	-9.5732***	
	(0.0889)	(0.1930)	(0.0659)	(0.6433, 0.8331)	(0.1841)	(42.17%, 60.01%)	(1.2633)	
GRE Quantitative Percentile							0.1156***	
							(0.0138)	
Gender							0.2266**	
							(0.1080)	
Demographic 1							-0.3082**	
							(0.1533)	
Demographic 2							0.0055	
							(0.1579)	
Advanced Math		0.6789***					0.5999***	
		(0.2009)					(0.1366)	
Committee Score							0.0496	
							(0.0382)	
Missing Committee Score							0.2197	
							(0.1818)	
Year Transition More	0.0542	0.1004	-0.0430	0.7013	0.5323**	64.05%	-0.0098	
	(0.1261)	(0.2067)	(0.0679)	(0.6129, 0.8024)	(0.2407)	(54.67%, 72.46%)	(0.1720)	
Program Rank							-0.0310***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9514***						
		(0.0674)						
Sigma								0.1387***
								(0.0190)
Obs | Param | LL | AIC	1000	21	-2694.3	5430.7				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.0122	4.8050***	-0.2369***	0.7891	0.0217	50.54%	-9.5133***	
	(0.0891)	(0.1878)	(0.0811)	(0.6731, 0.9250)	(0.2029)	(40.71%, 60.33%)	(1.3931)	
GRE Quantitative Percentile							0.1081***	
							(0.0149)	
Gender							0.2007*	
							(0.1204)	
Demographic 1							-0.5140***	
							(0.1790)	
Demographic 2							-0.0254	
							(0.1847)	
Advanced Math		0.7027***					0.5083***	
		(0.1961)					(0.1484)	
Committee Score							0.1894***	
							(0.0385)	
Missing Committee Score							0.2765	
							(0.1884)	
Year Transition More	0.0511	0.0095	-0.1323*	0.6913	0.4816*	62.32%	-0.0638	
	(0.1262)	(0.2016)	(0.0716)	(0.5895, 0.8107)	(0.2684)	(51.71%, 71.88%)	(0.1759)	
Program Rank							-0.0305***	
							(0.0019)	
$\sqrt{\text{MSE}}$		1.9289***						
		(0.0684)						
Sigma								0.1794***
								(0.0308)
Obs | Param | LL | AIC	1000	21	-2754.8	5551.6				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 32: Simulated Unrestricted Matriculation-Dynamic Estimates

(a) Optimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	4.8637***	-11.5885***	-2.2558***	-13.1577***	-2.3739***	-9.4807***	-1.3248***	-4.6066	-0.6127
	(0.7564)	(0.1830)	(1.3238)	(1.0232)	(0.1454)	(2.0081)	(0.2886)	(1.3435)	(0.1562)	(4.6480)	(0.6187)
GRE Quantitative Percentile	0.0208**	0.0051**	0.0008	0.1236***	0.0241***	0.1372***	0.0248***	0.1150***	0.0161***	0.0593	0.0079
	(0.0090)	(0.0022)	(0.0150)	(0.0118)	(0.0018)	(0.0223)	(0.0033)	(0.0145)	(0.0016)	(0.0459)	(0.0061)
Gender	-0.2335*	-0.0572*	-0.0783	0.2266	0.0441	0.1289	0.0233	0.2078	0.0290	0.4885	0.0650
	(0.1309)	(0.0319)	(0.2113)	(0.1464)	(0.0283)	(0.2502)	(0.0451)	(0.1766)	(0.0246)	(0.5063)	(0.0657)
Demographic 1	0.3691**	0.0905**	-0.3430	-0.3014	-0.0587	-0.2640	-0.0476	-0.3380	-0.0472	0.3587	0.0477
	(0.1732)	(0.0421)	(0.2885)	(0.1988)	(0.0386)	(0.3744)	(0.0672)	(0.2456)	(0.0342)	(0.7459)	(0.1001)
Demographic 2	0.0925	0.0227	-0.1806	0.1538	0.0299	0.1708	0.0308	-0.1354	-0.0189	0.1834	0.0244
	(0.1911)	(0.0468)	(0.3001)	(0.2215)	(0.0431)	(0.3844)	(0.0693)	(0.2652)	(0.0371)	(0.7962)	(0.1069)
Advanced Math			0.6870***			0.7737***	0.1396***	0.6827***	0.0954***	0.8106	0.1078
			(0.2024)			(0.2535)	(0.0443)	(0.1770)	(0.0237)	(0.5773)	(0.0733)
Committee Score						0.0783	0.0141	0.0467	0.0065	-0.7402***	-0.0985***
						(0.0602)	(0.0108)	(0.0672)	(0.0094)	(0.2064)	(0.0235)
Missing Committee Score								0.2177	0.0304		
								(0.1896)	(0.0264)		
Year Transition/Matriculate More	0.0664	0.0163	0.0932	0.9840***	0.1915***	-0.9251***	-0.1669***	-0.0466	-0.0065	3.3280***	0.4427***
	(0.1281)	(0.0314)	(0.2069)	(0.1472)	(0.0263)	(0.2513)	(0.0427)	(0.1740)	(0.0243)	(0.5674)	(0.0331)
Program Rank								-0.0309***	-0.0043***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.9465***								
			(0.0673)								
Obs | Param | LL | AIC	1000	46	-2708.1	5508.3							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	Advanced Math		Committee Score	Transition		Accept		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-1.9027**	-0.4663**	5.2717***	-11.2295***	-2.1801***	-6.7296***	-1.3296***	-9.7587***	-1.3686***	-1.0400	-0.1351
	(0.7564)	(0.1830)	(1.2701)	(1.0144)	(0.1474)	(1.7376)	(0.3202)	(1.3345)	(0.1538)	(3.6083)	(0.4721)
GRE Quantitative Percentile	0.0208**	0.0051**	-0.0016	0.1226***	0.0238***	0.0550***	0.0109***	0.1175***	0.0165***	0.0129	0.0017
	(0.0090)	(0.0022)	(0.0144)	(0.0117)	(0.0018)	(0.0191)	(0.0036)	(0.0145)	(0.0016)	(0.0391)	(0.0051)
Gender	-0.2335*	-0.0572*	-0.1023	0.2575*	0.0500*	0.0098	0.0019	0.1984	0.0278	0.1298	0.0169
	(0.1309)	(0.0319)	(0.2051)	(0.1469)	(0.0283)	(0.2368)	(0.0468)	(0.1766)	(0.0247)	(0.4717)	(0.0608)
Demographic 1	0.3691**	0.0905**	-0.4893*	-0.8293***	-0.1610***	-0.3340	-0.0660	-0.3897	-0.0547	0.6327	0.0822
	(0.1732)	(0.0421)	(0.2701)	(0.1966)	(0.0371)	(0.3394)	(0.0669)	(0.2462)	(0.0343)	(0.6281)	(0.0823)
Demographic 2	0.0925	0.0227	-0.2744	0.0797	0.0155	0.1764	0.0348	-0.2068	-0.0290	-0.2222	-0.0289
	(0.1911)	(0.0468)	(0.2740)	(0.2135)	(0.0414)	(0.3346)	(0.0659)	(0.2636)	(0.0370)	(0.6864)	(0.0880)
Advanced Math			0.7219***			0.3580	0.0707	0.6387***	0.0896***	0.5877	0.0763
			(0.1966)			(0.2375)	(0.0465)	(0.1763)	(0.0238)	(0.5321)	(0.0674)
Committee Score						0.2962***	0.0585***	0.0626	0.0088	-0.6092***	-0.0791***
						(0.0592)	(0.0106)	(0.0657)	(0.0092)	(0.1637)	(0.0176)
Missing Committee Score								0.2535	0.0356		
								(0.1896)	(0.0264)		
Year Transition/Matriculate More	0.0664	0.0163	0.0031	1.1108***	0.2156***	-0.7185***	-0.1420***	-0.0473	-0.0066	3.4846***	0.4526***
	(0.1281)	(0.0314)	(0.2010)	(0.1479)	(0.0257)	(0.2397)	(0.0455)	(0.1757)	(0.0247)	(0.6046)	(0.0349)
Program Rank								-0.0303***	-0.0043***		
								(0.0019)	(0.0002)		
$\sqrt{\text{MSE}}$			1.9186***								
			(0.0682)								
Obs | Param | LL | AIC	1000	46	-2749.9	5591.9							

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 33: Simulated Restricted Matriculation-Dynamic Estimates

(a) Optimal Committee

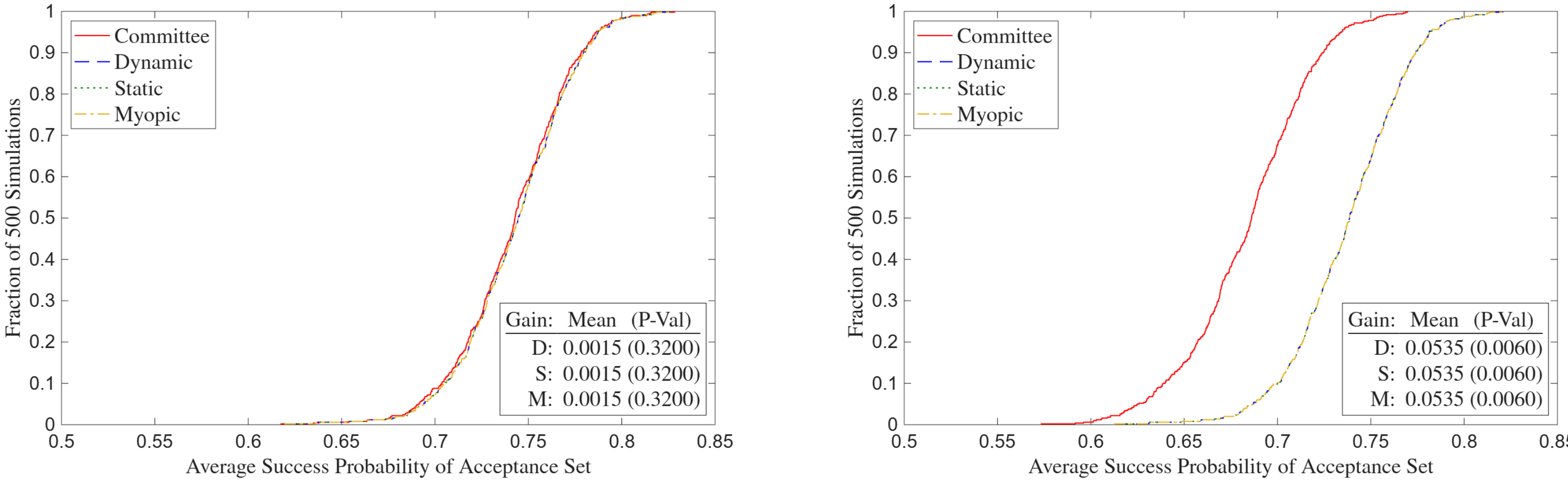
Intercept	-2.1963***	5.4976***	-1.8249***	0.1612	0.0339	50.85%	-9.4675***	-9.8786**	
	(0.7585)	(1.3429)	(0.1759)	(0.1142, 0.2276)	(0.2008)	(41.11%, 60.53%)	(1.2874)	(4.0907)	
GRE Quantitative Percentile	0.0244***	-0.0072					0.1153***	0.1186***	
	(0.0090)	(0.0153)					(0.0140)	(0.0397)	
Gender	-0.2210*	-0.0965					0.1845	0.7496*	
	(0.1307)	(0.2096)					(0.1213)	(0.4548)	
Demographic 1	0.3598**	-0.3130					-0.3019*	0.1162	
	(0.1729)	(0.2878)					(0.1729)	(0.7211)	
Demographic 2	0.0946	-0.1832					-0.0093	0.0811	
	(0.1907)	(0.2982)					(0.1766)	(0.7755)	
Advanced Math		0.6485***					0.6475***	1.0534*	
		(0.2032)					(0.1382)	(0.5829)	
Committee Score							0.0265	-0.6809***	
							(0.0400)	(0.1999)	
Missing Committee Score							0.2380		
							(0.1861)		
Year Transition/Matriculate More	0.0710	0.1591	-1.1147***	0.0529	0.5592**	64.41%	-0.0010	3.0532***	
	(0.1280)	(0.2057)	(0.2799)	(0.0296, 0.0946)	(0.2553)	(54.59%, 73.15%)	(0.1732)	(0.5423)	
Program Rank							-0.0309***		
							(0.0019)		
$\sqrt{\text{MSE}}$		1.9457***							
		(0.0672)							
Sigma									0.1457***
									(0.0234)
Obs | Param | LL | AIC	1000	37	-2755.2	5584.5					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Matriculate	Sigma
Intercept	-2.0816***	5.5356***	-1.4144***	0.2431	0.1778	54.43%	-9.6441***	-6.4840*	
	(0.7582)	(1.2750)	(0.2209)	(0.1576, 0.3748)	(0.2485)	(42.33%, 66.04%)	(1.3938)	(3.3823)	
GRE Quantitative Percentile	0.0230**	-0.0050					0.1098***	0.0774**	
	(0.0090)	(0.0145)					(0.0150)	(0.0363)	
Gender	-0.2276*	-0.1115					0.1637	0.3953	
	(0.1310)	(0.2034)					(0.1438)	(0.4595)	
Demographic 1	0.3530**	-0.4625*					-0.4614**	0.0566	
	(0.1729)	(0.2685)					(0.2103)	(0.6572)	
Demographic 2	0.0922	-0.2781					-0.0923	-0.4297	
	(0.1908)	(0.2722)					(0.2189)	(0.7030)	
Advanced Math		0.7079***					0.5798***	0.7404	
		(0.1960)					(0.1558)	(0.5503)	
Committee Score							0.1840***	-0.5475***	
							(0.0433)	(0.1601)	
Missing Committee Score							0.2500		
							(0.1936)		
Year Transition/Matriculate More	0.0723	0.0357	-1.3730***	0.0616	0.5357	67.12%	-0.0723	3.1502***	
	(0.1280)	(0.2000)	(0.3748)	(0.0274, 0.1385)	(0.3413)	(53.66%, 78.25%)	(0.1795)	(0.5830)	
Program Rank							-0.0305***		
							(0.0019)		
$\sqrt{\text{MSE}}$		1.9147***							
		(0.0677)							
Sigma									0.2346***
									(0.0484)
Obs | Param | LL | AIC	1000	37	-2818.0	5710.0					

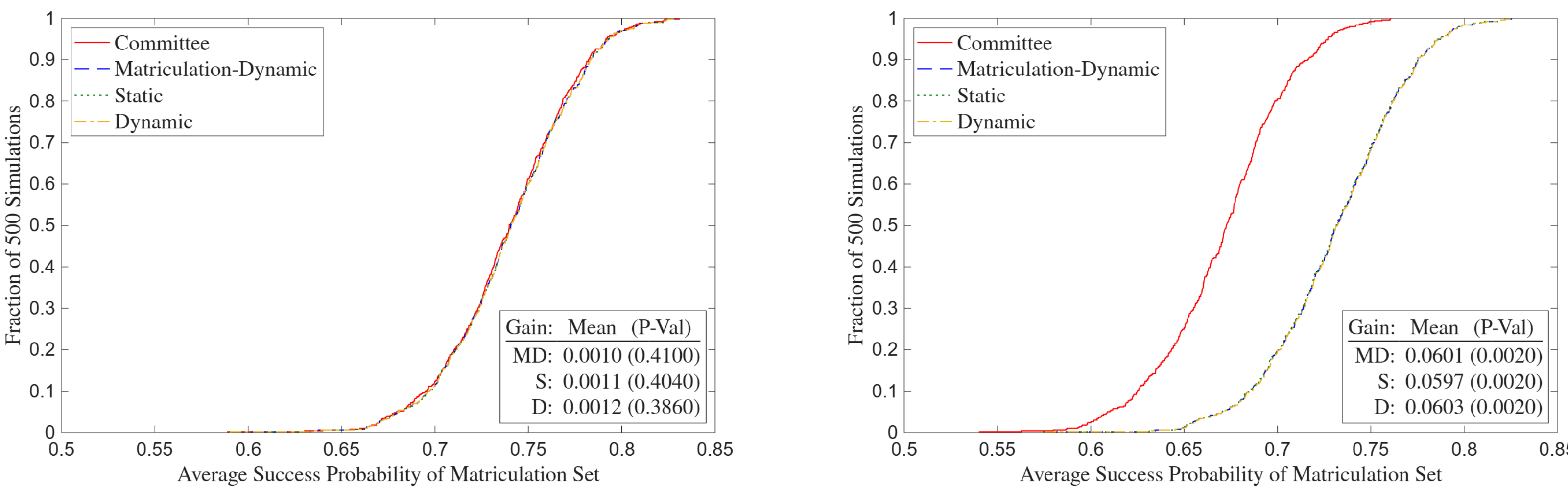
Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.



(a) Optimal Committee

(b) Suboptimal Committee

Figure 7: Simulated Deviation Gains



(a) Optimal Committee

(b) Suboptimal Committee

Figure 8: Simulated Matriculation Deviation Gains

E Two-Round Portfolio Choice

E.1 Without Matriculation

In Section 4.1, I show that the committee's portfolio choice problem reduces to a cutoff rule problem, keeping it to one round for simplicity. Then in Sections 4.1.1 and 4.1.2, I write out dynamic and static two-round cutoff rule problems with Round 1 filtering and Round 2 final selections. In this appendix, I solve a two-round portfolio choice problem by backward induction, showing that each round's solution takes the cutoff rule form in the paper. I start with Round 2:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n=S|x_n,z_n)\mathbb{1}(d_{2,n}=A), \text{ such that: } \sum_{n=1}^{N_T}\mathbb{1}(d_{2,n}=A)\leq N_A. \tag{35}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{2,n}=A$, which means that the constraint binds. The Lagrangian with multiplier λ_2 is:

$$\begin{aligned}\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n=S|x_n,z_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\left\{N_A-\left[N_T-\sum_{n=1}^{N_T}\mathbb{1}(d_{2,n}=R)\right]\right\}\\ &= \sum_{n=1}^{N_T}\left[P(y_n=S|x_n,z_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]+\lambda_2(N_A-N_T) \end{aligned}\tag{36}$$

By the same logic in Section 4.1, the Round 2 solution is as follows with set-identified cutoff probability $\lambda_2^*=P(y_2^*=S)\in\left[P(y_{N_A+1}=S|x_{N_A+1},z_{N_A+1}),P(y_{N_A}=S|x_{N_A},z_{N_A})\right]$:

$$\forall n\in\{1,\ldots,N_T\},\, d_{2,n}(x_n,z_n|\lambda_2^*)=\begin{cases}A & \text{if } P(y_n=S|x_n,z_n)>\lambda_2^*\\ \{A,R\} & \text{if } P(y_n=S|x_n,z_n)=\lambda_2^*\\ R & \text{if } P(y_n=S|x_n,z_n)<\lambda_2^*.\end{cases}\tag{37}$$

When the committee solves Round 1, it must anticipate λ_2^*, which is a function of the $(\mathbf{x},\mathbf{z})$ of the entire transitioned pool. I make two assumptions so that this anticipation is tractable. First, conditional on $\mathbf{x}$, I assume that $\mathbf{z}$ is independent across applicants. Second, I assume that N is large. Under these assumptions, the same Glivenko-Cantelli argument as in the proof of Theorem 1, with $\mathbf{z}$ in place of the preference shocks, makes λ_2^* deterministic for large N.

Denote $\hat{G}_{N_T}(c)=\frac{1}{N_T}\sum_{n=1}^{N}\mathbb{1}\{P(y_n=S|x_n,z_n)\leq c\}\mathbb{1}\{d_{1,n}=T\}$ as the empirical distribution of the success probabilities of the N_T transitioned applicants. The cutoff λ_2^* is the $(1-N_A/N_T)$ quantile of this distribution, since the committee accepts the N_A such applicants with the highest success probabilities. Because only $\mathbf{x}$ is observed in Round 1, transition depends on x_n alone; the Round 1 cutoff rule derived below transitions applicant n when x_n lies in transition region $\mathcal{X}_T$. For large N, the cutoff λ_1^* converges to a constant by the same argument, so $\mathcal{X}_T$ is deterministic.

Since the pairs (x_n,z_n) are iid and membership in $\mathcal{X}_T$ depends only on x_n, the transitioned success probabilities are themselves iid, with conditional distribution $G(c)=P(P(y=S|x,z)\leq$

$c|x \in \mathcal{X}_T)$. So by the Glivenko-Cantelli theorem, $\hat{G}_{N_T}$ converges almost surely to G for large N_T. Therefore, as $N_A/N_T \to q_2$, $\hat{G}_{N_T}$'s $(1-N_A/N_T)$ quantile converges to G's deterministic $(1-q_2)$ quantile. In turn, the set-identified interval for λ_2^* collapses to a point, denoted $\overline{\lambda_2^*}$. As in the proof of Theorem 1, independence is essential; a common component across applicants would slide $\hat{G}_{N_T}$ in unison and leave λ_2^* random in the limit. This convergence is a fixed point because the transition region $\mathcal{X}_T$ is pinned down by the Round 1 cutoff λ_1^*, which ranks applicants by their anticipated Round 2 payoff, which itself embeds $\overline{\lambda_2^*}$. The pair $(\lambda_1^*, \overline{\lambda_2^*})$ is jointly determined. Because $\overline{\lambda_2^*}$ is deterministic, no individual applicant's signal moves the cutoff, the committee's anticipated Round 2 payoff separates across applicants, and Round 1 reduces to a cutoff rule.

Let $EV_{2,n}(x_n) = \mathbb{E}[\max\{P(y_n = S|x_n, z_n), \overline{\lambda_2^*}\}|x_n] = \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), \overline{\lambda_2^*}\} dF_{z|x}(\tilde{z}_n|x_n)$. Then the Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{38}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned}\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} \{ EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \} + \lambda_1 (N_T - N). \end{aligned} \tag{39}$$

By the same logic in Section 4.1, the Round 1 solution is as follows with set-identified cutoff level $\lambda_1^* = L(y_1^* = S) \in \left[EV_{2,N_T+1}(x_{N_T+1}), EV_{2,N_T}(x_{N_T}) \right]$:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{40}$$

There is one discrepancy between $\lambda_1^* = L(y_1^* = S)$ in this appendix vs. Section 4.1.1. Here, λ_1^* is bounded above by 1 because for all n, it is the case that $\max\{P(y_n = S|x_n, \tilde{z}_n), \overline{\lambda_2^*}\} \leq 1$ pointwise in $\tilde{z}_n$. Therefore, $EV_{2,n}(x_n) \leq \int_{\tilde{z}_n} dF_{z|x}(\tilde{z}_n|x_n) = 1$. But in Section 4.1.1, the type 1 extreme value preference shocks increase this upper bound to $1 + \sigma \log 2$, where I now denote $\overline{\lambda_2^*}$ by $P(y_2^* = S)$:

$$\begin{aligned}\mathbb{E}[v_2(x_n, z_n, \varepsilon_{2,n})|x_n] &= \int_{\tilde{z}_n} \sigma \log\left(e^{P(y_n = S|x_n, \tilde{z}_n)/\sigma} + e^{P(y_2^* = S)/\sigma} \right) dF(\tilde{z}_n|x_n) \\ &\leq \int_{\tilde{z}_n} \sigma \log\left(2e^{\max\{P(y_n = S|x_n, \tilde{z}_n), P(y_2^* = S)\}/\sigma} \right) dF(\tilde{z}_n|x_n) \\ &= \int_{\tilde{z}_n} \max\{P(y_n = S|x_n, \tilde{z}_n), P(y_2^* = S)\} dF(\tilde{z}_n|x_n) + \sigma \log 2 \\ &\leq 1 + \sigma \log 2. \end{aligned} \tag{41}$$

In the static version of the two-round portfolio choice model, the Round 1 problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \le N_T. \tag{42}$$

Because success probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian with multiplier λ_1 is:

$$\begin{aligned}\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1\left\{N_T - \left[N - \sum_{n=1}^{N}\mathbb{1}(d_{1,n} = R)\right]\right\} \\ &= \sum_{n=1}^{N}\left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1\mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N). \end{aligned}\tag{43}$$

By the same logic in Section 4.1, the solution is as follows with set-identified cutoff probability $\lambda_1^* = P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^*. \end{cases} \tag{44}$$

E.2 With Matriculation

In Section 4.2, I show that because the committee's capacity constraint is on the number of matriculations rather than acceptances, a one-round portfolio choice problem with matriculation reduces to a cutoff rule problem on success probabilities. However, a two-round portfolio choice problem with matriculation is less straightforward because matriculation ends up mattering in Round 1. In this appendix, I present such a model with Round 1 filtering and Round 2 final selections. For readability, I keep finite-sum notation rather than the integrals in the proof of Theorem 3 in Appendix H. But I invoke that theorem's large-N framework throughout because it removes the finite-N edge cases and, as in Appendix E.1, collapses λ_2^* to the deterministic $\overline{\lambda_2^*}$. Working backward, I start with Round 2, where for brevity, I denote $w = (x,z)$:

$$\begin{gathered}\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A), \\ \text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \le N_M. \end{gathered}\tag{45}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{2,n} = A$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned}\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) &= \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\left\{N_M - \left[\overline{N_M} - \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = R)\right]\right\} \\ &= \sum_{n=1}^{N_T}\left[P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)\right]P(m_n = M|w_n) + \lambda_2(N_M - \overline{N_M}), \end{aligned}\tag{46}$$

where $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$ is the expected number of matriculants if everyone is accepted.

Edge cases aside, the Round 2 solution, with cutoff probability $\lambda_2^* = P(y_2^* = S)$, is:

$$\forall n \in \{1,\ldots,N_T\},\, d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^*. \end{cases} \tag{47}$$

Let $EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y_n = S|x_n,\widetilde{z}_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|x_n,\widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$. This netted form, and not $\int_{\widetilde{z}_n} \max\{P(y_n = S|x_n,\widetilde{z}_n), \overline{\lambda_2^*}\} P(m_n = M|x_n,\widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$, is the correct transition payoff. To see why, evaluate the Round 2 Lagrangian at the cutoff rule solution with deterministic cutoff $\overline{\lambda_2^*}$, where $P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)$ becomes $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\}$:

$$\mathcal{L}(d_2^*; \overline{\lambda_2^*}) = \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n), \overline{\lambda_2^*}\} P(m_n = M|w_n) + \overline{\lambda_2^*}(N_M - \overline{N_M}). \tag{48}$$

By $\max\{P(y_n = S|w_n), \overline{\lambda_2^*}\} = \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} + \overline{\lambda_2^*}$, and $\overline{N_M} = \sum_{n=1}^{N_T} P(m_n = M|w_n)$:

$$\begin{aligned} \mathcal{L}(d_2^*; \overline{\lambda_2^*}) &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n) + \overline{\lambda_2^*}\,\overline{N_M} + \overline{\lambda_2^*}(N_M - \overline{N_M}) \\ &= \sum_{n=1}^{N_T} \max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n) + \overline{\lambda_2^*} N_M. \end{aligned} \tag{49}$$

Although $\overline{N_M}$ varies with the Round 1 transition decisions, the $\overline{\lambda_2^*}\,\overline{N_M}$ terms cancel, leaving the constant $\overline{\lambda_2^*} N_M$. The Round 1 transition payoff is therefore the per-applicant netted term $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\} P(m_n = M|w_n)$, whose expectation over $\widetilde{z}_n$ defines $EV_{2,n}$.

The netting distinguishes the matriculation case from the case without it. Without matriculation, the netted and non-netted forms differ by $\int_{\widetilde{z}_n} \overline{\lambda_2^*} dF_{z|x}(\widetilde{z}_n|x_n) = \overline{\lambda_2^*}$, a constant across applicants that leaves the Round 1 ordering unchanged. With matriculation, they differ by $\int_{\widetilde{z}_n} \overline{\lambda_2^*} P(m_n = M|x_n,\widetilde{z}_n) dF_{z|x}(\widetilde{z}_n|x_n)$, which varies with x_n. The non-netted form would therefore order applicants differently, and incorrectly. The Round 1 portfolio choice problem is:

$$\max_{(d_{1,1},\ldots,d_{1,N}) \in \{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T. \tag{50}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned} \mathcal{L}(d_{1,1},\ldots,d_{1,N}; \lambda_1) &= \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \left\{ N_T - \left[N - \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = R) \right] \right\} \\ &= \sum_{n=1}^{N} \left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R) \right] + \lambda_1 (N_T - N). \end{aligned} \tag{51}$$

Edge cases aside, the Round 1 solution with cutoff level $\lambda_1^* = L(y_1^* = S, m_1^* = M)$ is:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^*. \end{cases} \tag{52}$$

The Round 1 transition payoff with Type 1 extreme value shocks is therefore:

$$\mathbb{E}[v_2(x_n, z_n, \varepsilon_{2,n})|x_n] = \int_{\widetilde{z}_n} \sigma \log\left(e^{[P(y_n=S|x_n,\widetilde{z}_n)-P(y_2^*=S)]/\sigma}+1\right) P(m_n = M|x_n,\widetilde{z}_n)dF(\widetilde{z}_n|x_n). \tag{53}$$

The committee prefers to transition likely matriculants so that it can be more selective in Round 2. As a simple example, suppose that $N = 4$, $N_T = 2$, and $N_M = 1$, and suppose that recommendation letters matter more for success than GRE scores. Suppose also that Applicant 1 has a high GRE and strong recommendation letter, Applicant 2 has a high GRE and weak letter, Applicant 3 has a low GRE and strong letter, and Applicant 4 has a low GRE and weak letter. If the committee ignores matriculation probabilities in Round 1, it will transition Applicants 1 and 2 and then accept Applicant 1. But suppose that because Applicant 1 is so strong, they choose a higher-ranked university. Then the committee has to settle for Applicant 2. Whereas if the committee had transitioned Applicants 3 and 4, it could have accepted and landed Applicant 3.

In the static version of the model with matriculation, the Round 1 problem, with $\alpha \in [0,1]$, is:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha}\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n} = T) \le N_T. \tag{54}$$

Because success and matriculation probabilities are never negative, the unconstrained objective increases every time $d_{1,n} = T$, which means that the constraint binds. The Lagrangian is:

$$\begin{aligned}\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) &= \sum_{n=1}^{N} P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha}\mathbb{1}(d_{1,n} = T) + \lambda_1\left\{N_T - \left[N - \sum_{n=1}^{N}\mathbb{1}(d_{1,n} = R)\right]\right\} \\ &= \sum_{n=1}^{N}\left[P(y_n = S|x_n)^{\alpha} P(m_n = M|x_n)^{1-\alpha}\mathbb{1}(d_{1,n} = T) + \lambda_1\mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N). \end{aligned} \tag{55}$$

Edge cases aside, the solution is as follows with cutoff probability $\lambda_1^* = P(y_1^* = S, m_1^* = M)$:

$$\forall n \in \{1,\ldots,N\},\, d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n)^{\alpha}P(m_n = M|x_n)^{1-\alpha} > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n)^{\alpha}P(m_n = M|x_n)^{1-\alpha} = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n)^{\alpha}P(m_n = M|x_n)^{1-\alpha} < \lambda_1^*. \end{cases} \tag{56}$$

But per the dynamic $EV_{2,n}$ integrand without shocks, $\max\{P(y_n = S|w_n) - \overline{\lambda_2^*}, 0\}P(m_n = M|w_n)$, matriculation probabilities matter less than success probabilities. For applicants with $P(y_n = S|w_n) \le \overline{\lambda_2^*}$, matriculation has a weight of zero. Even for the other applicants, the weight is only large when $P(y_n = S|w_n) \gg \overline{\lambda_2^*}$. Thus, the static model without matriculation approximates the matriculation-dynamic model well enough empirically that α can be set to 1. Doing so also aligns with committee members indicating that they do not consider matriculation in Round 1.

F Waitlist Models

In the matriculation-dynamic model in Section 4.2.1, the committee considers matriculation probabilities in Round 1. However, the degree to which it does so is small enough that the static model without matriculation serves as a static approximation for the matriculation-dynamic model. Committee members at the university featured in this paper also indicate that the committee does not consider matriculation probabilities in Round 1. The reason is that this university is ranked highly enough that it can fill its cohort with applicants who have strong objective and subjective variables, leaving Round 2 with enough slack to be selective on success regardless of Round 1's matriculation weighting. However, this university does consider matriculation probabilities after Round 1, where additional variables, such as *Letters Terms (%) PC1*, predict matriculation.[20]

Therefore, in this appendix, I introduce the waitlist and expand the cutoff rule framework from two rounds to three. Success probabilities equal, the committee favors likely matriculants in Round 2 to avoid losing high-quality applicants who would have matriculated by suboptimally keeping those applicants waiting until Round 3 for an acceptance. I also make the simplifying assumption that matriculation probabilities do not enter the Round 1 objective.

F.1 Asymmetric Waitlist Models

An admissions committee does not send out all its acceptances at once, but rather in batches. It starts by accepting its most preferred applicants, and then it puts the others who survived Round 1 on a waitlist. Then provided not all its early acceptances matriculate, it accepts applicants off the waitlist and ultimately moves down the waitlist until it has filled its cohort. I model this process as concisely as possible by increasing the number of rounds to three. As shown in Figure 9, the committee observes only the objective variables in Round 1 and either transitions the applicant to Round 2 or rejects them out of courtesy. But in Round 2, instead of accepting or rejecting the survivors based on the objective and subjective variables, it accepts or waitlists them.

Then in Round 3, it accepts or rejects the waitlisted applicants. Between Rounds 2 and 3, it observes how many of the early acceptances matriculate and makes its Round 3 decisions accordingly. If more early acceptances matriculate, it accepts fewer applicants off the waitlist because there are fewer openings left to fill. Intuitively, this situation is better than the other way around. More early acceptances matriculating means the committee gets more of its top choices. In contrast, if the committee has to accept many applicants off the waitlist, it may have to lower its standards because applicants it could have gotten with early offers may have already committed to other schools. Therefore, it should err on the side of giving Round 2 offers to applicants who are relatively likely to matriculate. That said, it would be incorrect to have the committee value

[20]Per Table 2b of Section 3, the matriculation-predicting variables in x are *Winsorized Age* (-), *U.S.* (+), *International Asian* (+), *GRE Quantitative Percentile* (-), and *Work Experience* (+). The matriculation-predicting variables in z are *Advanced Math* (-), *Letters Terms (%) PC1* (+), *CV Coding* (-), and *Essay Economics Terms (%)* (+).

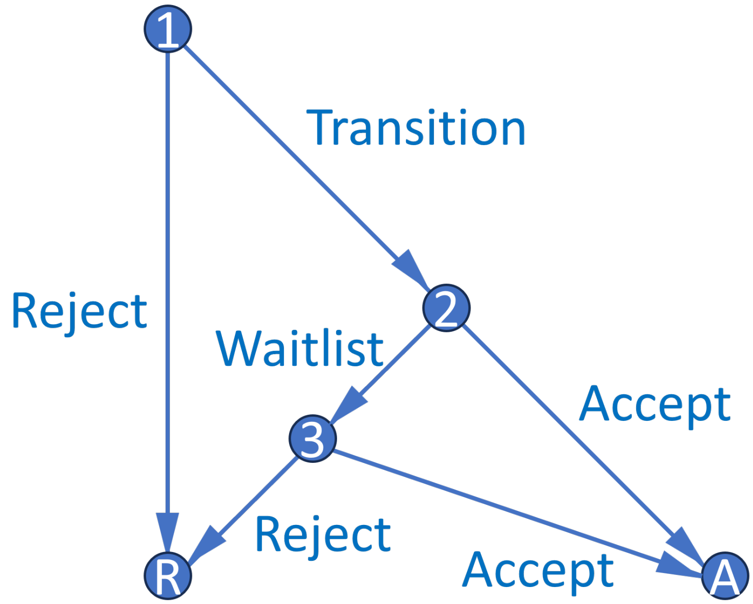


Figure 9: Waitlist Decision Graph

applicants by a combination of their success and matriculation probabilities, as that would be the cutoff rule solution to a portfolio choice problem with a constraint on the number of acceptances, not matriculants. The matriculation probabilities must instead appear elsewhere in the objective.

As such, I set up the model by letting the Round 3 cutoff probability be $P(y_3^* = S|\overline{m_{2A}})$ instead of $P(y_3^* = S|t)$. Here, $\overline{m_{2A}}$ is the average early accepted applicant's matriculation probability in a given year, which measures how likely the the applicant is to matriculate if accepted. $P(\text{Cutoff})_3$ increases in $\overline{m_{2A}}$ because a larger $\overline{m_{2A}}$ means there will be fewer openings for waitlisted applicants. Given that, what information available at the start of Round 2 determines $\overline{m_{2A}}$? For all applicants in each year, I define $\ddot{m}_2$ as $\overline{m_2}$ but with m_2 excluded, where m_2 is the applicant's own matriculation probability. Within each year, a low m_2 necessarily implies a high $\ddot{m}_2$, since $\overline{m_2}$ is a within-year average. I then estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; the parameter estimate should be positive provided there exists variation in $\overline{m_2}$ across years. For example, by decreasing acceptance probabilities, Covid may have made applicants more likely to matriculate if accepted. I assume that the effect of the average matriculation probability of the applicants the committee can accept ($\ddot{m}_2$) on that of the applicants it accepts ($\overline{m_{2A}}$) is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ resulting from changes in d_2.

The model implies that the optimal dynamic decision in Round 2 may not be the same as the optimal myopic one. Specifically, a committee may waitlist, as opposed to accept, an applicant with a low m_2 even if their success probability is greater than an applicant with a higher m_2. The reason is that compared with a high m_2 applicant, a low m_2 applicant has a relatively high $\ddot{m}_2$. That implies a high $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which in turn implies a high Round 3 cutoff probability, Round 3 value function, and payoff for waitlisting the low m_2 applicant in Round 2. Suppose that a hypothetical committee does not think dynamically; that is, removing these memory parameters increases the AIC. However, if $\ddot{m}_2$ affects $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$ and in turn $\mathbb{E}[P(\text{Cutoff})_3]$, perhaps the committee should.

That said, why is the committee not able to accept the low m_2 applicant in Round 2 and then, if this applicant does not matriculate, fill the additional opening with the high m_2 applicant that it necessarily waitlisted? There are two reasons why such a strategy may not pay off. First, all else equal, matriculation probabilities are expected to decline between Rounds 2 and 3 because applicants are more likely to go where they feel wanted; whether and by how much they decline is an empirical matter. If matriculation probabilities were to decline uniformly for likely vs. unlikely matriculants, the committee could still get away with accepting unlikely matriculants and filling any extra openings with likely matriculants. However, if applicants with high matriculation probabilities have more room for those matriculation probabilities to fall when the committee keeps them waiting until Round 3, then the better decision is to accept them in Round 2.

Second, matriculation probabilities can change unpredictably between Rounds 2 and 3 because applicants receive acceptances, waitlists, and rejections from multiple universities, communicate with faculty and graduate students at universities where they were accepted or waitlisted, and attend open houses. These events, which are unknown to the committee at the start of Round 2, make it so that waitlisting a likely matriculant under the belief that they will still matriculate if waitlisted is a risky endeavor; by Round 3, the committee may find out too late that this applicant is now unlikely to matriculate. Therefore, it is safer to accept this applicant in Round 2.[21]

As such, it is clear that accepting applicants who want to matriculate benefits the program. I also discount the expected Round 3 value with $\beta \in (0,1)$ so that the payoff from accepting an applicant in Round 3 is less than that from accepting them in Round 2. The discount factor captures the committee's perceived cost of deferring the final decision to Round 3 and can be interpreted as the chance of losing a representative applicant to another university; loosely, it reflects the ratio of expected matriculation probabilities for Round 3 vs. Round 2 acceptances. Of course, without a capacity constraint, the committee could give early acceptances to the low and high m_2 applicants, but because of the constraint, it must avoid overshooting its target cohort size. I now present the value functions for the waitlist-dynamic model, starting with Round 3:

$$v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x,z,t,\bar{k},\overline{m_{2A}},d_3) + \varepsilon_3(d_3), \text{ such that:} \tag{57}$$

$$u_3(x,z,t,\bar{k},\overline{m_{2A}},d_3) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_3=A) + P(y_3^*=S|\overline{m_{2A}})\mathbb{1}(d_3=R).$$

The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x,z,t,\bar{k},\ddot{m}_2,d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{58}$$

[21] In Appendix F.3, I present a more general version of the model in which the Round 3 cutoff probability is a function of $\overline{m_{2A}}$ and $\overline{m_3}$, which is the average matriculation probability of the waitlisted applicants and depends on m_3, the *ex-ante* probability of matriculating if accepted in Round 3. In this model, the effect of matriculation probabilities on Round 2 acceptance probabilities is theoretically ambiguous and depends on the parameter values.

$$u_2(x,z,t,\bar{k},\ddot{m}_2,d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]\mathbb{1}(d_2=W), \text{ such that:}$$

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\bar{k},\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2).$$

The Round 1 value function depends on whether I include a conditional expectation, as in the dynamic model, or whether I do not, as in the static model. I refer to the three-round model without a Round 1 conditional expectation as the waitlist-hybrid model, since it has a static Round 1 and dynamic Round 2. The waitlist-dynamic model's Round 1 value function is:

$$v_1(x,t,\bar{k},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{59}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T) + L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,W\}} e^{u_2(x,\widetilde{z},t,\bar{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t).$$

In contrast, the utility function in the waitlist-hybrid's Round 1 value function is:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R).$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r\in D_r}\exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}. \tag{60}$$

The waitlist-dynamic model's unrestricted CCPs, success probabilities, and cutoff probabilities are $P_{d_1} = P\big(d_1=T\big|x,t;\theta_T\big)$, $P_{d_2} = P\big(d_2=A\big|x,z,\ddot{m}_2;\theta_{A_2}\big)$, $P_{d_3} = P\big(d_3=A\big|x,z,\overline{m_{2A}};\theta_{A_3}\big)$, $P_y = P\big(y=S\big|x,z,t,\bar{k};\theta_S\big)$, $L_{y_1^*} = L\big(y_1^*=S\big|t;\theta_{S_1^*}\big)$, and $P_{y_3^*} = P\big(y_3^*=S\big|\overline{m_{2A}};\theta_{S_3^*}\big)$. These probabilities have the same specifications as in the dynamic model (see Equations (11)–(13) for the functional forms).[22] Let applicants 1 to N_W be waitlisted, N_W+1 to N_T accepted in Round 2, and N_T+1 to N rejected in Round 1. Then the unrestricted waitlist-dynamic likelihood is:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big) f\big(\overline{m_{2A,n}}|\ddot{m}_{2,n};\theta_{F_m}\big)$$

$$P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)} \prod_{n=1}^{N_W} P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}$$

$$\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \tag{61}$$

[22] $L_{y_1^*}$ is parameterized as exponential because it can exceed 1. However, in the waitlist-hybrid model, it is denoted as $P_{y_1^*}$ and parameterized as logistic. The waitlist-hybrid model also includes $P_{y_1}\big(y=S\big|x,t,\bar{k};\theta_{S_1}\big)$.

As with the dynamic and matriculation-dynamic likelihoods, a Heckman correction is necessary for $f_{z|x,t}$. I also estimate the parameters of $m_2 = P_{m_2}(\theta_{M_2})$, and therefore $\ddot{m}_2$, in a first-stage likelihood where I restrict the sample to Round 2 acceptances. The dependent variable equals 1 if the applicant matriculates and 0 otherwise, and the independent variables are x, z, and t. Because $\ddot{m}_2$ is a generated regressor in both the unrestricted and restricted likelihoods, I estimate standard errors per Murphy and Topel (1985) and Hardin (2002). The first-stage likelihood is:

$$\mathcal{L}^{M_2}_{N_T-N_W}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}. \tag{62}$$

To show that the waitlist models' dynamics work as intended, I prove the following theorem:

Theorem 6. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. See Appendix H.

F.2 Asymmetric Waitlist Results

Unlike the two-round models, I do not provide estimation results for the waitlist models on actual data. The reason is that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is only identified by variation across years, given that $\overline{m_{2A}}$ is constant within years. I have only five years of data, and the Covid shock, which may provide variation in $\overline{m_2}$ across years, is not part of the sample. As such, I have insufficient statistical power to estimate the F$\overline{m_{2A}}$ block of the likelihood, at least not until a potential future date when applicants from multiple application years after 2019 have completed their PhDs. Therefore, I stick to estimating the waitlist models on the same optimal and suboptimal simulated datasets that I use to demonstrate that the deviation gains test for the two-round models rejects optimality only when it is supposed to. In the simulated data, there are only two years, but the data are generated such that the relationship between $\overline{m_{2A}}$ and $\ddot{m}_2$ is identified.[23] The purpose of this simulated data estimation is to demonstrate that there exists a plausible data-generating process in which the waitlist models' parameters are identified.

The unrestricted waitlist-dynamic and hybrid estimates are in Tables 34 and 36. As expected, the $\ddot{m}_2$ parameter in the F$\overline{m_{2A}}$ block is positive and significant at the 1% level in both datasets. In addition, the $\ddot{m}_2$ parameter in the *Accept 2* block is negative in both datasets, which is also as expected. The reason is that the committee is more likely to give an early offer to an applicant with a high m_2 and therefore a low $\ddot{m}_2$. Finally, the $\overline{m_{2A}}$ parameter in the *Accept 3* block is negative and significant at the 1% level because a high $\overline{m_{2A}}$ means fewer spots will be open in Round 3 due to more Round 2 matriculants, and so it will be harder to get accepted off the waitlist.

[23] I set the weight that the committee puts on matriculation in Round 2 to be the same in both years, which keeps $\overline{m_{2A}}$ and $\ddot{m}_2$ strongly correlated. In the actual data, there is some variation in that weight across years, which along with only five years of data makes the correlation between $\overline{m_{2A}}$ and $\ddot{m}_2$ insignificantly different from zero.

Tables 35 and 37 contain restricted estimates for the waitlist-dynamic and hybrid models. In the waitlist-hybrid model, as with the two-round static model, the Round 1 cutoff probability is higher than the Round 3 cutoff probability in the year that is not *Year Transition/Matriculate More*. As such, the marginal applicant rejected in Round 1's $P(y=S|x,t,\bar{k})$ is greater than than the marginal applicant accepted into the program's $P(y=S|x,z,t,\bar{k})$, which may be undesirable for the committee. Moreover, in *Year Transition/Matriculate More*, the Round 3 cutoff probability is higher than the Round 1 cutoff probability not only because there are more Round 1 survivors, but also because those applicants have higher matriculation probabilities on average.

Finally, I run the deviation gains test for the waitlist models. For the waitlist-dynamic model, I rank the applicants in each year by $\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log\Big(e^{P(y=S|x,\tilde{z},t,\bar{k})/\sigma} + e^{\beta\mathbb{E}[v_3(x,\tilde{z},t,\bar{k},\widetilde{\overline{m_{2A}}},\tilde{\varepsilon}_3)|x,\tilde{z},t,\bar{k},\ddot{m}_2(x,\tilde{z},t)]/\sigma}\Big) dF_{z|x,t}(\tilde{z}|x,t)$ and transition the number of transitioned applicants in the data. Next, I rank the surviving applicants by the payoff difference $P(y=S|x,z,t,\bar{k}) - \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2] = P(y=S|x,z,t,\bar{k}) - \beta\int_{\widetilde{\overline{m_{2A}}}} \sigma\log\Big(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_3^*=S|\widetilde{\overline{m_{2A}}})/\sigma}\Big) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m}}$ accept the number of early acceptances in the data, and waitlist the rest. Each early acceptance matriculates with early matriculation probability $P(m_2=M|x,z,t)$. After that, I rank the waitlisted applicants by $P(y=S|x,z,t,\bar{k})$ and accept them in descending order, each matriculating with late matriculation probability $P(m_3=M|x,z,t)$, until the number of early plus late matriculants equals that in the data. Lastly, I compute the average success probability of the matriculation set and compare it to the committee's set. For the waitlist-hybrid model, I instead rank the applicants in Round 1 by $P(y=S|x,t,\bar{k})$. The results are in Figure 10. Given the difficulty of creating a dataset that does not reject optimality at the 5% level for both the two-round and waitlist models, the optimal dataset's p-values are 0.0180 and 0.0200 for the waitlist-dynamic and hybrid models, respectively, albeit with a deviation gain of just 1 percentage point. Meanwhile, the suboptimal dataset's p-values are less than 0.0002, and its deviation gains are 5 percentage points.

F.3 Symmetric Waitlist Models

In Appendix F.1, I present waitlist models in which success probabilities equal, the committee is more likely to give Round 2 acceptances to likely matriculants. Intuitively, this primary channel holds because likely matriculants decrease the expected number of spots to fill in Round 3, letting the committee be more selective. But there is also a secondary channel: if the committee accepts likely matriculants in Round 2, then the unlikely matriculants it waitlists end up in the Round 3 pool. As such, while there are fewer expected spots to fill, it is easier to fill those spots. For tractability, I do not estimate this secondary channel in the main specification. But tractability aside, there are two economic reasons why the primary channel is dominant. First, matriculation probabilities decline between Rounds 2 and 3, and for likely matriculants, they have more room to fall. Second, matriculation probabilities can change unpredictably between those rounds.

Still, I present symmetric waitlist models that account for both the primary and secondary

Table 34: Simulated Unrestricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0249	-11.5885***	-2.2558***	-11.9709***	-1.6176***	-13.2754***	-1.6998***	-8.1374***	-1.1678***	-9.4807***	-1.3248***	-5.9143	-0.6836
	(0.0516)	(1.0232)	(0.1454)	(2.5515)	(0.3154)	(2.4119)	(0.2845)	(1.1381)	(0.1344)	(1.3435)	(0.1562)	(6.4966)	(0.7703)
GRE Quantitative Percentile		0.1236***	0.0241***	0.1097***	0.0148***	0.1387***	0.0178***	0.1079***	0.0155***	0.1150***	0.0161***	0.0842	0.0097
		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0265)	(0.0032)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0645)	(0.0075)
Gender		0.2266	0.0441	0.0234	0.0032	0.3040	0.0389	0.1399	0.0201	0.2078	0.0290	0.4647	0.0537
		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3288)	(0.0421)	(0.1739)	(0.0249)	(0.1766)	(0.0246)	(0.9007)	(0.1052)
Demographic 1		-0.3014	-0.0587	0.5534	0.0748	-0.8832*	-0.1131*	-0.2533	-0.0364	-0.3380	-0.0472	-0.2627	-0.0304
		(0.1988)	(0.0386)	(0.4942)	(0.0665)	(0.4623)	(0.0578)	(0.2359)	(0.0338)	(0.2456)	(0.0342)	(0.7742)	(0.0879)
Demographic 2		0.1538	0.0299	0.5137	0.0694	-0.0523	-0.0067	-0.1179	-0.0169	-0.1354	-0.0189	-0.3955	-0.0457
		(0.2215)	(0.0431)	(0.5205)	(0.0700)	(0.4528)	(0.0580)	(0.2581)	(0.0371)	(0.2652)	(0.0371)	(1.2095)	(0.1391)
Advanced Math				0.5067*	0.0685*	0.8622***	0.1104***			0.6827***	0.0954***	1.1444	0.1323
				(0.2907)	(0.0390)	(0.3306)	(0.0421)			(0.1770)	(0.0237)	(0.9852)	(0.1080)
Committee Score				0.1061	0.0143	0.0378	0.0048			0.0467	0.0065	-0.9116**	-0.1054***
				(0.0689)	(0.0093)	(0.0672)	(0.0086)			(0.0672)	(0.0094)	(0.3783)	(0.0317)
Missing Committee Score										0.2177	0.0304		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		0.9840***	0.1915***					-0.0587	-0.0084	-0.0466	-0.0065	3.3941***	0.3923***
		(0.1472)	(0.0263)					(0.1690)	(0.0242)	(0.1740)	(0.0243)	(0.7689)	(0.0698)
Program Rank								-0.0310***	-0.0045***	-0.0309***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	1.0896***			-1.5992**	-0.2161**								
	(0.0851)			(0.7875)	(0.1049)								
$\overline{m_{2A}}$						-2.0178***	-0.2584***						
						(0.7700)	(0.0964)						
$\sqrt{\text{MSE}}$	0.0015***												
	(0.0005)												
Obs | Param | LL | AIC	1000	42	191.4	-298.8									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0496*	-11.2295***	-2.1801***	-6.7113***	-1.0637***	-7.1639***	-0.8002***	-8.2825***	-1.1913***	-9.7587***	-1.3686***	-1.5054	-0.2179
	(0.0259)	(1.0144)	(0.1474)	(2.0007)	(0.3043)	(2.5070)	(0.2796)	(1.1392)	(0.1341)	(1.3345)	(0.1538)	(3.8095)	(0.5534)
GRE Quantitative Percentile		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1097***	0.0158***	0.1175***	0.0165***	0.0362	0.0052
		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0435)	(0.0063)
Gender		0.2575*	0.0500*	-0.2767	-0.0439	0.4102	0.0458	0.1355	0.0195	0.1984	0.0278	-0.3519	-0.0509
		(0.1469)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.1741)	(0.0250)	(0.1766)	(0.0247)	(0.5435)	(0.0788)
Demographic 1		-0.8293***	-0.1610***	0.2577	0.0408	-1.1548**	-0.1290**	-0.2776	-0.0399	-0.3897	-0.0547	0.3291	0.0476
		(0.1966)	(0.0371)	(0.3774)	(0.0594)	(0.5110)	(0.0563)	(0.2344)	(0.0337)	(0.2462)	(0.0343)	(0.7092)	(0.1031)
Demographic 2		0.0797	0.0155	0.5369	0.0851	-0.3696	-0.0413	-0.1809	-0.0260	-0.2068	-0.0290	-0.7626	-0.1104
		(0.2135)	(0.0414)	(0.3886)	(0.0605)	(0.4652)	(0.0521)	(0.2566)	(0.0369)	(0.2636)	(0.0370)	(0.7932)	(0.1090)
Advanced Math				0.2363	0.0375	0.4866	0.0544			0.6387***	0.0896***	0.3977	0.0576
				(0.2639)	(0.0417)	(0.3566)	(0.0399)			(0.1763)	(0.0238)	(0.6444)	(0.0922)
Committee Score				0.3737***	0.0592***	0.0653	0.0073			0.0626	0.0088	-0.6714***	-0.0972***
				(0.0674)	(0.0096)	(0.0837)	(0.0093)			(0.0657)	(0.0092)	(0.1978)	(0.0217)
Missing Committee Score										0.2535	0.0356		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		1.1108***	0.2156***					-0.0767	-0.0110	-0.0473	-0.0066	2.9420***	0.4259***
		(0.1479)	(0.0257)					(0.1688)	(0.0243)	(0.1757)	(0.0247)	(0.6724)	(0.0579)
Program Rank								-0.0304***	-0.0044***	-0.0303***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	0.8882***			-0.5496	-0.0871								
	(0.0488)			(0.5706)	(0.0900)								
$\overline{m_{2A}}$						-2.9219***	-0.3264***						
						(0.9936)	(0.1102)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	374.8	-665.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 35: Simulated Restricted Waitlist-Dynamic Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0249	0.6925***	66.65%	0.4942	70.64%	-8.0628***	-9.5820***	-5.9143		
	(0.0516)	(0.1818)	(58.33%, 74.05%)	(0.3513)	(57.55%, 81.02%)	(1.1622)	(1.3341)	(6.4966)		
GRE Quantitative Percentile						0.1062***	0.1154***	0.0842		
						(0.0134)	(0.0146)	(0.0645)		
Gender						0.1999**	0.1696	0.4647		
						(0.1018)	(0.1358)	(0.9007)		
Demographic 1						-0.2482*	-0.3098	-0.2627		
						(0.1353)	(0.1912)	(0.7742)		
Demographic 2						0.0407	-0.0318	-0.3955		
						(0.1511)	(0.1962)	(1.2095)		
Advanced Math							0.6616***	1.1444		
							(0.1424)	(0.9852)		
Committee Score							0.0560	-0.9117**		
							(0.0425)	(0.3783)		
Missing Committee Score							0.2167			
							(0.1838)			
Year Transition/Matriculate More		-0.8491***	46.09%		88.08%	-0.0604	-0.0289	3.3941***		
		(0.2179)	(39.64%, 52.68%)		(65.79%, 96.60%)	(0.1667)	(0.1727)	(0.7689)		
Program Rank						-0.0310***	-0.0310***			
						(0.0019)	(0.0019)			
$\tilde{m}_2$	1.0896***									
	(0.0851)									
$\overline{m_{2A}}$				2.3665*						
				(1.2823)						
$\sqrt{\text{MSE}}$	0.0015***									
	(0.0005)									
Beta									0.9490***	
									(0.0391)	
Sigma										0.1926***
										(0.0246)
Obs | Param | LL | AIC	1000	26	186.8	-321.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0496*	0.8342***	69.72%	0.8216	80.71%	-8.6331***	-8.3628***	-1.5054		
	(0.0259)	(0.2171)	(60.07%, 77.90%)	(0.7603)	(61.39%, 91.67%)	(1.1831)	(1.4735)	(3.8095)		
GRE Quantitative Percentile						0.1147***	0.0947***	0.0362		
						(0.0137)	(0.0156)	(0.0435)		
Gender						0.2301**	0.1269	-0.3519		
						(0.1099)	(0.1412)	(0.5435)		
Demographic 1						-0.5688***	-0.3599*	0.3291		
						(0.1473)	(0.2046)	(0.7092)		
Demographic 2						-0.0225	-0.0641	-0.7626		
						(0.1609)	(0.2095)	(0.7932)		
Advanced Math							0.5402***	0.3977		
							(0.1471)	(0.6444)		
Committee Score							0.1927***	-0.6714***		
							(0.0410)	(0.1978)		
Missing Committee Score							0.1774			
							(0.1951)			
Year Transition/Matriculate More		-1.0924***	43.58%		96.46%	-0.0704	-0.0765	2.9420***		
		(0.2510)	(37.09%, 50.29%)		(13.07%, 99.98%)	(0.1661)	(0.1767)	(0.6724)		
Program Rank						-0.0307***	-0.0305***			
						(0.0019)	(0.0019)			
$\tilde{m}_2$	0.8882***									
	(0.0488)									
$\overline{m_{2A}}$				4.3874						
				(5.6885)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8225***	
									(0.0434)	
Sigma										0.2234***
										(0.0318)
Obs | Param | LL | AIC	1000	26	351.0	-650.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Table 36: Simulated Unrestricted Waitlist-Hybrid Estimates

(a) Optimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0249	-11.5885***	-2.2558***	-11.9709***	-1.6176***	-13.2754***	-1.6998***	-8.1374***	-1.1678***	-9.4807***	-1.3248***	-5.9143	-0.6836
	(0.0516)	(1.0232)	(0.1454)	(2.5515)	(0.3154)	(2.4119)	(0.2845)	(1.1381)	(0.1344)	(1.3435)	(0.1562)	(6.4966)	(0.7703)
GRE Quantitative Percentile		0.1236***	0.0241***	0.1097***	0.0148***	0.1387***	0.0178***	0.1079***	0.0155***	0.1150***	0.0161***	0.0842	0.0097
		(0.0118)	(0.0018)	(0.0271)	(0.0034)	(0.0265)	(0.0032)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0645)	(0.0075)
Gender		0.2266	0.0441	0.0234	0.0032	0.3040	0.0389	0.1399	0.0201	0.2078	0.0290	0.4647	0.0537
		(0.1464)	(0.0283)	(0.2899)	(0.0392)	(0.3288)	(0.0421)	(0.1739)	(0.0249)	(0.1766)	(0.0246)	(0.9007)	(0.1052)
Demographic 1		-0.3014	-0.0587	0.5534	0.0748	-0.8832*	-0.1131*	-0.2533	-0.0364	-0.3380	-0.0472	-0.2627	-0.0304
		(0.1988)	(0.0386)	(0.4942)	(0.0665)	(0.4623)	(0.0578)	(0.2359)	(0.0338)	(0.2456)	(0.0342)	(0.7742)	(0.0879)
Demographic 2		0.1538	0.0299	0.5137	0.0694	-0.0523	-0.0067	-0.1179	-0.0169	-0.1354	-0.0189	-0.3955	-0.0457
		(0.2215)	(0.0431)	(0.5205)	(0.0700)	(0.4528)	(0.0580)	(0.2581)	(0.0371)	(0.2652)	(0.0371)	(1.2095)	(0.1391)
Advanced Math				0.5067*	0.0685*	0.8622***	0.1104***			0.6827***	0.0954***	1.1444	0.1323
				(0.2907)	(0.0390)	(0.3306)	(0.0421)			(0.1770)	(0.0237)	(0.9852)	(0.1080)
Committee Score				0.1061	0.0143	0.0378	0.0048			0.0467	0.0065	-0.9116**	-0.1054***
				(0.0689)	(0.0093)	(0.0672)	(0.0086)			(0.0672)	(0.0094)	(0.3783)	(0.0317)
Missing Committee Score										0.2177	0.0304		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		0.9840***	0.1915***					-0.0587	-0.0084	-0.0466	-0.0065	3.3941***	0.3923***
		(0.1472)	(0.0263)					(0.1690)	(0.0242)	(0.1740)	(0.0243)	(0.7689)	(0.0698)
Program Rank								-0.0310***	-0.0045***	-0.0309***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	1.0896***			-1.5992**	-0.2161**								
	(0.0851)			(0.7875)	(0.1049)								
$\overline{m_{2A}}$						-2.0178***	-0.2584***						
						(0.7700)	(0.0964)						
$\sqrt{\text{MSE}}$	0.0015***												
	(0.0005)												
Obs | Param | LL | AIC	1000	42	191.4	-298.8									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.0496*	-11.2295***	-2.1801***	-6.7113***	-1.0637***	-7.1639***	-0.8002***	-8.2825***	-1.1913***	-9.7587***	-1.3686***	-1.5054	-0.2179
	(0.0259)	(1.0144)	(0.1474)	(2.0007)	(0.3043)	(2.5070)	(0.2796)	(1.1392)	(0.1341)	(1.3345)	(0.1538)	(3.8095)	(0.5534)
GRE Quantitative Percentile		0.1226***	0.0238***	0.0387*	0.0061*	0.0717***	0.0080**	0.1097***	0.0158***	0.1175***	0.0165***	0.0362	0.0052
		(0.0117)	(0.0018)	(0.0215)	(0.0034)	(0.0278)	(0.0031)	(0.0133)	(0.0015)	(0.0145)	(0.0016)	(0.0435)	(0.0063)
Gender		0.2575*	0.0500*	-0.2767	-0.0439	0.4102	0.0458	0.1355	0.0195	0.1984	0.0278	-0.3519	-0.0509
		(0.1469)	(0.0283)	(0.2746)	(0.0431)	(0.3417)	(0.0380)	(0.1741)	(0.0250)	(0.1766)	(0.0247)	(0.5435)	(0.0788)
Demographic 1		-0.8293***	-0.1610***	0.2577	0.0408	-1.1548**	-0.1290**	-0.2776	-0.0399	-0.3897	-0.0547	0.3291	0.0476
		(0.1966)	(0.0371)	(0.3774)	(0.0594)	(0.5110)	(0.0563)	(0.2344)	(0.0337)	(0.2462)	(0.0343)	(0.7092)	(0.1031)
Demographic 2		0.0797	0.0155	0.5369	0.0851	-0.3696	-0.0413	-0.1809	-0.0260	-0.2068	-0.0290	-0.7626	-0.1104
		(0.2135)	(0.0414)	(0.3886)	(0.0605)	(0.4652)	(0.0521)	(0.2566)	(0.0369)	(0.2636)	(0.0370)	(0.7932)	(0.1090)
Advanced Math				0.2363	0.0375	0.4866	0.0544			0.6387***	0.0896***	0.3977	0.0576
				(0.2639)	(0.0417)	(0.3566)	(0.0399)			(0.1763)	(0.0238)	(0.6444)	(0.0922)
Committee Score				0.3737***	0.0592***	0.0653	0.0073			0.0626	0.0088	-0.6714***	-0.0972***
				(0.0674)	(0.0096)	(0.0837)	(0.0093)			(0.0657)	(0.0092)	(0.1978)	(0.0217)
Missing Committee Score										0.2535	0.0356		
										(0.1896)	(0.0264)		
Year Transition/Matriculate More		1.1108***	0.2156***					-0.0767	-0.0110	-0.0473	-0.0066	2.9420***	0.4259***
		(0.1479)	(0.0257)					(0.1688)	(0.0243)	(0.1757)	(0.0247)	(0.6724)	(0.0579)
Program Rank								-0.0304***	-0.0044***	-0.0303***	-0.0043***		
								(0.0019)	(0.0002)	(0.0019)	(0.0002)		
$\dot{m}_2$	0.8882***			-0.5496	-0.0871								
	(0.0488)			(0.5706)	(0.0900)								
$\overline{m_{2A}}$						-2.9219***	-0.3264***						
						(0.9936)	(0.1102)						
$\sqrt{\text{MSE}}$	0.0010***												
	(0.0002)												
Obs | Param | LL | AIC	1000	42	374.8	-665.5									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 37: Simulated Restricted Waitlist-Hybrid Estimates

(a) Optimal Committee

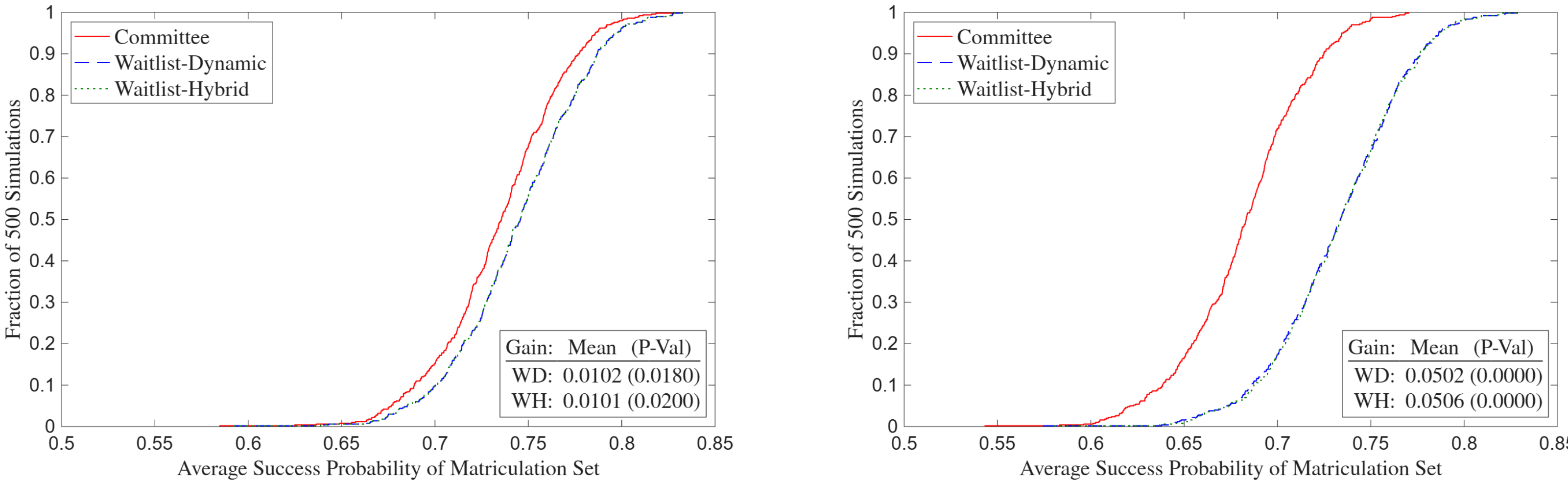
	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0249	0.6925***	66.65%	0.4942	70.64%	-8.0628***	-9.5820***	-5.9143		
	(0.0516)	(0.1818)	(58.33%, 74.05%)	(0.3513)	(57.55%, 81.02%)	(1.1622)	(1.3341)	(6.4966)		
GRE Quantitative Percentile						0.1062***	0.1154***	0.0842		
						(0.0134)	(0.0146)	(0.0645)		
Gender						0.1999**	0.1696	0.4647		
						(0.1018)	(0.1358)	(0.9007)		
Demographic 1						-0.2482*	-0.3098	-0.2627		
						(0.1353)	(0.1912)	(0.7742)		
Demographic 2						0.0407	-0.0318	-0.3955		
						(0.1511)	(0.1962)	(1.2095)		
Advanced Math							0.6616***	1.1444		
							(0.1424)	(0.9852)		
Committee Score							0.0560	-0.9117**		
							(0.0425)	(0.3783)		
Missing Committee Score							0.2167			
							(0.1838)			
Year Transition/Matriculate More		-0.8491***	46.09%		88.08%	-0.0604	-0.0289	3.3941***		
		(0.2179)	(39.64%, 52.68%)		(65.79%, 96.60%)	(0.1667)	(0.1727)	(0.7689)		
Program Rank						-0.0310***	-0.0310***			
						(0.0019)	(0.0019)			
$\hat{m}_2$	1.0896***									
	(0.0851)									
$\overline{m_{2A}}$				2.3665*						
				(1.2823)						
$\sqrt{\text{MSE}}$	0.0015***									
	(0.0005)									
Beta									0.9490***	
									(0.0391)	
Sigma										0.1926***
										(0.0246)
Obs | Param | LL | AIC	1000	26	186.8	-321.6						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

(b) Suboptimal Committee

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0496*	0.8342***	69.72%	0.8216	80.71%	-8.6331***	-8.3628***	-1.5054		
	(0.0259)	(0.2171)	(60.07%, 77.90%)	(0.7603)	(61.39%, 91.67%)	(1.1831)	(1.4735)	(3.8095)		
GRE Quantitative Percentile						0.1147***	0.0947***	0.0362		
						(0.0137)	(0.0156)	(0.0435)		
Gender						0.2301**	0.1269	-0.3519		
						(0.1099)	(0.1412)	(0.5435)		
Demographic 1						-0.5688***	-0.3599*	0.3291		
						(0.1473)	(0.2046)	(0.7092)		
Demographic 2						-0.0225	-0.0641	-0.7626		
						(0.1609)	(0.2095)	(0.7932)		
Advanced Math							0.5402***	0.3977		
							(0.1471)	(0.6444)		
Committee Score							0.1927***	-0.6714***		
							(0.0410)	(0.1978)		
Missing Committee Score							0.1774			
							(0.1951)			
Year Transition/Matriculate More		-1.0924***	43.58%		96.46%	-0.0704	-0.0765	2.9420***		
		(0.2510)	(37.09%, 50.29%)		(13.07%, 99.98%)	(0.1661)	(0.1767)	(0.6724)		
Program Rank						-0.0307***	-0.0305***			
						(0.0019)	(0.0019)			
$\hat{m}_2$	0.8882***									
	(0.0488)									
$\overline{m_{2A}}$				4.3874						
				(5.6885)						
$\sqrt{\text{MSE}}$	0.0010***									
	(0.0002)									
Beta									0.8225***	
									(0.0434)	
Sigma										0.2234***
										(0.0318)
Obs | Param | LL | AIC	1000	26	351.0	-650.0						

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.



(a) Optimal Committee

(b) Suboptimal Committee

Figure 10: Waitlist Deviation Gains

channels. There is are two additional states: $\overline{m_3}$, which is a given year's average waitlisted applicant matriculation probability, and m_3, which is the *ex-ante* probability that an applicant will matriculate if waitlisted. By *ex-ante*, I mean that m_3 depends only on information in the application files and not events that happen between Rounds 2 and 3. Therefore, the committee knows m_3 in Round 2. As with $\overline{m_{2A}}$, the Round 3 cutoff probability is increasing in $\overline{m_3}$. In addition, $\overline{m_3}$ is increasing in m_3, analogously to how $\overline{m_{2A}}$ is increasing in $\ddot{m}_2$. Whether the committee will be more or less likely to accept likely matriculants in Round 2 is theoretically ambiguous and depends on the parameter values. The equations are as follows, starting with Round 3:

$$v_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3) = \max_{d_3\in\{A,R\}} u_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},d_3) + \varepsilon_3(d_3), \text{ such that:} \tag{63}$$

$$u_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},d_3) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_3=A) + P(y_3^*=S|\overline{m_{2A}},\overline{m_3})\mathbb{1}(d_3=R).$$

The Round 2 value function is:

$$v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2) = \max_{d_2\in\{A,W\}} u_2(x,z,t,\bar{k},\ddot{m}_2,m_3,d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{64}$$

$$u_2(x,z,\ddot{m}_2,m_3,d_2) = P(y=S|x,z,t,\bar{k})\mathbb{1}(d_2=A) + \beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\overline{m_3},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2,m_3]\mathbb{1}(d_2=W), \text{ such that:}$$

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m_{2A}},\widetilde{\overline{m_3}},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2,m_3] = \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma\log\left(\sum_{\tilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\bar{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\tilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2).$$

Finally, the Round 1 value function is:

$$v_1(x,t,\bar{k}) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{k},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{65}$$

$$u_1(x,t,\bar{k},d_1) = \mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t,\bar{k}]\mathbb{1}(d_1=T) + L(y_1^*=S|t)\mathbb{1}(d_1=R), \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,t,\bar{k},\ddot{m}_2,m_3,\varepsilon_2)|x,t\bar{k}] = \int_{\tilde{z}} \sigma \log\left(\sum_{\tilde{d}_2\in\{A,R\}} e^{u_2(x,\tilde{z},t,\bar{k},\ddot{m}_2(x,\tilde{z},t),m_3(x,\tilde{z},t),\tilde{d}_2)/\sigma}\right) dF_{z|x,t}(\tilde{z}|x,t).$$

In the symmetric waitlist-hybrid model, the Round 1 utility function is instead:

$$u_1(x,t,\bar{k},d_1) = P(y=S|x,t,\bar{k})\mathbb{1}(d_1=T) + P(y_1^*=S|t)\mathbb{1}(d_1=R). \tag{66}$$

Expressed as simply as possible, the restricted CCPs are:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\tilde{d}_r\in D_r}\exp(u_r(\text{states},\tilde{d}_r)/\sigma)}. \tag{67}$$

The unrestricted CCPs, success probabilities, and cutoff level/probability are $P_{d_1} = P\big(d_1 = T\,\big|x,t;\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,\ddot{m}_2,m_3;\theta_{A_2}\big)$, $P_{d_3} = P\big(d_3 = A\big|x,z,\overline{m_{2A}},\overline{m_3};\theta_{A_3}\big)$, $P_y = P\big(y = S\big|x,z,t,\bar{k};\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $P_{y_3^*} = P\big(y_3^* = S\big|\overline{m_{2A}},\overline{m_3};\theta_{S_3^*}\big)$. They have the same specifications as the probabilities and levels earlier in the paper; see Equations (11)–(13) in Section 4.1.2.

Let applicants 1 to N_W be waitlisted, N_W+1 to N_T be accepted in Round 2, and N_T+1 to N be rejected in Round 1. Then the unrestricted symmetric waitlist-dynamic likelihood is:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big) f\big(\overline{m_{2A,n}}|\ddot{m}_{2,n};\theta_{F_{m_2}}\big) f\big(\overline{m_{3,n}}|m_{3,n};\theta_{F_{m_3}}\big)$$

$$P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)} \prod_{n=1}^{N_W} P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}$$

$$\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \tag{68}$$

In addition, m_2 and m_3 are parameterized as follows: $m_2 = P_{m_2} = P\big(m_2 = M\big|x,z,t;\theta_{M_2}\big)$ and $m_3 = P_{m_3} = P\big(m_3 = M\big|x,z,t;\theta_{M_3}\big)$. To estimate θ_{M_2} and θ_{M_3}, I would use the first-stage likelihoods: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$ and $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A,m_{3,n}=M)}(1-P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A,m_{3,n}=D)}$. The samples for estimating m_2 and m_3 are applicants accepted in Rounds 2 and 3 respectively. Since m_2 and m_3 are generated regressors, the standard errors are those from Murphy and Topel (1985) and Hardin (2002).

Finally, in Appendix G, I present two different infinite-horizon multiyear models. A limitation of the waitlist models is that while they give early acceptances to applicants with higher matriculation probabilities all else equal, matriculation probabilities do not affect the probability of a late acceptance. For this reason, the first multiyear model is constructed so that an applicant's overall acceptance probability is increasing in their matriculation probability, the logic being that more matriculants this year will give the committee the option to be more selective next year. Meanwhile, the second multiyear model rewards more diverse applicants in a scenario where diversity increases an applicant's success probability. For example, by accepting more theory applicants this year, the committee may have to worry less next year about having too many applied students relative to applied faculty advisors. But this is just one example of such a scenario.

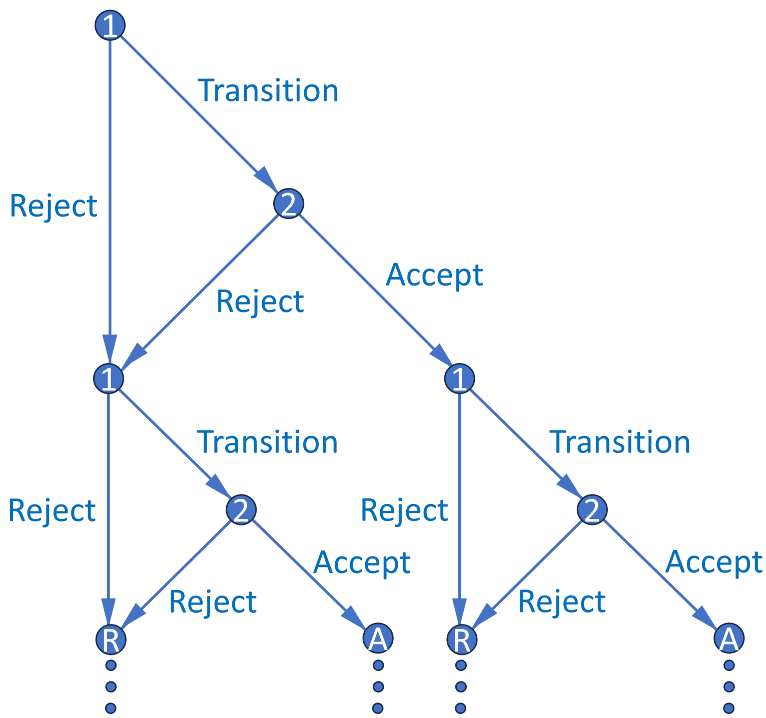


Figure 11: Multiyear Decision Graph

G Multiyear Models

There are two different multiyear models. In the first model, the memory variable is the size of the previous year's cohort. The committee accepts applicants more likely to matriculate because it wants to ensure that next year's memory variable, which is this year's cohort size, is sufficiently large that it can avoid accepting too many relatively weak applicants next year. In the second model, the memory is a diversity variable that increases an applicant's success probability; for example, the previous year's cohort's main area of interest. Applicants with a diverse area of interest may be more likely to succeed because their advisor, with fewer other students to advise, can give them more individualized attention. The committee will therefore accept applicants with a less common area of interest to help balance out the program for next year's class.

G.1 Matriculation Model

In the waitlist models in Appendix F, the committee benefits from giving early acceptances to applicants who are more likely to matriculate. Doing so helps prevent the program from losing out on good applicants who would likely have matriculated if given an early offer but matriculate elsewhere due to events that occurred while on the waitlist. However, the waitlist models do not imply that the committee should also give more late acceptances to likely matriculants. To model that phenomenon, a multiyear model is necessary, as there must be a period after the final round in any given year for dynamics to matter in that round. Figure 11 is a decision graph of the simplest possible multiyear model, in which each year has only two rounds, as in the dynamic model. This model is infinite horizon, and future years are discounted with factor $\beta \in (0,1)$. To avoid the considerations of two-stage estimation, I suppose for simplicity that matriculation probabilities m come from the committee's assessment of each applicant in Round 2. I also suppose that success probabilities depend only on applicant characteristics and not program rank.

In each year's Round 2 value function, I specify the cutoff probability as $P(y_2^* = S|c_{-1})$ instead of $P(y_2^* = S|t)$, where c_{-1} is the previous year's cohort size. The cutoff probability increases in c_{-1} because the committee must keep the total number of students in the program across years relatively constant. So if c_{-1} is large, the committee must now accept fewer students.

Given that, what information available at the start of Round 2 determines c? If and only if the applicant is accepted, their own matriculation probability m does. Now suppose the committee maximizes discounted "lifetime" cohort success by solving an infinite-horizon dynamic programming problem across years. As in the waitlist models, the optimal dynamic decision may not match the optimal myopic one. Instead, the committee may accept an applicant with a high m even if their expected success probability is less than what the Round 2 cutoff level (which need not be between 0 and 1) would otherwise be. The reason is that accepting a high m applicant increases c's distribution, which increases next year's Round 2 cutoff and value.

Intuitively, even if the committee can accept more applicants next year to make up for a smaller cohort this year, they will do so out of necessity, for example, to avoid future TA shortages. This situation will be undesirable provided next year's marginal applicant is of lower quality than this year's above-marginal applicant. Also, as in the waitlist models, even if the data show that the committee does not think dynamically, if c_{-1} affects the Round 2 cutoff, perhaps it should. Finally, estimating this model is more computationally challenging since it does not have a closed-form solution. Nested fixed point algorithms suffer from the curse of dimensionality, and the number of state variables in application files, in particular those from textual documents, is large. But principal component analysis can reduce the size of the state space.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,m,c_{-1},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,m,c_{-1},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{69}$$

$$\begin{aligned} u_2(x,z,t,m,c_{-1},d_2) = &\{P(y=S|x,z,t) + \beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|m]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|c_{-1}) + \beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log\left(\sum_{d_1{}' \in \{T,R\}} e^{u_1(x',t',c,d_1{}')/\sigma}\right) dF_x(x')dF_t(t')dF_{c|(m)}(c|(m))$$

$$= \int_c \int_{t_1{}'} \cdots \int_{t_{J_t}{}'} \int_{x_1{}'} \cdots \int_{x_{J_x}{}'} \Big[\cdot\Big] dF_{x,1}(x_1{}'|x_2{}',\ldots,x_{J_x}{}') \cdots dF_{x,J_x}(x_{J_x}{}') dF_{t,1}(t_1{}'|t_2{}',\ldots,t_{J_t}{}') \cdots dF_{t,J_t}(t_{J_t}{}') dF_{c|(m)}(c|(m)).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,m,c_{-1}) = \frac{\exp(u_2(x,z,t,m,c_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2}\in\{A,R\}}\exp(u_2(x,z,t,m,c_{-1},\widetilde{d_2})/\sigma)}. \tag{70}$$

Each year's Round 1 value function is:

$$v_1(x,t,c_{-1},\varepsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,c_{-1},d_1)+\varepsilon_1(d_1), \text{ such that:} \tag{71}$$

$$u_1(x,t,c_{-1},d_1) = \mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}]\mathbb{1}(d_1=T)+\{L(y_1^*=S|t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1{}')]\}\mathbb{1}(d_1=R), \text{ such that:}$$

$$\begin{aligned}\mathbb{E}[v_2(x,z,t,m,c_{-1},\varepsilon_2)|x,t,c_{-1}] &= \int_{\widetilde{m}}\int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d_2}\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d_2})/\sigma}\right)dF_{z|x,t}(\widetilde{z}|x,t)dF_{m|x,t}(\widetilde{m}|x,t)\\ &= \int_{\widetilde{m}}\int_{\widetilde{z_1}}\cdots\int_{\widetilde{z_{J_z}}}\left[\,\cdot\,\right]dF_{z|x,t,1}(\widetilde{z_1}|\widetilde{z_2},\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)dF_{m|x,t}(\widetilde{m}|x,t).\end{aligned}$$

And as such, Round 1's conditional choice probabilities (CCPs) are:

$$P(d_1|x,t,c_{-1}) = \frac{\exp(u_1(x,t,c_{-1},d_1)/\sigma)}{\sum_{\widetilde{d_1}\in\{T,R\}}\exp(u_1(x,t,c_{-1},\widetilde{d_1})/\sigma)}. \tag{72}$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1 = T\,\big|\,x,t,c_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,m,c_{-1};\theta_A\big)$, $P_y = P\big(y=S\big|x,z,t;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^*=S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^* = S\big|c_{-1};\theta_{S_2^*}\big)$, which aside from the cutoffs have the usual logistic specification. In this model, the cutoffs are not in a function that restricts their range. Given that, the unrestricted likelihood is:

$$\begin{aligned}\mathcal{L}_N^U(\theta) = &\prod_{n=1}^{N_A} f\big(c_n|m_n;\theta_{F_c}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(m_n|x_n,t_n;\theta_{F_m}\big)P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}\\ &\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}.\end{aligned} \tag{73}$$

As with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 7. *Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. See Appendix H.

G.2 Diversity Model

I remove the state variable c_{-1} and add in its place two new states a and $\overline{a}_{-1}$. Let $a = \mathbb{1}(\text{Diverse})$, where a diverse characteristic is one whose probability of appearing among next year's applicants is bounded above by some $p \in (0,1)$. For example, I could have $a = \mathbb{1}$(TheoryField), where in most years, theory applicants are less common than applied applicants. Also, let $\overline{a}_{-1}$ equal the percentage of last year's acceptance set that is theory. If $|a - \overline{a}_{-1}|$ shrinks, then the applicant's expected success probability decreases, as their future advisor may have less time for them.[24]

As such, the optimal dynamic decision may not match the optimal myopic one. In particular, the committee may accept a diverse applicant it would not have otherwise because doing so increases the success probability of next year's applicant pool. Accepting a diverse applicant increases the distribution of $\overline{a}$ conditional on a, which because diverse applicants are less common increases the distribution of $|a' - \overline{a}|$, which increases next year's expected success probability, which increases next year's expected value. That said, I can also estimate a myopic model with memory variable $\overline{a}_{-1}$. The committee will still prioritize diverse applicants because diversity increases applicants' success probabilities, but there will be no forward-looking dynamic effect.

I now present the value functions, starting with each year's Round 2 value:

$$v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x,z,t,a,\overline{a}_{-1},d_2) + \varepsilon_2(d_2), \text{ such that:} \tag{74}$$

$$\begin{aligned} u_2(x,z,t,a,\overline{a}_{-1},d_2) = &\{P(y = S|x,z,t,|a-\overline{a}_{-1}|) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|a] = \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\overline{a}},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}|a}(\widetilde{\overline{a}}|a) =$$

$$\int_{\widetilde{\overline{a}}}\int_{\widetilde{t_1'}}\cdots\int_{\widetilde{t_{J_t}'}}\int_{\widetilde{x_1'}}\cdots\int_{\widetilde{x_{J_x}'}}\left[\cdot\right] F_{x,1}({x_1}'|{x_2}',\ldots,{x_{J_x}}')\cdots dF_{x,J_x}({x_{J_x}}')dF_{t,1}({t_1}'|{t_2}',\ldots,{t_{J_t}}')\cdots dF_{t,J_t}({t_{J_t}}')dF_{\overline{a}|a}(\widetilde{\overline{a}}|a).$$

As such, Round 2's conditional choice probabilities are:

$$P(d_2|x,z,t,a,\overline{a}_{-1}) = \frac{\exp(u_2(x,z,t,a,\overline{a}_{-1},d_2)/\sigma)}{\sum_{\widetilde{d_2} \in \{A,R\}} \exp(u_2(x,z,t,a,\overline{a}_{-1},\widetilde{d_2})/\sigma)}. \tag{75}$$

Each year's Round 1 value function is:

$$v_1(x,t,\overline{a}_{-1},\varepsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,\overline{a}_{-1},d_1) + \varepsilon_1(d_1), \text{ such that:} \tag{76}$$

$$\begin{aligned} u_1(x,t,\overline{a}_{-1},d_1) = &\mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}]\mathbb{1}(d_1 = T) \\ &+ \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:} \end{aligned}$$

[24] If the program never gets too imbalanced, this relationship will not be identifiable, in which case estimation will not work for this choice of a. However, there may also be different suitable diversity variables.

$$\mathbb{E}[v_2(x,z,t,a,\overline{a}_{-1},\varepsilon_2)|x,t,\overline{a}_{-1}] = \int_{\widetilde{a}}\int_{\widetilde{z}}\sigma\log\left(\sum_{\widetilde{d}_2\in A,R} e^{u_2(x,\widetilde{z},t,\widetilde{a},\overline{a}_{-1},\widetilde{d}_2)/\sigma}\right)dF_{z|x,t}(\widetilde{z}|x,t)dF_a(\widetilde{a})$$

$$= \int_{\widetilde{a}}\int_{\widetilde{z}_1}\cdots\int_{\widetilde{z}_J}\left[\cdot\right]dF_1(\widetilde{z}_1|\widetilde{z}_2,\ldots\widetilde{z}_J,x,t)\cdots dF_J(\widetilde{z}_J|x,t)dF_a(\widetilde{a})$$

$$\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')] = \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}}\sigma\log\left(\sum_{d_1'\in\{T,R\}} e^{u_1(\widetilde{x'},\widetilde{t'},\widetilde{\overline{a}},d_1')/\sigma}\right)dF_{x'}(\widetilde{x'})dF_{t'}(\widetilde{t'})dF_{\overline{a}}(\widetilde{\overline{a}}) =$$

$$\int_{\widetilde{\overline{a}}}\int_{\widetilde{t_1'}}\cdots\int_{\widetilde{t_{J_t}'}}\int_{\widetilde{x_1'}}\cdots\int_{\widetilde{x_{J_x}'}}\left[\cdot\right]F_{x,1}({x_1}'|{x_2}',\ldots,{x_{J_x}}')\cdots dF_{x,J_x}({x_{J_x}}')dF_{t,1}({t_1}'|{t_2}',\ldots,{t_{J_t}}')\cdots dF_{t,J_t}({t_{J_t}}')dF_{\overline{a}}(\widetilde{\overline{a}}).$$

And as such, Round 1's conditional choice probabilities are:

$$P(d_1|x,t,\overline{a}_{-1}) = \frac{\exp(u_1(x,t,\overline{a}_{-1},d_1)/\sigma)}{\sum_{\widetilde{d}_1\in\{T,R\}}\exp(u_1(x,t,\overline{a}_{-1},\widetilde{d}_1)/\sigma)}. \quad (77)$$

The unrestricted CCPs, success probabilities, and cutoff levels are $P_{d_1} = P\big(d_1 = T\big|x,t,\overline{a}_{-1};\theta_T\big)$, $P_{d_2} = P\big(d_2 = A\big|x,z,t,a,\overline{a}_{-1};\theta_A\big)$, $P_y = P\big(y = S\big|x,z,t,|a-\overline{a}_{-1}|;\theta_S\big)$, $L_{y_1^*} = L\big(y_1^* = S\big|t;\theta_{S_1^*}\big)$, and $L_{y_2^*} = L\big(y_2^* = S\big|t;\theta_{S_2^*}\big)$; all but the cutoffs are logistic. Given that, the unrestricted likelihood is:

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} f\big(\overline{a}_n|a_n;\theta_{F_{\overline{a}}}\big)\prod_{n=1}^{N_T} f\big(z_n^c|z_n^{\neg c},x_n,t_n;\theta_{F_z}\big)f\big(a_n;\theta_{F_a}\big)P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}$$

$$\prod_{n=N_T+1}^{N}\big(1-P_{d_{1,n}}(\theta_T)\big)\prod_{n=1}^{N} f\big(x_n;\theta_{F_x}\big)f\big(t_n;\theta_{F_t}\big)f\big(z_n^{\neg c}|x_n,t_n;\theta_{F_z}\big)P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}. \quad (78)$$

I estimate $f_{\overline{a}|a}(\overline{a}|a)$ on the N_A accepted applicants only (those from N_A+1 to $N_{T,R}$ are rejected in Round 2) because only accepted applicants can influence the acceptance set's area of interest. Also, as with the dynamic models in the paper, I use (θ_F^U,θ_S^U) as an initial guess of (θ_F^R,θ_S^R). To show that the dynamics of this model work as intended, I prove the following theorem:

Theorem 8. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. See Appendix H.

H Proofs of Theorems

Theorem 1. *For large N, the committee's portfolio choice problem with preference shocks has a cutoff rule solution with logistic conditional choice probabilities.*

Proof. The portfolio choice problem with type 1 extreme value preference shocks is:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+\varepsilon_n(R)\mathbb{1}(d_n=R)\right], \text{ such that } \sum_{n=1}^{N}\mathbb{1}(d_n=A)\leq N_A. \quad (79)$$

Because success probabilities are never negative, the unconstrained objective increases in expectation every time $d_n = A$, and so the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(d_1,\ldots,d_N;\lambda) &= \sum_{n=1}^{N}\left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+\varepsilon_n(R)\mathbb{1}(d_n=R)\right]+\lambda\left\{N_A-\left[N-\sum_{n=1}^{N}\mathbb{1}(d_n=R)\right]\right\}\\ &= \sum_{n=1}^{N}\left[(P(y_n=S|x_n,z_n)+\varepsilon_n(A))\mathbb{1}(d_n=A)+(\lambda+\varepsilon_n(R))\mathbb{1}(d_n=R)\right]+\lambda(N_A-N). \quad (80)\end{aligned}$$

Suppose without loss of generality that the applicants are sorted in decreasing order of their shock-perturbed success probabilities such that $P(y_1=S|x_1,z_1)+\varepsilon_1(A)-\varepsilon_1(R)\geq\cdots\geq P(y_N=S|x_N,z_N)+\varepsilon_N(A)-\varepsilon_N(R)$. To ensure that the committee accepts exactly N_A applicants, it must be the case that $\lambda^*=\lambda^*(\boldsymbol{\varepsilon}(A),\boldsymbol{\varepsilon}(R))\in\left[P(y_{N_A+1}=S|x_{N_A+1},z_{N_A+1})+\varepsilon_{N_A+1}(A)-\varepsilon_{N_A+1}(R),P(y_{N_A}=S|x_{N_A},z_{N_A})+\varepsilon_{N_A}(A)-\varepsilon_{N_A}(R)\right]$. That is, in this setup, λ^* is set identified. Then the solution is to accept all applicants whose shock-perturbed success probability exceeds λ^* and reject the rest:

$$\forall n\in\{1,\ldots,N\},\, d_n(x_n,z_n,\varepsilon_n|\lambda^*)=\begin{cases}A & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)>\lambda^*+\varepsilon_n(R)\\ \{A,R\} & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)=\lambda^*+\varepsilon_n(R)\\ R & \text{if } P(y_n=S|x_n,z_n)+\varepsilon_n(A)<\lambda^*+\varepsilon_n(R).\end{cases} \quad (81)$$

This cutoff rule holds conditional on the realized shocks. Given the shocks, the committee computes λ^* and applies the cutoff, so there is no difficulty for the committee. But there is a complication regarding estimation, namely that unlike the committee, the econometrician does not observe the shocks. Therefore, the conditional choice probabilities used in estimation require taking expectations across each applicant's shocks, which would be fine except λ^* is a function of each applicant's shocks simultaneously. Suppose for now that λ^* is instead deterministic. Then:

$$\begin{aligned}P(d_n=A|x_n,z_n) &= P(P(y_n=S|x_n,z_n)+\varepsilon_n(A)\geq\lambda^*+\varepsilon_n(R))\\ &= P(\varepsilon_n(R)-\varepsilon_n(A)\leq P(y_n=S|x_n,z_n)-\lambda^*)\\ &= \frac{\exp(P(y_n=S|x_n,z_n)/\sigma)}{\exp(P(y_n=S|x_n,z_n)/\sigma)+\exp(\lambda^*/\sigma)},\end{aligned} \quad (82)$$

and the likelihood becomes the product of individual binary logistics. The third equality holds because the difference of two independent T1EV random variables has a logistic distribution.

That said, λ^* is not deterministic for finite N, so the logistic conditional choice probabilities do not hold exactly. But for large N, λ^* converges to a constant. Denote $\hat{F}_N(c) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\{P(y_n = S|x_n, z_n) + \varepsilon_n(A) - \varepsilon_n(R) \le c\}$ as the empirical distribution of the shock-perturbed success probabilities. The cutoff λ^* is the $(1 - N_A/N)$ quantile of this distribution. Because the shocks are iid, the perturbed indices in $\hat{F}_N$ are iid. So by the Glivenko-Cantelli theorem, $\hat{F}_N$ converges almost surely to the population distribution F for large N. Therefore, as $N_A/N \to q$, $\hat{F}_N$'s $(1 - N_A/N)$ quantile converges to F's deterministic $(1-q)$ quantile. The iid assumption is essential, and not only because each applicant's shock individually moves the cutoff by $O(1/N)$. A common perturbation component across applicants would have the same vanishing per-applicant influence, but it would slide the entire empirical distribution in unison, leaving λ^* random in the limit. Independence makes the simultaneous fluctuations of the N shocks cancel, so $\hat{F}_N$ and λ^* converge. With λ^* deterministic, no individual applicant's shock affects λ^*, the expectation defining each conditional choice probability factors across applicants, and the above logistic form holds.

Theorem 2. *For simplicity, suppose there is one round. The maximum likelihood estimate $\hat{P}_{y^*} = \hat{P}(y^* = S|t)$ is such that the number of acceptances N_A equals the expected N_A. Also, for any $\hat{P}_{y^*}$ and N_A, $\mathcal{L}^R$ is highest when the committee accepts the N_A highest $P_{y_n} = P(y_n = S|x_n, z_n, t_n, \bar{k}_n)$.*

Proof. Supposing one round, the restricted likelihood is:

$$\mathcal{L}^R = \prod_{n=1}^{N} \left[\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=A)} \left[\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right]^{\mathbb{1}(d_n=R)} \left[\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=S)} \left[\frac{1}{1+e^{P_{y_n}}}\right]^{\mathbb{1}(y_n=F)} \tag{83}$$

When an applicant is accepted, the resulting term in $\mathcal{L}^R$ decreases in P_{y^*}, providing downward pressure on $\hat{P}_{y^*}$. Also, when one is rejected, the term increases in P_{y^*}, providing upward pressure.

Since $\mathbb{1}(d_n = A) + \mathbb{1}(d_n = R) = \mathbb{1}(y_n = S) + \mathbb{1}(y_n = F) = 1$, the restricted log-likelihood is:

$$\begin{aligned}
\ell^R = \sum_{n=1}^{N} \Bigg[& \mathbb{1}(d_n = A)\log\left(\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right) + \mathbb{1}(d_n = R)\log\left(\frac{e^{P_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}}\right) \\
& + \mathbb{1}(y_n = S)\log\left(\frac{e^{P_{y_n}}}{1+e^{P_{y_n}}}\right) + \mathbb{1}(y_n = F)\log\left(\frac{1}{1+e^{P_{y_n}}}\right)\Bigg] \\
= \sum_{n=1}^{N} \Bigg[& \frac{1}{\sigma}\left(\mathbb{1}(d_n = A)P_{y_n} + \mathbb{1}(d_n = R)P_{y^*}\right) - \log\left(e^{P_{y_n}/\sigma} + e^{P_{y^*}/\sigma}\right) + \mathbb{1}(y_n = S)P_{y_n} - \log\left(1+e^{P_{y_n}}\right)\Bigg].
\end{aligned} \tag{84}$$

Since ℓ^R is strictly concave in P_{y^*}, I find the unique maximizer $\hat{P}_{y^*}$ by the first-order condition:

$$\frac{1}{\sigma}\left[N - N_A - \sum_{n=1}^{N}\frac{e^{\hat{P}_{y^*}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}}\right] = 0 \quad \Rightarrow \quad \sum_{n=1}^{N}\frac{e^{P_{y_n}/\sigma}}{e^{P_{y_n}/\sigma} + e^{\hat{P}_{y^*}/\sigma}} = N_A. \tag{85}$$

The left side of the latter equation is each applicant's acceptance probability summed over the number of applicants; that is, expected N_A. Therefore, $\hat{P}_{y^*}$ is such that the expected N_A equals N_A.

Given that $\hat{P}_{y^*}$ is identified, I will show that ℓ^R, and therefore $\mathcal{L}^R$, are highest when the committee accepts the N_A applicants with the highest P_{y_n}. The only part of ℓ^R that depends on d_n is

$\sum_{n=1}^{N}[\mathbb{1}(d_n=A)P_{y_n}+\mathbb{1}(d_n=R)P_{y^*}]=NP_{y^*}+\sum_{n=1}^{N}\mathbb{1}(d_n=A)(P_{y_n}-P_{y^*})$; this equality holds since $\mathbb{1}(d_n=A)+\mathbb{1}(d_n=R)=1$. As such, the only part depending on d_n is $\sum_{n=1}^{N}\mathbb{1}(d_n=A)(P_{y_n}-P_{y^*})$.

This expression is highest when $d_n=A$ for the N_A highest values of P_{y_n}, so the likelihood is highest when the committee accepts the N_A highest P_{y_n}. Intuitively, the committee's selectivity identifies $\hat{P}_{y^*}$, and it is not necessary for identification for the committee to accept applicants with higher P_{y_n}. However, if the committee engages in positive sorting, the model's fit will improve.

Theorem 3. *For large N, the committee's portfolio choice problem with matriculation probabilities has a cutoff rule solution on success probabilities.*

Proof. For brevity, let $w=(x,z)$. Then for large N, the portfolio choice problem is:

$$\max_{\delta\in\{\delta|\delta:\mathbb{R}^K\to[0,1]\}}\int_{w\in\mathbb{R}^K}P(y=S|w)P(m=M|w)\delta(w)f(w)dw,$$

$$\text{such that: }\int_{w\in\mathbb{R}^K}P(m=M|w)\delta(w)f(w)dw\leq N_M, \tag{86}$$

where δ is the committee's decision rule, and f is the density of w. Because success and matriculation probabilities, as well as densities, are never negative, the unconstrained objective increases in δ, which means that the constraint binds. The Lagrangian with multiplier λ is:

$$\begin{aligned}\mathcal{L}(\delta;\lambda)&=\int_{w\in\mathbb{R}^K}P(y=S|w)P(m=M|w)\delta(w)f(w)dw\\&\quad+\lambda\left\{N_M-\left[\int_{w\in\mathbb{R}^K}P(m=M|w)f(w)dw-\int_{w\in\mathbb{R}^K}P(m=M|w)(1-\delta(w))f(w)dw\right]\right\}\\&=\int_{w\in\mathbb{R}^K}\left[P(y=S|w)\delta(w)+\lambda(1-\delta(w))\right]P(m=M|w)f(w)dw+\lambda\left[N_M-\int_{w\in\mathbb{R}^K}P(m=M|w)f(w)dw\right].\end{aligned} \tag{87}$$

In the discrete case, there is not necessarily a cutoff rule solution. But in the continuous case, the committee can accept the necessary fraction of applicants such that $\int_{w\in\mathbb{R}^K}P(m=M|w)\delta(w|\lambda^*)f(w)dw=N_M$, which results in a cutoff rule solution on success probabilities alone:

$$\forall w\in\mathbb{R}^K,\ \delta(w|\lambda^*)=\begin{cases}1 & \text{if } P(y=S|w)>\lambda^*\\ [0,1] & \text{if } P(y=S|w)=\lambda^*\\ 0 & \text{if } P(y=S|w)<\lambda^*.\end{cases} \tag{88}$$

The committee accepts all applicants with success probabilities $P(y=S|w)$ strictly greater than the cutoff probability λ^*, rejects all applicants with $P(y=S|w)$ strictly less than λ^*, and may accept or reject any applicants with $P(y=S|w)$ equal to λ^*, if such a mass of applicants exists. All that remains is to identify λ^*. Like Bai et al. (2022), I define the following two functions:

$$\begin{aligned}M^H(\lambda)&=\int_{\{w\in\mathbb{R}^K|P(y=S|w)\geq\lambda\}}P(m=M|w)f(w)dw\\M^L(\lambda)&=\int_{\{w\in\mathbb{R}^K|P(y=S|w)>\lambda\}}P(m=M|w)f(w)dw.\end{aligned} \tag{89}$$

The upper bound function M^H is the fraction of applicants, scaled by their matriculation probabilities, whose success probabilities are greater than or equal to some λ, and the lower bound function M^L is the fraction whose success probabilities are strictly greater than λ. Both functions are between 0 and 1, both are decreasing in λ, and for any λ, it must be that $M^H(\lambda) \geq M^L(\lambda)$.

Then it is the case that $\lambda^* = \inf_{\lambda}\{\lambda | M^L(\lambda) \leq N_M\}$. Intuitively, λ^* is the largest cutoff probability where the fraction of matriculation-probability-scaled applicants with $P(y=S|w) > \lambda$ does not exceed the maximum number of matriculants N_M. If there is a positive mass of matriculation-probability-scaled applicants with $P(y=S|w) = \lambda^*$ (that is, if $M^H(\lambda^*) - M^L(\lambda^*) > 0$), then the committee will randomly accept the exact fraction $\alpha^* \in [0,1]$ of that mass of applicants such that $M^L(\lambda^*) + \alpha^*[M^H(\lambda^*) - M^L(\lambda^*)] = N_M$. That is, the committee will accept every applicant with $P(y=S|w) > \lambda^*$ and accept just enough applicants with $P(y=S|w) = \lambda^*$ to fill the cohort.

Theorem 4. *Let d^R be the model's decision rule with $\sigma = 0$ in the CCPs, and let d^U be the committee's decision rule. By optimality:*

$$\frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S \Big| x_n, z_n, t_n; \hat{\theta}^U_S\Big)\mathbb{1}\Big\{d_{1,n}\Big[P_{y_n}\Big(\hat{\theta}^U\Big)\Big] = T, d^R_{2,n}\Big[P_{y_n}\Big(\hat{\theta}^U_S\Big)\Big] = A\Big\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S \Big| x_n, z_n, t_n; \hat{\theta}^U_S\Big)\mathbb{1}\Big\{d_{1,n}\Big[P_{y_n}\Big(\hat{\theta}^U\Big)\Big] = T, d^U_{2,n}\Big[P_{y_n}\Big(\hat{\theta}^U_A\Big)\Big] = A\Big\} \quad (90)$$

Proof. This theorem states that conditional on a fixed Round 1 decision rule $d_1[P_y(\hat{\theta}^U)]$ and corresponding transition set of exogenously-determined size N_T, the sample average success probability of the also-exogenous N_A applicants that the model accepts in Round 2 weakly exceeds the sample average success probability of the N_A applicants that any actual committee accepts in Round 2. The inequality holds since the model's Round 2 decision rule is to accept the N_A applicants with the highest success probabilities. Therefore, the sample average success probability of the model's acceptances must weakly exceed that of any other set of N_A acceptances.

However, the inequality would not always hold if it were written as follows (changes in bold):

$$\frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S \Big| x_n, z_n, t_n; \hat{\theta}^U_S\Big)\mathbb{1}\Big\{\mathbf{d^R_{1,n}}\Big[\mathbf{P_{y_n}}\Big(\hat{\theta}^{\mathbf{U}}_{\mathbf{S}}\Big)\Big] = \mathbf{T}, d^R_{2,n}\Big[P_{y_n}\Big(\hat{\theta}^U_S\Big)\Big] = A\Big\}$$
$$\textbf{vs.} \ \frac{1}{N_A}\sum_{n=1}^{N} P\Big(y_n = S \Big| x_n, z_n, t_n; \hat{\theta}^U_S\Big)\mathbb{1}\Big\{\mathbf{d^U_{1,n}}\Big[\mathbf{P_{y_n}}\Big(\hat{\theta}^{\mathbf{U}}_{\mathbf{T}}\Big)\Big] = \mathbf{T}, d^U_{2,n}\Big[P_{y_n}\Big(\hat{\theta}^U_A\Big)\Big] = A\Big\} \quad (91)$$

The difference is that the Round 1 transition set is no longer fixed. Now, the model's transition set is based on choosing the N_T applicants in each year with the highest $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}] = \int_{\tilde{z}} \sigma \log(e^{P(y=S|x,z,t,\bar{k})/\sigma} + e^{P(y_2^*=S)/\sigma}) dF(\tilde{z}|x,t)$. In contrast, the committee may have a different Round 1 decision rule. Crucially, although the model is choosing applicants akin to those with the highest expected success probabilities, it is ignoring any value added from upside potential.

For example, suppose $N_A = 1$, and there is an applicant in Round 1 with a low expected success probability but a known high degree of variance in the distribution of z conditional on their (x_n,t_n). That is, $F(z|x,t)$ is heteroskedastic. Also suppose that their low expected success probability yields an $\mathbb{E}[v_2(x,z,t,\bar{k},\varepsilon_2)|x,t,\bar{k}]$ too low to survive Round 1, even though their actual success probability ends up being the highest of any applicant once z is realized in Round 2. If the committee's decision rule considers upside potential, then the committee may accept this applicant, while the model will not. In most cases, the model's Round 1 decision rule should lead to a better acceptance set than the committee's rule. But such an outcome is not guaranteed.

Theorem 5. *For large B, the fraction of $\tilde{\theta}^U_{S,b}$ with gain $g(\tilde{\theta}^U_{S,b}) \leq 0$ is a p-value for $H_0\colon g_0 \leq 0$.*

Proof. Define deviation gain $g(\theta_S) = \frac{1}{N_A}\sum_{n=1}^N P\Big(y_n = S\Big|x_n,z_n,t_n,\bar{k}_n;\theta_S\Big)\mathbb{1}(d^R_{1,n} = T, d^R_{2,n} = A) - \frac{1}{N_A}\sum_{n=1}^N P\Big(y_n = S\Big|x_n,z_n,t_n,\bar{k}_n;\theta_S\Big)\mathbb{1}(d^U_{1,n} = T, d^U_{2,n} = A)$. That is, the gain is the average success probability of the model's acceptance set minus that of the committee's acceptance set. Since the model and committee's sets are determined before performing the test, the proof works the same way for matriculation sets as acceptance sets. Suppose the maximum likelihood estimate $\hat{\theta}^U_S \overset{a}{\sim} \mathcal{N}(\theta^0_S,\Sigma_0)$, and each perturbation draw $\tilde{\theta}^U_{S,b} \sim \mathcal{N}(\hat{\theta}^U_S,\hat{\Sigma})$, where $\hat{\Sigma}$ is consistent for Σ_0.

Because g is a smooth function of θ_S, it is true by the delta method that $\hat{g} = g(\hat{\theta}^U_S) \overset{a}{\sim} \mathcal{N}(g_0,\sigma^2_g)$ and $\tilde{g} = g(\tilde{\theta}^U_{S,b}) \mid \hat{g} \overset{a}{\sim} \mathcal{N}(\hat{g},\sigma^2_g)$, where $g_0 = g(\theta^0_S)$ and $\sigma^2_g = \nabla g(\theta^0_S)'\Sigma_0\nabla g(\theta^0_S)$. As the number of perturbations B becomes large, it is true by the law of large numbers that the fraction of perturbations with $\tilde{g} \leq 0$ converges to the probability that a single perturbation is weakly negative:

$$\hat{p} = P(\tilde{g} \leq 0 \mid \hat{g}) = \Phi\left(\frac{0-\hat{g}}{\sigma_g}\right) = \Phi(-\hat{g}/\sigma_g). \tag{92}$$

For $\hat{p}$ to be a p-value, it must equal the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain. Denote $\hat{g}' = \sigma_g Z$ as a hypothetical estimated gain generated under $g_0 = 0$, where $Z \sim \mathcal{N}(0,1)$. Then, using $P(Z \geq c) = \Phi(-c)$:

$$P(\hat{g}' \geq \hat{g} \mid g_0 = 0) = P(Z \geq \hat{g}/\sigma_g) = \Phi(-\hat{g}/\sigma_g) = \hat{p}. \tag{93}$$

That is, the probability, if the null is true with $g_0 = 0$, of an estimated gain at least as large as the observed gain equals the probability to which the fraction of perturbations with $\tilde{g} \leq 0$ converges. Finally, I show that when $g_0 \leq 0$, the probability of a Type 1 error is bounded above by the statistical significance threshold α. That is, $P(\hat{p} \leq \alpha) \leq \alpha$ when $g_0 \leq 0$. Because inverses of strictly increasing functions preserve inequalities, Φ is strictly increasing, and $\hat{p} = \Phi(-\hat{g}/\sigma_g)$; it is the case that $\hat{p} \leq \alpha$ holds exactly when $\hat{g} \geq -\sigma_g\Phi^{-1}(\alpha)$. Substituting $\hat{g} = g_0 + \sigma_g Z$:

$$P(\hat{p} \leq \alpha) = P(Z \geq -\Phi^{-1}(\alpha) - g_0/\sigma_g) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g). \tag{94}$$

For $g_0 = 0$, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha)) = \alpha$. For $g_0 < 0$, because Φ is strictly increasing, it is the case that $P(\hat{p} \leq \alpha) = \Phi(\Phi^{-1}(\alpha) + g_0/\sigma_g) < \alpha$. Therefore, rejecting H_0 when $\hat{p} \leq \alpha$ produces Type 1 errors with probability at most α everywhere that H_0 is true.

Theorem 6. *In the waitlist-dynamic and hybrid models, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t,\bar{k})=P(y=S|x^L,z^L,t,\bar{k})$, but $m_2^H > m_2^L$. That is, the applicants have equal success probabilities, but H has the higher matriculation probability. Since $\ddot{m}_2$ is the average matriculation probability of all applicants in t with m_2 excluded, $\ddot{m}_2^H < \ddot{m}_2^L$ by construction. Given this inequality, I will show $P(d_2=A|x^H,z^H,t,\bar{k},\ddot{m}_2^H) > P(d_2=A|x^L,z^L,t,\bar{k},\ddot{m}_2^L)$. It is the case that:

$$P(d_2=A|x,z,t,\bar{k},\ddot{m}_2)=\frac{\exp(P(y=S|x,z,t,\bar{k})/\sigma)}{\exp(P(y=S|x,z,t,\bar{k})/\sigma)+\exp(\beta\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]/\sigma)}, \text{ where}$$

$$\mathbb{E}[v_3(x,z,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x,z,t,\bar{k},\ddot{m}_2]=\int_{\widetilde{\overline{m}_{2A}}}\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2). \quad (95)$$

The only place $\ddot{m}_2$ appears in $P(d_2=A|x,z,t,\bar{k},\ddot{m}_2)$ is in the distribution of $\overline{m}_{2A}$ conditional on $\ddot{m}_2$. I parameterize $\overline{m}_{2A}|\ddot{m}_2 \sim \mathcal{N}(\ddot{m}_2\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2})$, where by assumption $\beta_{\ddot{m}_2}>0$. $\beta_{\ddot{m}_2}$ is positive because years when $\ddot{m}_2$ is high are years when matriculation probabilities are high in general. Using inter-year variation, this relationship results in a greater expected $\overline{m}_{2A}$ in those years. Then:

$$F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^H;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2})=\Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^H\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right)>\Phi\left(\frac{\overline{m}_{2A}-\ddot{m}_2^L\beta_{\ddot{m}_2}}{\sigma_{\ddot{m}_2}}\right)=F_{\overline{m}_{2A}|\ddot{m}_2}(\overline{m}_{2A}|\ddot{m}_2^L;\beta_{\ddot{m}_2},\sigma^2_{\ddot{m}_2}). \quad (96)$$

Therefore, $\overline{m}_{2A}|\ddot{m}_2^L$ first-order stochastically dominates $\overline{m}_{2A}|\ddot{m}_2^H$. To apply this stochastic dominance, I must show that $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$ is strictly increasing in $\overline{m}_{2A}$. But the only place $\overline{m}_{2A}$ appears in this expression is in $P(y_3^*=S|\overline{m}_{2A};\theta_{S_3^*})=\frac{\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}{1+\exp\left(f_{S_3^*}(\overline{m}_{2A})\theta_{S_3^*}\right)}$, where by assumption $f_{S_3^*}(\cdot)$ is strictly increasing and $\theta_{S_3^*}>0$. $\theta_{S_3^*}$ is positive because the expected Round 3 cutoff probability is higher when early acceptees are more likely on average to matriculate. As such, I have that $P(y_3^*=S|\overline{m}_{2A})$, and in turn $\sigma\log\left(e^{P(y=S|x,z,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\overline{m}_{2A})/\sigma}\right)$, are strictly increasing in $\overline{m}_{2A}$. Combining this result with $(\overline{m}_{2A}|\ddot{m}_2^L)\succ_{\text{FOSD}}(\overline{m}_{2A}|\ddot{m}_2^H)$, I obtain:

$$\begin{aligned}
\mathbb{E}[v_3(x^L,z^L,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x^L,z^L,t,\bar{k},\ddot{m}_2^L] &= \int_{\widetilde{\overline{m}_{2A}}}\sigma\log\left(e^{P(y=S|x^L,z^L,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L)\\
&= \int_{\widetilde{\overline{m}_{2A}}}\sigma\log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^L)\\
&> \int_{\widetilde{\overline{m}_{2A}}}\sigma\log\left(e^{P(y=S|x^H,z^H,t,\bar{k})/\sigma}+e^{P(y_3^*=S|\widetilde{\overline{m}_{2A}})/\sigma}\right)dF_{\overline{m}_{2A}|\ddot{m}_2}(\widetilde{\overline{m}_{2A}}|\ddot{m}_2^H)\\
&= \mathbb{E}[v_3(x^H,z^H,t,\bar{k},\overline{m}_{2A},\varepsilon_3)|x^H,z^H,t,\bar{k},\ddot{m}_2^H]. \quad (97)
\end{aligned}$$

The second equality holds since $P(y=S|x^H,z^H,t,\bar{k})=P(y=S|x^L,z^L,t,\bar{k})$. Therefore, by Equation (95) and Inequality (97), I have that $P(d_2=A|x^H,z^H,t,\bar{k},\ddot{m}_2^H)>P(d_2=A|x^L,z^L,t,\bar{k},\ddot{m}_2^L)$.

Theorem 7. *In the matriculation multiyear model, success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

Proof. Suppose there are two arbitrary applicants in year t indexed by H and L, such that $P(y=S|x^H,z^H,t)=P(y=S|x^L,z^L,t)$, but $m^H>m^L$. That is, H has the higher matriculation probability. I will show that $P(d_2=A|x^H,z^H,t,m^H,c_{-1})>P(d_2=A|x^L,z^L,t,m^L,c_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y=S|x,z,t)+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m], \quad \pi_2^R \equiv L(y_2^*=S|c_{-1})+\beta\mathbb{E}[v_1(x',t',c,\varepsilon_1')],$$
$$\pi_1^A \equiv \mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c], \quad \pi_1^R \equiv L(y_1^{*\prime}=S|t')+\mathbb{E}[v_1(x'',t'',c',\varepsilon_1'')]. \tag{98}$$

It is the case that:

$$P(d_2=A|x,z,t,m,c_{-1})=\frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma)+\exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]=\int_c\int_{t'}\int_{x'}\sigma\log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right)dF_x(x')\,dF_t(t')\,dF_{c|m}(c|m), \text{ where:}$$

$$\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c]=\int_{\widetilde{m'}}\int_{\widetilde{z'}}\sigma\log\left(e^{\pi_2^{A\prime}/\sigma}+e^{\pi_2^{R\prime}/\sigma}\right)dF_{z|x,t}(\widetilde{z'}|x',t')dF_{m|x,t}(\widetilde{m'}|x',t'), \tag{99}$$

where $\pi_2^{A\prime}$ and $\pi_2^{R\prime}$ are defined analogously to Equation (98) at the next year's states.

Per-period utilities are bounded and $\beta\in(0,1)$, so the value functions exist and are unique. The only place m appears in $P(d_2=A|x,z,t,m,c_{-1})$ is in the distribution of c conditional on m. I parameterize $c|m\sim\mathcal{N}(m\beta_m,\sigma_m^2)$, where by assumption $\beta_m>0$. β_m is positive because higher matriculation probabilities lead to a greater expected cohort size. Multiple years of data are necessary to identify β_m, since cohort sizes vary only across years. Then:

$$F_{c|m}(c|m^H;\beta_m,\sigma_m^2)=\Phi\left(\frac{c-m^H\beta_m}{\sigma_m}\right)<\Phi\left(\frac{c-m^L\beta_m}{\sigma_m}\right)=F_{c|m}(c|m^L;\beta_m,\sigma_m^2). \tag{100}$$

Therefore, $c|m^H$ first-order stochastically dominates $c|m^L$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand consisting of all but the outermost integral is strictly increasing in c. But the only place c appears there is in $\mathbb{E}[v_2(x',z',t',m',c,\varepsilon_2')|x',t',c]$, and the only place c appears in that expectation is in $L(y_2^{*\prime}=S|c;\theta_{S_2^*})=f_{S_2^*}(c)\theta_{S_2^*}$.

By assumption, $f_{S_2^*}(\cdot)$ is strictly increasing and $\theta_{S_2^*}>0$. $\theta_{S_2^*,c}$ is positive because a greater cohort size this year means that the committee, needing to keep the total number of students in the program relatively constant across years, can afford to be more selective next year. As such, I have that $P(y_2^{*\prime}=S|c)$, and in turn $\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m]$'s integrand, are strictly increasing in c. Combining this result with $[c|m^H]\succ_{FOSD}[c|m^L]$, I obtain the inequality:

$$\begin{aligned}\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^H]&=\int_c\int_{t'}\int_{x'}\sigma\log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right)dF_x(x')\,dF_t(t')\,dF_{c|m}(c|m^H)\\&>\int_c\int_{t'}\int_{x'}\sigma\log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right)dF_x(x')\,dF_t(t')\,dF_{c|m}(c|m^L)\\&=\mathbb{E}[v_1(x',t',c,\varepsilon_1')|m^L].\end{aligned} \tag{101}$$

Therefore, by Equation (99), Inequality (101), and $P(y=S|x^H,z^H,t)=P(y=S|x^L,z^L,t)$, I have that $P(d_2=A|x^H,z^H,t,m^H,c_{-1})>P(d_2=A|x^L,z^L,t,m^L,c_{-1})$. Intuitively, if the committee rejects either applicant, their m does not matter. But since their success probabilities are equal, accepting applicant H yields a higher expected payoff than accepting applicant L.

Theorem 8. *Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

Proof. Suppose there are two arbitrary applicants in year t, one with $a = 1$ and the other with $a = 0$, but with $P(y = S|x^1, z^1, t, 1 - \overline{a}_{-1}) = P(y = S|x^0, z^0, t, \overline{a}_{-1})$. Unlike in the previous two proofs, these equal success probabilities imply that unless $\overline{a}_{-1} = \frac{1}{2}$, it must be that $(x^1, z^1) \neq (x^0, z^0)$. I will show that $P(d_2 = A|x^1, z^1, t, 1, \overline{a}_{-1}) > P(d_2 = A|x^0, z^0, t, 0, \overline{a}_{-1})$.

Define for each round the choice-specific payoffs for acceptance (A) and rejection (R):

$$\pi_2^A \equiv P(y = S|x, z, t, |a - \overline{a}_{-1}|) + \beta\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a], \quad \pi_2^R \equiv L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')],$$
$$\pi_1^A \equiv \mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}], \quad \pi_1^R \equiv L(y_1^{*\prime} = S|t') + \beta\mathbb{E}[v_1(x'', t'', \overline{a}', \varepsilon_1'')]. \tag{102}$$

It is the case that:

$$P(d_2 = A|x, z, t, a, \overline{a}_{-1}) = \frac{\exp(\pi_2^A/\sigma)}{\exp(\pi_2^A/\sigma) + \exp(\pi_2^R/\sigma)}, \text{ where:}$$

$$\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a] = \int_{\tilde{\overline{a}}} \int_{\tilde{t'}} \int_{\tilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma} + e^{\pi_1^R/\sigma}\right) dF_x(\tilde{x'})\, dF_t(\tilde{t'})\, dF_{\overline{a}|a}(\tilde{\overline{a}}|a), \text{ where:}$$

$$\begin{aligned}\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}] = {} & P(a' = 1) \int_{\tilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(1)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\tilde{z'}|x', t') \\ & + [1 - P(a' = 1)] \int_{\tilde{z'}} \sigma \log\left(e^{\pi_2^{A\prime}(0)/\sigma} + e^{\pi_2^{R\prime}/\sigma}\right) dF_{z|x,t}(\tilde{z'}|x', t'),\end{aligned} \tag{103}$$

where $\pi_2^{A\prime}(a')$ denotes next year's Round 2 acceptance payoff evaluated at $a' \in \{0, 1\}$.

Per-period utilities are bounded and $\beta \in (0, 1)$, so the value functions exist and are unique. The only two places a appears in $P(d_2 = A|x, z, t, a, \overline{a}_{-1})$ are in the success probabilities, which are assumed to be equal for the $a = 1$ and $a = 0$ applicants, and in the distribution of $\overline{a}$ conditional on a. I parameterize $\overline{a}|a \sim \mathcal{N}(a\beta_a, \sigma_a^2)$, where by assumption $\beta_a > 0$. β_a is positive because $a = 1$ applicants are more likely than $a = 0$ applicants to be in a year with a high percentage of accepted $a = 1$ applicants. As such, multiple years of data are necessary to identify β_a. Then:

$$F_{\overline{a}|a}(\overline{a}|1; \beta_a, \sigma_a^2) = \Phi\left(\frac{\overline{a} - \beta_a}{\sigma_a}\right) < \Phi\left(\frac{\overline{a}}{\sigma_a}\right) = F_{\overline{a}|a}(\overline{a}|0; \beta_a, \sigma_a^2). \tag{104}$$

Therefore, $\overline{a}|1$ first-order stochastically dominates $\overline{a}|0$. To apply this stochastic dominance, I must show that $\mathbb{E}[v_1(x', t', \overline{a}, \varepsilon_1')|a]$'s integrand consisting of all but the outermost integral is strictly increasing in $\overline{a}$. But the only place $\overline{a}$ appears there is in $\mathbb{E}[v_2(x', z', t', a', \overline{a}, \varepsilon_2')|x', t', \overline{a}]$, and the only place $\overline{a}$ appears in that expectation is in $P(y' = S|x', z', t', \overline{a}; \theta_S) = \frac{\exp(f_S(x', z', t', |a' - \overline{a}|)\theta_S)}{1 + \exp(f_S(x', z', t', |a' - \overline{a}|)\theta_S)}$.

By assumption, $f_S(\cdot)$ is strictly increasing and $\theta_{S,|a-\overline{a}_{-1}|} > 0$. $\theta_{S,|a-\overline{a}_{-1}|}$ is positive because a large value of $|a-\overline{a}_{-1}|$ means that applicant a is diverse relative to the previous year's accepted applicants and is therefore more likely to succeed. Recall $P(a'=1)=p$. Taking the limit $p \to 0$:

$$\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}] \\ \to \int_{\widetilde{z'}} \sigma \log(e^{\{P(y'=S|x',z',t',\overline{a})+\beta\mathbb{E}[v_1(x'',t'',\overline{a}',\varepsilon_1'')|0]\}/\sigma} + e^{\{L(y_2^{*\prime}=S|t')+\beta\mathbb{E}[v_1(x'',t'',\overline{a}',\varepsilon_1'')]\}/\sigma}) dF_{z|x,t}(\widetilde{z'}|x',t'). \tag{105}$$

$P(y'=S|x',z',t',\overline{a})$ is strictly increasing in $\overline{a}$, and by the continuity of $E[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$ in p, I have that it, and in turn $E[v_1(x',t',\overline{a},\varepsilon_1')|a]$'s integrand, are also strictly increasing in $\overline{a}$ for sufficiently small p. Combining this result with $\overline{a}|1 \succ_{FOSD} \overline{a}|0$, I obtain the inequality:

$$\begin{aligned}\mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|1] &= \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|1) \\ &> \int_{\widetilde{\overline{a}}}\int_{\widetilde{t'}}\int_{\widetilde{x'}} \sigma \log\left(e^{\pi_1^A/\sigma}+e^{\pi_1^R/\sigma}\right) dF_x(\widetilde{x'})\, dF_t(\widetilde{t'})\, dF_{\overline{a}|a}(\widetilde{\overline{a}}|0) \\ &= \mathbb{E}[v_1(x',t',\overline{a},\varepsilon_1')|0].\end{aligned} \tag{106}$$

Therefore, by Equation (103), Inequality (106), and $P(y=S|x^1,z^1,t,1-\overline{a}_{-1}) = P(y=S|x^0,z^0,t,\overline{a}_{-1})$, I have that $P(d_2=A|x^1,z^1,t,1,\overline{a}_{-1}) > P(d_2=A|x^0,z^0,t,0,\overline{a}_{-1})$. Intuitively, if the committee rejects either applicant, their a does not matter. But since their success probabilities are equal, accepting applicant $a=1$ yields a higher expected payoff than accepting $a=0$.

In this proof, I must assume $P(a'=1)$ is sufficiently small, and not just $P(a'=1) < \frac{1}{2}$. The reason is that if $P(a'=1)$ is too large, then $\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$ may not be strictly increasing in $\overline{a}$, considering $P(y'=S|x',z',t',1-\overline{a})$ is strictly decreasing in $\overline{a}$. This indeterminacy can be explored by calculating $\frac{\partial}{\partial \overline{a}}\mathbb{E}[v_2(x',z',t',a',\overline{a},\varepsilon_2')|x',t',\overline{a}]$. As such, to be sure the committee is more likely to accept the $a=1$ applicant, the committee must know that there will be sufficiently few $a'=1$ applicants. This assumption is strong, but possible if $a'=1$ is sufficiently rare for the committee to know this will happen. However, even without forward-looking dynamics, if $\overline{a}_{-1} < \frac{1}{2}$, then the committee will be more likely to accept an $a=1$ applicant than an $a=0$ applicant with identical x and z values. The reason is that $P(y=S|x,z,t,1-\overline{a}_{-1}) > P(y=S|x,z,t,\overline{a}_{-1})$.